UPSC CSE 2027 · GS Foundation

GEOGRAPHY

Geomorphology · Climatology · Oceanography · Indian geography · Human geography · Economic geography

Handwritten revision notes

Built from the Sarrthi IAS GS Foundation Geography handouts (pictures and the mind-map pages are in the textbook edition).
Bracket trees · diagrams · mnemonics · PYQ tags · current affairs
Right column and left binding margin are kept blank for your own use.

Personal study edition · for printing

INDEX

1 Geomorphology (universe to landforms) 2
2 Climatology (atmosphere to climate types) 20
3 Oceanography (relief, water, currents, tides, reefs) 29
4 Indian geography (location, physiography, drainage, climate, forests, soils) 36
5 Human geography (demography, settlements) 55
6 Economic geography (agriculture, resources, minerals, energy, industries, transport, communication) 59

Mind maps (from the handout)

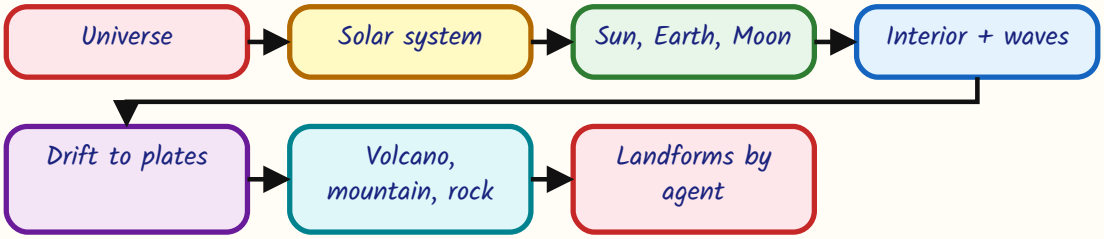
7 Mind maps · Geomorphology (36 maps) 74
8 Mind maps · Climatology (21 maps) 107
9 Mind maps · Oceanography (18 maps) 125
10 Mind maps · Indian Geography (29 maps) 139
11 Mind maps · Human Geography (6 maps) 162
12 Mind maps · Economic Geography (21 maps) 167

GEOMORPHOLOGY

PYQ Prelims 2025, 2024 ×3, 2023, 2022, 2019, 2018, 2015, 2014, 2013 ×2, 2012 ×2, 2011 + 4 with no year · Mains 2022 (primary rocks, 10m), 2017 (Juno mission, 10m)

✎ *From the Big Bang to the Sun, Earth, its interior, plates, rocks and every landform: 19 Prelims PYQs make this a HIGH chapter.*

At a glance: the Geomorphology route



Hot clusters

→ Earth–Sun geometry (pole star, day length, 21 June, time zones)
→ solar storm effects
→ P vs S waves
→ weathering by rain
→ drift evidences
→ volcanic products
→ chalk vs clay. No teacher ink: priority comes from handout PYQ counts only.

1 · UNIVERSE: ORIGIN THEORIES

Theory	By	Core idea	Drawback
Gaseous	Kant	Scattered primordial matter collided → heat → solid→liquid→gas = nebula; rotation → bulge; rings separated; residual centre = Sun, rings = planets; same process → moons	Source of primordial matter unexplained; collisions can't give rotation; how moons formed?
Nebular	Laplace, 1796 (modified Kant)	Faster spin → rings detach, each ring condenses into gas mass, cools → planets; remaining part = Sun	Source of nebula unexplained; Sun should spin fastest but doesn't
Planetesimal	Chamberlin–Moulton	Star passed close to young Sun → tidal bulges, solar prominences → ejected material; spiral arms; cooled → planetesimals + larger protoplanets → planets + moons; comets and asteroids = leftover debris	–
Accretion	Otto Schmidt 1944, Viktor Safronov 1969	Nebula → dust grains → boulders → planetesimals → planets (constant accretion)	–
Big Bang	Georges Lemaitre, 1920s	Universe began from a single primordial atom; violent explosion; dust scattered; denser regions pulled together over millions of years → gas clouds, stars, galaxies	Validated 1972

KLS – Kant = Kollision · Laplace = Loses heat (cools, shrinks, spins faster) · Schmidt–Safronov = Small Steps (dust → planet)

Proofs of Big Bang

→ Hubble's law (Hubble–Lemaitre): farther galaxies recede faster = expanding universe (basis: redshift); Edwin Hubble

→ CMB: 1960s discovery, Penzias & Wilson, 1965; oldest light, ~380,000 yrs after Big Bang; at ~3000 K electrons joined protons (recombination) so photons flew free = "baby photo of the universe"

→ Hotter body = shorter wavelength (more energy → faster motion → higher frequency)

→ Universe age 13.8 bn yrs

2 · DARK ENERGY & DARK MATTER

Together ≈ 95% of universe

→ Dark matter ≈ 27%: no light emitted, absorbed or reflected; inferred from gravity (galaxy rotation, gravitational lensing = light bending near massive object); invisible scaffold holding galaxies

→ Dark energy ≈ 68%: accelerating expansion; repulsive force pushing space apart (not matter or gravity)

3 · GALAXIES

Galaxy types

→ Spiral (Milky Way)

→ Elliptical (old, cool stars, little gas; minimal star formation)

→ Lenticular (bulge + disk, no arms)

→ Irregular (chaotic, gas-rich)

Active galaxies ~10%

→ Centre >100× brighter than all stars combined; supermassive black hole + accretion disk; radiation infrared → X-rays; jets near speed of light (AGN)

→ Seyfert (most common, lowest energy)

→ Quasar (brightest)

→ Blazar (jet points at Earth)

→ Milky Way is NOT active

4 · STARS

Lifecycle of stars



"Small stars retire quietly, big stars quit with a bang" ⭐

Mass	Path	End
🌐 < 8 × Sun	Main sequence → Red giant → Planetary nebula	White dwarf
🌐 > 8 × Sun	Red supergiant → Supernova (iron core)	Neutron star (8–20/25×) or Black hole (>25×)

Chandrasekhar limit

→ core < 1.44 × Sun = white dwarf
→ 1.44–3 = neutron star
→ >3 = black hole. Source quotes both 8–20 and 8–25× for neutron stars: use 25 as the cut-off to black hole.

Populations (by metallicity)

→ Pop III: hypothetical, very first stars, hundreds of millions of yrs after Big Bang, no heavy metals
→ Pop II: oldest observed, 5–10 bya, halo/globular clusters
→ Pop I: younger, metal-rich, 4–5 bya, disk and arms, our Sun

Neutron star

→ Collapsed core of massive star (8–25× Sun) after supernova; almost all neutrons; strong gravity + magnetic fields
→ Can accrete mass → collapse into black hole
→ Pulsar = rapidly rotating neutron star, beams from magnetic poles, regular pulses ("lighthouse")

🌐 Black hole (>25× Sun)

→ Nothing, not even light, escapes (Earth's escape velocity printed 11.8 m/s: unit looks a typo, don't memorise)
→ Event horizon: boundary of no return
→ Accretion disk: gas/dust spiralling in, hot from friction (disk → horizon → void)
→ Gravitational waves: ripples in spacetime; LIGO (Laser Interferometer Gravitational-Wave Observatory, Nobel 2017) and Virgo
→ Spaghettification: stretched by tidal forces
→ Hawking radiation: theoretical particle emission → slow evaporation

5 · SOLAR SYSTEM PYQ 2015 PYQ 2011

Components

→ **Sun**: central star, H + He, nuclear fusion

Planets

→ **Terrestrial**: Mercury, Venus, Earth, Mars
→ **Gas giants**: Jupiter, Saturn
→ **Ice giants**: Uranus, Neptune
→ **Planet Nine/Planet X**: hypothesised, explains Kuiper Belt orbits; gas or ice giant, **5–7× Earth mass**, ~200× farther than Earth

Dwarf planets (5)

→ **Pluto** (biggest), **Ceres** (biggest in asteroid belt), **Eris**, **Haumea**, **Makemake** (newest)
→ Criteria: orbits Sun · round (spherical) · not a moon · has not cleared its orbit

Body	Moons
Mercury, Venus	0 (Venus has quasi-satellite Zoozve)
Earth / Mars	1 / 2
Jupiter / Saturn	101 / 274 (Jupiter has biggest moon Ganymede)
Uranus / Neptune	28 / 16
Pluto	5

Handout conflict: Saturn's moons printed 274 in the moons list but 74 in the inner/outer planets table: use 274.

Region/body	Key facts
Quasi-moon	Asteroid orbiting Sun but staying close to a planet; orbit oblong, less stable, may drift away
Kuiper Belt	Beyond Neptune, 30–50 AU (1 AU = 149.6 million km); icy bodies, dwarf planets; rock, water ice, ammonia, methane; source of comets
Oort Cloud	Spherical shell of icy objects; starts 2,000–5,000 AU, outer edge 10,000–100,000 AU; long-period comets
Asteroids	Rocky; belt between Mars and Jupiter; Vesta largest ~530 km (329 miles); smallest under 10 m (33 feet); total mass < Earth's Moon; Trojans at L4/L5 Lagrange points
Comets	Icy; gas + dust → glowing coma and tail near Sun; from Kuiper Belt or Oort Cloud
Meteoroid → meteor → meteorite	Space particle (rocky/metallic) → burns in air ("shooting star") → survives, hits ground. Of 50,000+ meteorites 99.8% from asteroids; 0.2% split between Mars and Moon

Edge of the Sun's reach

Heliosphere (solar-wind bubble) → *Termination shock: wind supersonic → subsonic* → *Heliosheath: turbulent, field irregular* → *Heliopause: solar wind = interstellar pressure* → *Bow shock: existence contested by Voyager 1 and 2*

Goldilocks zone (2015, ans c): distance from a star where **water stays liquid**; search for Earth-like planets (Earth; Kepler-186f). **Asteroid vs comet (2011, ans B = 1 and 3):** asteroids rock/metal, between Mars and Jupiter (Jupiter–Mars), orbit like planets, no tail; comets ice + dust, tail. "Comets between Venus and Mercury" is wrong. Both formed ~4.5 bn yrs ago. ✅

6 · PLANETS

Planet facts

→ Venus and Uranus rotate East → West
→ Uranus axis parallel to its orbit
→ Jupiter: Great Red Spot 16,000 km, Little Red Spot 3,500 km (storms)

Inner (rocky)
→ dense, small, few/no moons, weak field, short orbits: Mercury 88 d, Earth 365, Mars 687

Outer (gaseous)
→ big, many moons, rings, strong fields: Jupiter 12 y, Saturn 29, Uranus 84, Neptune 165

Atmospheres
→ Venus CO_2, super-hot (greenhouse)
→ Mars thin CO_2
→ Jupiter/Saturn H + He
→ Uranus/Neptune methane + ammonia

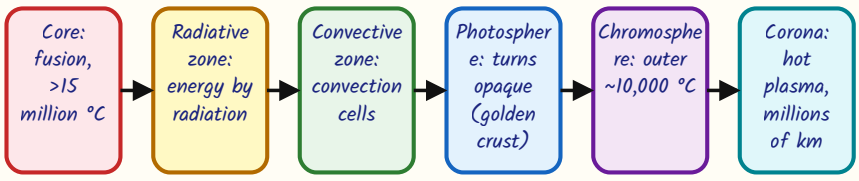
Handout alert: Planet Nine only hypothesised (5–7 × Earth mass).

7 · THE SUN PYQ 2022

Sun

→ Composition: 71% H, 27.1% He, 0.97% O, 0.4% C, traces
→ No "surface": gas gets denser toward the core
→

Sun = a samosa (inside → out)



Space-weather term	What it is	Remember
Solar wind	Charged particles streaming from corona	plasma too hot for gravity to hold
Solar flare	Explosion from twisted magnetic fields; X-rays + UV	classes A, B, C, M, X (X biggest); may or may not come with CME
CME (coronal mass ejection)	Plasma + magnetic field blobs from corona	15–18 h to reach Earth
Geomagnetic storm	Earth in path of a CME	Starlink: 40 satellites lost (2022) · Quebec blackout 1989 (GICs damage transformers) · GPS errors (aviation, defence) · polar flights rerouted (radiation) · ISS shielding · radar/missile-warning disruption
Aurora	Solar-wind particles excite O and N → photons	Borealis (N), Australis (S); at poles (magnetic poles near)
Van Allen belts	Charged particles trapped by Earth's field, doughnut-shaped; shield Earth, risk to spacecraft	inner: protons, 1,000–6,000 km · outer: electrons, 13,000–60,000 km
Sunspots	Darker, cooler areas	11-year (and 400-year) cycle; now in Maxima (Solar Maxima/Minima)

PYQ 2022 solar storm (ans c): effects = GPS/navigation failure, power-grid damage, intense auroras, satellite orbit disturbance, polar short-wave radio disruption. Not tsunamis (need vertical water shift) and not forest fires (flares don't raise Earth's temperature). May 2024 storm: auroras as far south as Leh. ✅

🎯 8 · EVOLUTION OF EARTH PYQ 2018 PYQ 2012

Order of formation

Crust → Atmosphere → Ocean → Life

📝 Atmosphere: 3 stages

→ (1) H and He, blown away by solar wind
→ (2) Degassing: volcanoes give CO_2, N and S oxides, water vapour → clouds → rain → oceans
→ (3) Life raises oxygen (photosynthesis)

4.5 bya	~4.2 bya	3.8 bya	1953
Earth forms	LUCA (as printed)	simple life near deep-sea vents (blue-green algae)	Miller–Urey: organic from inorganic

Source oddity: LUCA ~4.2 bya is printed but is earlier than the 3.8 bya first life (verify). 📌

PYQ 2018 (ans C = 1 and 3): magnetic field reversed every few hundred thousand yrs ✔; life modified early atmosphere ✔; "54% oxygen, no CO_2 at birth" ✘ (early air: water vapour, CO_2, N_2, no O_2). PYQ 2012 (ans B): elements behind origin of life = Carbon, Hydrogen, Nitrogen. ✅

🎯 9 · EARTH FACTS, MAGNETISM, DAY-NIGHT PYQ 2012 PYQ 2013 PYQ 2019 PYQ 2024

Earth fact	Value
Axial tilt	23.5° from vertical (towards Pole/North Star)
Rotation (sidereal day)	23 h 56 m 4.1 s, West → East (counter-clockwise from above North Pole); slowing (wind/wave friction, Moon receding, core slowdown)
Revolution (sidereal year)	365 d 5 h 48 m, elliptical orbit; aphelion = farthest from Sun, perihelion = nearest
Pole drift	Groundwater pumping (NW India, W North America) shifted pole ~80 cm east (0.00000719°), 1993–2010; ice melting + water moving to low latitudes also shifts axis
Magnetic field	Flips every 2–3 lakh years (chrons: normal/reversed); 183 reversals in 83 million yrs (random); magnetic north drifting Canada → Arctic → Siberia since 1990s

Why magnetic field matters

→ Blocks solar wind, stops atmospheric loss
→ Creates magnetosphere (holds Van Allen belts)
→ Causes auroras; helps compasses; guides migrating animals
→ Protects satellites and grids

Pole star 2012 (ans C): village 5 km east → keep pole star on left. Day length 2013 (ans B): revolution on a tilted axis + latitude; no tilt = no seasons, equal days. 21 June 2019 (ans A): Sun does not set at Arctic Circle; equator overhead only at equinoxes. 2024 (ans d): >12 h daylight at Tropic of Cancer and Arctic Circle (2 and 4). Northern days longest at June solstice, shortest at December. ✅

No-year PYQ, axis shift (ans b): melting polar ice → water moves toward equator (mass redistribution, "true polar wander") explains shift; solar flares/CMEs are true but don't affect tilt or rotation. ✅

Magnetic field shifts: the field keeps moving and **reverses about every 2,00,000 to 3,00,000 years**; caused by material inside the Earth, so reversal shows that this material can move (an indirect source on the interior). 💡

10 · MOON

Moon

→ Earth's only natural satellite; **384,400 km**; **27%** (a bit over one-fourth) of Earth's size; gravity drives tides

→ **Perigee** = closest, **Apogee** = farthest

→ **Tidally locked** (same face always)

🏛 Origins

→ **Fission** (George Darwin): rejected

→ **Giant Impact** (Theia, Mars-sized, 4.5 bya): accepted

→ **Binary-exchange capture**: proposed

Pear — Perigee = Pear-close · Apogee = Away

🎯 11 · ECLIPSES, LATITUDE, TIME PYQ 2024

Eclipses

→ **Lunar**: Earth between Sun and Moon; **Blood Moon** = whole Moon in umbra; **Blue Moon** = 2nd full moon in a month (no eclipse link)

🌐 Solar: Moon between Sun and Earth

→ **Annular**: Moon farther → ring

→ **Total**: Moon closer, diamond-ring effect

→ **Partial**: penumbra only

→ Not every full/new moon gives eclipse: Moon's orbit tilted **5°**

Latitude and longitude

→ Imaginary lines to locate places

→ **Latitudes**: parallel, east-west

→ **Longitudes**: north-south, meet at poles; also climatic zones and time zones

→ **Time**: **4 min per 1°** longitude; Earth rotates W → E so east is ahead; **International Date Line** keeps calendar day consistent

A-A-U — **Most time zones:** France **12** · Russia & USA **11** ("two elevens, a cricket team each") · A-A-U = Antarctica, Australia, UK **9** · Canada **6** · Denmark **5**

No-year PYQ	Answer
Anadyr (Siberia, UTC+12) vs Nome (Alaska, UTC-9), ~800 km apart across Bering Strait	**Only Statement I right** (ans a): different days at breakfast; but Monday in Anadyr is **Sunday** in Nome, not Tuesday
UK, Denmark, NZ, Australia, Brazil: more than 4 zones?	**Four** (ans b): Denmark 5 (mainland UTC+1 CET + Greenland), NZ 5 (mainland NZST/NZDT UTC+12/+13 + territories), Australia 6+, UK up to **9** with territories; **Brazil only 4** (UTC-2 to UTC-5; now 3 official)

PYQ: more than four time zones?	Time zones (handout answer: only four of the five)
United Kingdom	mainland GMT/BST (1 zone); with overseas territories (Bermuda, Falklands, Gibraltar) **up to 9**
Denmark	mainland UTC+1; Greenland: Danmarkshavn UTC+0, Ittoqqortoormiit UTC-1, Nuuk UTC-3, Thule UTC-4: **5**
🆕 New Zealand	mainland UTC+12/+13; Tokelau UTC+13, Cook Islands UTC-10, Niue UTC-11: **5**
Australia	mainland AWST UTC+8, ACST UTC+9:30, AEST UTC+10; Lord Howe UTC+10:30, Norfolk UTC+11, Cocos UTC+6:30, Christmas UTC+7: **6+**
Brazil	historically 4 zones (UTC-2 to UTC-5), now 3 official: **not more than 4**

🎯 12 · INTERIOR OF THE EARTH PYQ 2017

Why study interior

→ Changes over and below surface; volcanism, earthquakes
→ Internal structure of solar-system objects
→ Evolution of atmosphere; deep-sea mineral exploration

How we know

→ Direct: deepest mines (Mponeng, TauTona 3.6 km), deepest drill (Kola, 12 km), volcanic eruptions
→ Indirect: meteorites, gravity anomalies (heavy material = higher gravity), magnetic field shifts, best of all earthquake waves

Layer	Facts
🧍 Crust 30–50 km; 0.5–1.0% of volume	Oceanic (SiMa): denser, thinner, 5–30 km, basaltic · Continental (SiAl): lighter, thicker, 50–70 km (70–100 km under Himalaya), granitic · separated by Conrad discontinuity · outer = sedimentary cover, then crystalline igneous + metamorphic (acidic); lower = basaltic and ultra-basic
Mantle Moho → 2900 km; 83%	Density upper 2.9–3.3, lower 3.3–5.7, avg 4.5 g/cm³ · Lithosphere = crust + uppermost mantle, 10–200 km, brittle · Asthenosphere to 400 km, semi-solid, source of magma · lower mantle solid · upper/lower mantle by Repetti
Core 2900–6400 km; 16%	Heaviest, densest; nickel + iron [NiFe] · outer liquid, inner solid, separated by Lehmann's discontinuity

📌 **CoolMenReallyGetLove** — Conrad (continental/oceanic crust) · Moho / Mohorovičić (crust/mantle) · Repetti (upper/lower mantle) · Gutenberg (mantle/core, 2,900 km) · Lehmann (outer/inner core)

💡 Discontinuity: place where seismic waves change velocity and/or direction abruptly.
Earth = boiled egg: shell = crust, white = mantle, yolk = core.

⭐ Mains 2017 (10m): "How does NASA's Juno Mission help understand the origin and evolution of the Earth?" Handout prints the question only; only Jupiter notes (gas giant, storms) given.

🎯 13 · EARTHQUAKES AND SEISMIC WAVES PYQ 2023

Earthquake

→ Abrupt energy release along a fault (sharp break in crustal rock); friction holds, pressure builds, rock slips → shockwaves
→ Focus (hypocentre) = where energy released
→ Epicentre = surface point nearest focus

Distribution

→ Mostly plate boundaries; two belts: Circum-Pacific, Mid-continental
→ Deep-focus at convergent, shallow-focus at divergent boundaries

Scale	Measures
Richter	Magnitude from surface-wave amplitude, log scale 0–10 (5 → 6 = ×10 amplitude)
Mercalli	Intensity (damage) 1–12
Moment magnitude	Energy (fault area, slip, rock rigidity), log, most widely used now

Wave	Type	Motion	Medium	Order
P (primary)	Body, longitudinal	to-and-fro along travel	solid, liquid, gas (faster in solids)	1st, fastest
S (secondary)	Body, transverse	up-down, right angles	solids only	2nd
L (Love)	Surface, shear	horizontal shaking; fastest surface wave	–	3rd; collapses buildings

R (Rayleigh)	Surface, rolling ellipse	vertical + horizontal	–	slowest; tsunamis

Push-pull – P = Push-pull, Prompt. S = Sideways, Solids only. Love shakes Left-right; Rayleigh Rolls like a surf.

Wave behaviour

→ Recorded by seismographs; body waves (P, S) from focus; they generate surface waves at surface rocks

→ Velocity rises with elasticity; direction changes (reflect/refract) at different densities; cooler rigid areas transmit faster than hotter

Shadow zones

→ P: 103°–143° band round Earth

→ S: everything beyond 103°, much larger, a little over 40% of surface

→ S vanish → liquid outer core; P-shadow clue → solid inner core

PYQ 2023 (ans c): P waves recorded earlier than S; P particles vibrate along direction of travel, S at right angles. Both correct. ✅

🎯 14 · EARTH MOVEMENTS AND WEATHERING PYQ 2024

	Endogenic (radioactivity, tidal/rotational friction, primordial heat)	Exogenic (Sun-driven: temperature + rain)
Slow	Diastrophism (bending, folding, warping, fracturing): Orogenic = mountain-building, horizontal force → vertical effect, long narrow belts; Epeirogenic = continent-building, vertical force → horizontal effect, radial up/down	Weathering · Erosion · Deposition · Mass movement = denudation
Sudden	Earthquakes, volcanism	–

Mass movement: only under gravity. Denudation gives: soil formation, ore concentration (iron, manganese, aluminium, copper), landforms. Orogeny, epeirogeny, earthquakes, plate tectonics → faulting and fracturing. 💡

🎯 Epeirogenic examples

→ Uplift: Colorado Plateau, Brazilian Highlands, Australian Shield, Siberian Platform

→ Subsidence: Siberian Basin, Konkan

→ Sudden: 2004 quake submerged Indira Point; NZ 1885 uplift up to 3 m; Japan 1891 subsidence 6 m

Weathering

👥 Chemical (composition changes)

→ Solution/hydration

→ Carbonation (CO_2 + water → carbonic acid)

→ Oxidation (iron, manganese, sulphur rocks)

→ Reduction (red iron oxide → green/blue-grey)

Physical

→ unloading

→ granular disintegration

→ exfoliation (diurnal temperature, "onion peeling", smooth rounded, dry/high places)

→ block separation (jointed rocks)

→ shattering (angular pieces)

→ frost wedging (water freezes, expands)

→ salt weathering (crystallisation, deserts)

Biological

→ burrowing, roots

→ decaying matter gives humic, carbonic and other acids

→ minerals removed by algae and plants

S-C-O-R – "Some Chefs Often Roast" = Solution/hydration · Carbonation · Oxidation · Reduction

PYQ 2024 (ans a): rain weathers rock because rainwater carries dissolved CO_2 and O_2; both statements explain it. ✅

🎯 15 · DRIFT AND SEA-FLOOR SPREADING PYQ 2014

1858 Pellegrini
1912 Wegener drift
1930s Holmes convection
1960 Hess spreading
1967 McKenzie & Parker
1968 Morgan

💡 Earlier ideas

→ Antonio Pellegrini 1858: identical fossil plants in N American and European coal → once joined
→ Edward Suess (late 19th c.): big ancient continents split into present ones
→ F. B. Taylor: arcuate (bow-shaped) mountain belts of Asia, Europe from creep of continents toward Equator

Wegener 1912

→ One landmass Pangaea + one ocean Panthalassa
→ Tethys Sea split it: Laurentia (Laurasia) north, Gondwanaland south
→ Forces: tidal (westward) + gravity and polar-fleeing (equator-ward)
→ Break-up in Triassic (251–199.6 mya); ~200 mya all continents one supercontinent

Evidence	Detail
Jigsaw fit	Brazil bulge ↔ Gulf of Guinea; Greenland ↔ Ellesmere/Baffin; India–Madagascar–Africa
Rocks/ fossils	Ghana gold placers from Brazil plateau; Mesosaurus fossils only in South Africa and Iraver formations of Brazil (4,800 km apart); 2-bn-yr-old rocks on both coasts
Climate	Tillite; India's Gondwana sediments have counterparts in six southern landmasses
Mountains	Caledonian/Hercynian (Europe) continue as Appalachians

⚠️ Drawbacks

→ No free float of SiAl on SiMa: lithosphere moves over asthenosphere
→ Ocean floor isn't passive; tides/gravity too weak (convection currents the engine)
→ Only equator/west movement (but Antarctica moved away, Australia east, India NE)
→ No reason for Triassic break-up

Hess 1960: sea-floor spreading

→ War-time research: ocean floor full of relief (ranges, trenches)
→ Mid-oceanic ridges most volcanically active; ocean floors not static
→ Rocks equidistant either side of ridge crest similar in constituents and age
→ Shallow-focus quakes at ridges (deep-focus at Himalayan belt)
→ Oceanic rocks much younger than continental (subduct)

Holmes 1930s: convection

→ Radioactive heat in mantle (100–2900 km) seeks escape → convection currents
→ Rising limbs → divergence; falling limbs → convergence

Paleomagnetism

→ Rock minerals lock in direction + intensity of magnetic field (age and location of formation); basalt on sea floor

→ Ocean floor each side of ridges has magnetic stripes: normal, then reversed

→ New rock at ridge pushes older (reversed-era) rock away; polarity reversals recorded in magma

PYQ 2014 (ans c): continental drift and glacial cycles both influenced evolution (unique primates of Australia, S America; genus Homo in last 3 glacial cycles). No-year (ans c, all three): Brazil–W Africa rock belts, Ghana gold from Brazil plateau, Gondwana sediments in six southern landmasses (Africa, S America, Australia, Antarctica, Madagascar, NZ). ✅

16 · PLATE TECTONICS

Theory

→ Suggested McKenzie and Parker 1967; outlined Morgan 1968

→ Lithosphere broken into plates (7 major, many minor) floating on asthenosphere under convection currents (Holmes 1930s)

→ Oceanic: Simatic, thinner (Pacific); continental: Sialic, thicker (Arabian); both (Indo-Australian)

→ Minor plates from stress where majors converge (Mediterranean: Eurasian + African)

→ Explains earthquakes, volcanoes, fold and block mountains

7 major plates

→ Antarctica + ocean

→ North American

→ South American

→ Pacific

→ India-Australia-NZ

→ Africa + E. Atlantic floor

→ Eurasia + ocean

🗺️ Ring of Fire (Pacific boundary, NZ → Chile)

→ 70% earthquakes, 90% volcanoes (75% of active)

→ 20% earthquakes in Mid-continental belt

→ Slowest ridge Arctic (<2.5 cm/yr); fastest East Pacific Rise (>15 cm/yr, ~3,400 km west of Chile)

DCS – Divorce · Collide · Slide (divergent · convergent · transform)

Boundary	Plates...	Creates	Example
Divergent (constructive)	move apart	new crust, mid-ocean ridges, rift valleys, volcanic islands, shallow quakes	Mid-oceanic ridges, Great African Rift (Lebanon → Mozambique), Chagos
Convergent C–C	continent meets continent	fold mountains, no volcanoes (no subduction)	India–Eurasia → Himalayas
Convergent O–C	dense ocean plate subducts	trench, volcanoes, fold mountains (Cordilleran)	Pacific–S American → Andes
Transform (conservative)	slide past	only deformation; usually ⊥ ridges	San Andreas Fault

Himalayas have no volcanoes; Andes and Alps do.

~225 mya Tethys separates India from Asia

~200 mya moves north (Pangaea breaks)

~140 mya at 50°S

~65 mya Deccan Traps

40–50 mya collision (some say 60)

Indian plate

→ island off Australia first; Himalayas still rising. Margins: E Arakan Yoma → Java Trench

→ W Kirthar
→ N Himalaya. **Limits of theory:** doesn't explain hotspots; can't place ancient plates or predict future boundaries.

PYQ 2013 (ans d): electromagnetic radiation, geothermal energy, gravity, plate movement, rotation, revolution: all cause dynamic surface change (radiation → temperature change, gravity → downslope movement). ✅

🎯 17 · VOLCANISM PYQ 2024

Volcanism: molten rock from asthenosphere enters lithosphere, may or may not reach surface. **Volcano:** vent in crust erupting magma, gases, ash. Under Earth magma; after eruption lava. 💡

Acidic lava
→ Light-coloured, low density, high silica
→ Highly viscous, flows slowly, little spread
→ Solidifies in vent → explosive, pyroclastic material

Basic lava
→ Dark, high density, low silica
→ Highly fluid, spreads widely
→ Lava plateaus, gentle volcanic mountains

Vents and gases
→ Fumarole: steam + gas (SO_2, CO_2); sulphur-rich = solfatara, CO_2-rich = mofette
→ Gases: carbon, sulphur, nitrogen oxides, water vapour, H_2S, trace chlorine
→ Geysers only in Iceland, New Zealand, Yellowstone (USA)

Distribution
→ Only 10–20% of volcanic activity is above sea
→ Ring of Fire 75% active
→ Mid-Atlantic ridge: Iceland
→ Great African Rift: Kilimanjaro, Kenya
→ Cameroon (only active in W Africa); Pelée (West Indies); Mediterranean: Vesuvius, Stromboli, Ararat, Elburz
→ None in Australia; none in Himalaya

South Asia / India
→ Barren Island: only active lava volcano (last erupted 2022)
→ Narcondam extinct
→ Baratang: India's only mud volcano
→ Chandragup (Pakistan): mud, no lava/pyroclast, "not a true volcano"

Extrusive (surface)	Intrusive (inside; seen after erosion)
Conical and fissure vents, basalt plateaus · mid-ocean ridges · pyroclastic debris · caldera and crater lakes (Crater Lake, Oregon)	Batholith/pluton (huge; Mt Wicklow) · Sill (flat; Great Whin Sill) · Laccolith (dome; Henry Mts) · Lopolith (saucer; Transvaal) · Phacolith (wavy; Cordon Hill) · Dyke (vertical; Alaska)

Sill=sill — Sill = window-sill, flat · Dyke = wall, upright · Laccolith = Loaf (dome) · Lopolith = Lota (bowl/saucer) · Phacolith = wavy

I HaveSomeVada-Pav,Please — Eruption styles calm → crazy: Icelandic (fissure, basalt) → Hawaiian (central vent, most lava) → Strombolian (short bursts, pasty lava) → Vulcanian (dark cauliflower clouds) → Pelean (most dangerous, glowing clouds) → Plinian (ash up to 50,000 ft, deadly pyroclastic flows)

Types of volcano

→ By frequency: Active (St Helens, Merapi) · Dormant (Vesuvius before AD 79) · Extinct (Narcondam)

→ By shape: cinder cone · composite (strato) · shield

→ Vulcanian/Pelean/Plinian ash = hygroscopic nuclei → rain + ash → lahars (mudflows)

Mantle plume and hotspot

→ Rising heat arm in mantle; intra-plate (not at boundaries)

→ Plate moves over it → island chains (Hawaii); cut off from heat → extinct

→ Azores: hotspot + Mid-Atlantic ridge

→ Réunion hotspot: active >65 million yrs; Indian plate passing over it → Deccan Traps

PYQ 2024 (ans d): pyroclastic debris, ash and dust, nitrogen compounds, sulphur compounds: all four are volcanic products. ✅

18 · OROGENY: MOUNTAINS, ISOSTASY

Orogeny = mountain building. By location: continental, coastal, oceanic. By formation: fold, fault (block), volcanic, residual. 💡

Mountain	Formed by	Notes / examples
Fold	Compression of sedimentary strata; loftiest; young near converging boundaries	Himalaya (India–Eurasia), Alps and Atlas (Eurasia–Africa), Andes (S America–Pacific): quakes + volcanoes · old: Appalachians (USA), Aravallis
Block (fault)	Dip-slip faults (vertical displacement), tension/ compression; horst = raised, graben = lowered	Great African Rift (floor graben; Mitumba and Muchinga horsts), Rhine valley–Vosges
Volcanic	Lava, bombs, cinders, ash, dust, mud in layers around vent	"Mountains of accumulation"; Circum-Pacific: Fuji, Mayon, Merapi (Sumatra), Etna
Residual	Left after denudation of fold/fault/ volcanic mountains	Monadnocks (USA)

Isostasy

→ Equilibrium between lithosphere and asthenosphere (buoyancy vs gravity)

→ Crust subsides/uplifts when load added or removed (erosion, mountain building, snow/ice melt)

→ Causes: tectonic (subsidence, uplift) and climatic

→ Can cause earthquakes; ice melt → rebound → cracks → may reactivate Antarctic volcanoes

16-7-48 — Longest ranges (km): Mid-Atlantic Ridge 16,000 · Andes 7,000 · W. Cordillera 4,830 · Great Dividing Range & Trans-Antarctic 3,500 · Urals & Atlas 2,500 · Appalachians 2,414 · Himalayas 2,400 · Altai 2,000. Say "16, 7, 48, 35-35, 25-25, 24-24, 20".

🎯 19 · ROCKS AND ROCK CYCLE PYQ 2022

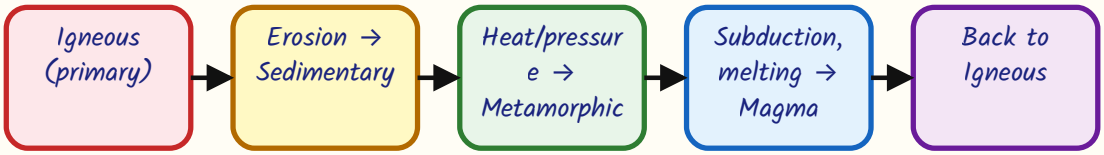
	Igneous (primary)	Sedimentary	Metamorphic
🏛️ Origin	Magma/lava cooling	Denudation → deposits → lithification	PVT (pressure, volume, temperature) change of existing rock; recrystallise
💡 Features	No fossils; plutonic/intrusive (slow, deep, big grains) vs extrusive (fast, surface, small smooth grains); acid (granite) light/less dense vs basic	Layered/strata, fossils, porous; 75% of crust area, 5% of volume; tillite (ice), loess (wind)	Foliation, lineation, banding; may/may not have strata and fossils; heat = thermal, pressure = dynamic types
🎯 Examples	Granite, gabbro, pegmatite, basalt	Sandstone, shale, limestone, coal, chalk, chert, halite	Gneiss, slate, schist, marble, quartzite
Value	Metal ores (iron, nickel, copper, lead, zinc, chromite, manganese, gold, diamond, platinum); building stone; black soil	Petroleum, gas, coal, phosphates, hematite iron ore, limestone; placers (bauxite, manganese, tin); rich soils	–

Sedimentary by mode

→ Mechanical: sandstone, conglomerate, shale, loess
→ Organic: chalk, limestone, coal, geyserite
→ Chemical: chert, limestone, halite, potash

Amygdales: almond bubbles in basalt filled with minerals (copper). Metamorphism: rocks forced down by tectonics or contacted by rising magma. 💡

Rock cycle (continuous; any rock can become any other)



Mineral	Mohs hardness (1 soft – 10 hard) and use
Feldspar	6–7 (glass, ceramics)
Quartz	7 (jewellery, electronics)
Bauxite	1–3 (aluminium)
Cinnabar	2–3 (mercury)
Dolomite	3.5–4
Hematite and Magnetite	5.5–6 (iron)

Mains 2022 (10m) "primary rocks": Intro: igneous from magma/lava, hence primary. Body: non-fossiliferous; plutonic vs volcanic; acid vs basic; examples; economic value (ores, building stone, black soil). Close: all other rocks derive via rock cycle. ✅

20 · EROSION, FLUVIAL CYCLE

Erosion

→ = removal + transportation + deposition (transport separates it from weathering). Agents: flowing water, underground water, wind, glaciers, coastal waves. Five cycles: fluvial
→ glacial
→ marine
→ arid
→ karst.

YMO – River life: Young – V, Mature – meander, Old – delta. Youth: fast, V-valley, waterfalls, rapids, headward erosion · Mature: lateral erosion, meanders, U-profile · Old: deposition, braiding, delta 📌

Fluvial erosion

→ Types: hydraulic action, corrosion (chemical), attrition (load vs load), corrasion/abrasion (load vs rock)

→ Downcutting deepens, lateral widens, headward lengthens → river piracy (Saraswati; Shyok captured Nubra)

→ Rejuvenation: stage change from tectonic (submergence/emergence) or climatic (sea level) activity → terraces

→ Valley: young = deep, narrow, V, vertical downcutting; mature = U, broad base, concave slope; tributary valley hangs above main valley (falls/rapids)

Erosional forms

→ Gorge (narrow, steep; Indus at Nanga Parbat, Brahmaputra at Namche Barwa: antecedent rivers) vs canyon (broader; Grand Canyon, Colorado, Arizona: 483 km long, 2.88 km deep)

→ Waterfall (youth; Jog/Gersoppa on Sharavati 260 m), rapids/cataracts

→ Potholes (swirling boulders drill)

→ Meander: outer = cliff-slope, erosion, concave; inner = slip-off, deposition, point bars

→ Ox-bow lake (cut-off meander; Ganga shifted south leaving ox-bows north)

→ Peneplain (end product) with monadnocks

Depositional forms

→ Alluvial fans/cones

→ Natural levees (breach = flood)

→ Delta: arcuate fan; bird's-foot weak tides; cuspate strong waves

→ Braided channel

→ Estuary: drowned valley; ports, fisheries

Drainage pattern	Description	Example
Dendritic/Pinnate	irregular tree	Indus, Godavari, Mahanadi, Cauvery, Krishna
Trellis	short tributaries at right angles; soft-rock erosion	Seine, Paris basin
Rectangular	main stream and tributaries bend at right angles	Colorado (USA); Mahanadi
Angular	tributaries at acute angles	Himalayan foothills
Parallel	parallel streams, uniform slope	rivers of lesser Himalayas
Radial	from a summit in all directions	Central French Plateau, Kilimanjaro, Amarkantak, Saurashtra
Annular	ring around summit (soft outer stratum)	Black Hills, S. Dakota
Centripetal	all converge into low basin	Ladakh, Tibet, Bagmati (Nepal)

Handout conflict: Mahanadi listed under both dendritic and rectangular. Take dendritic unless the question says rectangular.

🎯 21 · KARST CYCLE PYQ 2023

Groundwater words

→ Porous (pore spaces, sandstone) · Permeable (water passes; clay porous but not permeable)

→ Aquifer stores and transmits water; water table = top of saturation zone

→ Confined (impermeable cover, pressure) vs unconfined

→ Artesian well (rises unpumped); spring; perched water table

Karst

→ Named after a province of Yugoslavia (now Slovenia) on Adriatic coast

→ Thick limestone, dolomite or gypsum; rock dense, highly jointed, thickly bedded

Erosional

→ Sinkholes/swallow holes (3–9 m; dolines, uvalas, polje)
→ Lapies (grooves and ridges)
→ karst window, arch/bridge, caves, blind valley

Depositional (speleothems, in caves)

→ Stalactite hangs from roof
→ Stalagmite rises from floor
→ Column when joined
→ flowstone, cave pearls, draperies

Ceiling-Ground — Stalactite = Ceiling (holds tight to the roof). Stalagmite = Ground (might reach the roof).

PYQ 2025 (ans c): chalk = porous and permeable; clay = porous but not permeable (USGS: very high porosity, low permeability). Statement III ("clay not porous at all") wrong; II right and explains I. ✅

22 · MARINE CYCLE AND COASTS

Marine erosion

→ Waves aided by winds, currents, tides, storms; depends on wave size, slope, shore height, coast shape, rock, depth
→ Wave pressure compresses air in fissures → rupture; waves use debris as tools (corrasion); solvent action in limestone and chalk

Cats Always Sit Still: C-A-S-S



💡 Erosional features

→ Chasm: narrow, deep, headward erosion along vertical weakness; mouth widens → bay
→ Wave-cut platform: sea cliffs retreat; notch at base (maximum erosion)
→ Sea cave (differential erosion) → sea arch
→ Stack: pillars offshore; further eroded (below low tide) → stumps
→ Blow/spouting hole: air compressed in cave; noise; widen, walls collapse → geo/gloup

💡 Depositional features

→ Beach: temporary debris cover on wave-cut platform
→ Bar: submerged, offshore; encloses lagoon; barrier = overwater counterpart
→ Spit (one end tied to headland) → hook (curved end)
→ Tombolo: bar joining island
→ Marine dunes: by wind, not waves

Coast type	Cause	Examples
Emergence	land up / sea down	bars, spits, lagoons, salt marshes, beaches, cliffs, arches
Submergence	land sinks / sea up	Ria (river valleys drowned, irregular; SW Ireland) · Fjord (glacial U-trough drowned, long steep inlets; Norway) · Dalmatian (ridges parallel to coast; Yugoslavia) · Drowned lowland (low, no indentations; Baltic coast of E. Germany)
Neutral	new material built out; no net relative change	alluvial fan, delta, volcano, coral-reef coasts

Compound	submergence + emergence (either order)	Norway and Sweden
Fault	down-thrown block submerged, steep fault face to sea	Konkan coast (India)

23 · GLACIAL CYCLE

Glacier

→ Moving ice mass, few metres/day; loaded with rock debris
→ Types: continental, ice cap, piedmont, valley

Erosional

→ Cirque/Corrie/Cwm: hollow basin in ridge, steep on 3 sides, open on 1; tarn lake
→ Horn/Pyramidal peak: cut from >2 sides
→ Arête: knife-edge between two cirques
→ Bergschrund: crevasse between moving glacier and stationary ice/rock at upper end
→ Glacial trough: U-shaped valley (uniform horizontal + vertical erosion); flooded = fjord
→ Hanging valley: tributary valley left high, discordant
→ Roche moutonnée ("sheep rock"): smooth side faces ice advance, rough steep side plucked
→ Craig and tail: resistant mass + softer ridge downstream

Depositional

→ Outwash/till plain: stratified debris, clay, sand, gravel
→ Esker: winding ridge of unassorted deposits, like a levee, road-making
→ Kame terraces: broken ridges, humps
→ Drumlin: inverted boat; steep glacier side faces glacier, lee side tapers (shows direction)
→ Kettle holes: depressions in till plain
→ Moraine: ground, lateral, end

24 · ARID CYCLE, LAKES, PLATEAUS

H-R-E — Hamada = rocky (large smoothened rocks) · Reg/Serir = stony gravel and pebbles, easier to cross · Erg/Koum = sandy · Cold deserts (high altitude/latitude). H-R-E: Hard → Rubble → Ergs of sand

Water forms (flash floods + wind)

→ Bolson: intermontane basin
→ Gully (ditch-like, metres–tens of metres); Ravine (narrower than canyon, larger than gully, smaller than valley)
→ Badland: rills erode weak sediments, no vegetation, no agriculture (Chambal ravines)
→ Playa: temporary lake; salt-covered = Salina
→ Pediment: rock-cut surface at mountain foot (erosional)
→ Bajada: moderately sloping depositional plain

💰 Wind processes

→ Deflation: lifting and carrying away dry dust
→ Abrasion: sand-loaded wind grinds rock
→ Attrition: wear of sand particles in transport

Wind erosional

→ Blowouts/deflation hollows (to several km)
→ Mushroom rock (cap on narrow stem)
→ Zeugen = Zzz, lies flat (table-shaped, horizontal layers) · Yardang = Y stands up (vertical layers)

→ Wind window (holes widen) → wind bridge (arch)
→ Inselberg (= monadnock of deserts)
→ Pediplain (= peneplain of deserts)

Wind depositional

→ Ripple marks: saltation
→ Dunes: heaps, few m to 20 m (some hundreds of m high, 5–6 km long); deserts and coasts
→ Barchan: crescent, windward convex, leeward concave and steep
→ Longitudinal: parallel to wind (Sahara, Australian, Libyan, S African, Thar)
→ Transverse: ⊥ wind
→ Parabolic: U-shaped, longer and narrower than barchan, windward concave
→ Star: central peak, 3+ arms, multi-direction wind
→ Loess (ex-situ): fine desert dust carried far (Loess Plateau near Gobi, China)

Lakes

→ Larger, deeper than ponds; most have natural outflow; endorheic = no outflow (lose water by evaporation/seepage)
→ Tectonic: Titicaca, Caspian
→ Rift valley: Tanganyika, Malawi, Dead Sea
→ Glacial: tarns; Scandinavia, Canada
→ Volcanic crater/caldera: Lonar, Krakatoa
→ Erosional: karst, playa, wind-deflated (Great Basin, Utah)
→ Depositional: ox-bow, lagoon (Chilika), moraine-dammed
→ Man-made

Plateaus

→ Flat tablelands, a third of land; mineral-rich (mining); eroded → outliers (old dense rock; iron ore, coal)

Tectonic

→ Volcanism: Deccan
→ Crustal shortening: Tibet (intermontane)
→ Thermal expansion: Ethiopian Highlands

→ Erosional: Allegheny (Appalachians)

TRAPS

"Himalayas have volcanoes" ✘ · "P waves can't pass liquids" ✘ (that's S) · "orogeny = continent-building" ✘ (that's epeirogeny) · "solar flares cause tsunamis/forest fires" ✘ · "Blue Moon = eclipse" ✘ · "Sun overhead on equator on 21 June" ✘ (equinoxes).

Connects to: Himalaya and Deccan story → Indian Geography; tilt and 21 June → Climatology; ridges, trenches, coral reefs → Oceanography; Indus/Brahmaputra gorges → Indian drainage. ✅

Recall drill: cover and answer

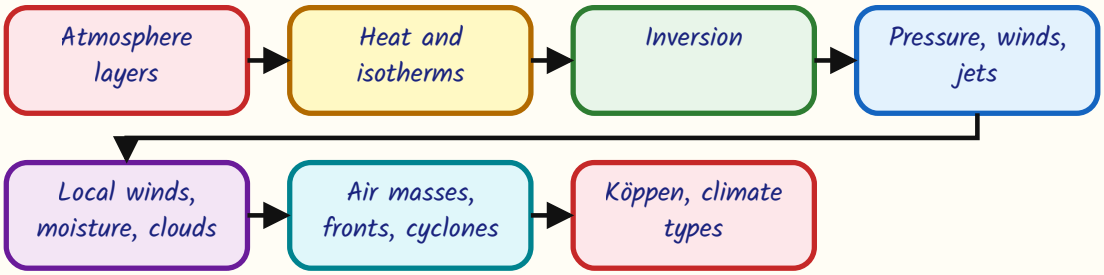
CMB found by whom, when?	Penzias & Wilson, 1965; ~380,000 yrs after Big Bang	White dwarf / neutron star / black hole mass?	<1.44 · 1.44–3 · >3 (Chandrasekhar)
Moons: Jupiter, Saturn, Uranus, Neptune?	101, 274, 28, 16	Earth tilt; sidereal day?	23.5°; 23 h 56 m 4.1 s
Earth volume: crust / mantle / core?	0.5–1% / 83% / 16%	Boundaries top to bottom?	Conrad, Moho, Repetti, Gutenberg, Lehmann
P shadow / S shadow?	103–143° / beyond 103°	Drift / spreading / plates?	Wegener 1912 / Hess 1960 / McKenzie-Parker 1967
Only active lava volcano in South Asia?	Barren Island (India)	Fastest / slowest ridge?	East Pacific Rise / Arctic

CLIMATOLOGY

PYQ Prelims 2011 ×5, 2012 ×3, 2013, 2014, 2015 ×3, 2018, 2019 ×2, 2020, 2021 ×2, 2022, 2023 ×3, 2024 ×6, 2025 + 2 with no year · Mains 2022 (troposphere, 15m)

✎ *Atmosphere, heat, pressure, winds, jet streams, clouds, air masses, cyclones and climate types: 32 Prelims PYQs make this the most-asked chapter.*

At a glance: the Climatology route



Hot clusters

→ heat budget and lapse rate (2012, 2023, 2024)
→ isotherm bending (2024, 2025)
→ westerlies, Coriolis, jet streams (2011, 2015, 2020, 2024)
→ clouds and dew (2019, 2022)
→ cyclone genesis (2015)
→ climate-type identification (2011–2024). **Prepare:** Statement I/II cause-effect pairs (2023, 2024, 2025); Mains: troposphere, jet streams and climate change, tropical vs temperate cyclones. No teacher ink: priority = handout PYQ count.

1 · ATMOSPHERE: LAYERS AND GASES PYQ 2011 PYQ 2012 PYQ 2024 PYQ 2022

Layer (bottom to top)	Height	Key facts
1 Troposphere (kitchen)	8–18 km (avg 13)	All weather, clouds, jet streams; 18 km at equator (strong convection), 8 km at poles; cools ~1 °C per 165 m
2 Stratosphere (AC floor)	to 50 km	Ozone (lower part) so temp rises with height; almost cloud-free = jets fly here
3 Mesosphere (freezer)	50–80 km	Coldest; meteors burn; noctilucent clouds
4 Thermosphere (antenna)	80–400 km	Temp rises again (UV, X-ray); ISS and satellites; Kármán line 100 km; ionosphere bounces radio
5 Exosphere (terrace)	> 400 km	Particles up to 1,000 °C yet feels cold (too thin to pass heat)

Gases

→ N_2 78% · O_2 21% (≈99%) · Argon 0.93% · CO_2 ~0.04%
→ Water vapour 0–5%: >90% below 5 km; falls equator to poles
→ CO_2 and water vapour only up to 90 km; ozone layer 20–25 km
→ By composition: homosphere (<80 km, uniform) · heterosphere (>80 km, layered by molecular mass)
→ Aerosols/salt = hygroscopic nuclei; microplastics seed ice 5–10 °C warmer than usual
→ Origin of gases: volcanic eruptions, hot springs, chemical breakdown of solids, biosphere

Source conflicts: lapse rate quoted as 1 °C/165 m and 6.5 °C/1,000 m (about equal). Ionosphere printed 75–1,000 km (PYQ note) vs thermosphere 80–400 km. Water vapour 0–5% (layers) vs "around 1% of air" (PYQ note).

PYQs: 2012 (C) cooler aloft because air is heated from the ground up + thinner air; NOT more moisture. 2012 (D) CO_2 absorbs infrared: transparent to incoming, opaque to outgoing radiation. 2023 (C) water vapour absorbs infrared (also: equator insolation 10× poles ✘; IR two-thirds of insolation ✘; IR not visible spectrum ✘). 2024 (a) troposphere thicker at equator because strong convection lifts heat. 2024 (d) atmosphere heated more by terrestrial (long-wave) than incoming radiation; only Statement II right. Undated (d, II and III): without atmosphere nights freeze but days get extremely hot, so "below freezing everywhere" ✘. Mains 2022 (15m) troposphere: clouds, air masses, jet streams sit here; terrestrial radiation is prime heat source. ✅

Ionosphere 2011 (D): charged particles reflect radio waves; NOT ozone; radio wavelength 1 mm–100 km, not "very long". Ionised by solar and cosmic radiation; D, E, F regions; electron density highest in F (day and night; day by solar radiation, night by cosmic rays); D vanishes at night, E weakens; VLF waves bounce in an Earth–ionosphere waveguide; Earth radius 6,370 km. ✅

CO_2: about 0.038% (now 0.04%) of dry air; product of combustion; greenhouse gas that helps keep the heat budget. Ionosphere: ionised by solar and cosmic radiation; lies 75–1000 km above the Earth (thin against the Earth's radius of 6370 km). 💡

🎯 2 · HEAT, ALBEDO, ISOTHERMS

PYQ 2013 · PYQ 2023 · PYQ 2024 · PYQ 2025

Temperature controls

- → Latitude (slant rays) · duration of daylight · sunspot cycle
- → Transparency (cloudy day cooler, cloudy night warmer)
- → Altitude · mountain orientation · vegetation (transpiration cools)
- → Distance from sea · ocean currents · local winds/urban

4 ways heat moves

- → Conduction: through solid, molecule to molecule
- → Convection: fluid movement (liquid/gas)
- → Advection: horizontal movement of air/water
- → Radiation: electromagnetic waves, no medium

📌 **Con-Con-Ad-Rad** — Conduction = Contact (spoon in hot chai) · Convection = Currents (boiling milk) · Advection = Aside, sideways (Loo carries heat in) · Radiation = Rays, no medium (Sun to Earth)

💰 Heat budget

- → Earth gets insolation (short-wave) and returns terrestrial radiation (long-wave) = constant temperature
- → Hotter body = shorter wavelength: Sun short-wave, Earth long-wave

Albedo (reflectivity)

- → High: snow/ice = white shirt (reflects, cools)
- → Low: ocean, forest, city = black shirt (absorbs, warms)

- → Energy: 0–40° N/S surplus; beyond 40° deficit; winds and currents carry heat poleward, mostly across 30°–50° (hence storms there)

🛡️ Isotherms = lines of equal temperature (vertical distribution: see layers)

- → January, NH: bend poleward over oceans (warmer), equatorward over land (cools fast)
- → July: reverse (south over oceans, north over land)
- → SH: shift less evident (little land) and opposite in season: NH January = SH July
- → Cause: water has high specific heat; land heats and cools fast

Handout conflict: isotherm text first says "July: isotherms shift towards equator on land", then summarises "July: north over land, south over oceans". Trust the summary and the January rule.

PYQs: 2013 (A) interiors have bigger annual range due to land–water thermal difference only (not altitude, winds, rain; interiors are dry). Maritime influence = moderating; continental = extreme. 2023 (D) land–sea contrast greater in winter; water has higher specific heat. 2024 (a) January: isotherms north over ocean, south over continent; statement 2 (Gulf Stream, N Atlantic Drift as "cold currents") ✘. 2025 (a) both right and II explains I. ✅

3 · TEMPERATURE INVERSION

Inversion: temperature rises with height

→ Needs clear, calm, long nights, no mixing, valleys/basins
→ Common at fronts, in winter, in mountains and valleys

Good
→ Frost protection for plants

Bad
→ Pollution dome
→ Fog: poor visibility, transport
→ Pests and diseases thrive

Pollution dome

→ Pollutants trapped under warm air; cold ground in cold weather cools air; no vertical mixing
→ Effects: poor air quality; respiratory, cardiovascular problems
→ Mitigation: cut emissions, air-quality monitoring/forecasting, public awareness
→ Beijing: China launched US$100-bn drive in 2013 (vehicle rules, relocate industry, clean energy): PM2.5 down 35% by 2017

🎯 4 · PRESSURE, PRESSURE BELTS, FORCES PYQ 2024

Air pressure

→ Weight of air column per unit area; maximum at sea level
→ Heat = expand, rise = low; cool = sink = high
→ Isobars = lines of equal pressure
→ 7 belts = 3 thermal + 4 dynamic; not uniform (land–water)
→ Equatorial low (10° N–S) = ITCZ = Doldrums (calm, trades converge)

Lazy Hens Lay Happily — Low equator (ITCZ/Doldrums 10° N–S) → High ~30° → Low ~60° → High poles

Belt shift (Earth's 23.5° tilt)

→ 5–10° north (up to 25° over India) when Sun over Tropic of Cancer 21 June
→ 5–10° south when over Tropic of Capricorn 22 December
→ Balanced at equinoxes 21 March, 23 September
→ At 30°–40° = Mediterranean pattern: winter westerlies = rain; summer dry trades offshore

3 forces on wind

→ Pressure gradient: speed + direction (high to low; close isobars = fast)
→ Coriolis: direction only; deflects right in N, left in S; zero at equator, max at poles; grows with wind speed
→ Friction: both; only a few thousand metres up; less over sea so faster
→ Winds are named for where they come from (polar easterlies come from east)

Types of winds

→ Planetary/permanent: all forces, all year (trades, westerlies, polar easterlies)

→ Geostrophic: tropopause level, no friction, parallel to isobars (jet streams)
→ Seasonal: monsoon
→ Diurnal: sea/land breeze, mountain–valley breeze
→ Local: specific place, all year or one season

Planetary wind	From to	Notes
Trade winds	subtropical high to equator, easterly	helped trade routes; push ocean currents
Westerlies	subtropical high to sub-polar low, W to E	strongest in S. Hemisphere (little land): Roaring 40s, Furious 50s, Shrieking 60s
Polar easterlies	polar high to mid-latitudes	cold, weaker

Hadley Hot — Hadley (equator–30°, biggest, trades) · Ferrel Fill-in (30°–60°, middle, runs opposite) · Polar Puny (60°–pole, smallest, polar easterlies). Tricellular meridional circulation = 3 cells per hemisphere.

PYQs: 2011 (A) southern westerlies stronger because less land (not higher Coriolis). ✅ 2011 (C) jets fly in lower stratosphere: no clouds/vapour and no vertical wind shear. 2015 (B) westerlies blow 30°–60° (not "30°N–60°S"); moist air for NW India's winter rain is westerlies: they start in horse latitudes, steer extratropical cyclones; Mediterranean disturbance dragged by the jet stream; J&K and NW India get the most. 2024 (c) Coriolis rises with wind velocity and is max at poles, absent at equator.

Pressure belts in both hemispheres stay balanced when the Sun is vertical over the Equator: 21 March and 23 September (equinoxes).

🎯 5 · GEOSTROPHIC WINDS, JET STREAMS PYQ 2020 PYQ 2024

Geostrophic wind: blows parallel to isobars; pressure gradient force balanced by Coriolis (right in N, left in S).

Permanent
→ Polar: 50°–60° N/S, boundary cold polar/warmer subtropical air, strongest in winter
→ Subtropical: 20°–30° N/S, weaker, less consistent

Temporary
→ Tropical Easterly
→ Somali

Jet streams: narrow fast bands, 8–15 km
→ Both hemispheres (2020: "NH only" ✘) · formed by temperature gradients of air masses
→ Wavy jets = Rossby waves
→ Steer storms; carry western disturbances (Mediterranean extra-tropical cyclones, by subtropical westerly jet); control South Asian monsoon; mix troposphere–stratosphere (ozone-depleting substances)

Climate change and jets
→ Arctic warms faster = smaller temperature contrast = weaker, more meandering jets
→ Long heatwaves, coldwaves, droughts or floods

Heat dome

→ Stagnant high pressure; air sinks, compresses, warms
→ Mid/high latitudes; continental climates
→ Heatwaves, heatstroke; agriculture, energy hit
→ Climate change: more frequent, intense, longer

Polar vortex (= DANA, Depresion Aislada en Niveles Altos, isolated depression/cut-off low)

→ Cold air held by polar jet; both poles, N pole matters most
→ Arctic Amplification + meandering jets = cold air spills to low latitudes

Stratospheric cooling

→ CO_2 radiates heat to space in the stratosphere (traps it in troposphere)
→ Ozone loss (CFCs, poles) = less UV absorption = less heating
→ GHGs affected stratosphere by ~1885, five decades before troposphere warming was detected
→ Impact: slows ozone recovery, strengthens polar vortex, disrupts jets and weather, extreme events, affects aviation and satellite orbits, key climate-change indicator ("warm troposphere + cooling stratosphere"); more PSCs

🎯 6 · LOCAL AND MONSOONAL WINDS PYQ 2014

Wind	Where	Hook
Chinook	E. Rockies (USA, Canada)	warm dry, "snow eater"
Foehn	Alps (leeward)	warm dry descending
Sirocco (Ghibli, Chili, Khamsin)	Sahara to Mediterranean	hot, dry; rainy in S. Europe
Harmattan	W. Africa (Sahara to Guinea)	dry, dusty NE trade; "Doctor" wind
Loo	N. India, Pakistan, May–June	furious, hot, heatstroke
Bora	Adriatic (Italy, Slovenia, Croatia)	extremely cold dry NE, katabatic
Mistral	Rhône valley (France)	cold NW
Santa Ana	S. California	hot dry; worsens wildfires
Khamsin	Egypt	dusty spring wind, up to 50 days
Purga	Siberia, Russian steppes	icy NE blizzard wind
Pamperos	Pampas (Argentina, Uruguay, Brazil)	violent cold SW

Brrr — Hot/warm: Chinook, Foehn, Sirocco, Harmattan, Loo, Santa Ana · Cold: Bora, Mistral, Purga, Pamperos. Mistral and Bora = "Brrr" winds.

Source note: Khamsin appears twice: as a Sirocco name (Sahara to Mediterranean) and as an Egyptian dusty spring wind.

Monsoon winds: seasonal, tropics, reverse with the Sun's apparent position (detail in Indian Geography). PYQ 2014 (C): seasonal reversal = monsoon climate; "mausim" (Arabic) = season; ~6 months NE, 6 months SW; South Asia, Africa, Australia, Pacific coast of Central America. ✅

🎯 7 · HUMIDITY, CLOUDS, PRECIPITATION PYQ 2019 PYQ 2022 PYQ 2024

Humidity (condensation = vapour to liquid water)

→ Absolute (g/m³) · relative (% of capacity) · specific (g per kg)
→ Dew point = saturation temperature
→ Dew: surface cools below dew point; none on cloudy nights (clouds send radiation back)
→ Frost = deposition (vapour to ice) · Fog = surface cloud · Smog = smoke + fog (sulphurous, photochemical) · Haze = dry particles
→ Atmospheric rivers: narrow moisture corridors (more water than Amazon), pushed by westerlies to mid-latitudes = heavy rain, floods

Clouds	Height	Note
High (Cirrus)	> 6,000 m	wispy ice "chicken feathers"; thin ones trap heat
Mid (Alto-)	2,000–6,000 m	
Low	< 2,000 m	Stratus, stratocumulus, nimbostratus; thick ones reflect
Cumulonimbus	vertical tower	thunder, hail, tornado

PYQs: 2022 (D) high thin clouds trap heat, low thick clouds reflect/cool: both statements ✘. 2019 (D) cirrus thinning + sulphate aerosols in stratosphere = geo-engineering to reduce global warming (cirrus trap long-wave; aerosols reflect sunlight). 2019 (B) no dew on cloudy night: clouds reflect back Earth's radiation. 2024 (a) water vapour decreases with altitude; not max at poles. ✅

Polar stratospheric clouds (PSCs)

→ 15–25 km, polar winter, below −78 °C
→ "Nacreous / mother-of-pearl"; bright at twilight
→ Surfaces for chlorine and bromine reactions = ozone destruction, Antarctic ozone hole

Cloud seeding (weather modification)

→ Agents: silver iodide, potassium iodide, salt (NaCl), dry ice = hygroscopic nuclei
→ Delivery: aircraft or ground generators (rockets, flares)
→ Uses: rain, snowpack, fog dispersal at airports, hail suppression, pollution control
→ Limits: needs moisture; variable, debated; silver iodide build-up; costly

Precipitation	How	Where
Convectional	heated air rises = cumulonimbus; short, intense, thunder	tropics 0–30°
Orographic	air forced up mountains; rain shadow on lee side	mountains
Frontal	warm over cooler: warm front long steady; cold front short intense	mid-latitudes 30–60°
Cyclonic	convection/frontal on a large scale around a low	cyclones

Thunderstorm

→ Stages: cumulus, mature, dissipating
→ Collisions of ice/water: lighter + at top, heavier − at bottom = voltage = lightning (excites N_2, O_2), thunder
→ Tall objects hit most: lightning conductors

Tornado

→ Violently rotating column from thunderstorm to ground
→ Over 300 km/h (long text: 300 miles/h)
→ Tornado Alley (Texas to Oklahoma, Kansas, Nebraska): warm moist Gulf air meets cold dry Canadian + warm dry desert air = supercells
→ Also Europe, Asia, Africa, Australia

Waterspout

→ Air + water droplets over water; less intense
→ Fair-weather (calm, little damage) vs tornadic (dangerous)

🎯 8 · AIR MASSES AND FRONTS PYQ 2020

Air mass: homogeneous temp + humidity, no vertical movement; surface to lower stratosphere; thousands of km; source = high-pressure, calm regions; jets carry them

🧑‍🤝‍🧑 Types (Tropical warm, Polar cold, continental dry, maritime moist)

→ mT: Amazon, SE Asia rain; heavy rain and thunderstorms on coasts, cyclones
→ cT: Sahara, central Australia; heatwaves, arid
→ mP: W Europe (UK, Ireland), Japan Sea coast (from Okhotsk) snow; cool, damp
→ cP: Siberia, Canadian Prairies; cold dry winters
→ cA: N Canada, Greenland, Scandinavia; frigid, polar vortex
→ mE (equatorial): Indonesia, Malaysia, W Africa (Gulf of Guinea); heavy rain

🗺️ Fronts: 3-D boundary of two contrasting air masses; mid-latitudes, rare in tropics/poles

→ Frontogenesis = formation (face-off); Frontolysis = dissipation
→ Temperate cyclones owe genesis to frontogenesis; bigger contrast = thinner front; cold fronts move faster

Occluded: cold front overtakes warm

→ Warm occlusion: air ahead of warm front cooler; clouds cumulus, stratocumulus, nimbostratus; moderate-heavy rain, thunder
→ Cold occlusion: air behind cold front colder; stratocumulus, nimbostratus, altostratus; rain/snow, cooler

→ Stationary = stalemate (light rain for days)

Cold front = Crash

→ (cumulonimbus, thunder, hail)
→ Warm front = Wet blanket (stratus/cirrostratus, steady drizzle, fog)
→ Occluded = cold overtakes warm.

🎯 9 · CYCLONES PYQ 2015 PYQ 2020

🗺️ Cyclone: low-pressure centre; winds counterclockwise in N, clockwise in S

→ Types: tropical, extratropical, subtropical
→ Names: Cyclone (Indian Ocean) · Hurricane (Atlantic) · Typhoon (W Pacific, S China Sea) · Willy-willy (W Australia)

	Tropical	Temperate (extra-tropical)
🏛️ Origin	Thermal	Dynamic (frontal)
Latitude	10–30°	35–65° (more in N. Hemisphere)
Forms over	warm seas only	land and sea
👥 Size	100–500 km	300–2,000 km

Rain	heavy, a few hours	slow, many days/weeks
Wind	100–250 km/h; wind, surge, rain damage	30–150 km/h; flood damage
Life	not more than a week	2–3 weeks
Fronts	absent	created by fronts
India	both coasts, east coast hot-spot	Western Disturbances

Tropical cyclone needs

→ Sea-surface ≥ 26.5 °C, moisture, low wind shear, pre-existing disturbance, Coriolis
→ Depression ≤ 62 km/h → cyclone ≥ 64 knots (74 mph)
→ Structure: eye (calm, clear), eyewall (worst), rainbands
→ Impacts: wind, rain, storm surge, tornadoes

Source conflict: cyclone sea temperature given as ≥ 26.5 °C in the notes and "> 27 °C" in the PYQ note.

Climate change

Tropical

→ Warmer seas = more intense, more Cat 4–5 (Arabian Sea, N Atlantic), fewer overall (N Atlantic slightly more active)
→ Poleward shift; slower, prolonged rain; rapid intensification; sea-level rise worsens surge

Temperate

→ Tracks move poleward (less cold air at low latitudes)
→ Gradient/jet changes alter intensity; more moisture = heavier rain, floods, landslides

PYQs: 2015 (B) no cyclones in S Atlantic / SE Pacific: ITCZ seldom occurs (also strong wind shear, cooler SST, no pre-existing vorticity). 2020 (C) only some cyclones develop an eye; eye is NOT 10 °C cooler (Erin's eye ~21 °F warmer; above 7.5 miles same as outside); jets in both hemispheres. ✅

🎯 10 · KÖPPEN AND CLIMATIC REGIONS PYQ 2011 PYQ 2024

AllBigCitiesDeserveExtraHalwa — A tropical humid · B dry · C warm temperate · D cold snow-forest · E polar · H highland. Second letters: f no dry season · w winter dry · s dry summer · m monsoon.

Handout conflict: Köppen group table labels Df = humid continental, Dw = subarctic; regional table labels Taiga = Df, Manchurian = Dw. Remember names and places, not D-codes. Rainfall numbers in the regional table (e.g. 150–1000 cm for Af) look scrambled: do not memorise.

Climate (code)	Where	Hooks
Equatorial (Af)	10° N–S: Amazon, Congo, Indonesia, PNG	24–27 °C, tiny range; 4 pm convectional rain; evergreen selvas, mahogany, rosewood, ebony; leached soil; slash-and-burn
Savannah (Aw)	10–30°, interiors: Sudan, Venezuela, Brazil, N Australia	hot rainy summer, dry winter; elephant grass, baobab; Masai; "Big Game country"
Tropical monsoon (Am)	10–30°, E margins: India, Vietnam	3 seasons; 200–250 cm; teak, sal; paddy
Hot desert (BWh)	W margins, subtropical high, cold currents: Sahara, Kalahari, Mojave, Atacama	xerophytes (waxy/thorny leaves, deep roots, succulents), date palms at oases; Bedouin, Tuareg, Bushmen
Cold desert (BWk)	interiors: Gobi, Patagonia	wide range, cold winters; bunchgrasses
📝 Steppe (BS)	45–60°, mid-continental: Prairies, Steppes, Downs	Chernozem; wheat/maize, ranching

Mediterranean (Cs)	30–45°, W margins: California, Chile, SW S Africa, SW Australia	hot dry summer, winter rain; citrus, olives, vineyards; cork oak, eucalyptus
👫 Marine West Coast (Cfb, British type)	45–60°, W margins: NW Europe, SW S America, NZ	mild, rain all year, small range; deciduous forest; dairy
China type (Cw/Cfa; Gulf, Natal)	30–45°, E margins: SE China, Japan	warm moist summer, cool dry winter; typhoons; most productive mid-latitude farming
Taiga/Siberian (Df)	50/60–70°: N Canada, Eurasia	–30 to –40 °C winters; conifers (softwood, pure stands); fur, lumbering
Manchurian/ Laurentian (Dw)	E margins, N Hemisphere only: Manchuria, Japan, NE America	cold dry winters, wet warm summers
Polar (ET, EF)	Greenland, N Canada, Alaska, Antarctica	summer < 10 °C; mosses, lichens; Eskimos, Lapps, Yakuts

Climate PYQ	Answer / rule
👫 2011 rainforest regrows slowly	Soil poor in nutrients: leached, nutrients stored in plants, rapid decay (thin humus)
2011 African–Eurasian desert belt	Subtropical high-pressure cells (not warm currents); rain shadow (Gobi), cold currents (Namib, Atacama) also cause deserts; hot deserts 20°–30°, W coasts
2012 savannah	Definite dry and wet season; 800–1,600 mm rain; less than monsoon
2015 clear morning, thunder, short rain	Equatorial: 26–28 °C, annual range as low as 3 °C, ≥ 2,000 mm, ITCZ two rainfall peaks
2018 desert leaf traits	Hard waxy, tiny, thorns: all three (succulents; cacti spines)
👫 2021 keeps savannah forest in check	Fire, grazing herbivores, seasonal rainfall (2, 3, 4); acacia; maize, millets, sorghum, cassava; groundnuts, cotton, E Africa coffee
👫 2021 fast litter decay, bare soil, epiphytes	Tropical rain forest (canopy 75 m+, 3 layers)
👫 2023 rainforest soil rich?	Statement I ✘ (leached); II ✔
🆕 2024 low annual/daily range, rain all year, 50–250 cm	Marine West Coast
👫 No year dust in subtropical/temperate	I ✔ (deserts: Sahara, Arabia, Central Asia, China; scarce vegetation, high wind); II ✘ (strong dry winds); equatorial humid/ forested, polar ice-covered

11 · TRAPS AND CONNECTIONS

Traps

- → jet streams only in NH ✘
- → land–sea contrast greater in summer ✘ (winter)
- → westerlies 30°N–60°S ✘
- → ozone reflects radio ✘
- → more moisture aloft = colder ✘
- → Gulf Stream is cold ✘
- → eye 10 °C cooler ✘
- → rainforest soil rich ✘.

Connects to

- → pressure belts and 23.5° tilt / 21 June → Geomorphology
- → Western disturbances, monsoon, Loo → Indian climate
- → ocean currents shape isotherms and deserts → Oceanography
- → climate regions → Economic Geography.

Recall drill: cover and answer

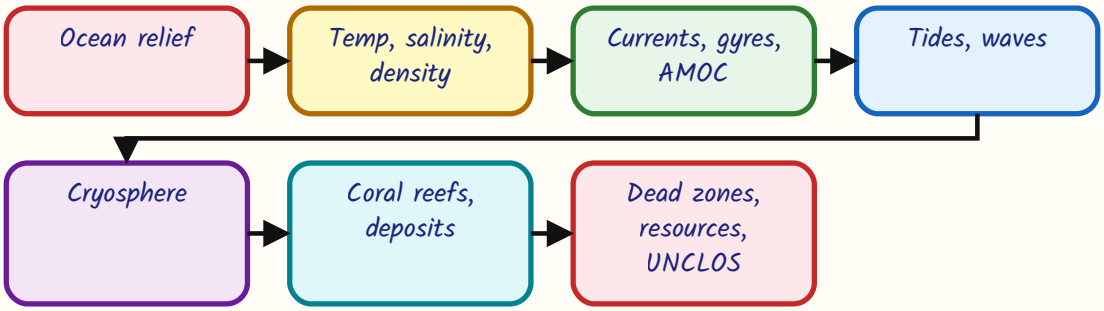
Troposphere height; lapse rate?	8–18 km (avg 13); 1 °C per 165 m (≈6.5 °C/km)	Ozone layer height; coldest layer; jet-flight layer?	20–25 km; Mesosphere; lower stratosphere
Kármán line; energy surplus/deficit boundary?	100 km; 40° N/S	Coriolis: zero, max, direction?	Equator; poles; right in N, left in S
Roaring / Furious / Shrieking?	40s / 50s / 60s (S. Hemisphere westerlies: less land)	Polar vs subtropical jet?	50°–60° (strong in winter) / 20°–30°; both hemispheres
Cold front vs warm front clouds?	Cold: cumulonimbus; Warm: stratus, cirrostratus	Cyclone sea temperature; why none in S Atlantic?	≥ 26.5 °C; ITCZ seldom occurs
Chinook: where, nickname?	E. Rockies; snow-eater	PSC altitude and temperature?	15–25 km; below −78 °C
Cloud seeding agents?	Silver iodide, potassium iodide, salt, dry ice	Main cause of desert belts?	Subtropical high-pressure cells (plus rain shadow, cold currents)

OCEANOGRAPHY

PYQ Prelims 2013 ×2, 2015 ×2, 2021 ×2 · Mains 2017 (oceanic salinity, 15m)

✎ *Ocean floor, temperature-salinity-density, currents, tides, waves, corals, deposits, resources and UNCLOS: 6 Prelims PYQs, heavy on current-affairs themes.*

At a glance: the Oceanography route



Pattern: the exam asks why: counter-current flows east, tides occur, fishing grounds sit where warm and cold currents meet. **Current-affairs heavy:** AMOC weakening, coral bleaching, dead zones, marine heatwaves, Southern Ocean carbon sink (good Mains material). Priority printed as MEDIUM–HIGH (6 Prelims, 1 Mains). No teacher ink. ⭐

1 · OCEANS AND RELIEF

Basics

- → Hydrosphere ~71% of Earth's surface
- → 5 oceans: Pacific, Atlantic, Indian, Arctic, Southern
- → Average depth 3,800 m vs lithosphere average height 840 m
- → Relief from tectonic, volcanic, erosional, depositional processes

Part (house)	Facts
Continental shelf (front porch)	gentle 1–2°; to ~200 m (up to 500 m); continental crust + sediment (sand, silt, clay); oil and gas, fisheries; North Sea, Gulf of Mexico; Great Barrier Reef (>2,300 km); width few km to hundreds of km
Continental slope (steep stairs)	3–6° (up to 25°); ~200–3,000 m; submarine canyons (Monterey, Hudson); Blake Plateau; deep-sea ecosystems (pressure, dark, cold)
Continental rise (landing)	gentle sediment apron at slope base; deep-sea fans, turbidity currents; Amazon rise

Abyssal/deep-sea plain (floor)	>2,000 m (2,000–6,000+); flat, fine sediment (clay, silt, sand); polymetallic nodules: Clarion–Clipperton Zone (Pacific, Hawaii–Mexico)
Trench (cellar)	>6,000 m; convergent boundaries, subduction, quakes; Mariana

Source overlap: slope runs to ~3,000 m but abyssal plain starts >2,000 m, and the notes also print "deep ocean 2–4 °C": treat depths as approximate.

Other features

→ Seamounts: ≥ 1,000 m underwater volcanoes (Davidson)

→ Guyots (table mounts): flat-topped seamounts, wave-eroded then submerged (Emperor Seamounts)

→ Hydrothermal vents: mineral-rich water; black smokers; life may have begun here

Term	Handout example
Strait (narrow waterway joining two waters)	Palk (India–Sri Lanka)
Isthmus (land strip joining landmasses)	Kra (Malay Peninsula–Asia)
Archipelago	Japanese (W Pacific)
Lake	Baikal (Russia), Wular (India)
Bay	Bengal
Cape	Comorin = Kanyakumari (southernmost tip of Peninsula); Good Hope (Africa)
Lagoon (behind barrier island/reef)	Chilika (brackish, Odisha)
Island	Lakshadweep (Arabian Sea)
Gulf (constricted entrance vs bay)	Kutch (Gujarat)
Peninsula (3 sides water)	Indian (Arabian Sea, Bay of Bengal, Indian Ocean)

2 · TEMPERATURE, SALINITY, DENSITY PYQ 2017

	Temperature	Salinity	Density
Key numbers	26.7 °C equatorial mean; 12.7 °C at 45° N/S; poles freezing; deep 2–4 °C; mixed layer to 200–400 m; thermocline = rapid fall	deep ocean ≈ 34.7 ppt (avg ~35); Great Salt Lake 220 · Dead Sea 240 · Lake Van 330	cold > warm; high salinity > low; deep > surface (pressure)
Pattern	warmest at equator; Indian Ocean warmest; N Hemisphere warmer (more land)	highest at ~20°–30° (high evaporation, little rain); lower at equator and poles	temperature matters more than salinity
Factors	latitude; currents (Gulf Stream, Kuroshio warm); atmospheric circulation; continentality; depth; season	evaporation/precipitation; rivers; sea-ice (freezing leaves salt); currents (Gulf Stream saline, Labrador cold fresher); winds	temperature, salinity, pressure

Sun + No Rain = Salty – Subtropics (hot, dry) saltiest; equator and poles diluted. Enclosed seas keep getting saltier: 220 – 240 – 330 = Great Salt Lake → Dead Sea → Lake Van. Continentality: sea ringed by land is warmer.

Mains 2017 (15m) salinity: Intro: dissolved salts, avg ~35 ppt, deep 34.7. Variation: latitude (subtropical high), evaporation vs rain, rivers, sea ice, currents, inland seas (220/240/330). Effects: density → thermohaline circulation, stratification (halocline), nutrients, habitats and productivity. Close: melting-ice freshwater now alters salinity and circulation. ✅

Ocean acidification

→ pH 8.2 (pre-industrial) → 8.11 (1985) → 8.04 (2024); ~30% more acidic in 200 yrs

→ More in Arctic than tropics (cold water absorbs more CO_2)

→ Causes: CO_2 dissolving, volcanoes, agricultural runoff

→ Effects: coral bleaching, weaker shells/exoskeletons, phytoplankton changes

Deep ocean (abyssal zone, below the thermocline): uniform and cold, about 2°C to 4°C; unaffected by solar heating (volcanic hot spots are the exception).

🎯 3 · FRESHWATER PYQ 2013 PYQ 2021

Freshwater share

→ Only ~3% of Earth's water is fresh (of freshwater, other ~70% after ice: groundwater ~30%, rivers/lakes ~1%); 2.5% unavailable (ice, atmosphere, soil, polluted, too deep); 0.5% available

→ Ice caps/glaciers ~68–69% · groundwater ~30% (largest active resource; equals a lake 180 m deep over the Earth) · rivers/lakes ~1%

Handout conflict: surface water (lakes, rivers, swamps) given as 0.3% in 2013 note and ~1% in 2021 note. 2013 (C): after ice, most freshwater is groundwater. 2021 (B): ice caps/glaciers hold more than groundwater; rivers/lakes not more.

🎯 4 · OCEAN CURRENTS PYQ 2013 PYQ 2021

🆕 **Currents:** continuous flow of large water volumes in a definite direction, all year; move heat tropics to poles

→ Surface: top ~400 m, only ~10% of ocean water; wind-driven

→ Deep (thermohaline): density-driven; down-welling at poles, up-welling in tropics

Circumpolar

→ Continuous loop round poles, isolate polar waters

→ Antarctic Circumpolar Current, West Wind Drift

→ Factors: latitude, wind, Coriolis, gravity, temperature, salinity, continental margins

Equatorial system

Trades push water E to W (+ Earth's spin) → *Coriolis: N and S Equatorial Currents* → *Water piles up in the west* → *Gravity: flows back east* → *Equatorial Counter Current (3–10° N)*

2015 (B): counter-current flows east due to convergence/pile-up of the two equatorial currents, not Earth's rotation directly. NECC 100–150 m deep, in Atlantic, Indian, Pacific. Counter Equatorial Current (CEC). Not the Equatorial Undercurrent (EUC) (eastward along equator, ~200 m deep in W Pacific, 100 m in E Pacific). ✅

West = Warm & Wild – Western boundary currents (Gulf Stream, Kuroshio, Brazil, East Australian, Agulhas): warm, narrow, fast, strong. East = cold & Easygoing: Eastern boundary (Canary, Benguela, California, Peru, West Australian): cold, broad, slow, nutrient upwelling.

Gyres (wind + Coriolis + coastlines): N Hemisphere clockwise, S anticlockwise

→ N Atlantic (clockwise): Gulf Stream, N Atlantic Current, Canary, N Equatorial; Sargasso Sea in middle = only sea without a coastline (Sargassum weed)

→ S Atlantic (anti): Brazil, Benguela, S Equatorial

→ N Pacific (clockwise): Kuroshio, N Pacific, California

→ S Pacific (anti): East Australian, Peru, S Equatorial

→ Indian (anti): Agulhas, West Australian, S Equatorial

2021 (C), both right: tropics: west of ocean warmer (trade winds carry warm water west); temperate: east warmer (westerlies carry warm equatorial water to continents' west coasts). ✅

Water masses: big homogeneous bodies with own temp, salinity, density from a source region; keep properties far

→ Move by gravity, density, wind stress, Earth's rotation (3-D); backbone of global circulation

→ Set habitat: warm clear = coral; cold nutrient-rich = kelp, high phytoplankton and productivity; deep = specialised life

AMOC (Atlantic Meridional Overturning Circulation)

→ Warm salty surface water north (Gulf Stream) → cools, **sinks** in N Atlantic = **NADW** → flows south at depth

→ Regulates Atlantic circulation of heat, climate, weather, rain (also written AOC)

→ **Weaker than any time in past millennium** (warming; melting-ice and rain freshwater disrupt sinking)

→ If weak: cooler W Europe, more extreme events (heat, cold, storms), stronger US east-coast hurricanes, less CO_2 uptake, sea-level rise, fisheries shifts

Southern Ocean carbon anomaly

→ Absorbs **~40%** of CO_2 taken up by oceans (stronger sink than models)

→ Glacier-melt + rain freshwater = light "lid", blocks mixing, traps carbon-rich water **100–200 m** down

→ "**Fleeting reprieve**": lid weakens = CO_2 release, faster warming

NEW Ekman current

→ Wind-driven; water ~**45° right** (N) / left (S) of wind

→ Layers turn = **Ekman spiral**

→ Gives **coastal upwelling** (nutrients, fisheries), heat redistribution, El Niño/La Niña links

NEW Climate change on currents

→ Weaker thermal/salinity gradients = thermohaline slowdown

→ Less upwelling nutrients; less oxygen in downwelling zones; shifted winds; ice melting from below; biodiversity decline

Marine heatwaves: prolonged unusually warm sea

→ Episodes **doubled since 1982**; warming rate nearly doubled since **2005** (Copernicus)

→ Causes: climate change, high pressure, shifting currents, El Niño, feedback loops

→ Effects: coral bleaching, ecosystem disruption, current changes

🎯 5 · TIDES AND WAVES PYQ 2015

Tide: rise (high/flood) and fall (low/ebb) of seawater

→ Cause: gravity of **Moon (main)** and **Sun** + **centrifugal force** of Earth = bulges near and far side

Timing

→ Most places **semi-diurnal**: 2 highs, 2 lows a day; same tide every **12 h 26 min**

→ Moon takes **24 h 52 min** to return over a tide centre

→ **Diurnal**: Gulf of Mexico, Gulf of Thailand

→ Height depends on: Sun-Moon-Earth distances, **funnel-shaped bays/estuaries**, local winds, pressure (low pressure raises sea)

→ Use: nutrient circulation, temperature moderation, flushing pollution

2015 (D): tides = Sun's gravity + Moon's gravity + Earth's centrifugal force (all three). ✅

SPrung Straight – Spring tide = syzygy: new and full moon, Sun–Moon–Earth aligned; highest highs, lowest lows, ~20% higher; twice a month; nothing to do with the season.

Nearly right-angle – Neap tide = quadrature: quarter moons, forces oppose; lower highs, higher lows.

Tidal bore

→ Rare: incoming tide makes wave travelling upstream in narrow estuary/shallow river; absent at neaps

→ India: Hooghly; world's biggest: Qiantang (China)

Climate change on tides

→ Higher sea level = larger areas tidal

→ May amplify tidal ranges; threatens mangroves

📜 Waves: energy moves, not water; particles move in small circles

→ Wind gives energy; gravity pulls crests down; falling water lifts troughs

Tsunami: very long seismic sea waves

→ Causes: earthquakes at subduction zones (size ∝ vertical shift), volcanoes, landslides, meteorite impact, nuclear tests

→ 500–1,000 km/h in deep water, < 1 m high (invisible); slows and grows near coast; many waves

→ Warning: sea-level change, odd ocean behaviour

Rogue (freak) waves

→ Unusually large, spontaneous, even in calm seas

→ Constructive interference: wave systems in phase (rare, random)

→ Focusing by currents (Agulhas Current opposing waves) and topography (ridges, shelves)

Seiche: standing wave in lake/bay/reservoir

→ Causes: wind, pressure change, quakes/landslides

→ Resonance with natural rhythm grows waves

→ Basin size, shape, depth decide size and duration

Tsunami, human cause: nuclear explosions, e.g. US nuclear testing at the Marshall Islands (1940s and 1950s). 💡

6 · CRYOSPHERE

Cryosphere = Earth's ice

→ 69% of freshwater; albedo; stores methane/CO_2; feeds rivers; affects circulation; GLOFs in Himalaya, Alps, Andes

Melting effects

→ Thermohaline change; winds/jet streams
→ Freshwater (short-term up, long-term down)
→ Lower albedo; GHG release
→ Heavy elements and pathogens released
→ Isostatic rebound and volcanic activity

Arctic

→ Weaker polar vortex = cold spells in N America/Europe; more temperate storms
→ New shipping routes (thick ice drifts into Northwest Passage); resource access
→ Hundreds of Greenland lakes burst since 2008; trapped mercury leaks (reaches Arctic by air and ocean circulation)

Antarctica

→ Jet-stream weakening; winter heatwaves (mid-July −25 to −30 °C instead of −50 to −60 °C); wildlife and research

→ Oceans: acidification; SST rise 0.06 °C/decade (1980s) → 0.27 °C/decade now; circulation and nutrient changes

Climate change and cities: a quarter of the world's urban population faces a 0.5°C rise by 2040: health risks, lower labour productivity, more cooling demand, more extreme weather.

7 · CORAL REEFS

Corals: marine invertebrates, colonies of polyps; secrete calcium carbonate; "rainforests of the sea"

→ Symbiotic algae zooxanthellae give energy by photosynthesis

Kinds

→ Hard (stony, reef-builders): Acropora, Porites
→ Soft: octocorals, sea fans
→ Cold-water: deep, polar/temperate
→ Black corals: 30–300+ m, Gulf of Mannar

Warm, Clear, Shallow, Salt-stable, Light-snack, Mild-waves — 20–29 °C · clear sunlit water · <50 m deep · salinity 32–40 ppt · low nutrients (oligotrophic; runoff feeds algae that choke them) · moderate waves

Fringing = Friend — Fringing = Friend, hugs shore (boat channels) · Barrier = Bodyguard, stands off, lagoon between · Atoll = Wedding ring around lagoon (sinking volcanic island/seamount)

Bleaching: corals expel zooxanthellae (white; recover or die); more disease

→ Causes: heat, acidification, pollution, UV, tectonic exposure/drowning, overfishing (fish keep reefs clean), disease, tourism
→ Great Barrier Reef: 4th mass bleaching ongoing since early 2023
→ India: bleaching in Gulf of Mannar and Andamans; Lakshadweep, March 2024
→ Climate: acidification (harder to build skeletons); stronger storms (damage, smothering sediment)

Source conflict: 4th mass bleaching start printed as January 2023 once and February 2023 once.

8 · OCEANIC DEPOSITS

Land, Life, Liquid, Cosmos — Terrigenous (land: rivers, glaciers, wind; margins) · Biogenous (organisms): calcareous ooze $CaCO_3$ (warmer, shallower), siliceous ooze (diatoms, radiolarians; colder, deeper) · Hydrogenous (chemical): manganese nodules, phosphorite · Cosmogenous (meteorite dust)

🎯 9 · DEAD ZONES PYQ 2013

🧍‍♂️ Dead zones (hypoxic): oxygen too low for most life

→ Cause: nutrient pollution (N, P: farm runoff, sewage, industry, even wind-blown dust: Madagascar bloom)

→ Chain: algal bloom → dies, sinks → bacteria use up oxygen

→ Effects: marine life suffocates, fisheries/tourism lose, toxins and bacteria contaminate seafood

🎯 Examples

→ N Gulf of Mexico (Mississippi runoff, annual)

→ Baltic Sea (chronic, nutrient pollution)

→ Black Sea (natural, poor circulation)

→ Fix: cut nutrient pollution (farming rules, wastewater treatment), monitoring, public awareness

🎯 10 · OCEAN RESOURCES AND FISHING GROUNDS PYQ 2013

Resources

Biotic

→ Fish: tuna, salmon (Pacific); cod (N Atlantic); sardines (Mediterranean, Peru)

→ Shellfish; seaweed (Japan nori); marine pharma; pearls

Abiotic

→ Offshore oil/gas: North Sea, Gulf of Mexico, Persian Gulf

→ Tidal, OTEC, wave, offshore wind

→ Dissolved salts (NaCl); manganese nodules (source of "dark oxygen"); phosphorite (Morocco, Peru, USA)

Services

→ Absorb ~90% of excess heat, 25–30% of human CO_2 (blue carbon: mangroves, seagrass)

→ Phytoplankton make 50–60% of Earth's oxygen

2013 (C) fishing grounds: best where warm and cold oceanic currents meet (plankton mixing): Newfoundland banks, Sea of Japan. Helpers: coastal upwelling, indented coasts, shelf food from erosion + sunlight, nearby forests for boats, rugged land pushing people to fishing, temperate water below 20 °C. ✅

📜 11 · LAW OF THE SEA: UNCLOS 1982

12 - 24 - 200 — Territorial sea 12 nm (22.2 km) · Contiguous zone 24 nm (44.4 km; customs, fiscal, immigration, sanitary laws) · EEZ 200 nm (370.4 km; sovereign rights over resources) · High seas beyond (navigation, overflight, fishing, research). A dozen → a day → a double century.

UNCLOS (United Nations Convention on the Law of the Sea), adopted 1982: main legal framework on maritime boundaries, navigation, resources, protection of marine environment. 💡

12 · TRAPS AND CONNECTIONS

Traps

→ spring tide = in spring ✘
→ counter-current due to Earth's rotation ✘
→ salinity highest at equator ✘
→ density driven more by salinity ✘ (temperature)
→ tsunamis tall in deep ocean ✘
→ rivers/lakes hold most freshwater after ice ✘ (groundwater)
→ atolls form on continents ✘ (sinking volcanic islands).

Connects to

→ trenches, guyots, hotspots → Geomorphology
→ Coriolis, trade winds, westerlies, cyclones, El Niño → Climatology/ Indian climate
→ resources, ports, fisheries → Economic Geography
→ Palk Strait, Chilika, Gulf of Kutch, Cape Comorin → Indian Geography.

Recall drill: cover and answer

Average ocean depth; shelf and slope gradient?	3,800 m; 1–2° / 3–6°	Trench depth; example?	> 6,000 m; Mariana
Mean temperature equator vs 45°?	26.7 °C / 12.7 °C	Saltiest listed water bodies?	Lake Van 330 > Dead Sea 240 > Great Salt Lake 220
Ocean pH now?	8.04 (2024); pre-industrial 8.2	Why does the counter-current flow east?	Convergence/pile-up of the two equatorial currents
Only sea with no coastline?	Sargasso Sea (N Atlantic gyre)	Tide repeats after; spring tide when?	12 h 26 min; new and full moon (syzygy)
Tidal bore in India; world's biggest?	Hooghly; Qiantang (China)	Tsunami speed in deep ocean?	500–1,000 km/h, under 1 m high
Coral temperature / depth / salinity?	20–29 °C / <50 m / 32–40 ppt	UNCLOS zones?	12 · 24 · 200 nm

INDIAN GEOGRAPHY

PYQ Prelims 40 PYQs: physiography 15 (2012, 2014 x2, 2015, 2016, 2017 x2, 2018 x2, 2020, 2022, 2023 + 3 undated) · drainage 12 (2011, 2013, 2014, 2015, 2016 x2, 2017, 2021 x2, 2022, 2023 + 1 undated) · climate 3 (2011, 2017 + 1 undated) · forests and soils 10 (2011, 2013, 2015 x3, 2016, 2021, 2023 x2 + 1 undated) · Mains none quoted

✎ *Location, six physiographic units, rivers, climate and monsoon, forests and soils: the largest PYQ bank of the six chapters (40 Prelims), all map-and-list questions.*

At a glance: the Indian Geography route



Pattern: "which pairs / rivers / peaks / states?": tributaries of Godavari and Brahmaputra, hills of which range, which state has what, which forest has which tree. Place every fact on a map. Mains hooks (none quoted): Western Disturbances, cloudbursts, urban flooding, ELLA, river-linking, mangroves, social forestry. No teacher ink: priority = PYQ count. Image-only pages (peaks, glaciers, passes, waterfalls, lakes) are kept below as tables. ⭐

1 · LOCATION OF INDIA PYQ 2015 PYQ 2018 PYQ 2017

Fact	Number
Area	32,87,263 km² · 7th largest · 2.4% of world area
Extent	N–S (Indira Col, Kashmir → Kanyakumari) 3,214 km · E–W (Rann of Kutch → Arunachal) 2,933 km
Latitude / longitude	8°4′N (6°45′ for islands) to 37°6′N · 68°7′E to 97°25′E · wholly Northern Hemisphere
Southernmost point	Indira Point (Pygmalion Point), 6°45′N
Time	1° longitude = 4 min · E–W gap ≈ 2 hrs (~30 × 4) · Chai Bagan time = 1 hr ahead of IST (British, NE India, to raise productivity)
Tropic of Cancer	splits India in two; area north ≈ 2× south · 8 states: Gujarat, Rajasthan, MP, CG, Jharkhand, West Bengal, Tripura, Mizoram
Borders	land 15,106.7 km through 17 states · coastline 11,098 km, 13 states/UTs · longest = Bangladesh · shortest = Afghanistan (Wakhan)
No border / no coast	Telangana, MP, CG, Jharkhand, Delhi, Chandigarh, Haryana; all others = frontline states (border management)

U C My Old Aunt — Standard Meridian 82°30′E passes through 5 states: Uttar Pradesh · Chhattisgarh · Madhya Pradesh · Odisha · Andhra Pradesh (handout mnemonic).

Neighbours (clockwise) → Pakistan · Afghanistan · China · Nepal · Bhutan · Bangladesh · Myanmar · Sri Lanka · Maldives · Indonesia

Neighbour	Bordering Indian states/UTs
Pakistan	J&K, Ladakh, Punjab, Rajasthan, Gujarat
Afghanistan	Ladakh
China	Ladakh, HP, Uttarakhand, Sikkim, Arunachal
Nepal	UP, Uttarakhand, Bihar, Sikkim, West Bengal
Bhutan	Assam, Sikkim, Arunachal, West Bengal
Bangladesh	West Bengal, Assam, Meghalaya, Mizoram, Tripura
Myanmar	Arunachal, Nagaland, Manipur, Mizoram

Source doubt: coastline is 11,098 km here, but the cyclone note says "over 7,500 km" (as printed). Use 11,098 km for coastline.

2 · PHYSIOGRAPHY: SIX UNITS PYQ 2012 PYQ 2014 PYQ 2023

My Parents Do Play Cricket Indoors — handout's six units top to bottom: Mountains · Plains · Desert · Plateau · Coasts · Islands (Northern and North-eastern Mountains · Northern Plains · Indian Desert · Peninsular Plateau · Coastal Plains · Islands).

Northern and NE mountains

→ Origin: Indian + Eurasian plates converge (Tethys closes); Tethys sediments fold

→ Multiple upliftments = parallel ranges; steep south, gentle north (hogback); ridge–valley

→ 2,500 km long; width 500 km (Kashmir) → 200 km (Arunachal); 13 states/UTs

→ Main Himalaya: Nanga Parbat (Indus gorge) to Namche Barwa (Brahmaputra gorge) = two syntaxial (knee-joint) bends; then Purvanchals

→ Transhumance: seasonal herding to bugyals: Bakarwals, Gaddis, Bhotiyas, Drokpas

Belt	Also called	Notes
Trans-Himalaya	Karakoram, Ladakh, Zaskar	highest, cold dry · most glaciers in Karakoram · K2 (Godwin-Austen) · Ladakh = highest Indian plateau · Zaskar south of Indus · Kailash entirely outside India
Great Himalaya	Himadri, inner	avg 6,100 m; tallest peaks

Middle Himalaya	Himachal, Lesser	3,500–4,500 m · Pir Panjal, Dhauladhar (southernmost HP range), Kumaon, Mussoorie, Mahabharat · Karewas (clay–silt–sand–gravel of Pleistocene lakes, between Pir Panjal and Zaskar) → saffron (zafran), walnuts, almonds · Bugyals
Outer Himalaya	Shiwaliks	lowest; nearly unbroken, gap of 80–90 km at Teesta valley

Foothills (top to bottom)

→ Bhabar (stones, rivers vanish) → Terai (swamp, rivers reappear) → plains

- **Duns (W)**
  - → valleys between Shiwaliks and Himachals (HP, Uttarakhand); longitudinal
  - → Dehradun, Patli Dun, Kota Dun
- **Duars (NE)**
  - → floodplain foothills north of Brahmaputra valley (Assam, Arunachal)
  - → alluvial, fertile, dense vegetation
- **Chos / Choes / Raos**
  - → seasonal Shiwalik streams, NW Punjab foothills; sandy beds
  - → less fertile than Duars; need irrigation

Purvanchals

→ crescent, convex to west; perpendicular to Himalaya; rough, forested = hard to develop

→ Mishmi, Patkai Bum, Naga, Kohima, Manipur hills (Loktak), Barail, Mizo (Lushai)

→ Barail: separates Manipur and Naga hills; divides Brahmaputra and Barak basins

Regional Himalaya	Limits	Peaks / valleys (as printed)
👫 Punjab / Kashmir / Himachal (west)	Indus → Sutlej	Nanga Parbat 8,126 m · K2 8,611 m · Dhaulagiri I 8,167 m · Kashmir, Kullu, Kangra valleys
Kumaon (central)	Sutlej → Kali	Nanda Devi 7,816 m (highest entirely in India) · Nanda Kot 7,818 m · Trishul 7,120 m · Valley of Flowers (UNESCO) · Ganga source
Nepal (eastern)	Kali → Teesta	Everest 8,848 m · Kanchenjunga 8,586 m (3rd highest) · Lhotse 8,516 m · Annapurna I 8,091 m
👫 Assam (easternmost)	Teesta → Arunachal / Purvanchals	Namche Barwa 7,756 m · Manas NP (UNESCO)

Source conflicts: Nanda Devi 7,816 m vs Nanda Kot 7,818 m (so Nanda Devi is "highest entirely in India" only by that label; the 2022 PYQ note says Nanda Devi 7,817 m). Kamet is printed as "highest of Uttarakhand" though Nanda Devi (Uttarakhand) is higher. Learn ~7,816 and rely on the label "highest entirely in India".

Region	Peak: what the handout says
Trans / Himadri	Nanga Parbat (Ladakh Himadri, >8,000 m, syntaxial bend) · K2/Godwin-Austen (Karakoram, 2nd highest in world) · Broad Peak, Gasherbrum I and II (Gasherbrum group, all >8,000 m) · Nun (J&K), Kun (Ladakh): Himadri
HP	Reo Purgyil: highest of HP, Zaskar Range
Uttarakhand	Satopanth (lake at foot) · Kamet: highest of Uttarakhand and of Zaskar · Trishul: 3 peaks like a trident · Nanda Devi: Greater Himalaya, UNESCO World Heritage (with its NP)
Shiwalik peaks	Karoh: highest of Haryana · Amsot: highest of UP · Someshwar: highest of Bihar
👫 East Himalaya	Kanchenjunga: highest Himalayan peak in India, Sikkim, biosphere reserve + NP, India's first UNESCO mixed heritage site · Sandakphu: highest of West Bengal (Darjeeling, tea) · Tiger Hill (Himachal range) · Kangto: highest of Arunachal
Purvanchal / NE	Dapha Bum: highest of Purvanchals (Mishmi) · Saramati: Nagaland · Mt Esii/Iso: Manipur · Phawngpui (Blue Mountain): Mizoram · Betlingchhip (Thaidawr Tlang): Tripura · Nokrek: Garo, NP + biosphere reserve · Shillong Peak: Khasi, highest of Meghalaya · Tumjang/Sielkal/Thingtuibung: Barail, highest of Assam

Peninsula (north → south)	Parasnath: highest of Jharkhand, Jain pilgrimage, Chotanagpur · Guru Shikhar: highest of Aravallis and Rajasthan · Gorakhnath: highest of Gujarat, Girnar hills · Dhupgarh: highest of Satpuras and MP · Sadbhawna Shikhar (Goodwill Peak): highest of Vindhyas · Kalsubai: highest of Maharashtra (W. Ghats) · Mahabaleshwar: origin region of Krishna · Sonsogor: highest of Goa (Maharashtra–Goa–Karnataka trijunction)
Western Ghats south	Mullayanagiri: highest of Karnataka · Kudremukh (minerals, tiger reserve, NP) · Brahmagiri: source of Kaveri · Doddabetta: highest of Tamil Nadu and Nilgiris · Anaimudi: highest of Western Ghats and Kerala · Agasthyamalai: TN–Kerala border
Others	Saddle Peak: highest of Andaman and Nicobar · Mt Harriet (now Mt Manipur) · Bailadila: Chhattisgarh, Deccan · Gaurlata: highest of Chhattisgarh, Central Highlands · Doligutta: highest of Telangana · Mahendragiri: Eastern Ghats · peak No. 45: highest of Odisha, Eastern Ghats (name blank in handout) · Arma Konda / Jindhagada: highest of Eastern Ghats and Andhra Pradesh

Peak pegs — Trishul = trident (3 peaks) · Kamet = the "Kamet-ition winner" of Uttarakhand · Doddabetta = "Dodo-big-bat" of the Nilgiris · Anaimudi = "Anna" is tallest in the Ghats' family (Kerala) · Kalsubai = Maharashtra's "kal-subah" (tomorrow morning) climb.

Glaciers (colour = basin on handout map)

- **Tarim** → Yarkant, Skarmi
- **Indus system**
  - → Hispar, Chogo, Basha, Lungma, Rupal (Nanga Parbat)
  - → Hunza, Batura, Khurdopin, Biafo, Baltoro → Shigar
  - → Siachen, Rimo → Nubra/Shyok; Chong Kumdan, Drang Drung → Shyok/Stod
  - → Shafat · Sonapani → Kutle Nala · Barashigri → Chandra · Suru (Nanga Parbat side)
- **Ganga basin**
  - → Gangotri → Ganga · Chorabari → Mandakini · Milam → Gori Ganga · Pindari → Pindari · Kamet · Dhauliganga
  - → Yamunotri and Bandarpunch → Yamuna
- **Brahmaputra basin** → Zemu → Teesta

Pass	State/UT	Hook
Aghil	Ladakh	Karakoram; historic trade; military
Karakoram	Ladakh	ancient silk route
Khardung La	Ladakh	connects Leh and Siachen
Mig La	Ladakh	printed as "world's highest motorable pass"
Chang La	Ladakh	connects Pangong Tso to rest of Ladakh
Umling La	Ladakh	printed as 2nd highest motorable road
Zoji La	Ladakh–J&K	connects Ladakh and Kashmir valley; snow-blocked
Burzil La	J&K (now PoK)	trade
Banihal	J&K	Jawahar Tunnel (NH44)
Shinku La	HP	NPD route (Nimmu–Padum–Darcha), all-weather to Ladakh
Baralacha La	HP	Leh–Manali highway
Rohtang	HP	Atal Tunnel under it; Leh connectivity
Shipki La	HP	India–China border; Sutlej enters India
Mana	Uttarakhand	highest motorable pass of Uttarakhand
Niti	Uttarakhand	northernmost pass of Uttarakhand
Lipulekh	Uttarakhand	Nepal claims it, handout says India's; Mansarovar Yatra
Nathu La	Sikkim	Indo-China trade; Mansarovar Yatra
Jelep La	Sikkim	connects India and Bhutan
Bum La	Arunachal	Dalai Lama entered India 1959

Sela	Arunachal	Sela Tunnel; connects Tawang
Bomdila	Arunachal	Tawang connectivity; used in 1962 war
Yongyap	Arunachal	near entry of Dihang; British Stilwell Road
Diphu	Arunachal	tri-junction India–China–Myanmar
Chaukan, Pangsau	Arunachal	Indo-Myanmar; Stilwell Road passes Pangsau
Haldighati	Rajasthan	Aravallis; Maharana Pratap vs Mughals
Asirgarh / Burhanpur	MP	Satpuras; Asirgarh fort; key to the Deccan (north–south link)
Thalghat	Maharashtra	Mumbai–Nashik highway (Western Ghats)
Bhorghat	Maharashtra	Mumbai–Pune highway
Chorla Ghat	Goa	near Goa–Maharashtra–Karnataka junction
Palghat (Palakkad)	Kerala–TN	widest gap in Western Ghats; monsoon gateway to TN; between Nilgiris (north) and Annamalai (south)
Shenkotta	Kerala–TN	Western Ghats gap

NORTHERN PLAINS

Northern plains

→ Great depression south of Himalaya filled by Indus, Ganga, Brahmaputra alluvium; 3,200 km W to E

→ Flat, slight slope north to south; one of the most densely populated regions; rivers give irrigation, transport, hydropower; Delhi, Varanasi, Patna

→ Climate: semi-arid (west) → humid subtropical (east)

→ Problems: soil erosion, water pollution, groundwater depletion

📜 Four sections

→ Rajasthan plain: bagar, saline lakes

→ Punjab–Haryana: Bari and Bist doab, canals

→ Ganga plain: Yamuna eastward; Rohilkhand, Awadh, Mithila, Magadha, Bengal; delta

→ Brahmaputra plain: chars / point bars, floods, tea, Sundarbans delta

Term	Meaning
Bhabar	alluvial cones at foothills; rivers vanish
Terai	swampy; rivers re-emerge
Bhangar	older coarse alluvium away from channel; kankar nodules; less fertile
Khadar	newer alluvium near rivers; renewed yearly; very fertile
Reh / Kallar	barren saline patches (UP, Haryana); spread by over-irrigation
Bhur	elevated land along the Ganga (Ganga–Yamuna doab)
Bagar	semi-arid plain from Aravalli foothills, Rajasthan; Rohi = fertile patches in it
Doab	between two rivers: Bari, Bist, Rechna, Chaj
Betlands	Khadar plains of Punjab

Old Bhangar, New Khadar – Bhang-ar = bhang, old and coarse, kankar. Khad = manure: renewed every year, super fertile.

INDIAN DESERT: THAR

Thar (Great Indian Desert; local Marusthali)

→ Cause: sub-tropical high pressure + rain shadow of Aravallis; extension of Rajasthan plains

→ ~2,00,000 km²; 85% India, 15% Pakistan; Rajasthan, Gujarat, Punjab, Haryana

→ Borders: Indus plain (west) · Punjab plain (north, NE) · Aravalli (SE) · Rann of Kutch (south); natural India–Pakistan border

→ Features: dunes (some >150 m), sandy plains (Indira Gandhi Canal), low barren hills bhakars, seasonal "rivulets", saline lakes Sambhar, Didwana

→ Source of salt, limestone, gypsum

PENINSULAR PLATEAU PYQ 2023

Peninsular plateau

→ Oldest and largest unit; irregular triangle; broken from Gondwanaland; rich in metallic minerals; includes Deccan Traps

→ Meghalaya plateau cut off by Rajmahal–Garo (Malda) Gap / Ganga Suture Zone during northward drift

→ Two halves: Central Highlands and Deccan Plateau, split by the Narmada rift

Part	Where / what
Mewar / Marwar	NW, Rajasthan, near Aravallis; scrub
Malwa	Aravallis (NE) to Bundelkhand (E) to Vindhyas (S); Chambal badlands; black soil, cotton
Bundelkhand	between Vindhyas and Ganga plain (MP, UP); sal, teak, kheria
Baghelkhand	north Deccan, MP/CG; forests; coal
Chotanagpur (Eastern Highlands)	Hazaribagh, Ranchi, Rajmahal Hills; coal, mica; sal
Meghalaya (NE tip)	Garo, Khasi, Jaintia, Mikir, Rengma; Mawsynram wettest on Earth
Deccan	largest, central; avg 600 m; Deccan Traps; Chhattisgarh plateau and plains (Dandakaranya, Bastar; Mahanadi rice bowl)

Range	Facts (as printed)
Aravalli	relict of world's oldest mountains · NE edge = Delhi Ridge · highest Guru Shikhar · Gujarat, Rajasthan, Haryana, Delhi
Vindhyas	Jobat (Gujarat) → Sasaram (Bihar); east as Bharner and Kaimur · Gujarat, MP, UP, CG, Bihar
Satpuras	higher than Vindhyas · Rajpipla, Mahadev, Maikala · Dhupgarh highest · Gujarat, Maharashtra, MP, CG
Western Ghats (Sahyadris)	Satmala, Harishchandra, Balaghat, Ajanta, Baba Budan, Nilgiri, Annamalai, Cardamom, Palani · climatic AND drainage divide · 6 states: Gujarat, Maharashtra, Goa, Karnataka, Kerala, TN
Eastern Ghats	Garjhat, Mahendragiri, Eramalla, Nallamalla, Seshachalam, Velikonda, Palkonda, Javadi, Shevroy, Kalrayan, Biligiri Rangan · highly dissected by rivers · highest Jindhagada (Araku) · Odisha, Telangana, AP, Karnataka, TN
Meeting point	Eastern and Western Ghats meet at the Nilgiris

Thal-Bhor-Chorla-Pal-Shenkotta — Western Ghats gaps north → south: Thalghat · Bhorghat · Chorla · Palghat · Shenkotta. Ghats compare: Western = Wall (continuous, high, wet windward, divide); Eastern = Entry-gate pieces (broken by rivers, low, drier, scrub).

COASTAL PLAINS PYQ 2022

Coast	Facts
Western plain	mixed shoreline (emergence, submergence, stable)
Kachchh	emergence (tectonic + depositional) → Rann of Kachchh salt marsh; Banas and Luni inland drainage; Banni grassland (one of India's largest)
Kathiawar (Saurashtra)	Deccan Trap black soil; Gir: only Asiatic lions; Narmada, Tapi form estuaries
Konkan (Daman–Goa)	submergence; wider shelf; sea cliffs, marine erosion
Karnataka plain (Goa–Mangalore)	submergence
Malabar (Mangalore–Kanyakumari)	stable (deposition); kayals = lakes, lagoons, backwaters
Eastern plain	Subarnarekha → Kanyakumari; emergence by deposition (long rivers into Bay of Bengal); Northern Circars (Mahanadi–Krishna) + Coromandel (south of Krishna)
Utkal / Andhra / TN	Chilika (biggest lake, Utkal) · Kolleru (once lagoon, now inland) + Krishna–Godavari delta · plain ends at Pulicat → Kanyakumari

Eastern and western coasts meet at Kanyakumari. ⭐

ISLANDS PYQ 2014 PYQ 2018

Islands	Facts
Andaman and Nicobar	500+ islands, 6°–13°N, 92°–93°E, Bay of Bengal · extension of Arakan Yoma (plate collision; a few volcanic) · Saddle Peak highest · 10° Channel separates Andaman and Nicobar · 6° Channel: Great Nicobar–Sumatra · Barren Island = only active lava volcano in South Asia (Indian plate under Burmese); Narcondam extinct · Baratang = only active mud volcano · Havelock (Swaraj Dweep) largest
Lakshadweep	36 coral islands and reefs, ~400 km off Kerala, Arabian Sea · coral atolls on submerged volcanic Lakshadweep–Chagos Ridge; low, sandy, no rivers or permanent freshwater · Andrott largest · Minicoy (Maliku) southernmost, split by 9° Channel

10 – 9 – 6 — Channels = own number: 10° (Andaman | Nicobar) · 9° (Lakshadweep | Minicoy) · 6° (Great Nicobar | Sumatra): count down from the north.

🎯 PYQ BANK: PHYSIOGRAPHY

Year	Asked	Answer / trap
2012	Young fold mountains: deep gorges, U-turn courses, parallel ranges, steep gradients	D, all four (folding of crust; highest ranges; steep slopes = recent; old fold mountains are lower)
2014	Pairs: Cardamom–Coromandel? Kaimur–Konkan? Mahadeo–Central India? Mikir–NE?	C: Mahadeo and Mikir only. Kaimur = eastern Vindhya (~483 km, Katangi near Jabalpur → Sasaram, Rohtas; Rewa, Mirzapur). Cardamom (Yela Mala) = south W. Ghats, SE Kerala + SW TN, UNESCO
2014	Ten Degree Channel separates	A Andaman and Nicobar
2015	Easternmost and westernmost states	D Arunachal and Gujarat
2016	Place–region: Khajuraho–Bundelkhand ✔, Tirupati–Rayalaseema ✔; Bodhgaya–Baghelkhand ✘, Shirdi–Vidarbha ✘, Nashik–Malwa ✘	C (2 and 5). Malwa = W MP + SE Rajasthan; Baghelkhand = NE MP + small W UP; Vidarbha = E Maharashtra (Nagpur, Amravati divisions)
2017	Kohima → Kottayam: minimum states	B = 7 (Nagaland, Assam, WB, Odisha, AP, TN, Kerala)
2017	Himalaya in 5 states only? W. Ghats in 5 only? Pulicat in 2 only?	B: only 3 right. Himalaya >5 states (J&K, HP, Uttarakhand, Sikkim, Arunachal + hill parts of Assam, WB); W. Ghats 6 (30–50 km inland); Pulicat = AP (40%) + TN (60%), 2nd largest brackish lagoon after Chilika, Sriharikota barrier island (Satish Dhawan Space Centre)
2018	Barren Island: active ✔; ~140 km east of Great Nicobar ✘; last erupted 1991, inactive since ✘	D (1 only). Erupting again since 1991, last as recent as 2022
2018	City on longitude nearest Delhi	A Bengaluru (both 77°E)
2020	Siachen Glacier is…	D north of Nubra Valley (eastern Karakoram, ~70 km / 44 mi)
2022	Namcha Barwa–Garhwal ✘ · Nanda Devi–Kumaon ✔ · Nokrek–Sikkim ✘	B (2 only). Namcha Barwa in SE Tibet inside Great Bend of Yarlung Tsangpo; Nokrek biosphere on Tura Range, Meghalaya Plateau
–	Repeated falls in sea level → marshland	D Rann of Kutch (shoreline of emergence)
2023	Amarkantak = Vindhya + Sahyadri? Biligiri = easternmost Satpura? Seshachalam = southernmost W. Ghats?	D none: Amarkantak = Vindhya + Satpura; Biligiri Rangan in SE Karnataka; Seshachalam = Eastern Ghats
–	Peninsular Block: W-coast submergence ✔; residual hills (Velikonda, Mahendragiri) ✔; deep V-valleys ✘	B: ancient stable craton; Eastern Ghats denuded; peninsular rivers mature
–	UP has most neighbouring states · Rajasthan longest int'l border · Sikkim only one-neighbour state	A as printed (see conflict)

Source conflict: last PYQ prints answer A (statement 1 not correct), but its explanation says West Bengal has the longest international border and Meghalaya also has only one neighbouring state, which would make statements 2 and 3 wrong. Check atlas / class notes.

🎯 3 · DRAINAGE PYQ 2011 PYQ 2013 PYQ 2014 PYQ 2015 PYQ 2016 PYQ 2017 PYQ 2021 PYQ 2022 PYQ 2023

Factors governing drainage

→ Topography (slope, relief)
→ Nature of rocks (basalt low permeability; granite impermeable)
→ Amount of water (rain + snow)
→ Periodicity (perennial / seasonal)
→ Geological time
→ Anthropogenic: deforestation, urbanisation, dams

⏳ Terms

→ Catchment (drainage basin / watershed): whole land area draining to one point
→ Drainage divide: elevated landform separating catchments
→ Basin size: major >20,000 km² · medium 2,000–20,000 · minor <2,000
→ Inland drainage: does not reach sea (Kachchh, shoreline of emergence)

Type	Meaning	Example
Consequent	flows down original slope	Himalayan streams
Subsequent	tributary on softer rock	Appalachian
Antecedent	river older than the uplift; cuts gorges	Indus, Brahmaputra (through Himalaya); Ghaghara
Superimposed	pattern from older landscape laid onto new rocks	Subarnarekha
Concordant / Discordant	follows structure (stable regions) / does not (uplift, erosion past)	

Big numbers

→ 77% drains to Bay of Bengal · 23% to Arabian Sea
→ Divide: Delhi Ridge – Aravallis – Vindhyas – Satpuras – Western Ghats
→ Himalayan rivers 40–60 mn yrs; peninsular 200 mn yrs
→ Basin areas: Indus 11.78, Ganga 8.61, Brahmaputra 5.8 lakh km²; Godavari 3.12 lakh km² = largest peninsular basin

	HIMALAYAN rivers = teenagers	PENINSULAR rivers = grandparents
Age, valley	young, V-valleys; shift channels (avulsion, e.g. Kosi); origin: Shiwalik river divided into Indus and Brahmaputra	mature (~200 mn yrs), broad shallow graded valleys
Flow	perennial (snow + rain); meanders; big deltas; flooding	seasonal (except Kaveri); straight courses; small basins; Narmada and Tapi make estuaries, others deltas

HIMALAYAN RIVERS: INDUS PYQ 2021

Indus

→ Origin Bokhar Chu, Kailash; 3,180 km, rises W Tibet; in India only Ladakh
→ Riparian: Ladakh, J&K, HP, Punjab; gorge near Nanga Parbat
→ Panjnad = I–J–K–R–B–S (left tributaries, as printed)
→ Right bank: Shyok (Rimo glacier, Siachen tongue; joins at Keris, PoK), Gilgit · Others: Zanskar, Suru

River	Origin / hook
Jhelum	Verinag; meanders; Wular Lake; joins Chenab in Pakistan
Chenab	near Baralacha La (Lahaul–Spiti); joins Sutlej in Pakistan
Ravi	west of Rohtang, Kullu hills; joins Chenab in Pakistan
Beas	Beas Kund near Rohtang; gorges at Kati and Largi (Dhauladhar); entirely in India; meets Sutlej in Punjab
Sutlej	Rakshasthal near Mansarovar; enters India at Shipki La; joins Indus in Pakistan as Panjnad

GANGA SYSTEM

Ganga

→ Origin: Gangotri (Bhagirathi) + Satopanth (Alaknanda); Uttarakhand, UP, Bihar, Jharkhand, WB; Padma in Bangladesh
→ NW–1: Haldia → Prayagraj

Panch Prayag (up → down)

→ Vishnu · Nand · Karna · Rudra · Dev

Feeders

→ Dhauli (Ganga) · Nandakini · Pindar · Mandakini · Bhagirathi

Vasudev Ne Karna ko Rasta Dikhaya – Panch Prayag: Vishnuprayag · Nandprayag · Karnaprayag · Rudraprayag · Devprayag. Feeders: Dhul Nandakini ke Pair Mein Bikhri = Dhauli(Ganga) · Nandakini · Pindar · Mandakini · Bhagirathi.

Rama Ghoomte hue Ghaghara Ganda Kar Mat – LEFT-bank tributaries: Ramganga · Gomti · Ghaghara · Gandak · Kosi · Mahananda (+ Burhi Gandak between Gandak and Kosi).

Left bank	Facts
Ramganga	Chamoli; meets Ganga at Kannauj
Gomti	Pilibhit; meets Ganga at Ghazipur
Ghaghara	Tibet; antecedent; joins in Bihar; tribs Sarda, Saryu, Rapti, Choti Gandak
Gandak	Tibet–Nepal border; Kali Gandak, Mayangadi, Trishuli; joins near Patna
Burhi Gandak	West Champaran; Kanwar Tal (India's biggest oxbow; Sikrahna)
Kosi	Sapt Kosi (7 tribs: Arun, Tumar, Sun Kosi); Sorrow of Bihar (shifting course); Nepal; joins near Kursela
Mahananda	near Tiger Hill; "last major right-bank tributary" to join Ganga in India (handout wording)
🛡️ Right bank	Yamuna (Yamunotri; joins at Prayagraj; Chambal, Sind, Ken, Betwa) · Tons/Tamsa · Son (Amarkantak, radial drainage with Narmada, Johila; joins near Patna) · Phalgu (Gayaji)
Distributary	Hooghly (Giria → Sagar Island, estuary) · Damodar (Chotanagpur → Hooghly near Kolkata) · Mayurakshi, Ajay

Source conflict: Mahananda sits in the left-bank list but is described as the "last major right bank tributary". Treat it as printed (left list).

🎯 BRAHMAPUTRA AND BARAK PYQ 2011 PYQ 2014 PYQ 2016 PYQ 2017

Brahmaputra ("son of Brahma"; Luit, Dilao)

→ Source Chemayungdung glacier (also Angsi), SW Tibet, Kailash; near sources of Indus and Sutlej; Mariam La separates it from Manasarovar; Indus–Tsangpo Structure Zone between Great Himalaya and Kailas Range

→ U-turn at Namcha Barwa → enters as Dihang (Siang); Lohit and Dibang join; Sadiya → Dhubri

→ Name chain: Tsangpo – Dihang – Brahmaputra – Jamuna – Padma – Meghna

→ Majuli island where Subansiri joins; Assam floods

Tributaries

→ Right: Subansiri · Kameng · Manas · Beki · Raidak · Jaldhaka · Teesta (+ Sankosh)

→ Left: Lhasa · Nyang · Parlung Zangbo · Lohit · Dibang · Burhi Dihing · Dhansiri · Kolong

SKM-ST – Brahmaputra right-bank tributaries: Subansiri · Kameng · Manas · Sankosh · Teesta.

Teesta (2017)

→ 315 km; Pahunri (Teesta Kangse) glacier, above 7,068 m; Sikkim, West Bengal, Bangladesh, Bay of Bengal

→ Fed by Thangu, Yumthang, Donkha ranges; Rangpo joins; border of Sikkim–WB to Teesta Bazaar

→ Main tributary Rangeet (originates in Sikkim)

→ Source is not the same as Brahmaputra (originates in Sikkim)

Barak

→ 900 km; Manipur hills; enters plains near Lakhipur; Manipur, Nagaland, Mizoram, Assam

→ Splits into Surma + Kushiyara → Meghna (after meeting Padma, Ganga + Brahmaputra flow)

→ Separated from Brahmaputra by Barail Hills; inland waterway; rich flora and fauna

🎯 PENINSULAR DRAINAGE

PYQ 2013 · PYQ 2015 · PYQ 2021 · PYQ 2022 · PYQ 2023

Peninsular rivers

→ By direction: East-flowing Mahanadi, Godavari, Krishna, Kaveri · West-flowing Narmada, Tapi · NE-flowing Chambal, Ken

→ ~200 mn yrs, mature, broad shallow graded valleys

Seonath Man Hasdo On the Joke I Told — Mahanadi: left Seonath · Mand · Hasdo · right Ong · Jonk · Tel · and Ib (left). ("I" = Ib, "Told" = Tel.)

River	Facts
Mahanadi	Sihawa, Raipur (CG); CG + Odisha (53% / 47% as printed); Hirakud Dam; gorge near Tikkarpara; delta
Godavari	Longest peninsular; Vridha / Dakshin Ganga; 2nd longest river of India after Ganga; Trimbakeshwar (Nashik); Maharashtra, Telangana, CG, AP, Yanam · right: Manjara, Pravara, Manair (16.1% of basin) · left: Purna, Pranhita (Wainganga + Penganga), Indravati, Sabari (59.7%); below Rajahmundry splits into Gautami + Vashishtha; KG delta
Krishna	Mahabaleshwar; Maharashtra, Karnataka, Telangana, AP · right: Koyna, Panchganga, Dudhganga, Ghataprabha, Malaprabha, Tungabhadra · left: Bhima, Dindi, Halia, Musi
Kaveri	Brahmagiri (Kodagu); perennial (SW monsoon upper + NE monsoon lower); Kerala 3%, Karnataka 41%, TN 56% (as printed); delta; splits at Tiruchirapalli into Kollidam + Kaveri, rejoin at Srirangam island · right: Lakshmana Tirtha, Kabini, Suvarnavathi, Bhavani, Noyyal, Amaravati · left: Harangi, Hemavati, Shimsha, Arkavati
🧑‍🤝‍🧑 Others (east)	Brahmani (S. Koel + Sankh near Rourkela; rises near Nagri, Ranchi, ~600 m; Eastern Ghats start south of it) · Subarnarekha (Ranchi plateau, Nagri ~600 m; Odisha–WB boundary; estuary) · Pennеru (Nandi Durg, Chikballapur; Gandikota canyon, Kadapa, "Grand Canyon of India", Erramala hills; estuary) · Ponniyar, Vaigai (short) · Vamsadhara (Kalahandi, Eastern Ghats; between Rushikulya and Godavari; Kalingapatnam) · Nagavali (Lakhbahal, Kalahandi, 1,300 m; Langulya)
Narmada	Amarkantak (Maikala, ~1,080 m); rift valley between Vindhya and Satpura, few tributaries (short, trellis); Marble Rocks Gorge; estuary ~21 km wide below Bharuch, Gulf of Khambhat; Hiran, Kolar, Hatni, Choti Tawa; MP, Maharashtra, Gujarat; meander near Bharuch
Tapi	"Narmada's twin"; Multai (MP); Purna, Girna, Panjara; Satpura north, Mahadev east, Ajanta/Satmala south
Luni	west of Ajmer; Sagarmati → Luni after Sarsuti joins; ephemeral; best inland-drainage example in the world → Rann of Kachchh; Rajasthan's biggest system
Sabarmati	Aravallis → Gulf of Khambhat; gorge at Dharoi
Mahi	Vindhyas (MP) → Gulf of Khambhat; only river to cross the Tropic of Cancer twice
🧑‍🤝‍🧑 Short west rivers	parallel drainage off W. Ghats, no deltas: Gujarat (Auranga, Par, Daman Ganga) · Goa (Mandovi, Zuari, Rachol) · Maharashtra (Ulhas, Kalindi) · Karnataka (Kalinadi, Sharavati – Jog Falls, Tadri, Netravathi) · Kerala (Bharathapuzha, Periyar, Pamba)
Inland	Ghaggar-Hakra (Ghaggar in India, Hakra in Pakistan after Ottu barrage; ends in Thar) · Luni, West Banas (Rajasthan → Gujarat → Rann of Kachchh)

River-linking project

→ Idea: Sir Arthur Cotton (19th c.); revived 1960s, momentum 1980s (scarcity, floods)

→ NWDA set up 1982 to study feasibility; Supreme Court intervened several times

→ Himalayan component 14 links, peninsular 16 links; some states propose intra-state links; under debate

→ Ken-Betwa pilot: cabinet approval 2014, hurdles (environment, inter-state)

→ Pattiseema lift scheme (Godavari → Krishna; Polavaram right canal to Krishna at Vijayawada), inaugurated 2016

State	Waterfall (height m) – river
Karnataka	Kunchikal 455 (Varahi) · Barkana 259 (Sita) · Jog 253 (Sharavati) · Magod 198 (Bedti) · Shivanasamudra 98 (Kaveri) · Hebbe 168 (Thunga) · Shimsha 46

Odisha	Barehipani 399 (Budhabalanga) · Khandadhar 244 (Korapani Nala) · Koilighughar 61 (Ahiraj)
Kerala	Meenmutty 300 (Kallar) · Soochipara 200 (Chaliyar) · Athirapally 25 (Chalakuddy → Periyar)
Chhattisgarh	Amritdhara 90 (Hasdeo) · Teerathgarh 91 (Munga) · Chitrakote 29 (Indravati)
MP	Chachai 130 (Bihad → Tons) · Dhuandhar 30 and Kapildhara 30 (Narmada) · Sonemuda 15 (Son)
Jharkhand	Lodh 143 (Budha) · Hundru 99 (Subarnarekha) · Panchghagh 45 (Banai)
Others	Nohsngithiang (Seven Sisters) 315, Meghalaya · Vantawng 229, Mizoram · Kempty 40 (Yamuna, Uttarakhand) · Dudhsagar 320 (Mandovi, Goa) · Thalaiyar (Rat Tail) 297 (Manjalar/Vaigai, TN) · Maharashtra: Thoseghar, Someshwar 10 (Godavari)

Tallest first — Kunchikal 455 (Karnataka) > Barehipani 399 (Odisha) > Dudhsagar 320 > Nohsngithiang 315 > Meenmutty 300 > Thalaiyar 297 > Jog 253.

Lake	State/UT	Hook
Wular	J&K	freshwater, part of Jhelum
Dal	J&K	Srinagar
Pangong Tso	Ladakh	high-altitude saline; Indo-China border
Tso Moriri	Ladakh	high-altitude saline
Renuka	HP	largest natural lake of HP
Roopkund	Uttarakhand	near Trishul; skeletons at bottom
Nainital / Bhimtal	Uttarakhand	natural freshwater / artificial FW
Gobind Ballabh Pant Sagar	UP	Rihand dam; largest artificial lake of India
Sambhar	Rajasthan	playa lake
Pushkar, Udaipur lakes	Rajasthan	artificial; Pushkar religious
Kanwar	Bihar	largest oxbow lake of Asia; Burhi Gandak
Lhonak, Cholamu	Sikkim	high-altitude glacial
Gurudongmar	Sikkim	highest lake of India
Deepor Beel	Assam	Brahmaputra basin
Loktak	Manipur	between west and east Manipur hills; phumdis; Keibul Lamjao NP, sangai deer

Named on map, no notes: Parshuram Kund, Nal Sarovar, Bhojtal, Lonar, Salim Ali, Husain Sagar, Kolleru, Pulicat, Stanley, Vembanad, Ashtamudi, Chilika.

🎯 PYQ BANK: DRAINAGE

Year	Asked	Answer / rule
2011	Why Brahmaputra makes a U-turn	B syntaxial bending of young Himalaya. Tethys ~70 mn yrs ago, receded from early Eocene; maximum push from Aravalli (W) and Shillong Plateau (E); Arakan Yoma bends near Namcha Barwa, Sulaiman near Nanga Parbat (hairpin)
2013	Why Narmada flows west	A linear rift valley only (not "slope west")
2014	Through Arunachal: Barak, Lohit, Subansiri	B Lohit + Subansiri; Barak = Manipur, Nagaland, Mizoram, Assam
2015	Godavari tributaries: Vamsadhara, Indravati, Pranahita, Pennar	D Indravati + Pranahita
2016	Brahmaputra tributaries: Dibang, Kameng, Lohit	D all three
2016	Recent river linking	B Godavari–Krishna (Pattiseema)
2017	Teesta: same source as Brahmaputra? Rangeet a tributary? Flows into Bay on the border?	B only 2 right
2021	Which joins the Indus directly: Chenab, Jhelum, Ravi, Sutlej	D Sutlej (others join the Sutlej system first)
2021	Rise from Eastern Ghats: Brahmani, Nagavali, Subarnarekha, Vamsadhara	B Nagavali + Vamsadhara (Brahmani, Subarnarekha rise on Ranchi plateau)
2022	Gandikota canyon	C Pennar
2023	Jhelum through Wular ✔; Krishna feeds Kolleru ✘; Gandak meander → Kanwar ✘	A only one. Kolleru lies between Krishna and Godavari; Kanwar = Burhi Gandak (Sikrahna)
–	River: antecedent, 3 countries, Tibetan origin, irrigation, no distributaries	C Sutlej (Brahmaputra and Indus have distributaries; Teesta rises in Sikkim)

🎯 4 · INDIAN CLIMATE PYQ 2011 PYQ 2017

Monsoon climate

→ ~75% rain in monsoon months; tropical south to very cold north, desert west to heavy rain east

Controls

→ Latitude ~6°N–35°N (tropical)

→ Himalayas: block Central Asian cold; shape monsoon

→ Pressure and winds: summer low pulls SW winds; winter high → dry NE winds; sub-tropical westerly jet → Western Disturbances

→ Ocean currents (Indian Ocean Equatorial Current: warm, humid coasts)

→ Continentality (inland extremes)

→ Relief: W. Ghats barrier, rain shadows

WiSAR — Four seasons: Winter (mid-Nov–Feb) · Summer / pre-monsoon · Advancing monsoon (Jun–Sep) · Retreating monsoon (Oct–Nov).

Winter (mid-Nov–Feb)

→ Fog (can turn to smog), inversion, pollution domes, frost, cold waves in north

→ Offshore NE trade winds

→ WD rain = "Mawath / Mahawat" (Mewat) vital for Rabi

→ Fewest tropical cyclones (ITCZ shifts, high pressure on land and ocean)

Western Disturbances

→ Extratropical storms from the Mediterranean; steered east by the sub-tropical westerly jet; lose moisture on the Himalaya

→ Winter and pre-monsoon rain in Punjab, Haryana, UP, Delhi; Himalayan snow (feeds Ganga / Indus perennial flow); moderate winter cold; hail, thunderstorms

→ Climate change: possibly more intense (warmer, moister air), shifted timing/path, now seen in summers, interacting with cyclones and monsoon → extreme rain

Summer

→ Sun north → ITCZ shifts (~25°N)

→ Loo, dust storms (Aandhi; WD can bring showers); heat waves

→ Pre-monsoon showers (warm continental + maritime air meet): 1% of annual rain; lots of cyclonic activity

Pre-monsoon showers: what they help — Mango showers (Karnataka/Kerala) · Coffee / Cherry-blossom showers (Karnataka, Kerala) · Norwester / Kaal Baisakhi (WB, Bihar, Jharkhand, Odisha: rice, tea, jute) · Bardoli Chheerha / Bordoisila = tea showers (Assam).

Cyclones in India

→ Coast >7,500 km (as printed); season April–Dec, peaks May–June and Oct–Nov

→ Bay : Arabian Sea ≈ 4 : 1 (bigger warm-water area, lower wind shear); 5–6 per year, 2–3 severe

→ Climate change: intensity up, not necessarily frequency; Arabian Sea activity rising; 2022 study: Bay numbers may fall, intensity rise

Naming

→ WMO/ESCAP Panel on Tropical Cyclones, since September 2004

→ 13 countries × 13 names: Bangladesh, India, Iran, Pakistan, Oman, Maldives, Myanmar, Sri Lanka, Thailand, UAE, Yemen, Qatar, Saudi Arabia

→ RSMC New Delhi assigns; names retired after use; a South China Sea cyclone entering the Bay keeps its name

IMD colour warning	Lead time
Pre-cyclone watch (no colour)	72 h
Cyclone Alert: Yellow	≥ 48 h
Cyclone Warning: Orange	≥ 24 h (landfall point forecast)
🧍‍♂️ Post-landfall outlook: Red	≥ 12 h

Mnemonic (handout): 72 – 48 – 24 – 12, halving each step: Watch → Yellow → Orange → Red. ⭐

Urban heat island

→ Urban areas much hotter than rural surroundings

→ Causes: concrete/asphalt, less vegetation, vehicle/industrial/AC heat, anthropogenic energy, urban geometry ("heat canyons")

🌐 Advancing monsoon (Jun–Sep; "mausim"; seasonal wind reversal; 75% rain)

Drivers

→ Land–sea differential heating (primary)

→ ITCZ shifts north

→ Sub-tropical westerly jet (weaker/northward = stronger monsoon)

→ Tibetan Plateau heating → Tropical Easterly Jet (TEJ)

→ Somali jet (moisture across Arabian Sea)

→ ENSO · IOD · MJO

→ Arctic sea ice (less ice → more rain in central and NE India, less in west and south peninsula; mechanism unknown)

→ Monsoon trough = low over India attracting SW winds

→ Burst = sudden onset with thunder, lightning, downpour; ~1 June, Kerala

Two branches

→ Arabian Sea (stronger): Malabar, Konkan, Kathiawar

→ Bay of Bengal (blocked by Arakan): Utkal and North-Eastern

→ Meet around Agra–Ferozepur

→ Tamil Nadu dry: rain shadow of W. Ghats; branches parallel to its coast

→ Break: usually August: weak Tibetan Low + winds parallel to west coast; flooding common

Cloudbursts

→ ≥ 10 cm rain in an hour over a small area; flash floods, landslides

→ Orographic lift + rapid condensation + cold-air downdraft; more in mountains, also plains

→ Himalaya, Western Ghats; monsoon; hard to predict (dense radar, high-res models); more with warming (warmer air holds more moisture)

Urban flooding

→ Impervious surfaces (concrete, asphalt) + overwhelmed drains

→ Mumbai 2005 (944 mm in a day) · Chennai 2015 (drainage, water-body encroachment) · Bengaluru (unplanned growth) · Hyderabad (lake encroachment) · Mumbai mangrove loss · heavy rain in Mumbai/Kolkata linked to climate

El Nasty / La Nice — El Niño = El Nasty (warm east Pacific → weak SW monsoon, drought in India/Australia; NE monsoon stronger). La Niña = La Nice (cold east Pacific → good monsoon, Australian floods; NE monsoon weaker).

Phenomenon	What it is	India link
El Niño	trades weaken; warm water sloshes east; Humboldt (Peruvian) current replaced, less upwelling	weak SW monsoon; Australia drought/fire; Peru floods; California storms
La Niña	trades strengthen; cold east Pacific; more upwelling	better monsoon; floods in Australia, E. Africa, SE Asia
ELLA years	transition between El Niño and La Niña (IMD–IITM Pune study)	severe, prolonged summer heat; more heatwaves in NW and central India; early identification helps forecasting
El Niño Modoki	"similar but different" (Japanese); warmth in central Pacific (dateline El Niño), cooler east and west	different global pattern
PDO	North Pacific; each phase 20–30 yrs (ENSO = 2–7 yrs); positive = warm east	modulates ENSO links to monsoon; amplifies or weakens El Niño/La Niña
IOD	west–east SST gap in the Indian Ocean	Positive IOD (warm west near Arabian Sea, cool east) → stronger monsoon, even in El Niño years; negative → weaker, more break days
Atlantic Niño / Niña (Atlantic Ocean Dipole / Zonal Mode)	Niño: east equatorial Atlantic warms, west cools; Niña = reverse	not as wide-ranging as Pacific
Arctic Ocean Dipole	Arctic sea-level pressure pattern (Arctic Dipole Anomaly)	–
MJO	eastward cloud-rain pulse, 30–60 days; active (wet) vs suppressed (dry)	active phase enhances monsoon; modulates cyclone formation; shorter cycle = better monsoon impact

Retreating monsoon (Oct–Nov)

→ ITCZ shifts; pressure rises over north India; "October heat" (humid, clear)

→ NE monsoon rain: coastal Andhra + most of Tamil Nadu (NE trades over Bay); El Niño: NE stronger; La Niña: NE weaker, SW stronger

→ Season of most severe cyclones in Bay (hot sea, trough moves south, wind change)

Rainfall map

→ >200 cm: windward W. Ghats (S. Gujarat → Kerala), NE (Assam, Meghalaya, Arunachal; funnelling: Cherrapunji, Mawsynram), Andaman and Nicobar

→ 100–200: Himalayan foothills, east India, east TN

→ 50–100: parts of Deccan, NW India

→ <50: west Rajasthan, W. Ghats rain-shadow, Ladakh

Year	Asked	Answer
2011	La Niña vs El Niño: (1) La Niña = cold equatorial Indian Ocean? (2) La Niña no effect on monsoon?	D neither: La Niña = cooling of tropical west coast of S. America (Peru/Chile cold current returns); helps monsoon; Australia floods
2017	IOD: (1) W Indian Ocean vs Eastern Pacific? (2) can influence El Niño's effect	B only 2: IOD = W (near Arabian Sea) vs E (near Bay of Bengal) Indian Ocean
–	Andaman and Nicobar climate: humid tropical coastal ✔; SW + NE monsoon ✔; max rain Dec–May ✘	B (1 and 2); max rain May–Sept (SW monsoon); ocean winds onshore both ways

🎯 5 · FORESTS PYQ 2015 PYQ 2016 PYQ 2023 PYQ 2011

Forest	Climate	Where	Trees / traits
Wet evergreen	>200 cm; 25–27 °C	W. Ghats (south of Mumbai), NE, A&N	no undergrowth (dense canopy); jackfruit, rosewood, mahogany, aini, ebony, agarwood
Semi-evergreen	175–225; 24–27 °C	W. coast, Assam, lower E. Ghats, Odisha, Andamans	wet evergreen + moist deciduous mix; evergreen due to undergrowth; kadam, cashew, cedar, chestnut, thorny bamboo
Dry evergreen	~100 cm all year	TN coast	short; tamarind, neem, toddy palm
🧍‍♂️ Moist deciduous	100–200; ~27 °C	around W. Ghats, NE, Shiwalik (Terai-Bhabar), E. MP, CG, Chotanagpur, Odisha	shrubby undergrowth; teak, sal, mahua, tendu, bamboo (also shisham, sandalwood, khair, kusum, arjun, mulberry); leaf-fall 6–8 weeks; humidity 60–75%
Dry deciduous	75–100	rainier peninsula, plains of UP, Bihar, MP, Rajasthan	transition to thorn; teak, tendu, red sanders, sal, khair
Thorn	<50; 25–30 °C	Rajasthan, SW Punjab, W. Haryana, Kachchh, Saurashtra, some UP, lee of W. Ghats (MH, Karnataka, Telangana, AP, TN)	low scattered; acacia, babul, neem, khair, cacti, wild date
Littoral and swamp (mangroves)	brackish water	Sundarbans (Sundri), A&N, Mahanadi-Godavari-Krishna deltas	viviparity, pneumatophores; sundri, keora, agar, rhizophora; ~7% of world mangroves; fresh + brackish

Rainfall ladder (cm) – >200 wet evergreen → 175–225 semi-evergreen → 100–200 moist deciduous → 75–100 dry deciduous → ~100 spread through year dry evergreen (TN coast) → <50 thorn. Wetter = evergreener, taller; drier = shorter, thornier.

🛡️ Mangroves protect coast

→ Erosion, storm surge, tsunami (2004 PYQ); trap sediment (protect coral, seagrass)

→ Fisheries nurseries; rot-resistant timber, fodder, medicine; tourism

→ Handout alert: capacity to adapt to rising seas is being compromised; roots salt-tolerant (not salt-sensitive), do not store freshwater

Mountain forests

Himalaya (north)

→ base moist deciduous → 1,000–2,000 m wet temperate with pine (British replaced oak with pine for railway sleepers)

→ up to ~3,000 m blue pine, spruce, grassland

→ 3,000–4,000 m silver fir, juniper, rhododendron + bugyals → tree line → alpine tundra → snow line; northern slopes little vegetation

Western Ghats (south)

→ evergreen base → temperate evergreen "sholas" (Nilgiris, Anamalai, Palani) both slopes; no alpine tundra (height limit)

🎯 Andaman and Nicobar forests (2015 PYQ)

→ ~2,200 plant varieties, 200 endemic, 1,300 not on mainland; forest cover 86.2%

→ South Andaman: epiphytes (ferns, orchids) · Middle: moist deciduous · North: wet evergreen + woody climbers

→ North Nicobar: no evergreen; central/south Nicobars mostly evergreen; grasslands only in Nicobars

→ Mangroves 1,136 km² (Andaman) + 35 km² (Nicobar)

Forest status	Figure (handout)
🧑‍🤝‍🧑 Forest + tree cover	8,27,357 km² = 25.17% (target 33% for stability)
🧑‍🤝‍🧑 Forest / tree cover	7,15,343 km² = 21.76% / 1,12,014 km² = 3.41%
Change vs 2021	+1,445 km² (forest +156, tree +1,289); lower than +1,540 in 2021
Mangrove	4,992 km² (–7.43 km²) · bamboo area +5,227 km²
🗺️ States	>33% forest: 19 states/UTs · >75%: 8 (Mizoram, Lakshadweep, A&N, Arunachal, Nagaland, Meghalaya, Tripura, Manipur)
🧑‍🤝‍🧑 Highest % cover	Lakshadweep 91.33 · Mizoram 85.34 · A&N 81.62
🧑‍🤝‍🧑 Largest area	forest + tree: MP, Arunachal, Maharashtra · forest: MP, Arunachal, Chhattisgarh · bamboo: MP, Arunachal, Maharashtra
Max increase	total: Chhattisgarh, UP, Odisha · forest: Mizoram, Gujarat, Odisha · tree: Gujarat, Bihar, Kerala
Megacities	Delhi has the highest forest cover among the six
Worries	decline in Andhra Pradesh, Maharashtra · Western Ghats ESA –58.22 km² in 10 years · NE –327.30 km²
🆕 New in 2023 report	agroforestry chapter · "forest characteristics" chapter back after 10 yrs · bamboo in tree cover · first estimate of 5–10 cm DBH small trees · 751 districts (636 in 2021)
FAO	India +2,66,000 ha/yr 2010–2020, 3rd (after China, Australia) · GFRA 2025: 9th in total forest area, 3rd in annual gain · Asia the only continent with forest gain since 1990

Social forestry

→ Increase tree cover + rural livelihoods through local people; term first used in India in 1976 by the National Commission on Agriculture

→ Types: community, urban (Miyawaki = dense native planting, Japanese), extension, agroforestry (trees with crops), farm forestry

→ Benefits: jobs, fuel/fodder, NTFPs, carbon, erosion control, habitat, less pressure on natural forests

🎯 6 · SOILS PYQ 2013 PYQ 2021 PYQ 2023

Soil formation factors

→ Relief · parent rock · climate · biota · time · humans

A Red Black Arid Laterite Saline Peaty Forest — ICAR's 8 soils: Alluvial · Red & yellow · Black · Arid desert · Laterite · Saline · Peaty/marshy · Forest & mountain. Pairs: Black = cotton + cracks (self-ploughing); Red = iron; Laterite = tea–coffee–cashew (acidic, leached); Alluvial = the breadbasket.

Soil	Where	Rich / poor	Crops and hooks
Alluvial	N. plains, coastal plains; >40% of India, most common	rich: potash, lime, phosphoric acid; poor: nitrogen, humus	sugarcane, paddy, wheat, pulses; Bhangar / Khadar
Black (Regur)	Maharashtra, Saurashtra, MP, CG; Godavari–Krishna; 3rd most common	rich: Mg, potash, iron, lime, alumina; poor: phosphoric, N, organic	cotton, sugarcane, pulses; cracks when dry; high water retention; volcanic (Deccan / Rajmahal trap lava, fissure volcanic rock)
Red and yellow	low-rain E and S Deccan; W. Ghat piedmont; yellow in parts of Odisha/CG	rich: iron, potash; poor: N, phosphorus, humus	crystalline igneous rocks; yellow = hydrated iron; wheat, cotton, pulses, tobacco, oilseeds, potato
Laterite	wetter margins; south, W. Ghats of Maharashtra, Odisha, WB, NE; also Karnataka, TN, MP, Bihar, Assam, Meghalaya	iron oxide + potash high (notes); poor: organic, N, phosphate, calcium	acidic; tea, coffee, cashew (tapioca: PYQ); rusty red
Arid	Rajasthan / Thar	little humus, N; phosphorus adequate; kankar	millets, fodder, guar; Indira Gandhi Canal
Saline (Usara)	arid/semi-arid Rajasthan, Punjab, UP, Haryana, Kachchh, Maharashtra	salts of Na, K, Mg	some coconut, salt-resistant fodder (White Revolution)
Peaty / marshy	heavy rain: Odisha, Sundarbans, Bihar, UP, Kerala	humus/organic 40–50%	rice if drained
🧑‍🤝‍🧑 Forest and mountain	Himalaya, W and E Ghats	varies	valley sides loamy–silty (rice, wheat); upper slopes coarse (horticulture)

Source conflict: laterite is "rich in iron oxide and potash" in the soil table but "poor in nitrogen, potash, lime, organic matter; rich in iron and aluminium" in the 2013 explanation. Exam rule: "laterite rich in N?" = no; laterite is red, cashew + tapioca ✔.

Soil degradation

→ Erosion (deforestation, bad tillage, overgrazing)
→ Chemical (fertiliser/pesticide, salinisation, irrigation acidity)
→ Physical (mining, construction, machinery)
→ UNCCD: affects 15 million km² (larger than Antarctica), growing ~1 million km²/yr; excessive degradation → desertification

India: a sub-continent

→ Himalaya = natural border
→ Own plate: India = Gondwanaland; rest of Asia on Eurasian plate (Laurasia)
→ Almost all major climate and forest types (desert to tundra)
→ Distinct history and culture

🎯 PYQ BANK: FORESTS AND SOILS

Year	Asked	Answer / rule
2023	Rainforest soil nutrient-rich because of fast decomposition	D: I false, II true; soil leached and poor, trees take nutrients as fast as produced
2011	Mangroves as safety hedge (2004 tsunami)	D: extensive roots hold against storms and tides
2013	Laterite: red ✔; rich in N and potash ✘; Rajasthan/UP ✘; tapioca + cashew ✔	C (1 and 4)
2015	Region with mangrove + evergreen + deciduous	D Andaman and Nicobar
2015	Tropical wet evergreen in: Arunachal, HP, Mizoram	C Arunachal + Mizoram (also Assam, Meghalaya, Nagaland, Tripura, WB, A&N, W. Ghats slopes in TN, Karnataka, Kerala, Maharashtra)
2015	Teak dominant in	A tropical moist deciduous
2016	Red Sanders: found in a part of S. India ✔; important in tropical rainforest ✘	A (1 only): Pterocarpus santalinus, endemic to southern Eastern Ghats; wood not aromatic, not Santalum sandalwood; dry deciduous
2021	Black cotton soil forms from weathering of	B fissure volcanic rock
2023	Jackfruit, Mahua, Teak: how many deciduous?	B two (jackfruit evergreen)
–	Mangrove rationale for climate resilience	D bio-shield + livelihood; roots salt-tolerant, not salt-sensitive; do not store freshwater

Traps

→ "highest of..." lists are state-specific (Karoh Haryana, Amsot UP, Someshwar Bihar are Shiwalik; Kangto Arunachal; Shillong Peak Meghalaya)
→ Narmada flows west because of the rift, not slope (2013)
→ Nokrek = Meghalaya, not Sikkim; Namcha Barwa ≠ Garhwal (2022)
→ Amarkantak = Vindhya + Satpura (2023)
→ IOD is Indian Ocean W-E, not Pacific (2017)
→ La Niña helps the monsoon (2011)
→ Rainforest soil is poor (2023)
→ Red Sanders is not aromatic sandalwood. **Connects to:** Human Geography, Geomorphology, Climatology.

Recall drill: cover and answer

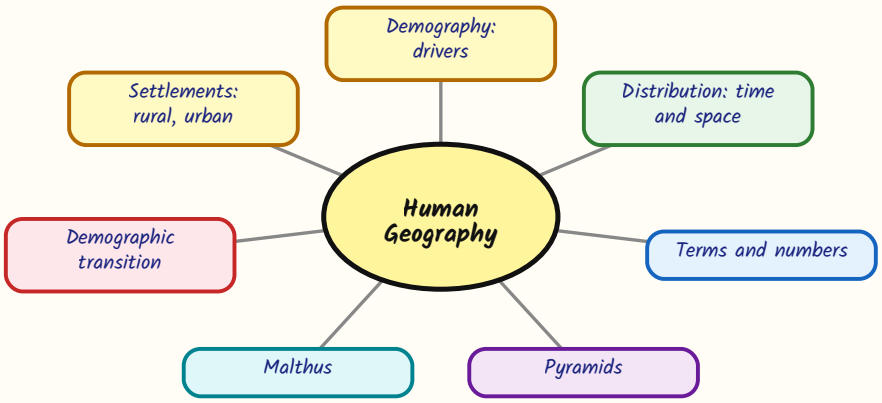
Standard meridian and 5 states?	82°30'E: UP, CG, MP, Odisha, AP (U C My Old Aunt)	Area, rank, coastline, land border?	32,87,263 km², 7th; 11,098 km; 15,106.7 km, 17 states
Channels 10, 9, 6?	Andaman|Nicobar · Lakshadweep|Minicoy · Great Nicobar|Sumatra	Highest: India Himalayan peak; entirely in India; Maharashtra; Kerala; TN?	Kanchenjunga; Nanda Devi; Kalsubai; Anaimudi; Doddabetta
Panch Prayag in order?	Vishnu, Nand, Karna, Rudra, Dev	Left-bank Ganga tributaries?	Ramganga, Gomti, Ghaghara, Gandak, Kosi, Mahananda
Why does Narmada flow west? Brahmaputra U-turn?	Linear rift valley; syntaxial bend at Namcha Barwa	Drainage split; river that crosses Tropic of Cancer twice?	77% Bay : 23% Arabian Sea; Mahi
Cyclone alert lead times; peak months?	72 – 48 – 24 – 12 h; May-June, Oct-Nov	IOD positive; El Niño; ELLA?	good monsoon; weak SW, stronger NE; heatwave years
Soils: alluvial share, black, laterite crops?	>40%; cotton, volcanic; tea, coffee, cashew	Forest + tree cover; target?	21.76% + 3.41% = 25.17%; 33%

HUMAN GEOGRAPHY

PYQ Prelims 2013 (tribe-state pairs), 2014 (Changpa) + 2 undated (TFR definition; low birth rate / ageing countries)

✎ *Population terms and numbers, pyramids, Malthus, demographic transition, tribes, rural and urban settlements: short, definition-heavy, 4 Prelims PYQs.*

At a glance: the Human Geography set



Pattern: definitions (TFR, sex ratio, dependency) and where-is-which-tribe; theory: Malthus, Demographic Transition. Prepare: five DTT stages as a birth/death table; state-wise tribes. Mains hooks (none quoted): demographic dividend, ageing, urbanisation, gentrification, climate-driven settlement shifts. No teacher ink: priority = PYQ count. ⭐

1 · DEMOGRAPHY

Demography = study of population size, structure, distribution, dynamics (birth, death, migration, age, fertility). 💡

Drivers of growth

- → Birth rate: live births per 1,000 people per period
- → Death rate: deaths per 1,000; low = growth
- → Migration: immigration (in) raises, emigration (out) lowers
- → Fertility rate: live births per 1,000 women of childbearing age; TFR = average children per woman
- → Age structure: youthful = rapid growth; ageing = slow/decline
- → Economic and social: health care, education, jobs, cultural norms

Temporal distribution

- → World: UN, peaks 10.3 billion in mid-2080s
- → 2025 shares: Asia 62.18% · Africa 19.36% · Europe 9.3% (-0.09% vs 2024) · N. America 7.71% · S. America 5.47% · Oceania 0.58%
- → India: most populous; 1.46 billion (UN); peaks early 2060s at 1.7 billion; latest SRS: TFR 1.9

Handout alert: "1.31 billion according to the 2011 census" is printed next to UN's 1.46 billion. Check the census figure before quoting. 📌

Spatial distribution: drivers

- → Geography/climate: Indo-Gangetic plain, temperate zones
- → Resources: river valleys (Nile, Ganga)
- → Economic opportunity: Mumbai, New York, Tokyo
- → Political/social stability: Canada, Australia
- → History: coasts, trade routes
- → Patterns: urban > rural; Asia most populous; coast > inland

Term	Meaning / India number
TFR	average children a woman has in her reproductive years at present birth rates (PYQ: "average live births a woman would have by end of child-bearing age") · 2.1 = replacement rate
IMR	infant (<1 yr) deaths per 1,000 live births · India 25.5 (2021)
Sex ratio	males per 1,000 females (world); India counts females per 1,000 males · 2011: 943 · NFHS-5: 1020
👫 Child sex ratio	juvenile sex ratio, age 0–6 · 2011: 914 · NFHS-5: 920
Life expectancy	average years if current mortality continues · 70.19 yrs (2022)
Wasting / Stunting / Underweight	low weight-for-height / low height-for-age / low weight-for-age
Dependency ratio	dependants (0–14 and >65) ÷ working-age (15–64)
Demographic dividend	working-age ≫ dependants · 56% in 2022–23 → peak 68.9% in 2030
Second dividend	lower fertility + longer life → older people with few dependants and big savings fund investment, boosting growth

Big Mummy's Up-to-Juggle Mania — Replacement 2.1 ("two parents → two kids + a little extra"). India's TFR: NFHS-5 2.0 → SRS 1.9 → projected 1.29 by 2050. Only 5 states have TFR > 2.1: Bihar · Meghalaya · UP · Jharkhand · Manipur.

Stunt = Short — Stunting = short for age; Wasting = thin for height (Waste = Weight-for-height); Underweight = light for age (Under = Weight-for-age).

2 · POPULATION PYRAMIDS

Pyramid = graph of age and sex distribution; shows growth pattern, development, future challenges. 💡

Shape	Features	Example
Expansive 🔺	broad base, narrow top; rapid growth; high birth rates and dependency; low life expectancy	Nigeria; India = expansive with a stabilising base
Constricted 🏺	narrow base (low birth rate), bulging middle (working age), narrow top; growth slowing; top wider than base = ageing, long life	Japan
Stationary ▮	even across ages, narrow only at the very old; birth and death rates low; stable or very slow growth	USA

3 · MALTHUS

Thomas Robert Malthus (English cleric)

→ "An Essay on the Principle of Population", 1798

→ Population grows geometrically (2–4–8–16); food arithmetically (1–2–3–4) → population outruns food

Checks

→ Preventive: moral restraint, delayed marriage, celibacy, contraception (under our control)

→ Positive: famine, disease, war (after the catastrophe, not in our control)

⚠ Criticisms

→ Technology (Industrial and Green Revolution)

→ Ignores social/economic complexity; "equated sociological and biological needs"

→ Modern reading: environmental sustainability and resource depletion, not just food

Multiply vs Add — Babies multiply (×2), bread adds (+1). Preventive = "Pre" = before birth; Positive checks = "Pos-t-mortem", death does the checking.

4 · DEMOGRAPHIC TRANSITION THEORY

Stage	Name	Birth	Death	Growth	Example
1	High stationary (pre-industrial)	High	High	slow, fluctuating	tribal societies, pre-18th c.
2	Early expanding (transitional)	High	Falling	rapid	many African countries
3	Late expanding (industrial)	Falling	Low	slowing	India, Brazil
4	Low stationary (post-industrial)	Low	Low	stable/declining	USA, much of Europe
5	Decline (proposed)	Very low	Low	declining	Japan, South Korea, Germany, Italy

Death Drops first, Birth Boards later — Stage 1: both high. Stage 2: death falls → boom. Stage 3: births fall → boom slows. Stage 4: both low. Stage 5: births below deaths.

Handout alert: the assumption line says "fertility and mortality reduce over time. Former before latter", but the stage table shows mortality falls first (Stage 2), fertility next (Stage 3). Exam rule: follow the stage table.

Till 1921	1921–51	1951–81	1981 on
stagnant	increasing	explosion	late expanding (growth with stability)

India's transition: 1921–81 = early expanding stage; 1981 onwards = late expanding.
Why DTT matters: development ↔ lower birth/death; ageing and pensions; urbanisation; women's education and rights lower fertility; environmental pressure vs labour shortages.

5 · TRIBES AND PYQ PYQ 2013 PYQ 2014

PYQ	Asked	Answer
2013	Limboo–Sikkim ✔ · Karbi–Himachal ✘ · Dongaria–Odisha ✔ · Bonda–Tamil Nadu ✘	A (1 and 3). Limbu = Kirati, Limbuwan (E Nepal, N Sikkim, W Bhutan). Dongria Kondh of Niyamgiri hills, Odisha; god Niyam Raja; Niyam Dongar ~4,000 m
2014	Changpa: live mainly in Uttarakhand ✘; rear pashmina goats ✔; Scheduled Tribe ✔	B (2 and 3). Semi-nomadic Tibetan people, Changtang, Ladakh; yaks, horses, sheep; pashmina/cashmere; smaller number in W Tibet, partly relocated for Changtang Nature Reserve; half a million nomads in Changtang (1989)
Undated	Total fertility rate is defined as	D average live births a woman would have by end of child-bearing age (not birth rate, not births per couple, not birth rate minus death rate)
Undated	Low birth rate / ageing / declining: Italy, Japan, Nigeria, South Korea, South Africa	A = Italy, Japan, South Korea (1, 2, 4)

Changpa = Chang-Pa-shmina — Ladakh's high plateau herders (Changtang). Dongria = "Dongar" (hill) in Niyamgiri hills. Karbi ≠ Himachal; Bonda ≠ Tamil Nadu.

6 · SETTLEMENTS

Rural: factors

- → Physical: relief, climate, water, soil, resources
- → Economic: farm productivity (Punjab), jobs, roads/rail
- → Social-cultural: kinship, ritual, land ownership (zamindari)
- → Political: policies, security; Green Revolution reshaped Punjab/Haryana
- → Environmental: floods, cyclones (coastal Odisha)
- → Climate change: coastal villages shift inland; drought/flood displacement (parts of Brazil, Burkina Faso)

Huddled vs scattered — Good relief + good climate → huddled together; bad → scattered. Compact (north Indian plains, wet points of Rajasthan, Bundelkhand, Nagaland) · Semi-compact (Indus basin, Gujarat plains, parts of Rajasthan) · Hamleted (socially separate units: panna, para, palli, nagla, dhani; Ganga plains, lower Himalayan valleys, Chhattisgarh plains (tribal), parts of south) · Dispersed (Uttarakhand, HP, Ladakh, Meghalaya).

Rural patterns (shape)

→ Linear: most common; roads, forest paths, rivers; T-shaped, star or checkerboard at crossings
→ Oval/circular: around temple, oasis, pond
→ Triangular: deltas, doabs
→ Rectangular/grid: plains
→ Step: hills

Urban term	Meaning	Handout example
Town	bigger than village, smaller than city; local administrative, commercial functions	–
City	large, dense, diverse economy, advanced infrastructure	Mumbai, Lucknow, Guwahati
Metropolis	major economic, political, cultural centre	New York, Delhi
Megacity	>10 million	Tokyo, Greater Mumbai, Greater Chennai
Conurbation	several towns/cities + suburbs merged	Delhi NCR
Megalopolis	several big cities grown together; labour-market separation	BosWash (Boston–Washington)
Suburb	residential, outskirts, less dense, depends on city	Bandra; Gurugram earlier
Urban sprawl	uncontrolled, low-density, car-based spread into rural land	–
Slum	densely populated, substandard housing, squalor, no basic services	Dharavi
CBD	commercial and business centre ("downtown"); high land values	Chandni Chowk, Connaught Place
Edge city	business, shopping, entertainment outside the CBD in former suburb/rural land	Gurgaon (for Delhi)
Gentrification	redevelopment → higher property values → lower-income residents displaced	Dharavi redevelopment
Satellite city	planned city to relieve a major city	–

Climb the ladder — Town → City → Metropolis → Megacity (>10 million) → Conurbation → Megalopolis. CBD = the city's wallet (downtown); Edge city = the mall that moved to the edge; Gentrification = the café that raised the rent.

🎯 MISSED PYQS (INDIAN GEOGRAPHY)

Asked	Answer / explanation
Andaman and Nicobar climate: humid tropical coastal ✔; SW + NE monsoon ✔; max precipitation Dec–May ✘	B (1 and 2): humid tropical maritime; both SW and NE winds onshore; max rain May–Sept (SW monsoon). (Heat = LP = rain; NH has heat Mar–Sept)
🧑‍🤝‍🧑 Peninsular Block: W-coast submergence by tectonics ✔; residual Veliconda, Mahendragiri hills ✔; deep V-shaped valleys ✘	B: ancient stable craton; Eastern Ghats denuded; peninsular rivers mature on old hard rock
Mangroves for climate resilience: store freshwater/paddy ✘; salt-sensitive roots ✘; bio-shield + livelihood ✔	D (3 only): roots are salt-tolerant, do not store salt water
🏛️ River: antecedent, 3 countries, Tibetan origin, irrigation, no distributaries	C Sutlej (Brahmaputra and Indus have distributaries; Teesta originates in Sikkim)
🗺️ States: UP most neighbours · Rajasthan longest int'l border · Sikkim only one-neighbour state: which not correct?	A as printed; explanation: West Bengal has the longest int'l border; Meghalaya also shares border with only 1 state

Source conflict

→ the last PYQ's printed answer (A, statement 1 wrong) disagrees with its own explanation (statements 2 and 3 wrong). Check atlas / class notes. Traps: TFR ≠ birth rate

→ Changpa = Ladakh, not Uttarakhand

→ sex ratio in India = females per 1,000 males. Connects to: Indian Geography (Changpa/Ladakh, Dongria/Niyamgiri, Thar/Himalaya settlements), Economic Geography.

Recall drill: cover and answer

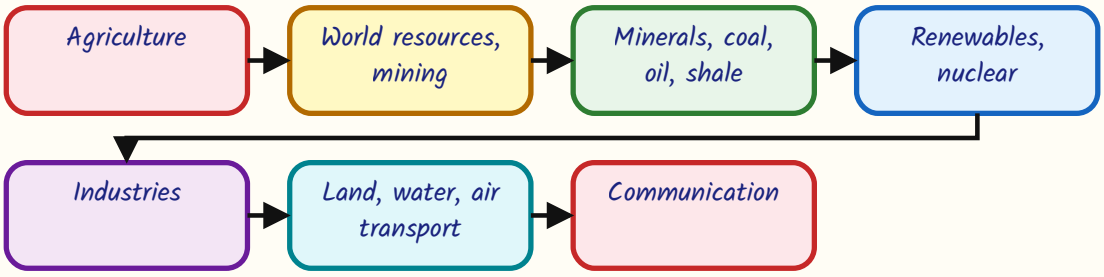
TFR replacement; India NFHS-5 and SRS?	2.1; 2.0 and 1.9	States with TFR above 2.1?	Bihar, Meghalaya, UP, Jharkhand, Manipur (Big Mummy's Up-to-Juggle Mania)
Sex ratio 2011; child sex ratio 2011?	943; 914 (NFHS-5: 1020 / 920)	Working age; dependants; dividend peak?	15–64; 0–14 and >65; 68.9% in 2030
Pyramids: Expansive, Constricted, Stationary?	Nigeria (India stabilising) · Japan · USA	Malthus: growth, work, year?	population geometric, food arithmetic; Essay on the Principle of Population, 1798
DTT: which falls first; India stage?	Death rate (Stage 2); late expanding (Stage 3) since 1981	Changpa; Dongria Kondh?	Changtang, Ladakh, pashmina; Niyamgiri, Odisha
Rural types and urban ladder?	Compact, semi-compact, hamleted, dispersed; town, city, metropolis, megacity (>10 m), conurbation, megalopolis	Low birth rate PYQ answer?	Italy, Japan, South Korea

ECONOMIC GEOGRAPHY

PYQ Prelims 31 PYQs: agriculture 19 (2011 ×4, 2012 ×2, 2013, 2014 ×2, 2015, 2017, 2018, 2019, 2020, 2021 ×2, 2022, 2024, 2025) · minerals and energy 9 (2012, 2013 ×2, 2016, 2018, 2022, 2023 ×2 + 1 undated on rare earths) · transport 3 (2014, 2015, 2023) · industries 0 · Mains none quoted

✎ *Which crop where, which mineral where, which energy source and how much, then industries, transport and communication: the heaviest handout (54 pages) and a VERY HIGH PYQ bank.*

At a glance: the Economic Geography route



Pattern: crop-to-region, mineral-to-state, "statement" questions on fertiliser, salinisation, permaculture, conservation agriculture, plus current affairs (cocoa, turmeric, REE, NCMM). No teacher ink: priority = PYQ count. Image-only pages (crop map, waterways) kept as tables.

1 · AGRICULTURE: SEASONS PYQ 2013

Arable land: India has the largest arable area in the world, ≈ 1.597 million km² (394.6 million acres).

	Kharif	Rabi	Zaid
Time	Jun–Oct; sow with SW monsoon, harvest Sep–Oct	Oct/Nov–Mar/Apr; sow Oct–Dec, harvest Mar–Apr	Mar–Jun; harvest Jun–Jul

Weather	hot, humid, heavy rain	cool dry growth, warm at harvest; Western Disturbance rain helps	warm, dry; irrigation needed
Crops	paddy, maize, jowar, bajra, pulses (moong, urad, arhar), cotton, groundnut, soybean, sugarcane	wheat, barley, oats, gram, mustard, peas, linseed	watermelon, muskmelon, cucumber, vegetables, fodder, some moong/urad

Kar-Rain · Rest-of-winter · Zara-garmi — Kharif = Kar-Rain (monsoon crops) · Rabi = Rest-of-winter (WDs help) · Zaid = Zara-garmi (summer gap).

Pairing pegs

- → Cotton + black soil (Maharashtra, Gujarat, Telangana)
- → Jute + West Bengal (wet, 120–200 cm)
- → Tea = acidic slopes (Assam, WB, TN, Kerala)
- → Coffee = Karnataka (shade; Arabica and Robusta)
- → Rubber = Kerala
- → Coconut = Kalpavriksha (Kerala)
- → Bajra = toughest (tolerates 42 °C), Rajasthan.

Trap (2013 PYQ): wheat is Rabi, not Kharif. Kharif = Arabic for autumn; generally Jun–Sep (state-wise differs); sown ~June, harvested Sep/Oct. Kharif also: rice, maize, bajra, ragi, soybean, groundnut, cotton.

CROP TABLE (temp °C · rain cm)

Crop	Temp · rain	Soil	Top Indian states	Top world	Remarks
Rice	20–32 · 150–300	deep clayey, alluvial	UP, Telangana, WB	India, China, Bangladesh, Indonesia, Vietnam	staple; Aus, Aman, Boro in WB; high humidity
Wheat	10–25 · 50–100	loamy, well-drained	UP, MP, Punjab	China, EU, India, Russia, USA	winter crop; bright sun at ripening
Jowar	20–27 · 50–75	light sandy-loamy	Maharashtra, Karnataka, MP	India, Niger, China, Mali, Nigeria	Kharif and Rabi; fodder + biofuel
Bajra	20–27 · 50–75	sandy loam	Rajasthan, UP, Gujarat	India, Niger, China, Nigeria	drought-resistant, up to 42 °C
Millets (jowar, bajra, ragi)	20–27 · 50–75	wide range	Rajasthan, Karnataka, MP	India, Niger, China, Mali, Nigeria	antioxidants, diabetes/heart, protein; low cost; marginal farmers
Maize	15–27 · 50–80	well-drained loam	MP, Karnataka, Bihar	USA, China, Brazil, EU, Argentina	food, alcohol, feed, biofuel
Cotton	21–30 · 50–100	black (regur)	Maharashtra, Gujarat, Telangana	China, India, Brazil, USA, Australia	cash crop; ≥ 210 frost-free days
Jute	24–35 · 120–200	alluvial	WB, Bihar, Assam	India, Bangladesh, China	retting; plastic alternative
Sugarcane	20–27 · 100–150	loamy	UP, Maharashtra, Karnataka	Brazil, India, China, Thailand, Pakistan	very water-intensive; ethanol + sugar
Pulses	15–27 · low rain	well-drained	MP, Maharashtra, UP	India, Canada, Myanmar, Brazil, Ethiopia	N-fixing; intercropping; India 766 kg/ha vs world 1,015
Oilseeds	15–30 · 30–75	loamy-sandy	Rajasthan, MP, Maharashtra	USA, China, Brazil, India	soybean, mustard, groundnut

Crop	Temp · rain	Soil	Top Indian states	Top world	Remarks
🎯 Palm oil PYQ 2021	25–28 · >200	tropical well-drained	AP, Telangana, Kerala	Indonesia, Malaysia, Thailand, Colombia, Nigeria	deforestation concern; push in NE
🎯 Tea PYQ 2022	16–32 · 150–300	acidic, slopes	Assam, WB, TN, Kerala (+ Tripura, HP)	China, India, Kenya, Sri Lanka, Vietnam	plantation; labour-intensive
Coffee	15–28 · 150–200	acidic, fertile	Karnataka, Kerala, TN	Brazil, Vietnam, Colombia, Ethiopia, Indonesia	shade-grown; Arabica and Robusta
🎯 Cocoa PYQ 2024	18–32 · 100–250	laterite	AP, Kerala, Karnataka, TN	Ivory Coast, Ghana, Indonesia, Nigeria, Ecuador	shade (under coconut/ areca); ~70% of world beans from W. Africa (Ivory Coast, Ghana, Nigeria, Cameroon)
Rubber	25–35 · 200–450	laterite, acidic	Kerala, TN	Thailand, Indonesia, Vietnam, India, China	latex
Cashew	20–30 · moderate	sandy loam	Maharashtra, AP, Odisha, Karnataka, Kerala	Vietnam, India, Ivory Coast, Brazil	export cash crop
🎯 Turmeric PYQ 2025	20–30 · >150	sandy loam	Maharashtra, Telangana, TN, Karnataka	India, China, Myanmar, Bangladesh	>30 varieties, >20 states
Mango	24–30 · 75–250	deep loamy	UP, AP, Telangana, Karnataka, Bihar	India, China, Thailand, Indonesia, Mexico	India largest; Alphonso, Dasheri, Langra
Coconut	20–32 · 100–200	sandy, coastal alluvial, laterite	Kerala, TN, Karnataka, AP	Indonesia, Philippines, India, Sri Lanka, Brazil	Kalpavriksha; humid coast
Red gram (tur)	20–30 · 60–100	loam–black	Maharashtra, Karnataka, UP, MP	India, Myanmar, Malawi, Tanzania, Kenya	Kharif; deep-rooted; N-fixing
Mustard	10–25 · 25–75	alluvial–loamy	Rajasthan, UP, Haryana, MP	Canada, China, India, Germany, France	Rabi; frost-sensitive; sarson, rai

📰 *In the news · 2025 (verify dates)* **National Turmeric Board:** HQ Nizamabad (Telangana), inaugurated June 2025.

Sugarcane moved south: tropical climate, higher sucrose, longer crushing season (Oct–May south vs Nov–Feb north; no extreme summer heat), cooperatives more successful in the south. ⭐

🎯 TYPES OF FARMING PYQ 2011 PYQ 2012

Type	Rainfall	Growing season	Focus
Dry farming	<75 cm	<75 days	moisture conservation, drought-resistant crops
Dryland	75–115 cm	75–120 days	soil moisture retention, drought-tolerant crops, adaptive strategies
Wetland	>115 cm	>120 days	rainwater harvesting, moisture conservation, crop selection
Irrigated	–	–	protective (saves crops from drying) vs productive (raises yield)

75–115–120 ladder. Dry/dryland farming gives ~80% maize, 90% bajra, ~95% pulses, 75% oilseeds. Groundnut, sesamum, pearl millet = predominantly rainfed (2012 PYQ, D): sesamum on marginal dry sloping land; ~80% groundnut area rainfed; main rainfed cereals sorghum, bajra, ragi. ⭐

Micro-irrigation PYQ 2011

→ Drip, sprinklers, foggers; improves water-use efficiency

→ Pros: saves water, stops soil erosion, no levelling, any soil, uniform watering, fertigation and chemigation, labour saving, low pressure (drip), higher yield, bigger fruit

→ Cons: high investment, pipe clogging, spacing needed, technical installation

→ 2011 (C): reduces nutrient loss ✔; checks falling groundwater table ✔; "only means of irrigation in dryland farming" ✘

→ Drip: water drop by drop at root zone; emitters inline, online, microtubes, pressure-compensated; for wide-spaced crops · Micro-sprinkler: sandy/loamy soils, horticulture, small grasses, nurseries and lawns

SRI (System of Rice Intensification)

→ Madagascar, 1980s; raises yield while using fewer resources

→ AWD (Alternate Wetting and Drying) instead of continuous flooding in nurseries

→ Organic manures; yields +30–50%; less methane

AGRICULTURE REVOLUTIONS

Revolution	When · who	Win	Cost
Green	late 1960s–early 1980s · M.S. Swaminathan	HYV seeds + fertiliser + irrigation; wheat/rice jump; food self-sufficiency	regional (Punjab, Haryana, W. UP), social (big farmers), pollution, water-table fall
White (Operation Flood)	1970s · Verghese Kurien	world's largest milk producer; cooperatives (Amul)	quality, cold chain, processing
Blue	1980s	fish output; ~17% of agri exports (2021–22)	overfishing, pollution
Yellow	1980s	oilseeds; fewer edible-oil imports	climate and price swings
Evergreen	Swaminathan's proposal	sustainable mix of traditional + modern (ecology + technology)	scale, policy support

Colour = crop

→ Green grains
→ White milk
→ Blue fish
→ Yellow oilseeds
→ Golden fruit/honey/horticulture
→ Golden fibre jute
→ Silver eggs
→ Silver fibre cotton
→ Pink onion/pharma/prawn
→ Red tomato/meat
→ Round potato
→ Brown coffee/leather/cocoa
→ Grey fertiliser
→ Protein lentils.

🌐 AGRICULTURAL REGIONS OF THE WORLD

Type	Key points
Nomadic herding	4th-world communities; camels–Bedouins · cattle–Masai · horses–Kazakhs · reindeer–Chukchis
Livestock ranching	subsistence: African savannah; commercial: Australia (sheep, cattle), Brazil (beef)
📚 Shifting agriculture	mostly 4th-world communities; sedentary = rotation of crops, not fields
Intensive subsistence	small holdings, multiple seasons; with paddy = wetter margins, without paddy = drier; primitive (manual, no surplus) vs non-primitive (tech, surplus)
🎯 Mixed subsistence / mixed farming PYQ 2012	crops + livestock together (2012, C); not "mixed cultivation"; Europe (Ireland–Russia), N. America east of 98° meridian, Argentine pampas, SE Australia, S. Africa, NZ; dense, urbanised, industrial societies
Extensive commercial grain	breadbasket of Anglo-America, Wheat Triangle of Eurasia; temperate grasslands
Commercial dairy	wetter temperate margins; India = tropical yet leading milk producer
Plantation	tropics; years of yield from same stand: coffee (Brazil), cocoa (Ghana), rubber (Indonesia, Malaysia), spices (India), tea (China); non-traditional (2–3 yrs): cotton, cane
Mediterranean	gardens and vineyards
📚 Horticulture / truck	Bulgaria (roses), Netherlands (tulips); now India (marigold, rose, lily, brinjal)
Cooperative	pool resources except land; Belgium, Netherlands, Denmark; IFFCO, Amul
Collective	USSR Kolkhoz; state owns land; free-riding problem

⏳ AGRICULTURAL TERMS PYQ 2017 PYQ 2020

Term	Meaning
Organic	no synthetic pesticides/fertilisers
📚 Permaculture	mimics natural ecosystems; mulching; multicropping; agroforestry example
Agroforestry	trees and shrubs in farmland
Regenerative	renew soil health and ecosystems
📚 Conservation agriculture	minimum tillage, residue cover, rotations
Hydroponics · Aquaponics · Aeroponics	nutrient water, no soil · fish + hydroponics · air/mist
Vertical farming	stacked layers, often indoors
Precision · Smart	GPS and IoT field management · AI, drones, sensors
Crop rotation · Intercropping	alternate crops for fertility · crops in proximity for pest control, resource use
Cover cropping	protects and enriches soil between crops: mustard, clover, radish
📚 Silviculture · Viticulture · Apiculture · Sericulture	forest plantations · grapes · bees · silkworms
📚 Horticulture · Floriculture	fruits, vegetables, flowers, ornamentals · flowers for the floral industry
Climate-smart	raises productivity and resilience, cuts emissions
Biopharming	GM plants/animals making pharmaceuticals
Agrivoltaic	solar power + farming: land use, rural energy access

Natural farming (Fukuoka method)

→ Minimal human intervention, works with nature; no synthetic chemicals, fertilisers, pesticides, excess tilling

→ Biodiversity, soil health, local natural resources

→ Masanobu Fukuoka (Japan), The One-Straw Revolution, 1975

→ ZBNF (Zero Budget Natural Farming): low-cost, local preparations, no external inputs; weeds, insects, animals not enemies; emulates natural conditions

💰 Land use: 9 categories (Land Revenue records)

→ Forests · Barren and wasteland · Non-agricultural use

→ Permanent pastures (mostly Panchayat/govt) · Misc. tree crops and groves (mostly private)

→ Culturable wasteland (fallow >5 yrs) · Current fallow (≤1 yr) · Fallow other than current (1–5 yrs) · Net sown area

→ Records: Land Revenue Dept (Survey of India); Digital India Land Records Modernization Programme; land degradation: see Soils of India

🎯 AGRICULTURE PYQ BANK

Year	Asked	Answer / rule
2011	Micro-irrigation statements	C (1 and 3)
2011	Lower Gangetic plain, humid-hot: best crop pair	Paddy and jute (C): jute >25 °C, humidity 70–90%; rice 21–37 °C, tolerates max 40–42 °C
2011	State: arid north, cotton centre, cash crops > food crops	Gujarat (B)
2011	Effect of salinisation	Makes some soils impermeable (B): salt breaks aggregates, pores collapse; advanced stage = barren, desertified land
2012	Mixed farming	Crops + livestock (C)
2012	Rainfed: groundnut, sesamum, pearl millet	All three (D)
2013	Kharif: cotton, groundnut, rice, wheat	1, 2, 3 (B); not wheat
2014	Kinnaur–areca? Mewat–mango? Coromandel–soya?	None (D); MP = soybean
2014	Maize: starch, biodiesel oil, alcohol	All three (D)
2015	Excess nitrogen fertiliser	Acidity ↑ and nitrate leaching (C); not more N-fixing microbes
2017	Conservation agriculture	Minimum tillage, residue cover, rotations (D: 2, 4, 5)
2018	Organic matter ↓ water holding? Soil no role in sulphur cycle? Irrigation salinises?	Only 3 (B)
2019	Agri soils emit nitrogen oxides; cattle ammonia; poultry reactive N	All three (D); ~90% N_2O from soils
2020	Permaculture vs chemical farming	1, 2, 4 (B); chemical farming not easy in semi-arid areas
2021	Palm oil: native to SE Asia ✘; lipstick/perfume raw material ✔; biodiesel ✔	B (2 and 3); native to hills of Guinea
2021	Least water-efficient crop	Sugarcane (A): 1,800–2,200 mm/season; sunflower 672.4; pearl millet 350
2022	Tea states: AP, Kerala, HP, Tripura	All four (D); Kerala major, rest minor
NEW 2024	Two largest cocoa producers	Côte d'Ivoire and Ghana (~70% of world cocoa)
NEW 2025	Turmeric 2022-23: India largest producer/ exporter; >30 varieties; Maharashtra, Telangana, Karnataka, TN	All three (D)

Source conflict: 2024 cocoa PYQ prints "Answer: D" but the correct pair is option (c) Côte d'Ivoire and Ghana (explanation: ~70% of world cocoa). Take the pair.

🌐 2 · WORLD RESOURCES, MINING

Resource = availability + utility

→ Anything usable from the environment for needs and wants

→ **Availability:** accessible (technology, economic feasibility, cultural acceptance)

→ **Utility:** some benefit

Classes

→ Origin: natural / human-made · biotic / abiotic

→ Exhaustibility: renewable / non-renewable

→ Ownership: individual / community / national / international (high seas, outer space)

Mining condition: cost of extraction + purification < value of final product. Types: **open-cast** (near surface), **shaft** (deep).

🎯 MINERALS OF INDIA PYQ 2013 PYQ 2018 PYQ 2022 PYQ 2023

State pegs

- → Iron = Odisha–Jharkhand–Chhattisgarh–Karnataka–Goa belt
- → Copper = Rajasthan (Khetri)
- → Bauxite = Odisha
- → Gold = Kolar and Hutti (Karnataka)
- → Diamond = Panna (MP) only
- → Mica = Jharkhand
- → Monazite/thorium = Kerala
- → Uranium = Jaduguda (Jharkhand)
- → Lead-zinc-silver = Rajasthan (Zawar, Rampura Agucha)
- → Chromite = Odisha >96%.

Mineral	Where in India	Hooks
Iron ore	Odisha (Mayurbhanj, Keonjhar, Sundargarh); Jharkhand (Singhbhum); Chhattisgarh (Durg, Dantewada, Bailadila); Karnataka (Ballari, Hospet, Kudremukh); Goa, Maharashtra, Telangana, AP, TN	hematite, magnetite, limonite, siderite; steel
Manganese	Odisha, Karnataka, MP, Maharashtra	steel, batteries
Copper	Rajasthan 52.25% (Jhunjhunu, Khetri, Alwar) · MP 23.28% (Balaghat) · Jharkhand 15.14% (Singhbhum)	chalcopyrite; world leaders Chile, DR Congo, Peru
Bauxite	Odisha (Kalahandi, Koraput), Gujarat, Jharkhand (Lohardaga), Maharashtra, Chhattisgarh (Amarkantak), MP; new finds in Meghalaya	aluminium smelters near power plants (huge electricity)
Gold	Karnataka (Kolar, Hutti), Jharkhand, Chhattisgarh, Rajasthan, AP (Jonnagiri); Bihar preliminary	Kolar to restart 24 yrs after closure
Silver	by-product; Rajasthan ≈ 60% (Zawar, Rampura Agucha)	lead–zinc by-product
Uranium	Jharkhand (UCIL Jaduguda); potential Rajasthan, Chhattisgarh, Maharashtra, HP, Andhra	natural U has 0.72% U-235; U-235 fissile, U-238 not; enrichment raises concentration
Thorium (monazite)	India ≈ 25% of global deposits; Palakkad and Kollam (Kerala), Visakhapatnam, Mahanadi delta	cleaner; U-233; reactors can use up to 99% of fuel
Tin	Chhattisgarh (Dandkaranya)	bronze
Lead / Zinc	Rajasthan largest (HZL mines: Rampura Agucha, Sindesar Khurd, Rajpura Dariba, Zawar, Kayad)	zinc: galvanising; Zn–S batteries as lithium alternative
Nickel	Odisha (Sukinda), Jharkhand, Nagaland; ~0.2% of global reserves; no active mine	stainless steel, batteries
Cobalt	potential: Jharkhand, Odisha, Rajasthan, Nagaland, MP; no domestic production	
Lithium	J&K, Rajasthan, Karnataka (Mandya, Yadgiri); first mine expected in Korba, Chhattisgarh; no operating mine	
🎯 Titanium PYQ 2023	Kerala, TN, AP, Odisha (ilmenite, rutile)	ilmenite $FeO.TiO_2$, rutile TiO_2
Rare earths	monazite, bastnäsite, xenotime; coastal placers in Kerala, TN, Odisha, AP; 13.07 million t monazite (5th largest)	17 elements = 15 lanthanides + scandium + yttrium; China near-monopoly → Minerals Security Partnership
Mica	Jharkhand (Hazaribagh, Koderma), AP (Nellore), Rajasthan	Feb 2025: Major Mineral with feldspar, barytes, quartz
Tantalum, silicon	no production; silicon from quartz in Haryana, Rajasthan, AP	

💰 Tungsten	Karnataka, Rajasthan (Degana)	Madurai mining scrapped after protest
Antimony, germanium, indium, helium	~100% import dependent (antimony complete; germanium 100%; indium complete; helium 100%)	China limited exports (antimony July 2024; gallium July 2023)
Gallium, tellurium, selenium	by-products of bauxite/copper	ALCO plans 10 tpy gallium plant at Damanjodi
Diamond	~90% Panna (MP); AP (Krishna basin), Chhattisgarh	production only in Panna
Limestone	Rajasthan, MP (Satna, Katni), AP, Gujarat	$CaCO_3$ sedimentary rock; world: China, India, USA, Russia, Brazil
Others	Chromite (>96% Odisha) · Pyrite (none since 2003) · Borax (small, Puga valley) · Cryolite (nil)	

2013 PYQ (B): natural gas in Gondwana beds ✘; mica abundant in Kodarma ✔; Dharwars famous for petroleum ✘ (Dharwars: iron, manganese, lead, zinc, gold; Champion reef, Kolar). Natural gas areas: KG basin, Assam, Gulf of Khambhat, Cuddalore (TN), Barmer. Kodarma = ruby mica. 2018 (C): Bhilwara (Rajasthan) has iron ore; MMDR Amendment Act 2015 (from Jan 2015): states auction non-coal mines, first auction Dec 2015; Rajasthan ~57 minerals, sole producer of lead-zinc, selenite, wollastonite; copper 809.09 mt = 54% of resources, 41% of output (2nd leading producer). ✅

Handout alert (2018 PYQ): explanation prints both "states can auction" and "states cannot", and says AP/Jharkhand "do not have gold mines" while the table lists them as gold states. Take answer C and the gold table.

2022 PYQ (b: 1, 2, 4): monazite = rare earths + thorium (prescribed substance, DAE); only govt bodies process/export: IREL (under DAE); in-situ monazite in Odisha, AP, TN, Kerala, WB, Jharkhand (AMD); not the entire coast. REE PYQ (a: 1, 3): NCMM launched 2025 ✔; China first but India not second ✘; REE = 17 elements, not 13. ✅

In the news · 2025 (verify dates) NCMM: National Critical Mineral Mission launched 2025 for critical-mineral self-reliance; REE and China export curbs are the context.

🎯 COAL, OIL, GAS, SHALE PYQ 2012 PYQ 2013 PYQ 2016

Coal

→ Plant remains → coalification; world: China (Shanxi, Inner Mongolia), India, USA, Australia, Indonesia

→ Indian coal: non-coking (non-metallurgical), high ash 40–45%, low sulphur ~0.5–1% (~0.7%), high ash-fusion ~1,500 °C; ~75% of electricity from coal thermal plants

Gondwana (bituminous)

→ Valleys: Damodar, Mahanadi, Godavari, Son

→ Jharkhand-Bengal: Raniganj, Jharia, Bokaro, Giridih, Karanpura

→ Singrauli (MP/UP) · Korba (CG) · Talcher, Rampur (Odisha)

→ Chanda-Wardha, Kamptee, Bander (Maharashtra) · Singareni (Telangana) · Pandur (AP)

Tertiary

→ Anthracite: Kalakot (J&K)

→ Assam (Makum, Jaipur, Nazira) · Arunachal (Namchik-Namphuk) · Meghalaya (Darangiri, Cherrapunji, Mewlong, Langrin)

→ Nagaland: rat-hole mining despite SC ban, Art. 371A

→ Lignite: Neyveli (TN), Gujarat, Puducherry

2012 (B): India imports coal because power plants lack domestic supply AND steel needs coking coal; not "saving reserves". 2013 (B: 1, 2): high ash ✔, low sulphur ✔, low ash-fusion ✘. Also: India 5th largest reserves, 3rd largest producer and importer; sulphur removed by washing. ✅

Oil and gas

→ Crude: Mumbai High, Barmer (Rajasthan), Cambay (Gujarat), Assam-Arakan; KG belt has reserves but no production; new find in Andaman Sea region (being evaluated)

→ Natural gas: Assam (Digboi, Rudrasagar), Mumbai High, Gujarat, Rajasthan, Tripura, KG; also Andaman Sea

🎯 Shale oil and gas PYQ 2016

→ Unconventional, in low-permeability shale; hydraulic fracturing; world: USA, China, Argentina, Canada, Russia

→ India basins: Cambay, Ganga valley, Assam-Arakan, KG, Kaveri

→ 2016 PYQ (D): Cambay, Cauvery, KG all ✔; six basins: Cambay, Assam-Arakan, Gondwana, KG onshore, Cauvery onshore, Indo-Gangetic

→ Problems: water contamination, induced seismicity, land disturbance, methane leaks

🎯 ENERGY: RENEWABLES, NUCLEAR PYQ 2023

Source	India facts	World / notes
All renewables	283.46 GW up to March 2026	
Solar	157 GW by May 2026; target 280 GW by 2030; top: Rajasthan, Gujarat, Maharashtra, TN, Karnataka	China, USA, India, Japan, Germany; thermal + electric; desert panels: warm area (darker than sand), logistics and maintenance costs, sand cover
Wind	56 GW; top: Gujarat, TN, Karnataka, Rajasthan, Maharashtra; no offshore farm	China, USA, India, Germany, Brazil; offshore winds faster (no friction) but cost cancels gain
Bio-energy	10.75 GW; Maharashtra, Karnataka, UP, Punjab, Bihar	USA, Brazil, Indonesia, China, India; waste-to-energy (incineration → steam), biogas (anaerobic digestion: methane + CO_2)
Geothermal	potential 10,600 MW; Manikaran (HP); target 10 GW by 2030; not yet tapped; SCCL 20 kW pilot, Manuguru, Bhadradri Kothagudem (Telangana)	USA, Indonesia, Philippines, Turkiye, New Zealand
Tidal	Kachchh (Gujarat) 50 MW; Durgaduani Creek (WB) 3.75 MW; started, stopped due to cost	S. Korea, France, Canada, China, Russia
OTEC	65 kW at Kavaratti (Lakshadweep) for desalination	only 2 pilot plants: Japan (Okinawa), USA (Makai, Hawaii)
Nuclear	8,180 MW → 22,480 MW by 2031–32; +70% in last 10 years	fusion and fission; only fission used; 2021: USA, China, France, Russia, S. Korea

📝 3 stages of India's nuclear programme

→ 1: PHWR, natural uranium (U-235)
→ 2: Fast Breeder, plutonium (Pu-239), Kalpakkam
→ 3: thorium, U-233, KAMINI, Kalpakkam

U → Pu → Th – Use uranium, Produce plutonium, Then thorium. Stages 2 and 3 are Kalpakkam-linked.

2023 PYQ (C): India depends on coal (~75% power from thermal) despite uranium ✔; "uranium enriched ≥60% needed" ✘, ~5% enrichment is enough. ✅

Source conflict (handout vs mind map): mind map prints older figures, renewables 220.10 GW and solar 105.65 GW (31 Mar 2025, MNRE); handout text says 283.46 GW (Mar 2026) and 157 GW (May 2026). Use the handout; 280 GW by 2030 is the same in both.

Why not cover deserts with panels? Just 1.2% of the Sahara could power the world, but dark panels absorb more heat, raise temperature and change rain and wind nearby; plus high logistics and maintenance costs (sand cover). 💡

🎯 4 · INDUSTRIES (no PYQ)

Sector	What	Examples
Primary	extraction/harvest	agriculture, fishing, forestry, mining
Secondary	processing/manufacture	light (consumer goods), heavy (capital goods: steel, ships), construction
Tertiary	services	trade, tourism, transport, finance, health, education, IT, professional
Quaternary	knowledge	R&D, information services, advanced training
Quinary	top decisions	government, NGOs, culture and arts

P-S-T-Q-Q – Dig → Make → Serve → Think → Decide.

👥 Size

→ (investment / turnover): Micro <₹2.5 cr / <₹10 cr
→ Small ₹2.5–25 cr / ₹10–100 cr
→ Medium ₹25–125 cr / ₹100–500 cr
→ Large above. Ownership: public, private, joint, cooperative. Use: consumer vs capital goods.

Location factors

→ Raw material · labour · market · transport/infrastructure · energy · capital · government policy

→ Climate/environment: circular-economy push, co-locate near recycling centres, "urban mining"

→ Social/cultural · technology: reshoring (robots replace cheap labour, plants return to high-income countries)

→ Global: nearshoring, friend-shoring · ecological sustainability

Weber's weight rule

→ Weight-losing (bulky raw material → light product, e.g. ore) → sit at the raw material

→ Weight-gaining (product heavier/bulkier than inputs) → sit near the market

→ Footloose: weight does not matter

→ Isotim = equal transport-cost line · Isodapane = equal total-cost line (transport of inputs + outputs + labour); best site on the lowest isodapane

World industrial regions	Handout lines
North America	NE USA (steel, autos) · West Coast (tech, aerospace) · South USA (oil, aerospace) · Ontario-Quebec
Europe	Ruhr, Alsace-Lorraine, Benelux · UK Midlands · Poland-Czechia-Slovakia-Hungary (Western, Northern, Eastern Europe)
Asia	East Asia (China, Japan, S. Korea, Taiwan) · South Asia (Maharashtra, Gujarat, TN, Karnataka) · SE Asia (Singapore, Malaysia, Thailand, Indonesia)
Africa	N. Africa · Gauteng and Western Cape (Southern) · W. Africa (Nigeria, Ghana)
Australia · S. America	east Australia · Brazil (São Paulo-Rio-Minas Gerais), Argentina, Chile

Indian region	Industries	Key cities
Mumbai-Pune	textiles, autos, chemicals, engineering, pharma, IT	Mumbai, Pune, Thane, Nashik
Kolkata-Hooghly	jute, textiles, chemicals, engineering, iron and steel, paper	Kolkata, Howrah, Durgapur, Haldia
Delhi-NCR	IT, electronics, autos, consumer goods	Delhi, Gurugram, Noida, Faridabad
Ahmedabad-Vadodara	textiles, chemicals, pharma, petrochemicals	Ahmedabad, Vadodara, Surat, Bharuch
Chennai-Bengaluru	autos, electronics, IT, heavy engineering	Chennai, Bengaluru, Hosur, Coimbatore
Kollam-Thiruvananthapuram	cashew, coir, software, biotech	Kollam, Thiruvananthapuram
Chotanagpur	iron and steel, coal, heavy engineering	Jamshedpur, Ranchi, Bokaro, Dhanbad
Visakhapatnam-Guntur	shipbuilding, petrochemicals, fertiliser, steel	Visakhapatnam, Guntur

🎯 5 · ROADS AND RAILWAYS PYQ 2014

Roadways

→ Network >63 lakh km (Jan 2025), among the largest in the world

→ NH = 2% (NHAI, MoRTH) · State highways 4% (state PWDs) · District roads 14% · rural roads (Gram Panchayats, PMGSY) · urban roads (municipalities)

→ NH44 (earlier NH7) Srinagar-Kanyakumari = longest · NH48 (earlier NH8) Delhi-Mumbai

→ Since 1 April 2018 length counts each lane separately

→ NH numbers: even = North-South (rise east → west: NH2 east, NH68 west) · odd = East-West (rise north → south: NH1 north, NH87 south) · sub-highway = prefix (NH101 from NH01); new scheme after 2010

Railways

→ >68,000 km; state-owned, 19 zones (General Manager each), Ministry of Railways

Gauges

→ **Broad 1,676 mm**: most used, long-distance, freight
→ **Metre 1,000 mm**: rural; Unigauge launched 1 April 1992
→ **Narrow 762 or 610 mm**: hills, being phased out
→ **Standard 1,435 mm**: metro, monorail, tram

DFCCIL dedicated freight corridors

→ **Eastern:** Ludhiana (Punjab) → Dankuni (near Kolkata), ~1,839 km (coal, steel)
→ **Western:** Dadri (UP) → JNPT (Mumbai), ~1,506 km
→ Future: East-West (Kolkata-Mumbai), North-South (Delhi-Chennai), East Coast (Kharagpur-Vijayawada), Southern (Chennai-Goa)

→ Also suburban (Mumbai, Delhi), metro (DMRC, BMRCL), proposed Mumbai–Ahmedabad high-speed rail

Metre = 1,000 — Metre = 1,000 mm (name = number). Broad = the biggest, 1,676. Standard 1,435 = city metros.

2014 PYQ (D): NH4 Chennai-Hyderabad? NH6 Mumbai-Kolkata? NH15 Ahmedabad-Jodhpur? All ✘: old NH4 became NH48; NH6 (Jorabat-Selling) is in the NE; NH15 starts at Baihata, Assam. ✅

Pipelines

→ Liquids and gases only; less flexible than rail and road; safe, cost-effective over long distances
→ **Naharkatia–Noonmati–Barauni** (crude) · **Mumbai High–Mumbai–Ankleshwar–Koyali** (crude + gas) · **Salaya–Koyali–Mathura** (crude; to Panipat, Jalandhar)
→ **Hazira–Bijapur–Jagdishpur (GAIL)**: largest underground pipeline in the world; extended to Delhi via Dadri
→ Also carry ores (slurry), water, local milk

Ropeways

→ Suspended cables for difficult terrain; early 20th c. (mining, logging); first major passenger ropeway **Auli, 1969**; also Darjeeling, Gulmarg, Vaishno Devi
→ **Parvatmala** (Budget 2022-23, PPP, MoRTH): 8 projects ~60 km initially; **Jan 2024: 400 projects, ₹1.25 lakh crore in 5 yrs**
→ Varanasi urban ropeway · Gaurikund–Kedarnath (9.7 km, 3,584 m)

📜 NH numbering (new system 2010; earlier under NH Act 1956)

→ **North-South = even numbers**, rising from **east to west** (NH2 east, NH68 west)
→ **East-West = odd numbers**, rising from **north to south** (NH1 north, NH87 south)
→ Branch (sub-highway): prefix on the parent number, e.g. **NH101** from NH01
→ **1 April 2018:** lane system, each lane's length counted separately

PYQ 2014 (answer D, none matched): **NH 4** = Andaman Trunk Road, **230.7 km** (old NH 4 Mumbai-Pune-Hubli-Bangalore-Chennai is now **NH 48**); **NH 6** = Jorabat (Meghalaya) to Selling (Mizoram); **NH 15** = Baihata (Assam) to Wakro (Arunachal).

🎯 6 · WATER TRANSPORT PYQ 2023

Why cheap: low friction, fuel efficient; no separate infrastructure except terminals. 💡
IWAI established **1986**; **National Waterway Act 1986**; **111** notified inland waterways (13 in operation).

Waterway	Route	Length
NW-1	Ganga–Bhagirathi–Hooghly (Haldia–Allahabad/Prayagraj)	1,620 km
NW-2	Brahmaputra (Dhubri–Sadiya)	891 km
NW-3	West Coast Canal: Champakara–Udyogmandal	205 km
NW-4	Krishna (Muktiyala–Vijayawada)	82 km
NW-73 / NW-100	Narmada / Tapi	226 / 436 km
NW-97	Sunderbans	172 km
NW-10, 68, 83, 85, 91, 111	Amba 45 · Mandovi 41 · Rajpuri creek 31 · Revdanda–Kundalika 31 · Shastri–Jaigad 52 · Zuari 50	km

Ocean transport

→ ~95% of foreign trade by volume, 70% by value

→ 13 major ports + 200+ minor/intermediate; newest major Galathea Bay (Great Nicobar, transshipment)

→ Vadhavan (14th, near Mumbai, Palghar): approved June 2024, ₹76,220 crore, yet to be built

→ Major ports: Port Trusts (Major Port Trusts Act 1963) · minor: State Maritime Boards

Kind Men Join Very Many New Clubs — West coast, north → south: Kandla (Deendayal) · Mumbai · JNPA (Nhava Sheva, leading container port) · Vadhavan · Mormugao (Goa, iron ore) · New Mangalore (only major port of Karnataka) · Cochin (Vallarpadam transshipment).

Very Clever Kids Visit Pretty Kolkata — East coast, south → north: V.O. Chidambaranar (Tuticorin) · Chennai · Kamarajar (Ennore) · Visakhapatnam (natural harbour) · Paradip (bulk) · Kolkata–Haldia (Syama Prasad Mookerjee Port, only riverine major port, Hooghly).

2023 PYQ (B, two pairs): Kamarajar = first major port registered as a company (Companies Act, started 2001) ✔; Mundra (Adani) = largest private port, also overtook JNPT as largest container port ✔; Visakhapatnam largest container port ✘. ✅

Handout alert: the 2023 PYQ explanation says "Kamarajar Port at Tuticorin", but the port list puts Kamarajar at Ennore and V.O. Chidambaranar at Tuticorin. Use the list.

7 · AIR TRANSPORT

18 Feb 1911 — first commercial flight, Allahabad–Naini

1948 — govt takes 49% of Tata-owned Air India

1953 — nationalised: Indian Airlines domestic, Air India international

1990s — liberalised

2021 — Air India sold to Tata Group (handed over Jan 2022)

157 operational airports, 133 with AAI (Airports Authority of India). >90% traffic with two companies (duopoly concern). Private operators: GMR, Adani, GVK (Delhi, Mumbai, Bengaluru). ⭐

8 · COMMUNICATION

1837 Indian Post Office Act

mid-19th c. Railway Mail Service

1851 telegraph Kolkata–Diamond Harbour

1854 Post Office Act, uniform rates; telegraph links Kolkata, Bombay, Madras, Peshawar

1881 first telephone exchange, Kolkata

1927 Indian Broadcasting Company (Mumbai, Kolkata)

1936 All India Radio

1959 Doordarshan

1982 colour TV

1986 Speed Post

1990s private telecom and TV

1995 Internet

2013 telegraph ended

2015 Digital India

Notes

→ **Post:** before it, communication went with transport or messengers; first organised system late 18th c. (British); Act 1854 set urban and rural post offices; Speed Post (1986) to compete with private couriers; network among largest after 1947

→ **Radio:** private clubs early 1920s; British took over 1936 as AIR (key in WWII); **Quit India** used radio; multilingual after independence

→ **TV:** private channels broke Doordarshan monopoly (1990s liberalisation)

→ **Telephone:** urban and government offices at first; early 2000s mobile revolution; one of the fastest 5G rollouts

→ **Digital India (2015):** infrastructure, connectivity, e-governance, online education, telemedicine

→ **Satellite:** ISRO communication satellites for TV, telephony, data, internet (see Science and Tech)

2015 PYQ (A: 1, 2, 3): IRS satellites used for crop productivity, groundwater, mineral exploration ✔; telecom, traffic ✘. IRS-1A launched 17 March 1988 (polar sun-synchronous, from Baikonur); data used under NNRMS. ✅

Traps

→ wheat is Rabi

→ palm oil native to Guinea hills, not SE Asia

→ excess N = acidity + nitrate leaching

→ chemical farming hard in semi-arid

→ uranium ~5% not 60%

→ REE = 17, China first not India second

→ Mundra/JNPT lead containers, not Visakhapatnam

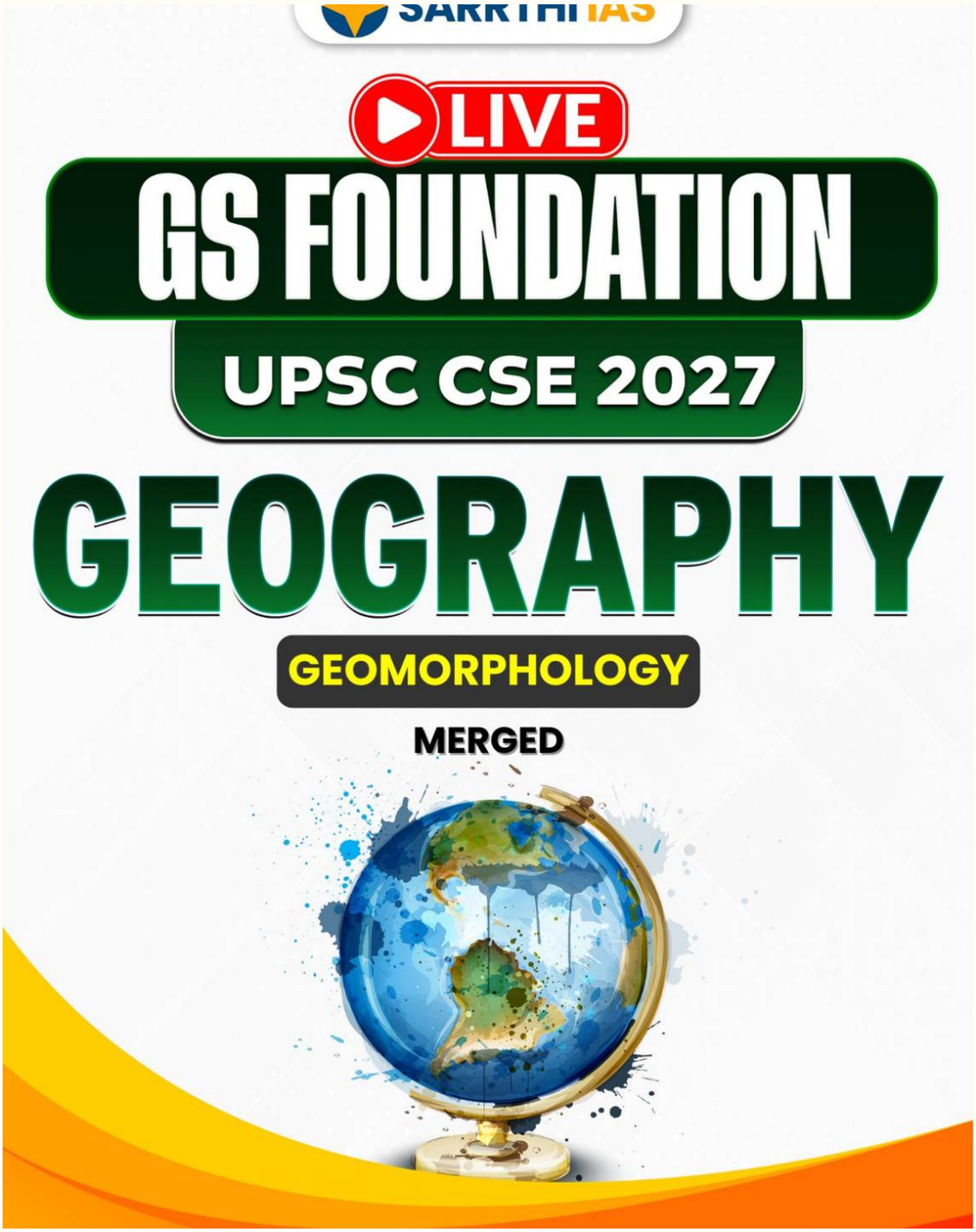
→ even NH = N–S, odd = E–W. **Connects to:** Indian Geography (soils, Gondwana coal, rivers, ports)

→ Climatology

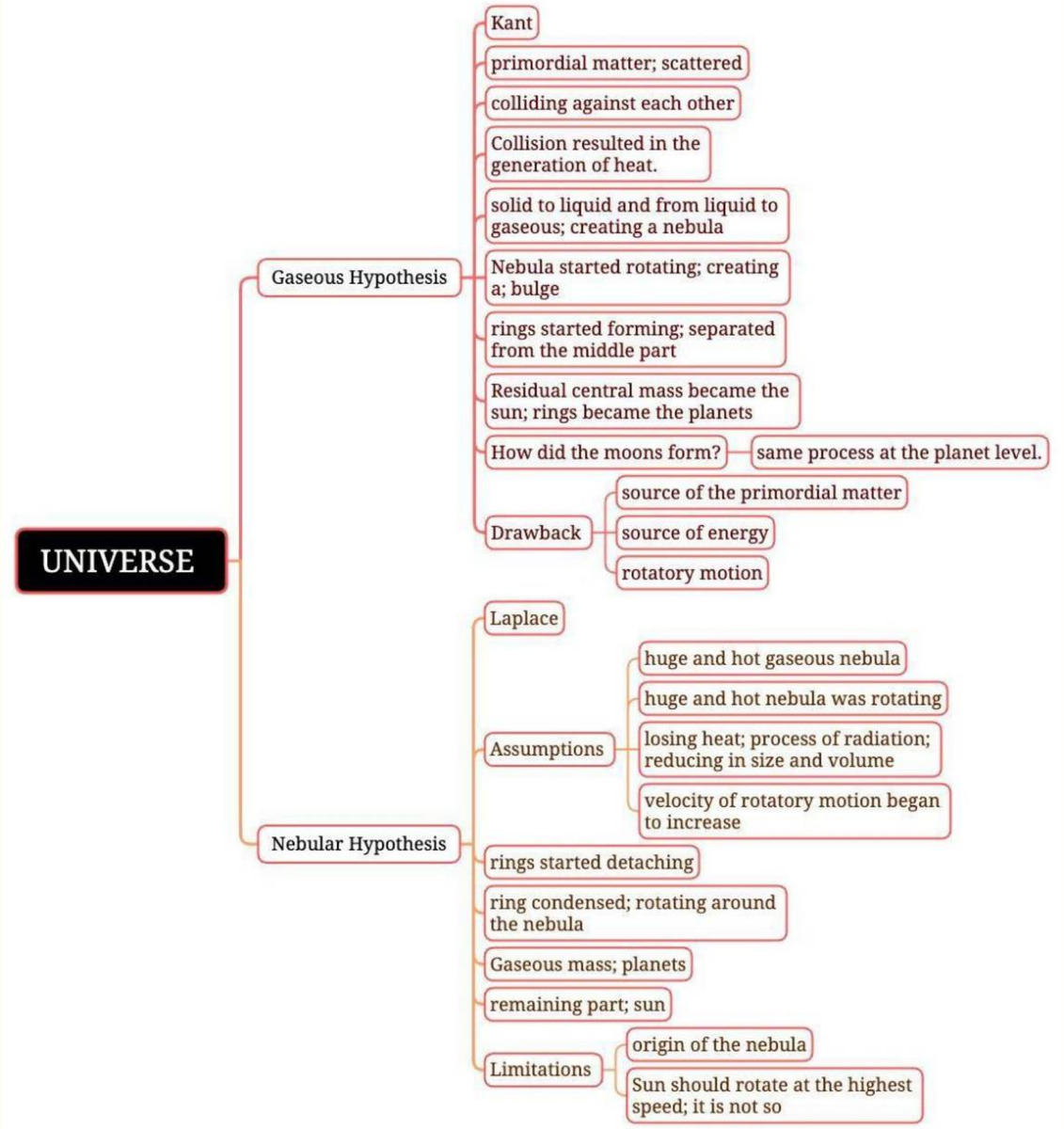
→ Human Geography.

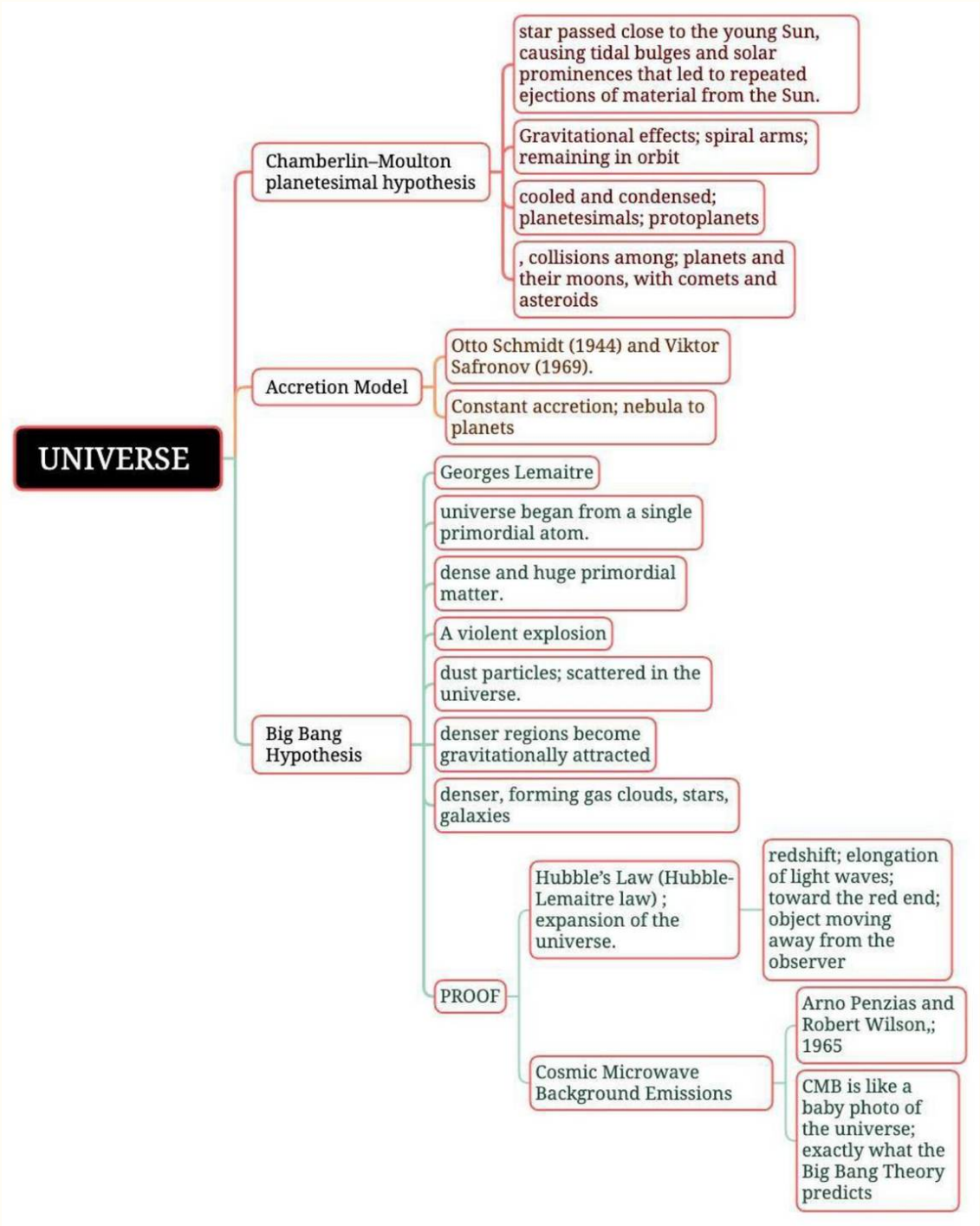
Recall drill: cover and answer

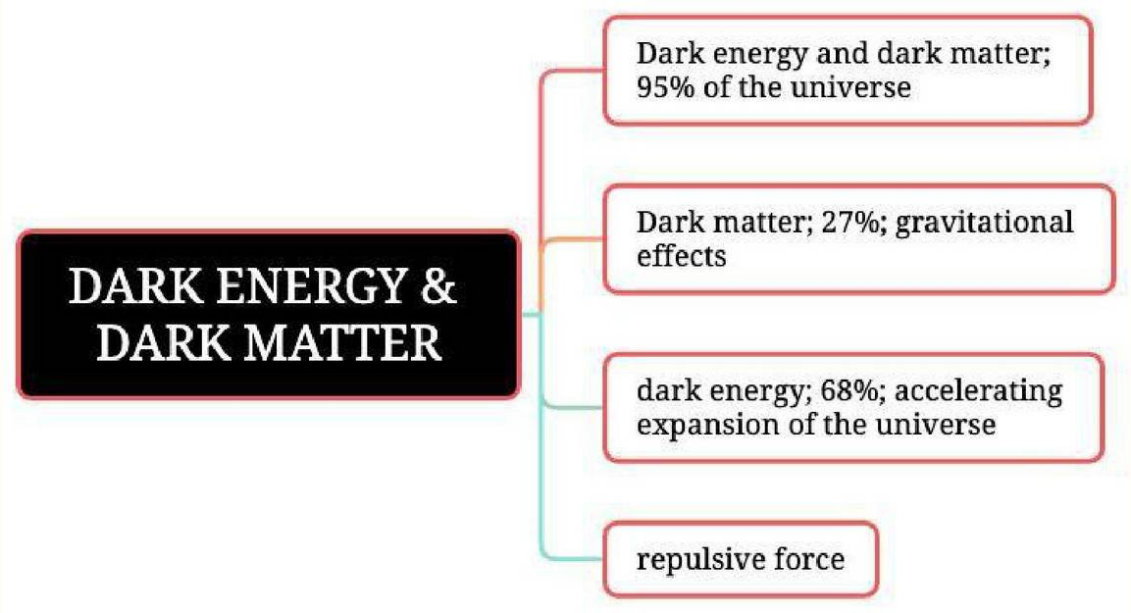
Kharif, Rabi, Zaid months?	Jun–Oct · Oct/Nov–Mar/Apr · Mar–Jun	Dry, dryland, wetland farming?	<75 cm/<75 d · 75–115/75–120 · >115/>120
Dry/dryland share: maize, bajra, pulses, oilseeds?	80% · 90% · 95% · 75%	Least water-efficient crop (2021)?	Sugarcane (1,800–2,200 mm)
Who: Green, White revolution?	M.S. Swaminathan; Verghese Kurien	Copper shares: Rajasthan, MP, Jharkhand?	52.25%, 23.28%, 15.14%
Indian coal traits; why import (2012)?	non-coking, high ash, low sulphur; short supply + coking coal	Renewables, solar, wind, nuclear?	283.46 GW; 157 GW (280 by 2030); 56 GW; 8,180 → 22,480 MW
Weber: weight-losing, isotim, isodapane?	at raw material; equal transport cost; equal total cost	Gauges; NH rule?	1,676 / 1,000 / 762-610 / 1,435 mm; even N–S, odd E–W
Ports: count, newest, only riverine?	13 major + Vadhavan; Galathea Bay; Kolkata–Haldia (SPM)		

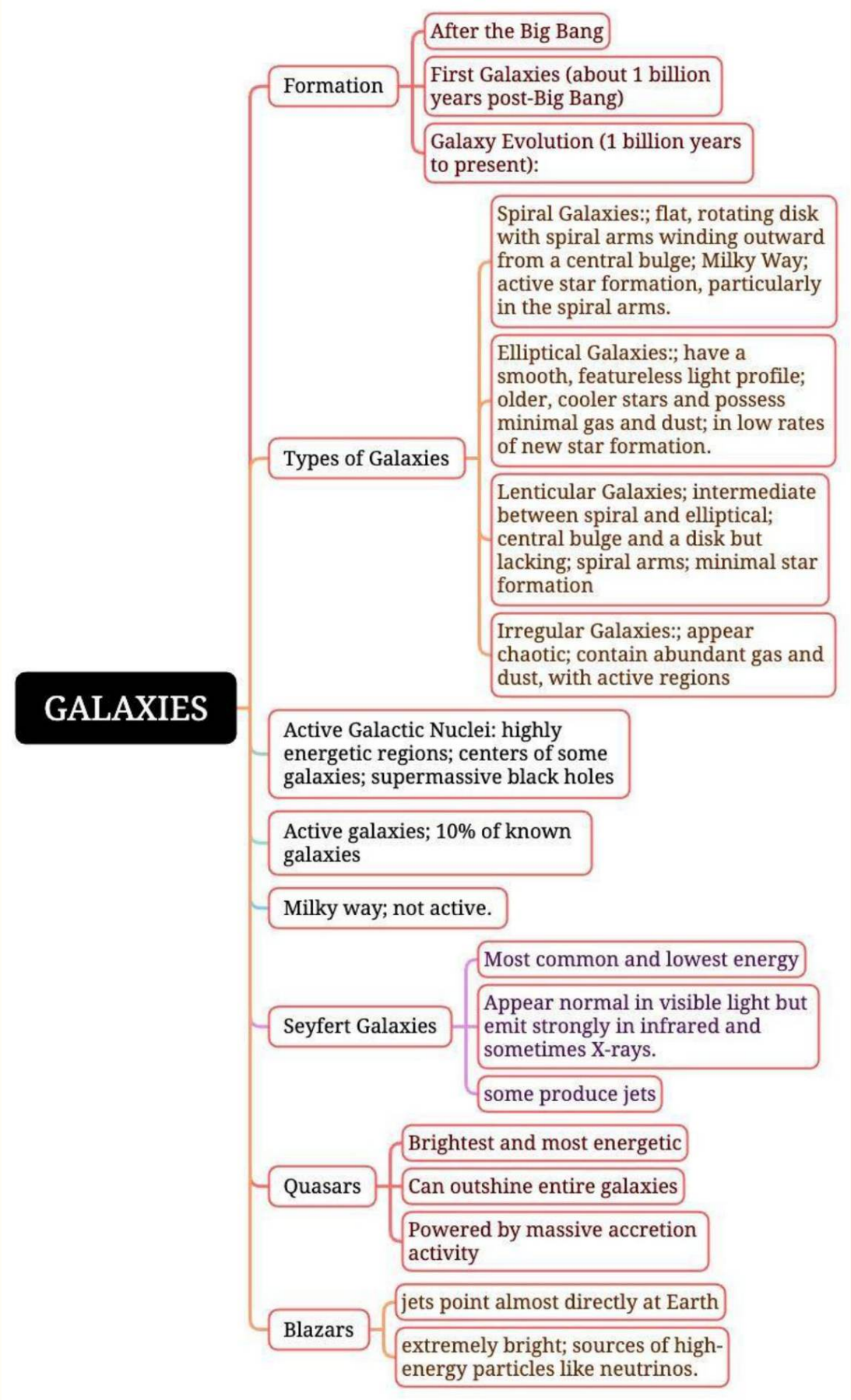

LIVE
GS FOUNDATION
UPSC CSE 2027
GEOGRAPHY
GEOMORPHOLOGY
MERGED

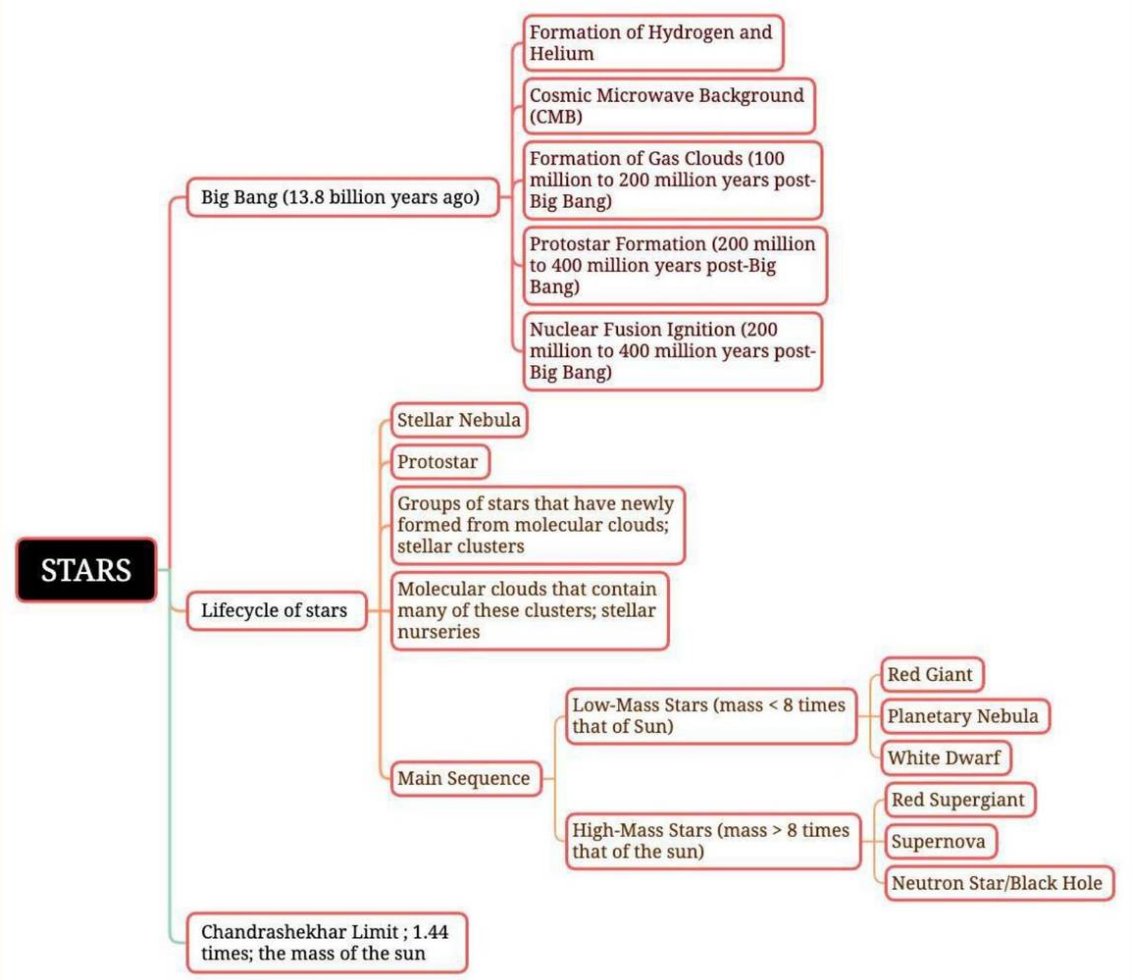
Universe & Earth Facts

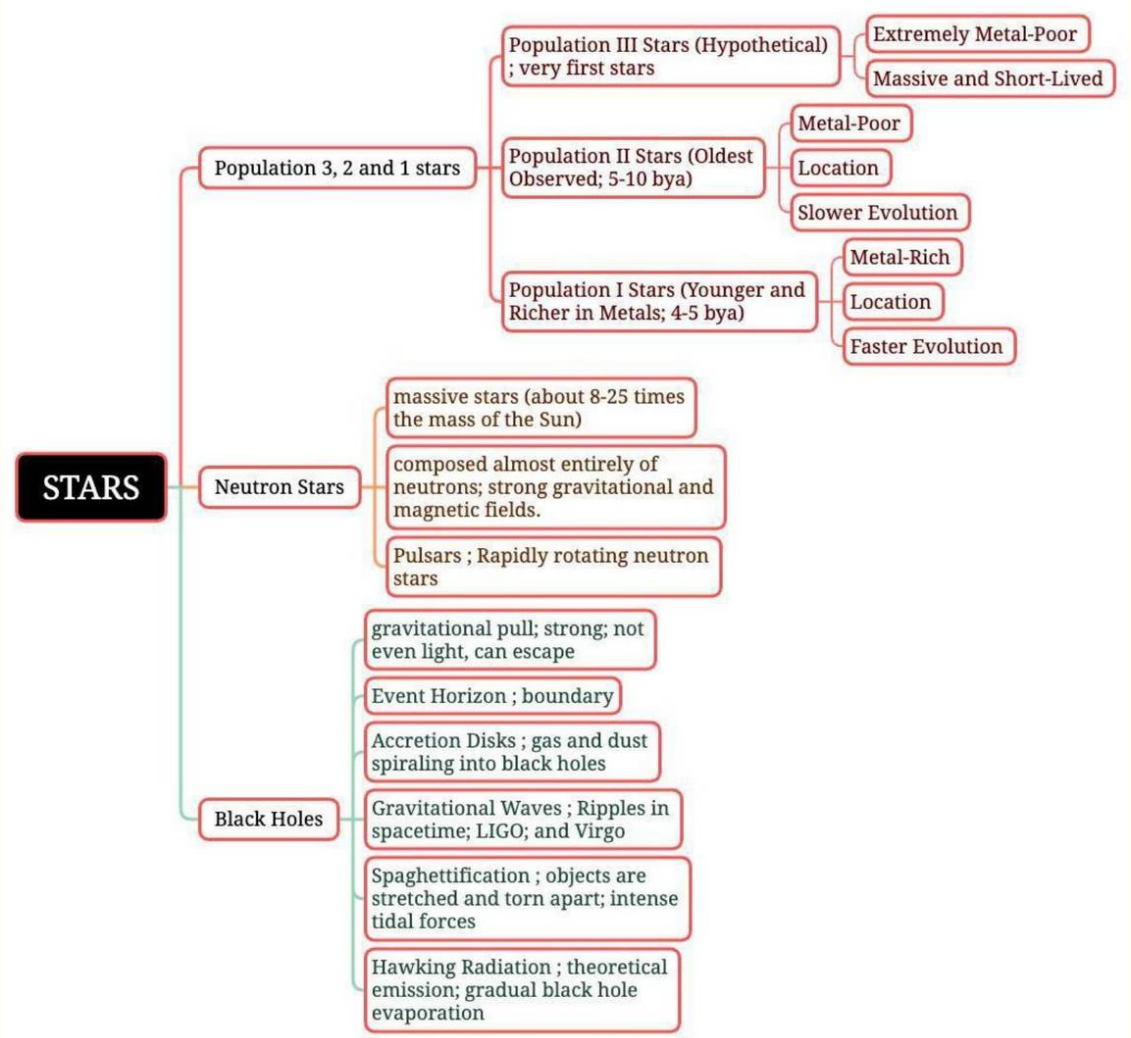


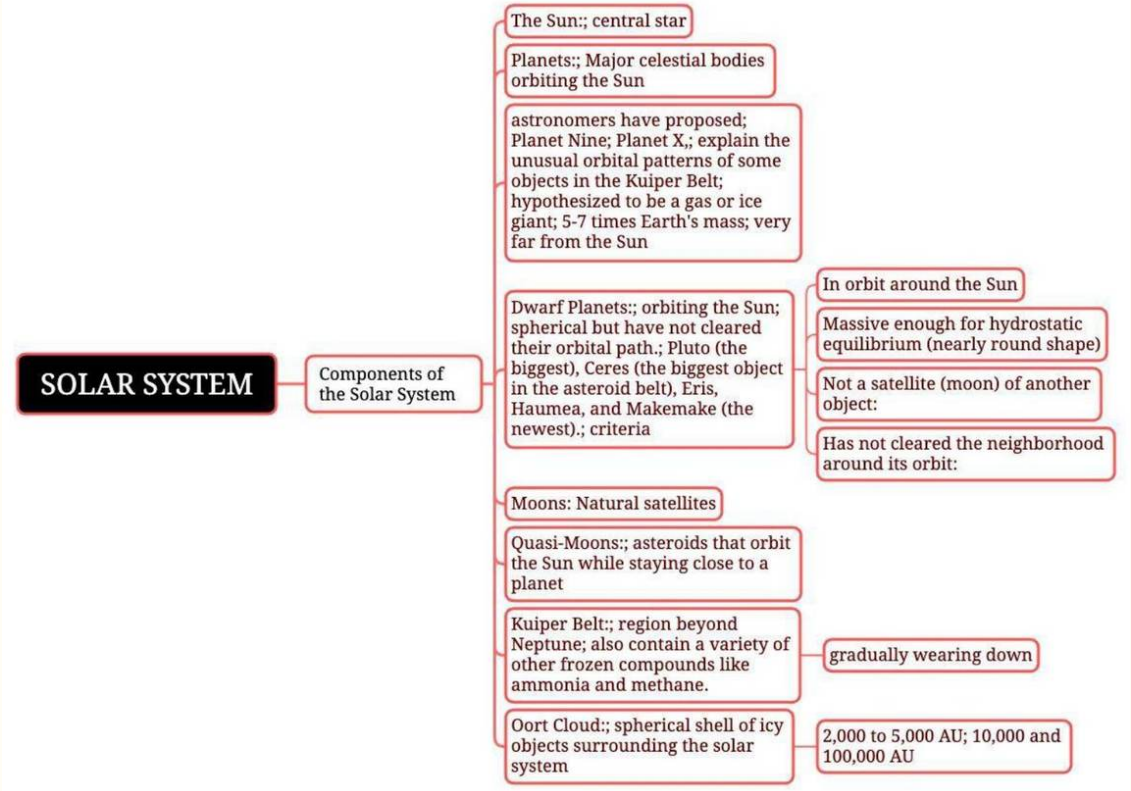

UNIVERSE
Chamberlin–Moulton planetesimal hypothesis
star passed close to the young Sun, causing tidal bulges and solar prominences that led to repeated ejections of material from the Sun.
Gravitational effects; spiral arms; remaining in orbit
cooled and condensed; planetesimals; protoplanets
, collisions among; planets and their moons, with comets and asteroids
Accretion Model
Otto Schmidt (1944) and Viktor Safronov (1969).
Constant accretion; nebula to planets
Big Bang Hypothesis
Georges Lemaitre
universe began from a single primordial atom.
dense and huge primordial matter.
A violent explosion
dust particles; scattered in the universe.
denser regions become gravitationally attracted
denser, forming gas clouds, stars, galaxies
PROOF
Hubble's Law (Hubble-Lemaitre law) ; expansion of the universe.
redshift; elongation of light waves; toward the red end; object moving away from the observer
Cosmic Microwave Background Emissions
Arno Penzias and Robert Wilson,; 1965
CMB is like a baby photo of the universe; exactly what the Big Bang Theory predicts

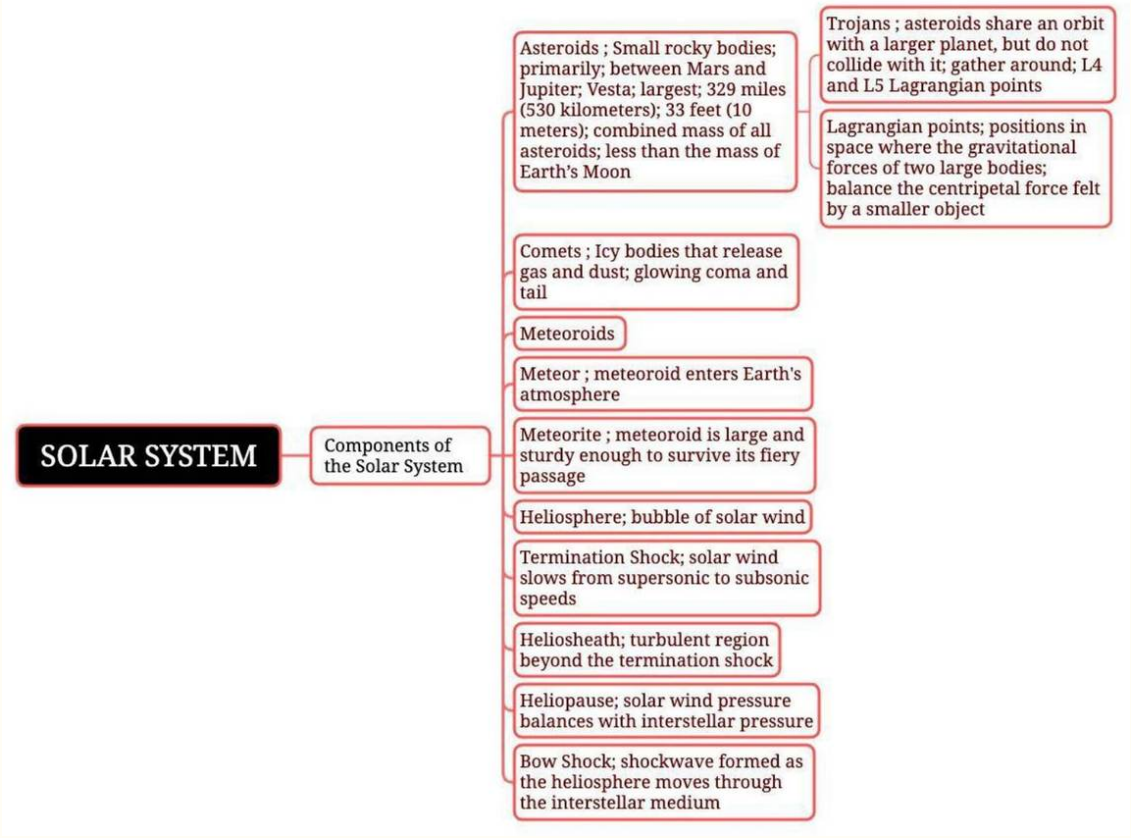

DARK ENERGY & DARK MATTER
Dark energy and dark matter; 95% of the universe
Dark matter; 27%; gravitational effects
dark energy; 68%; accelerating expansion of the universe
repulsive force

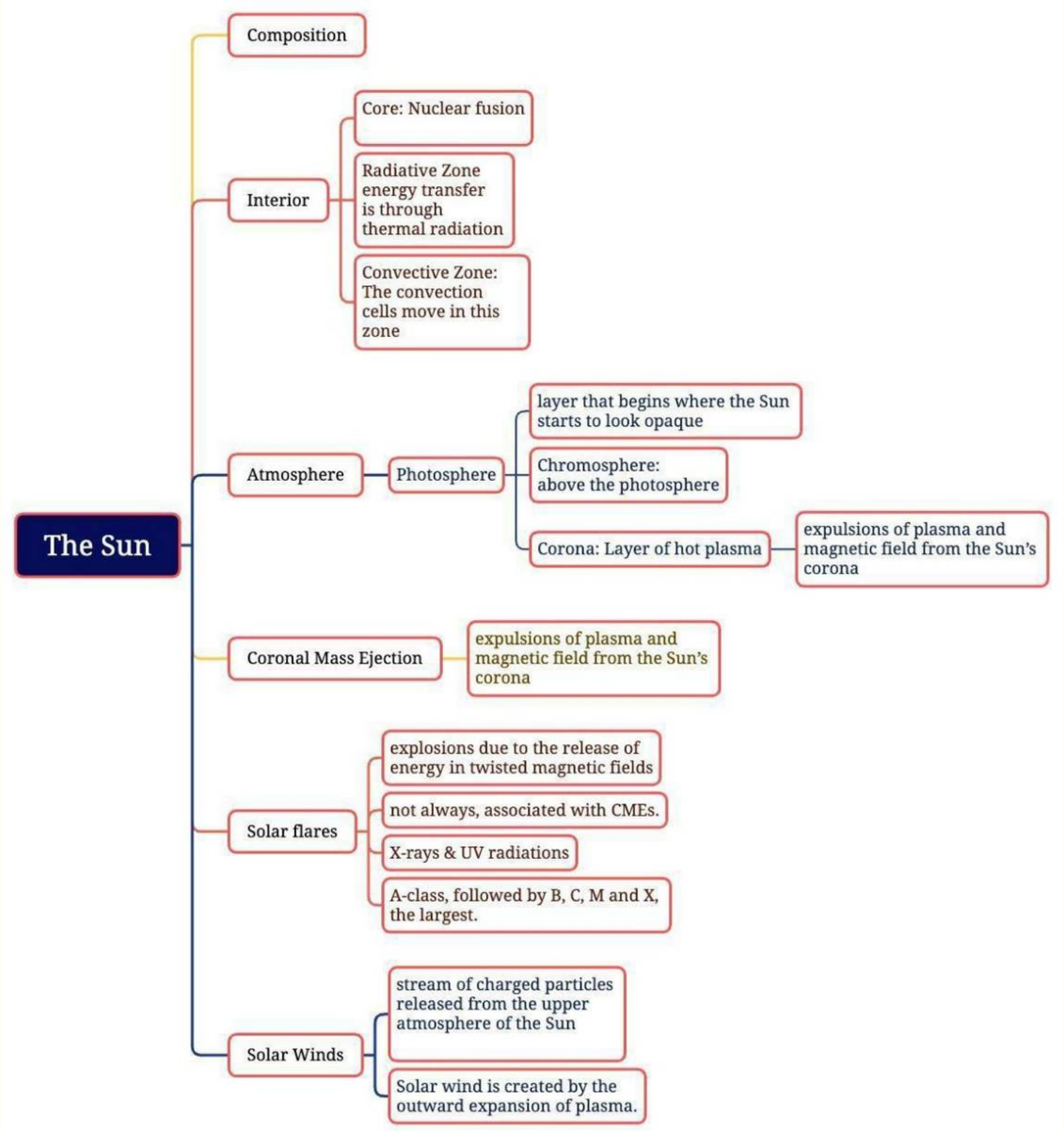

GALAXIES
Formation
After the Big Bang
First Galaxies (about 1 billion years post-Big Bang)
Galaxy Evolution (1 billion years to present):
Types of Galaxies
Spiral Galaxies:; flat, rotating disk with spiral arms winding outward from a central bulge; Milky Way; active star formation, particularly in the spiral arms.
Elliptical Galaxies:; have a smooth, featureless light profile; older, cooler stars and possess minimal gas and dust; in low rates of new star formation.
Lenticular Galaxies; intermediate between spiral and elliptical; central bulge and a disk but lacking; spiral arms; minimal star formation
Irregular Galaxies:; appear chaotic; contain abundant gas and dust, with active regions
Active Galactic Nuclei: highly energetic regions; centers of some galaxies; supermassive black holes
Active galaxies; 10% of known galaxies
Milky way; not active.
Seyfert Galaxies
Most common and lowest energy
Appear normal in visible light but emit strongly in infrared and sometimes X-rays.
some produce jets
Quasars
Brightest and most energetic
Can outshine entire galaxies
Powered by massive accretion activity
Blazars
jets point almost directly at Earth
extremely bright; sources of high-energy particles like neutrinos.

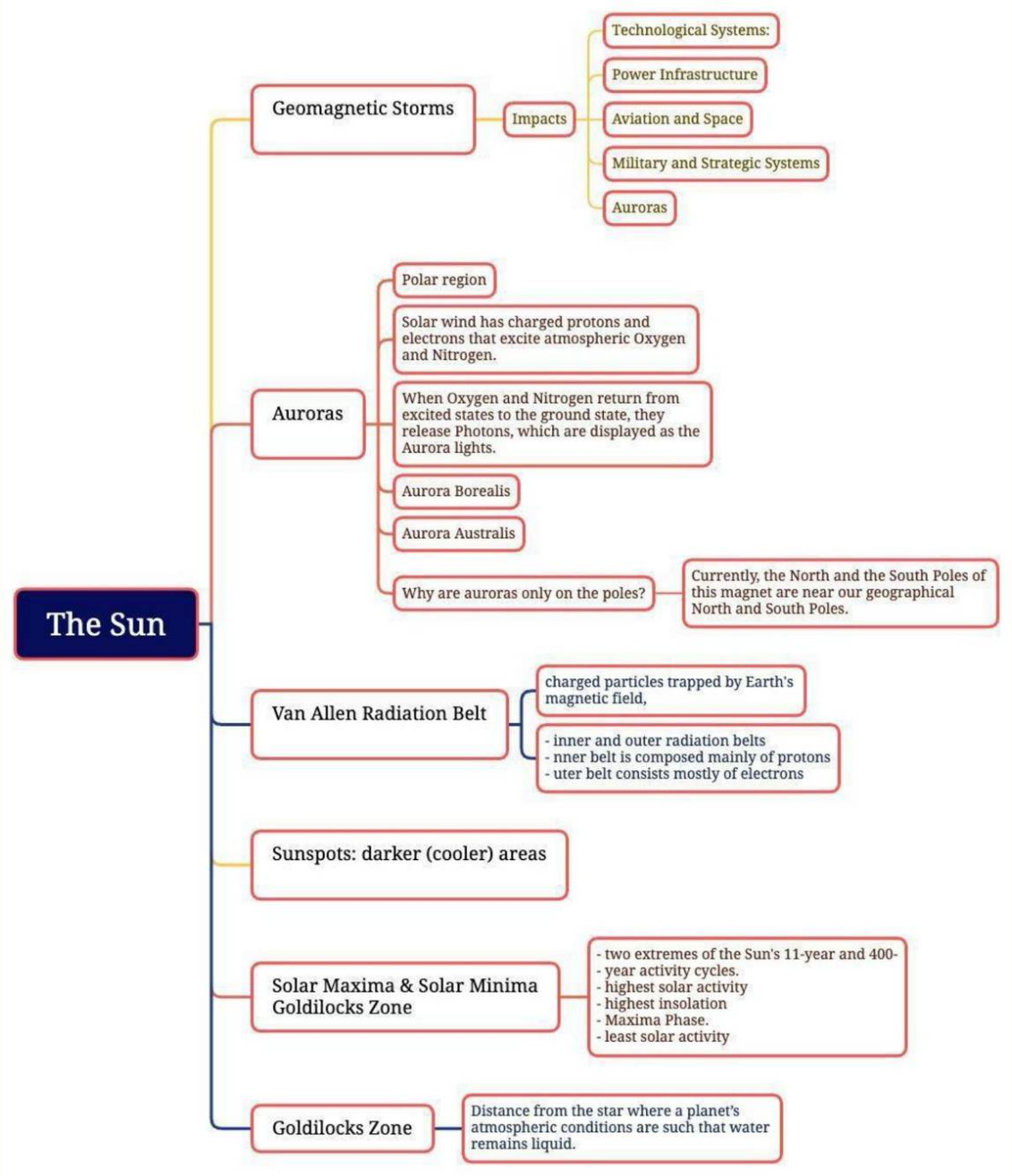

STARS
Big Bang (13.8 billion years ago)
Formation of Hydrogen and Helium
Cosmic Microwave Background (CMB)
Formation of Gas Clouds (100 million to 200 million years post-Big Bang)
Protostar Formation (200 million to 400 million years post-Big Bang)
Nuclear Fusion Ignition (200 million to 400 million years post-Big Bang)
Lifecycle of stars
Stellar Nebula
Protostar
Groups of stars that have newly formed from molecular clouds; stellar clusters
Molecular clouds that contain many of these clusters; stellar nurseries
Main Sequence
Low-Mass Stars (mass < 8 times that of Sun)
Red Giant
Planetary Nebula
White Dwarf
High-Mass Stars (mass > 8 times that of the sun)
Red Supergiant
Supernova
Neutron Star/Black Hole
Chandrashekhar Limit ; 1.44 times; the mass of the sun

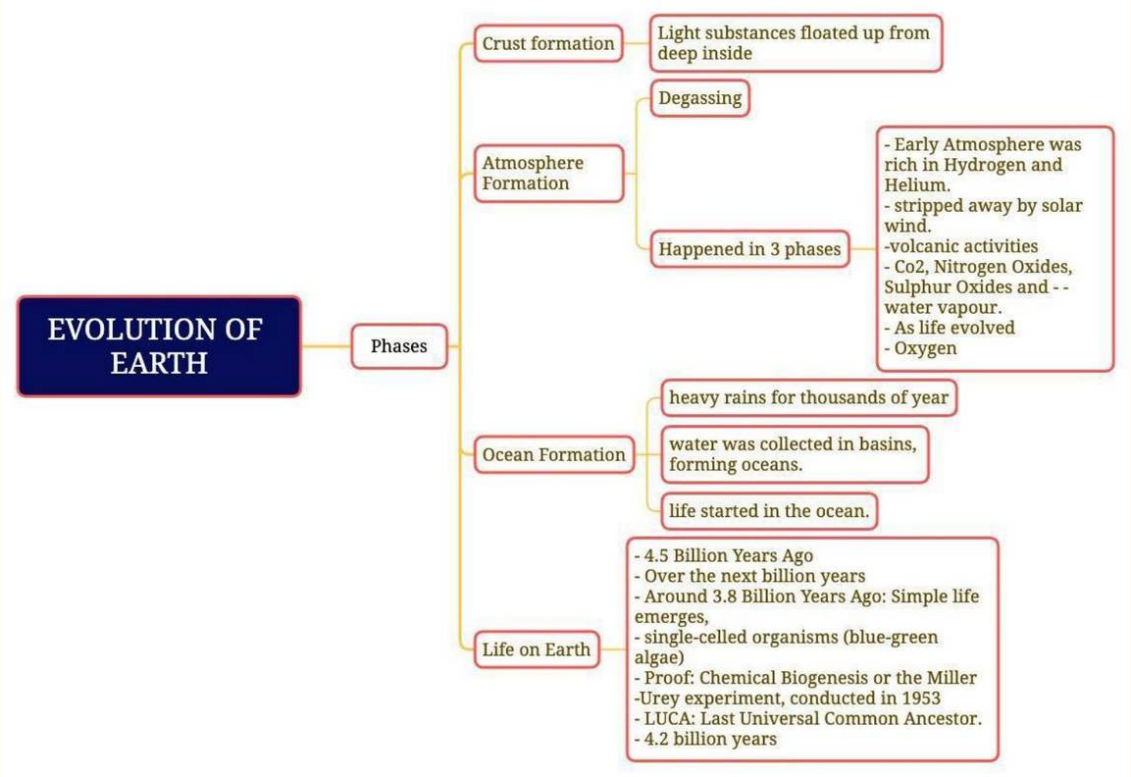

STARS
Population 3, 2 and 1 stars
Population III Stars (Hypothetical) ; very first stars
Extremely Metal-Poor
Massive and Short-Lived
Population II Stars (Oldest Observed; 5-10 bya)
Metal-Poor
Location
Slower Evolution
Population I Stars (Younger and Richer in Metals; 4-5 bya)
Metal-Rich
Location
Faster Evolution
Neutron Stars
massive stars (about 8-25 times the mass of the Sun)
composed almost entirely of neutrons; strong gravitational and magnetic fields.
Pulsars ; Rapidly rotating neutron stars
Black Holes
gravitational pull; strong; not even light, can escape
Event Horizon ; boundary
Accretion Disks ; gas and dust spiraling into black holes
Gravitational Waves ; Ripples in spacetime; LIGO; and Virgo
Spaghettification ; objects are stretched and torn apart; intense tidal forces
Hawking Radiation ; theoretical emission; gradual black hole evaporation

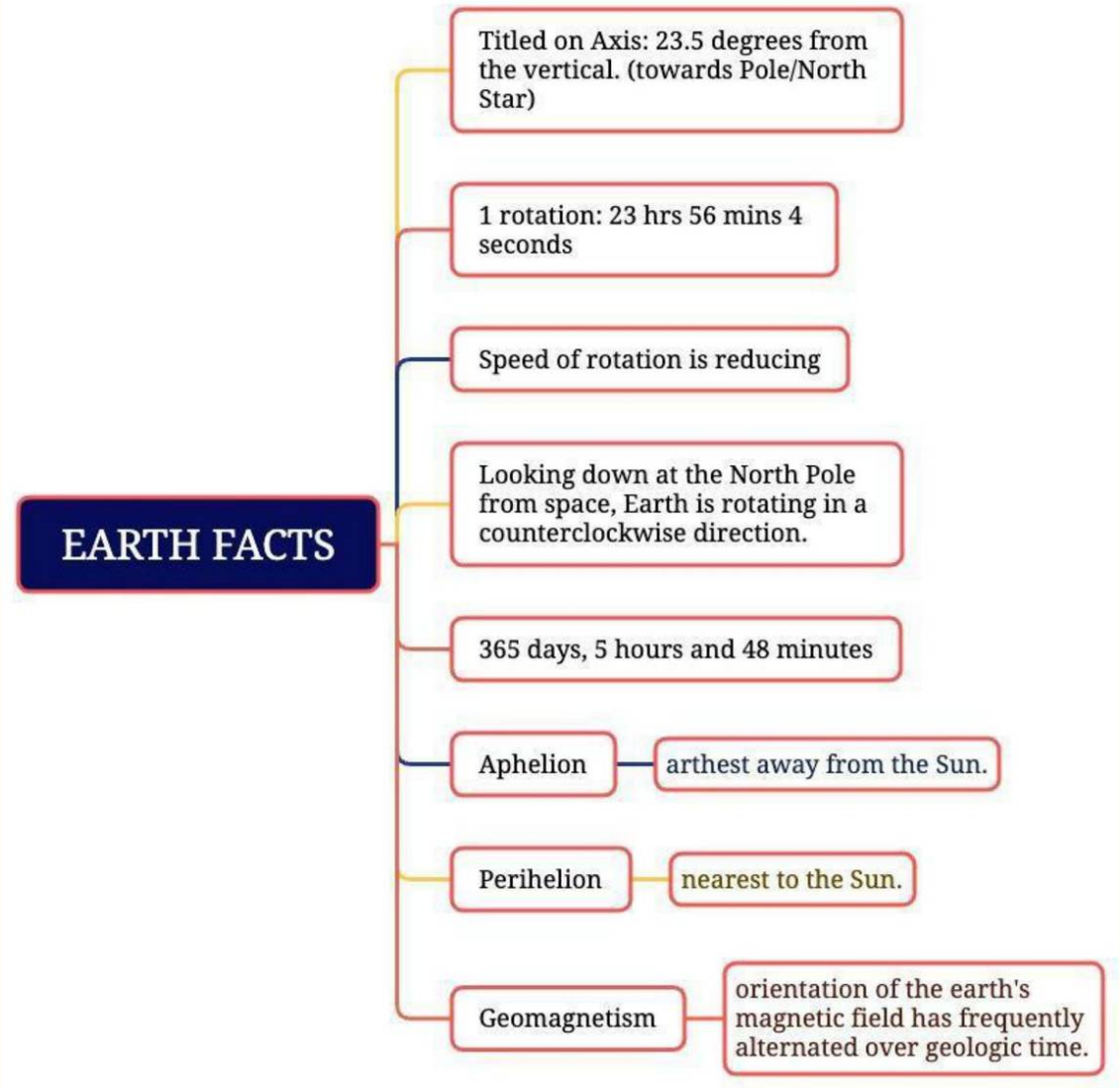

SOLAR SYSTEM
Components of the Solar System
The Sun:; central star
Planets:; Major celestial bodies orbiting the Sun
astronomers have proposed; Planet Nine; Planet X,; explain the unusual orbital patterns of some objects in the Kuiper Belt; hypothesized to be a gas or ice giant; 5-7 times Earth's mass; very far from the Sun
Dwarf Planets:; orbiting the Sun; spherical but have not cleared their orbital path.; Pluto (the biggest), Ceres (the biggest object in the asteroid belt), Eris, Haumea, and Makemake (the newest).; criteria
In orbit around the Sun
Massive enough for hydrostatic equilibrium (nearly round shape)
Not a satellite (moon) of another object:
Has not cleared the neighborhood around its orbit:
Moons: Natural satellites
Quasi-Moons:; asteroids that orbit the Sun while staying close to a planet
Kuiper Belt:; region beyond Neptune; also contain a variety of other frozen compounds like ammonia and methane.
gradually wearing down
Oort Cloud:; spherical shell of icy objects surrounding the solar system
2,000 to 5,000 AU; 10,000 and 100,000 AU

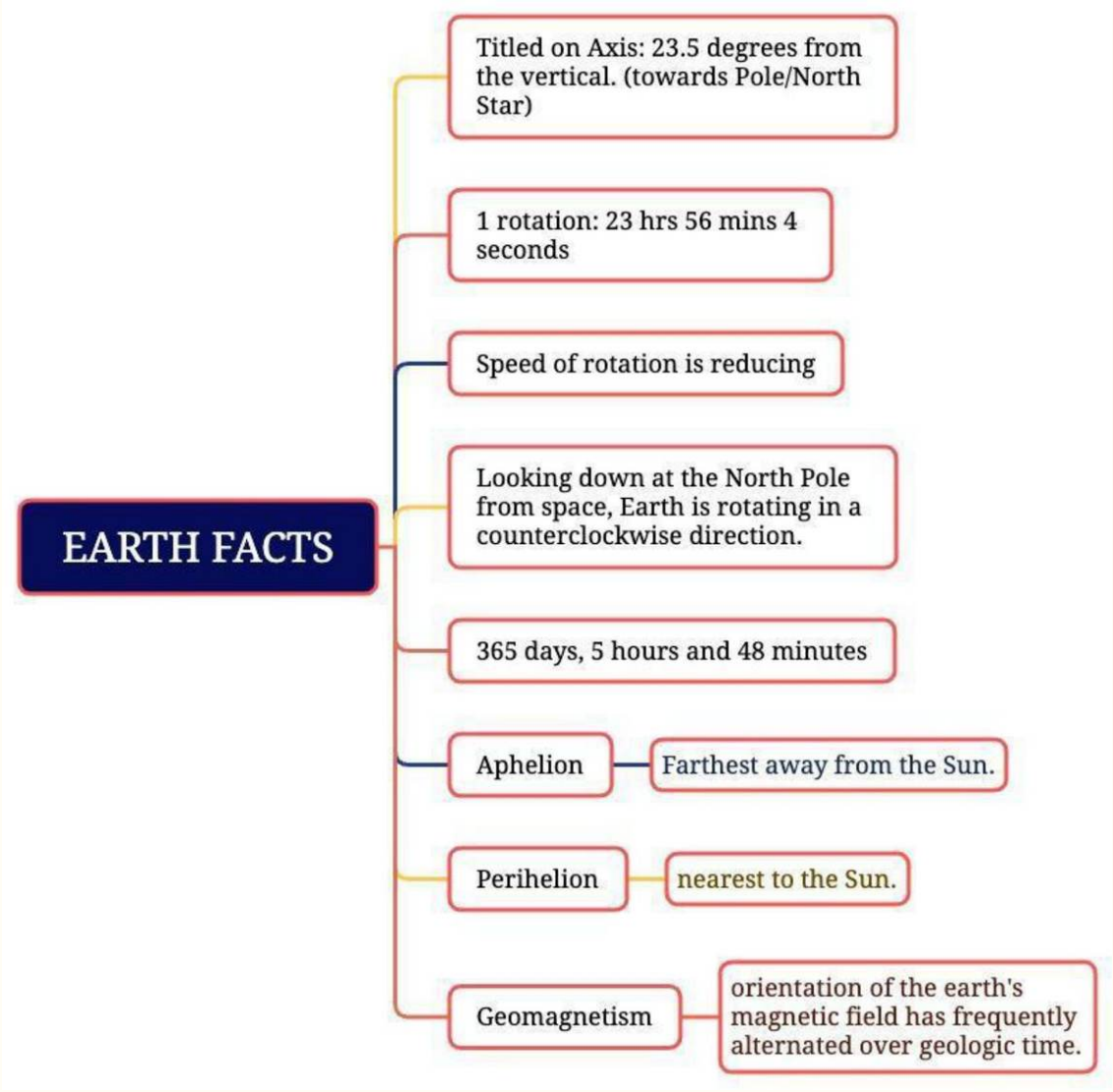

SOLAR SYSTEM
Components of the Solar System
Asteroids ; Small rocky bodies; primarily; between Mars and Jupiter; Vesta; largest; 329 miles (530 kilometers); 33 feet (10 meters); combined mass of all asteroids; less than the mass of Earth's Moon
Trojans ; asteroids share an orbit with a larger planet, but do not collide with it; gather around; L4 and L5 Lagrangian points
Lagrangian points; positions in space where the gravitational forces of two large bodies; balance the centripetal force felt by a smaller object
Comets ; Icy bodies that release gas and dust; glowing coma and tail
Meteoroids
Meteor ; meteoroid enters Earth's atmosphere
Meteorite ; meteoroid is large and sturdy enough to survive its fiery passage
Heliosphere; bubble of solar wind
Termination Shock; solar wind slows from supersonic to subsonic speeds
Heliosheath; turbulent region beyond the termination shock
Heliopause; solar wind pressure balances with interstellar pressure
Bow Shock; shockwave formed as the heliosphere moves through the interstellar medium

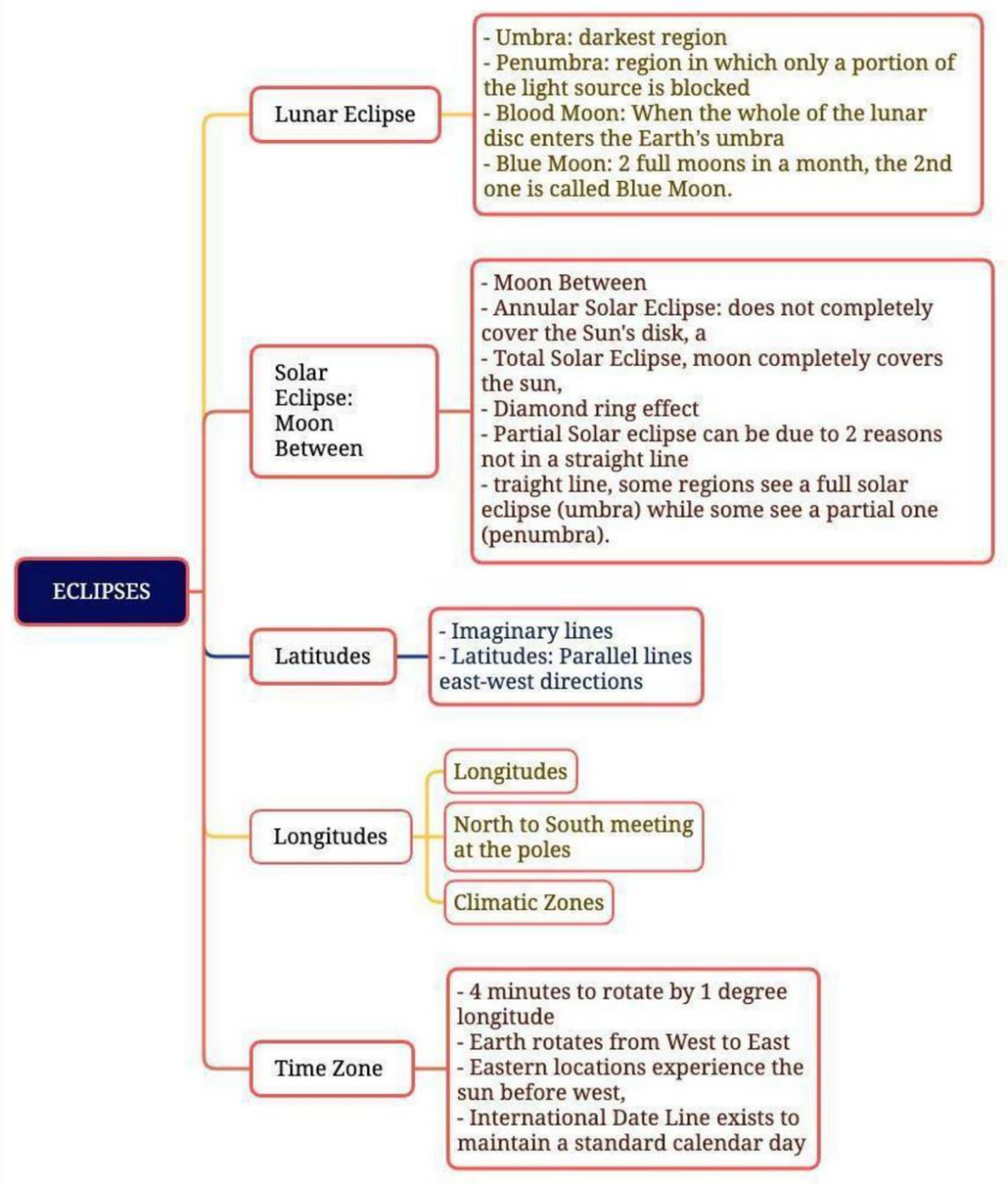

The Sun
Composition
Interior
Core: Nuclear fusion
Radiative Zone energy transfer is through thermal radiation
Convective Zone: The convection cells move in this zone
Atmosphere
Photosphere
layer that begins where the Sun starts to look opaque
Chromosphere: above the photosphere
Corona: Layer of hot plasma
expulsions of plasma and magnetic field from the Sun's corona
Coronal Mass Ejection
expulsions of plasma and magnetic field from the Sun's corona
Solar flares
explosions due to the release of energy in twisted magnetic fields
not always, associated with CMEs.
X-rays & UV radiations
A-class, followed by B, C, M and X, the largest.
Solar Winds
stream of charged particles released from the upper atmosphere of the Sun
Solar wind is created by the outward expansion of plasma.


The Sun
Geomagnetic Storms
Impacts
Technological Systems:
Power Infrastructure
Aviation and Space
Military and Strategic Systems
Auroras
Auroras
Polar region
Solar wind has charged protons and electrons that excite atmospheric Oxygen and Nitrogen.
When Oxygen and Nitrogen return from excited states to the ground state, they release Photons, which are displayed as the Aurora lights.
Aurora Borealis
Aurora Australis
Why are auroras only on the poles?
Currently, the North and the South Poles of this magnet are near our geographical North and South Poles.
Van Allen Radiation Belt
charged particles trapped by Earth's magnetic field,
- inner and outer radiation belts
- nner belt is composed mainly of protons
- uter belt consists mostly of electrons
Sunspots: darker (cooler) areas
Solar Maxima & Solar Minima
Goldilocks Zone
- two extremes of the Sun's 11-year and 400-
- year activity cycles.
- highest solar activity
- highest insolation
- Maxima Phase.
- least solar activity
Goldilocks Zone
Distance from the star where a planet's atmospheric conditions are such that water remains liquid.

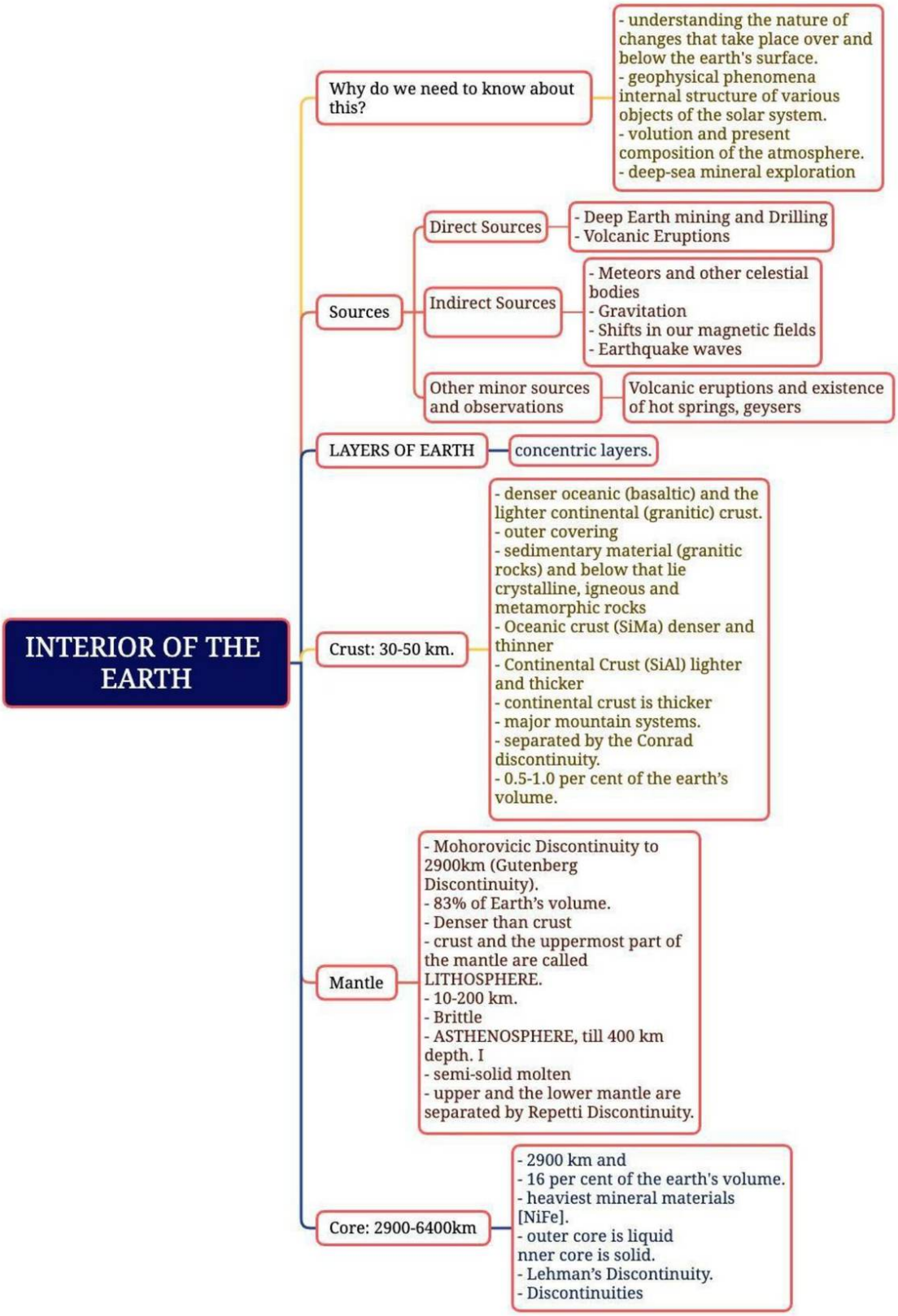

EVOLUTION OF EARTH
Phases
Crust formation
Light substances floated up from deep inside
Atmosphere Formation
Degassing
Happened in 3 phases
- Early Atmosphere was rich in Hydrogen and Helium.
- stripped away by solar wind.
-volcanic activities
- Co2, Nitrogen Oxides, Sulphur Oxides and - - water vapour.
- As life evolved
- Oxygen
Ocean Formation
heavy rains for thousands of year
water was collected in basins, forming oceans.
life started in the ocean.
Life on Earth
- 4.5 Billion Years Ago
- Over the next billion years
- Around 3.8 Billion Years Ago: Simple life emerges,
- single-celled organisms (blue-green algae)
- Proof: Chemical Biogenesis or the Miller -Urey experiment, conducted in 1953
- LUCA: Last Universal Common Ancestor.
- 4.2 billion years


EARTH FACTS
Titled on Axis: 23.5 degrees from the vertical. (towards Pole/North Star)
1 rotation: 23 hrs 56 mins 4 seconds
Speed of rotation is reducing
Looking down at the North Pole from space, Earth is rotating in a counterclockwise direction.
365 days, 5 hours and 48 minutes
Aphelion
arthest away from the Sun.
Perihelion
nearest to the Sun.
Geomagnetism
orientation of the earth's magnetic field has frequently alternated over geologic time.

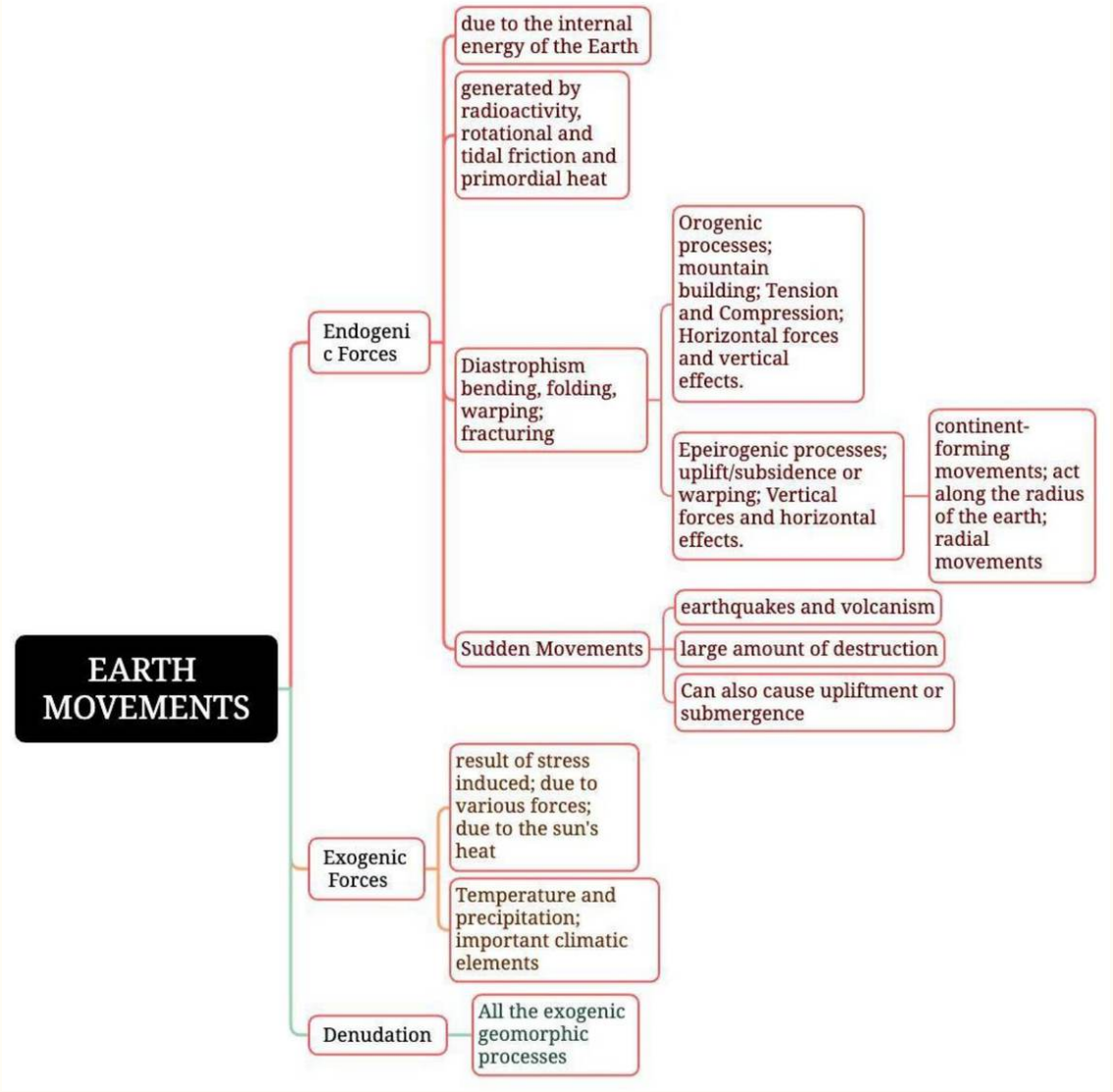

EARTH FACTS
Titled on Axis: 23.5 degrees from the vertical. (towards Pole/North Star)
1 rotation: 23 hrs 56 mins 4 seconds
Speed of rotation is reducing
Looking down at the North Pole from space, Earth is rotating in a counterclockwise direction.
365 days, 5 hours and 48 minutes
Aphelion
Farthest away from the Sun.
Perihelion
nearest to the Sun.
Geomagnetism
orientation of the earth's magnetic field has frequently alternated over geologic time.

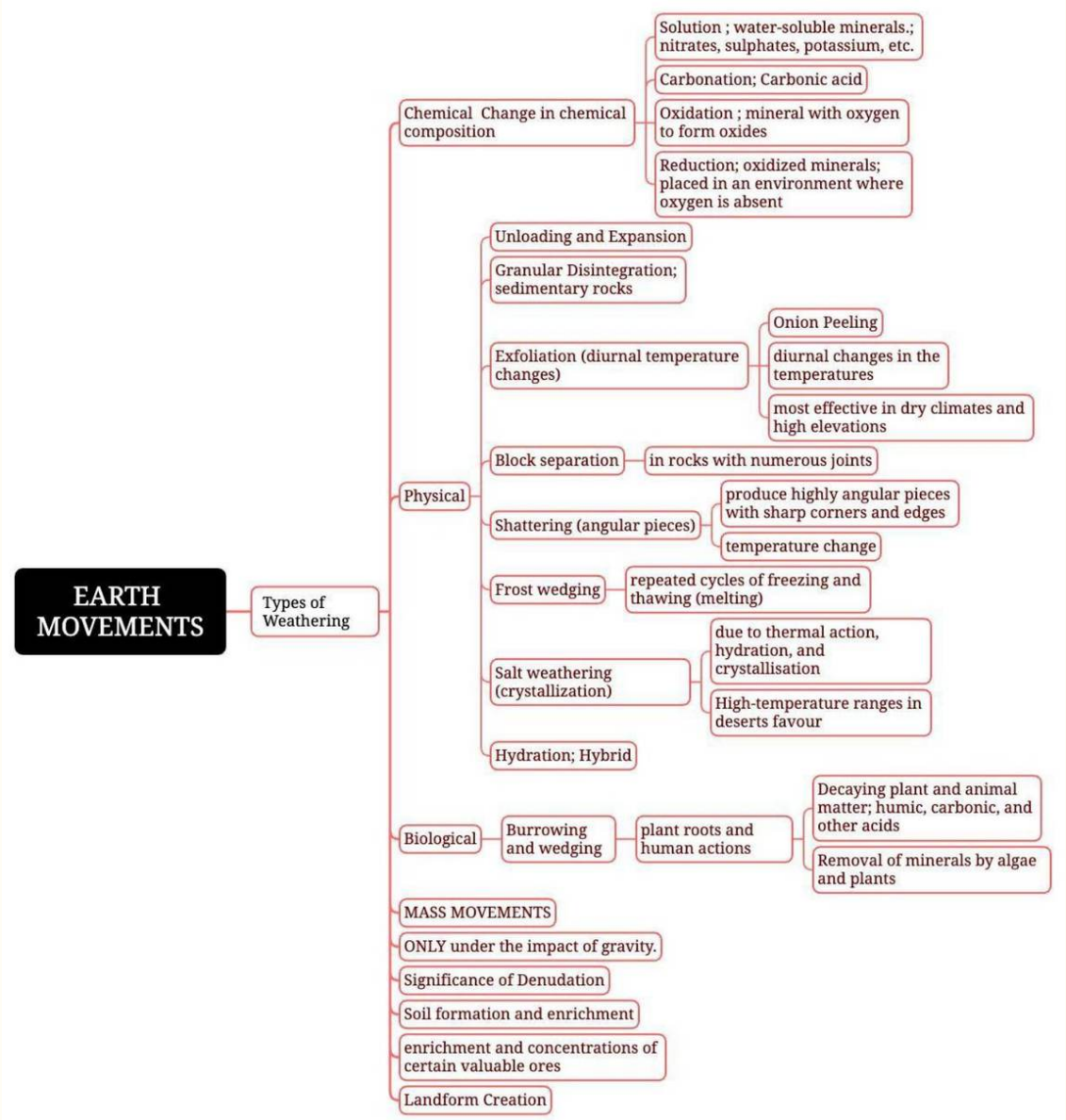

ECLIPSES
Lunar Eclipse
- Umbra: darkest region
- Penumbra: region in which only a portion of the light source is blocked
- Blood Moon: When the whole of the lunar disc enters the Earth's umbra
- Blue Moon: 2 full moons in a month, the 2nd one is called Blue Moon.
Solar Eclipse: Moon Between
- Moon Between
- Annular Solar Eclipse: does not completely cover the Sun's disk, a
- Total Solar Eclipse, moon completely covers the sun,
- Diamond ring effect
- Partial Solar eclipse can be due to 2 reasons not in a straight line
- traight line, some regions see a full solar eclipse (umbra) while some see a partial one (penumbra).
Latitudes
- Imaginary lines
- Latitudes: Parallel lines east-west directions
Longitudes
Longitudes
North to South meeting at the poles
Climatic Zones
Time Zone
- 4 minutes to rotate by 1 degree longitude
- Earth rotates from West to East
- Eastern locations experience the sun before west,
- International Date Line exists to maintain a standard calendar day

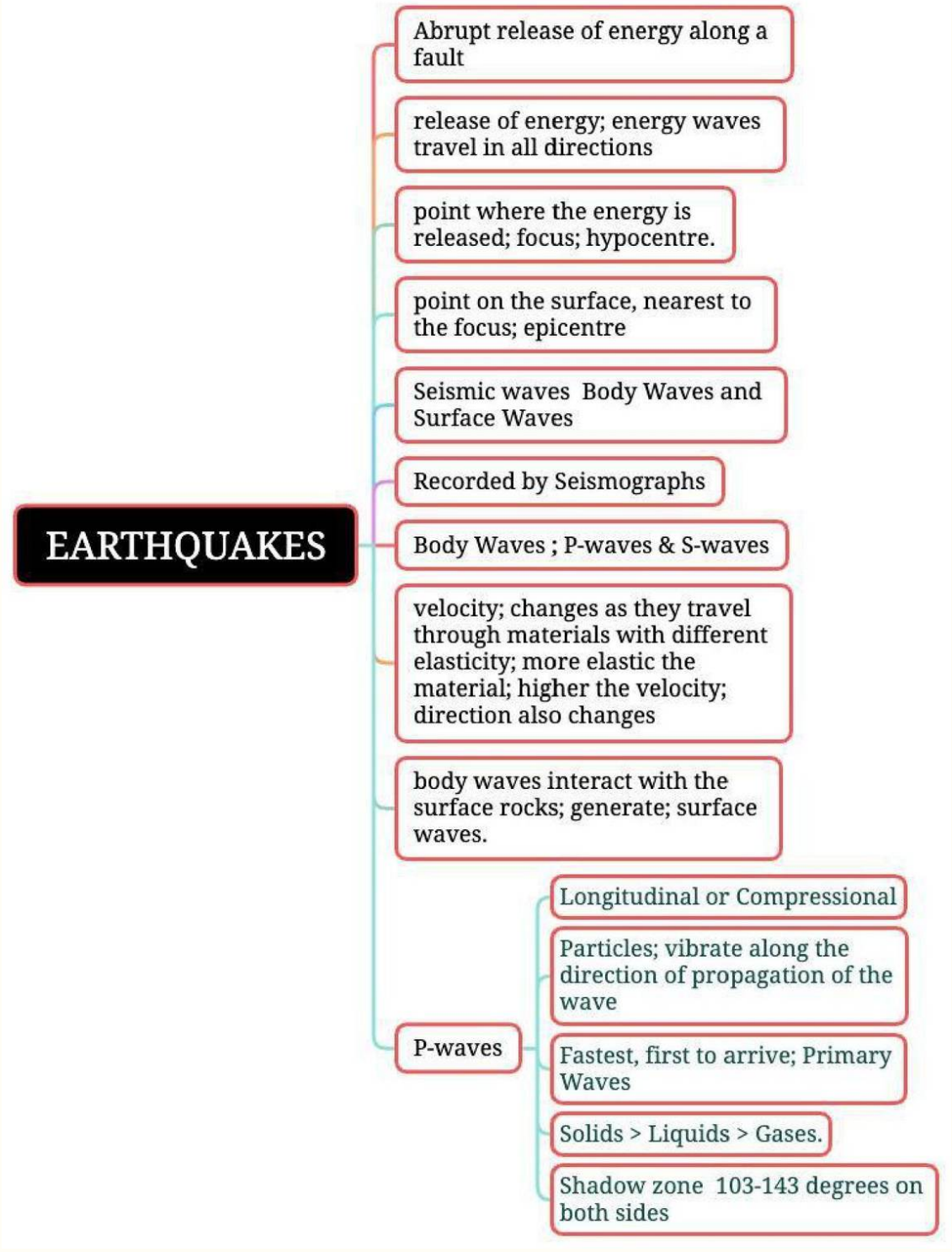
Interior of the Earth

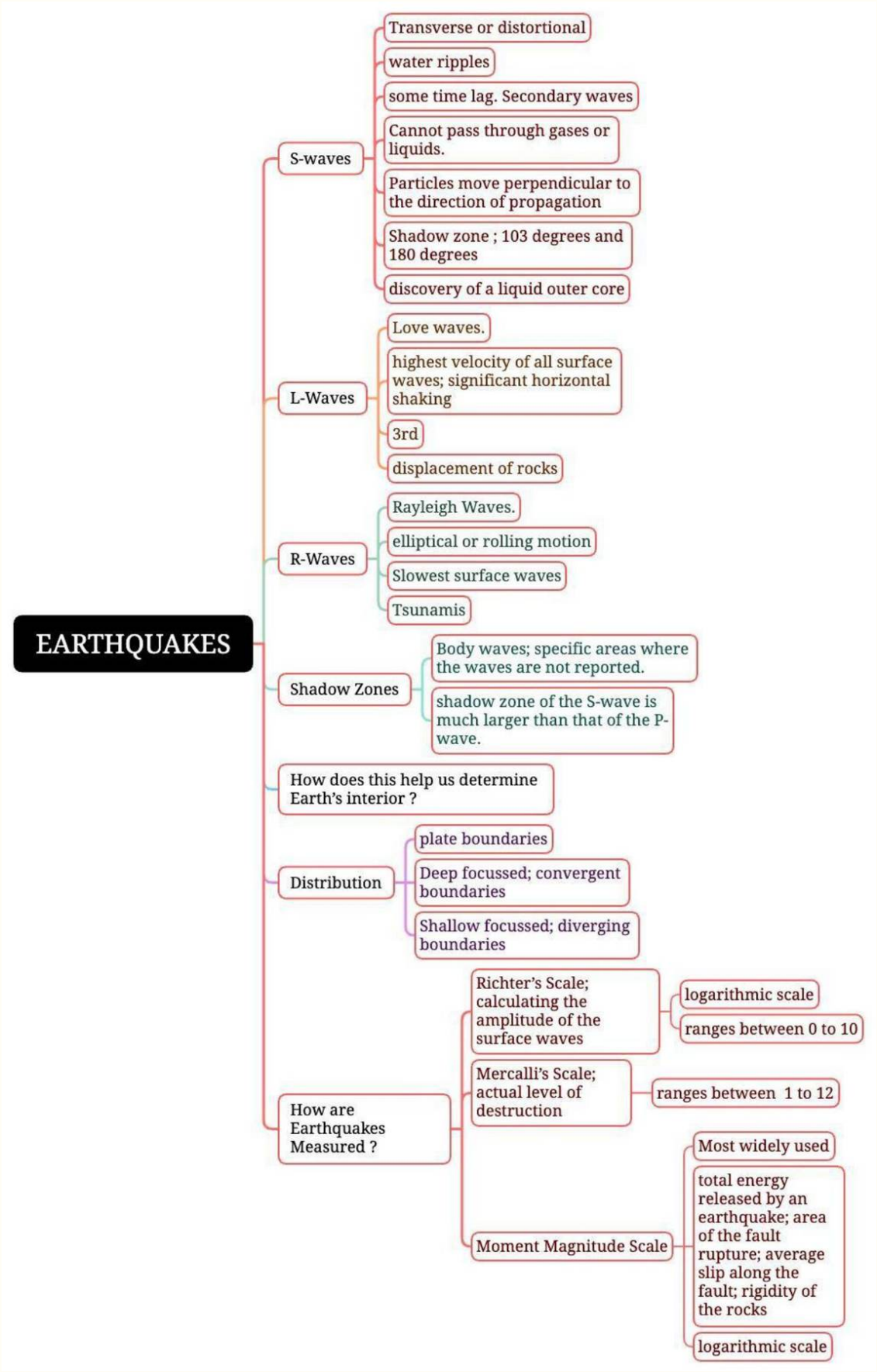


EARTHQUAKES & EARTH MOVEMENTS




EARTH MOVEMENTS
Types of Weathering
Chemical Change in chemical composition
Solution ; water-soluble minerals.; nitrates, sulphates, potassium, etc.
Carbonation; Carbonic acid
Oxidation ; mineral with oxygen to form oxides
Reduction; oxidized minerals; placed in an environment where oxygen is absent
Physical
Unloading and Expansion
Granular Disintegration; sedimentary rocks
Exfoliation (diurnal temperature changes)
Onion Peeling
diurnal changes in the temperatures
most effective in dry climates and high elevations
Block separation
in rocks with numerous joints
Shattering (angular pieces)
produce highly angular pieces with sharp corners and edges
temperature change
Frost wedging
repeated cycles of freezing and thawing (melting)
Salt weathering (crystallization)
due to thermal action, hydration, and crystallisation
High-temperature ranges in deserts favour
Hydration; Hybrid
Biological
Burrowing and wedging
plant roots and human actions
Decaying plant and animal matter; humic, carbonic, and other acids
Removal of minerals by algae and plants
MASS MOVEMENTS
ONLY under the impact of gravity.
Significance of Denudation
Soil formation and enrichment
enrichment and concentrations of certain valuable ores
Landform Creation


EARTHQUAKES
Abrupt release of energy along a fault
release of energy; energy waves travel in all directions
point where the energy is released; focus; hypocentre.
point on the surface, nearest to the focus; epicentre
Seismic waves Body Waves and Surface Waves
Recorded by Seismographs
Body Waves ; P-waves & S-waves
velocity; changes as they travel through materials with different elasticity; more elastic the material; higher the velocity; direction also changes
body waves interact with the surface rocks; generate; surface waves.
P-waves
Longitudinal or Compressional
Particles; vibrate along the direction of propagation of the wave
Fastest, first to arrive; Primary Waves
Solids > Liquids > Gases.
Shadow zone 103-143 degrees on both sides


EARTHQUAKES
S-waves
Transverse or distortional
water ripples
some time lag. Secondary waves
Cannot pass through gases or liquids.
Particles move perpendicular to the direction of propagation
Shadow zone ; 103 degrees and 180 degrees
discovery of a liquid outer core
L-Waves
Love waves.
highest velocity of all surface waves; significant horizontal shaking
3rd
displacement of rocks
R-Waves
Rayleigh Waves.
elliptical or rolling motion
Slowest surface waves
Tsunamis
Shadow Zones
Body waves; specific areas where the waves are not reported.
shadow zone of the S-wave is much larger than that of the P-wave.
How does this help us determine Earth's interior ?
Distribution
plate boundaries
Deep focussed; convergent boundaries
Shallow focussed; diverging boundaries
How are Earthquakes Measured ?
Richter's Scale; calculating the amplitude of the surface waves
logarithmic scale
ranges between 0 to 10
Mercalli's Scale; actual level of destruction
ranges between 1 to 12
Moment Magnitude Scale
Most widely used
total energy released by an earthquake; area of the fault rupture; average slip along the fault; rigidity of the rocks
logarithmic scale

CONTINENTAL DRIFT & SEA FLOOR SPREAD THEORIES



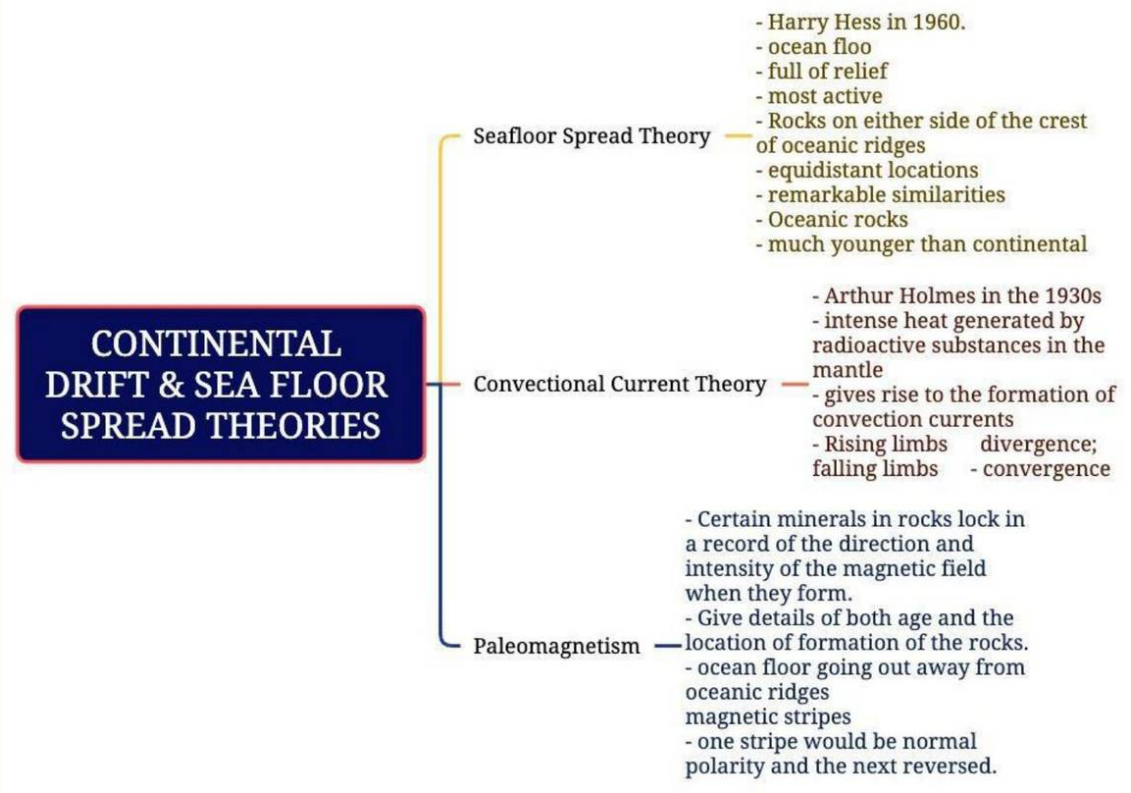
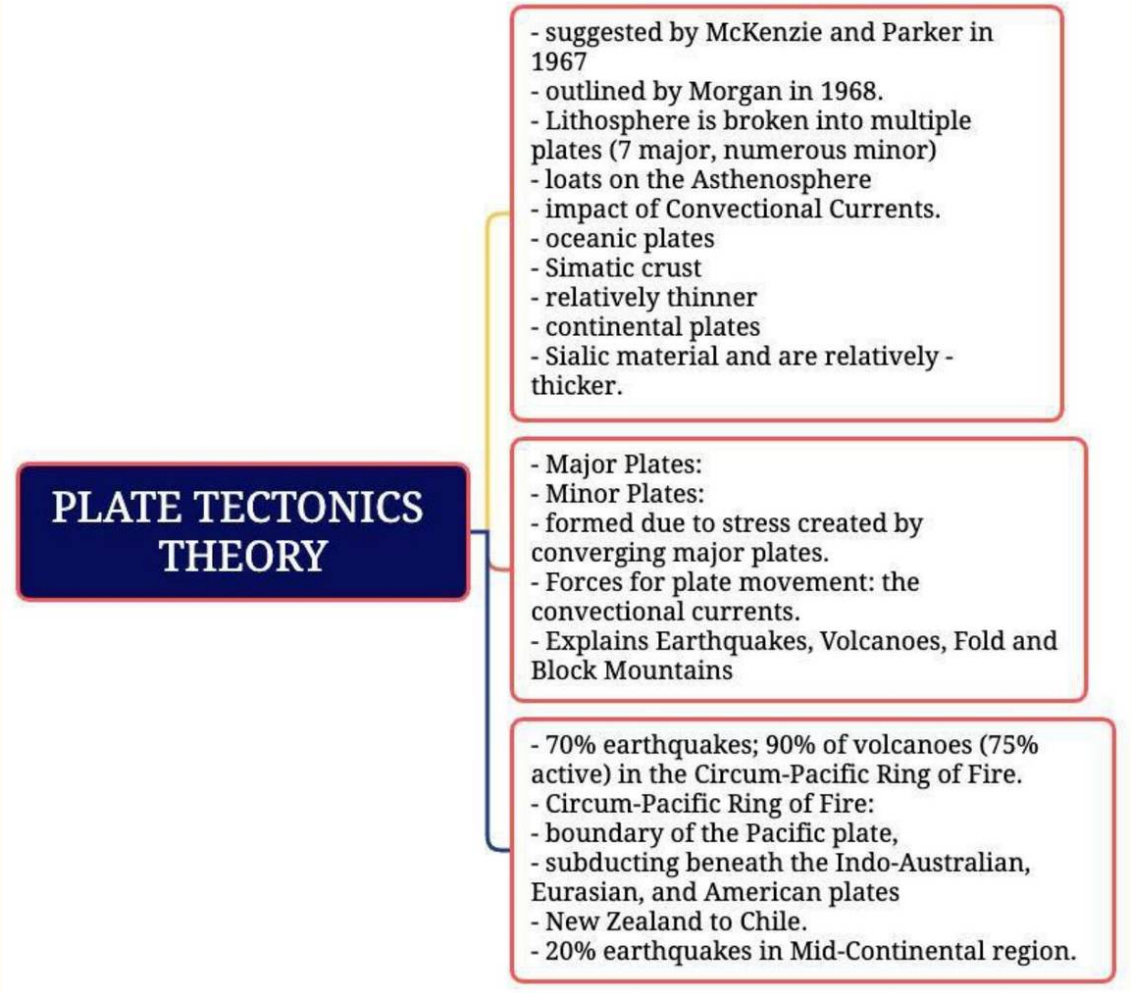
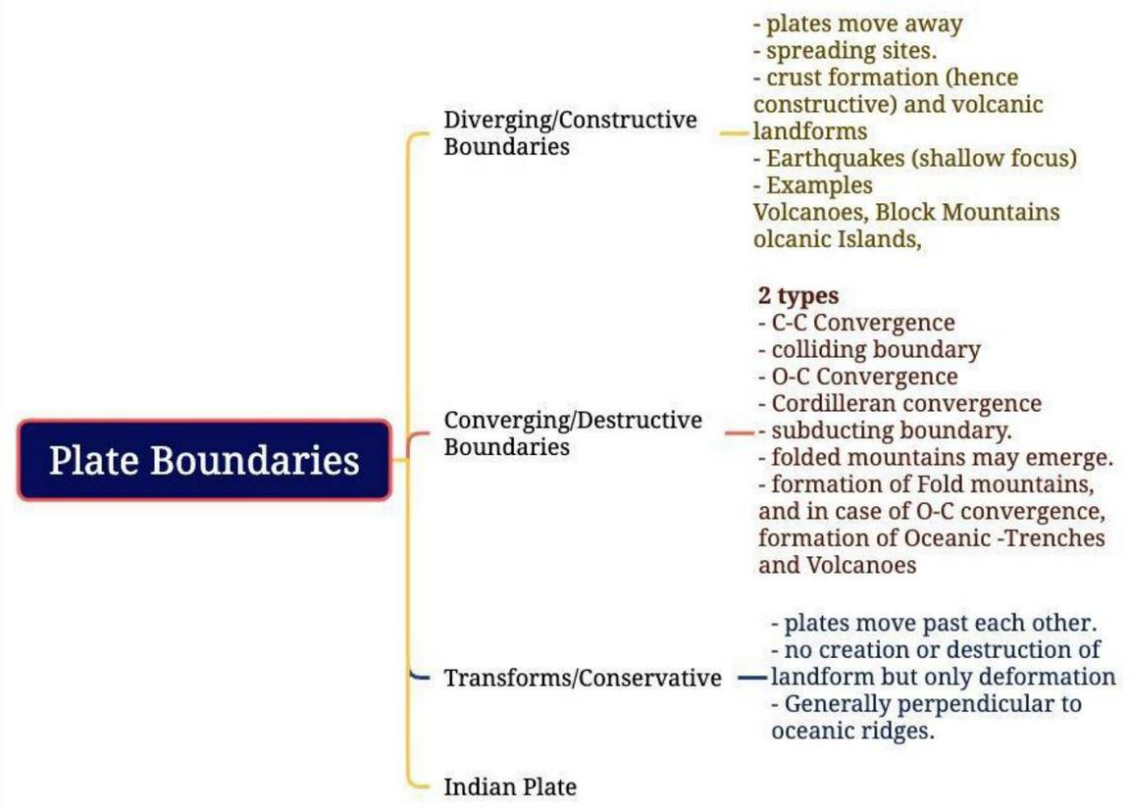
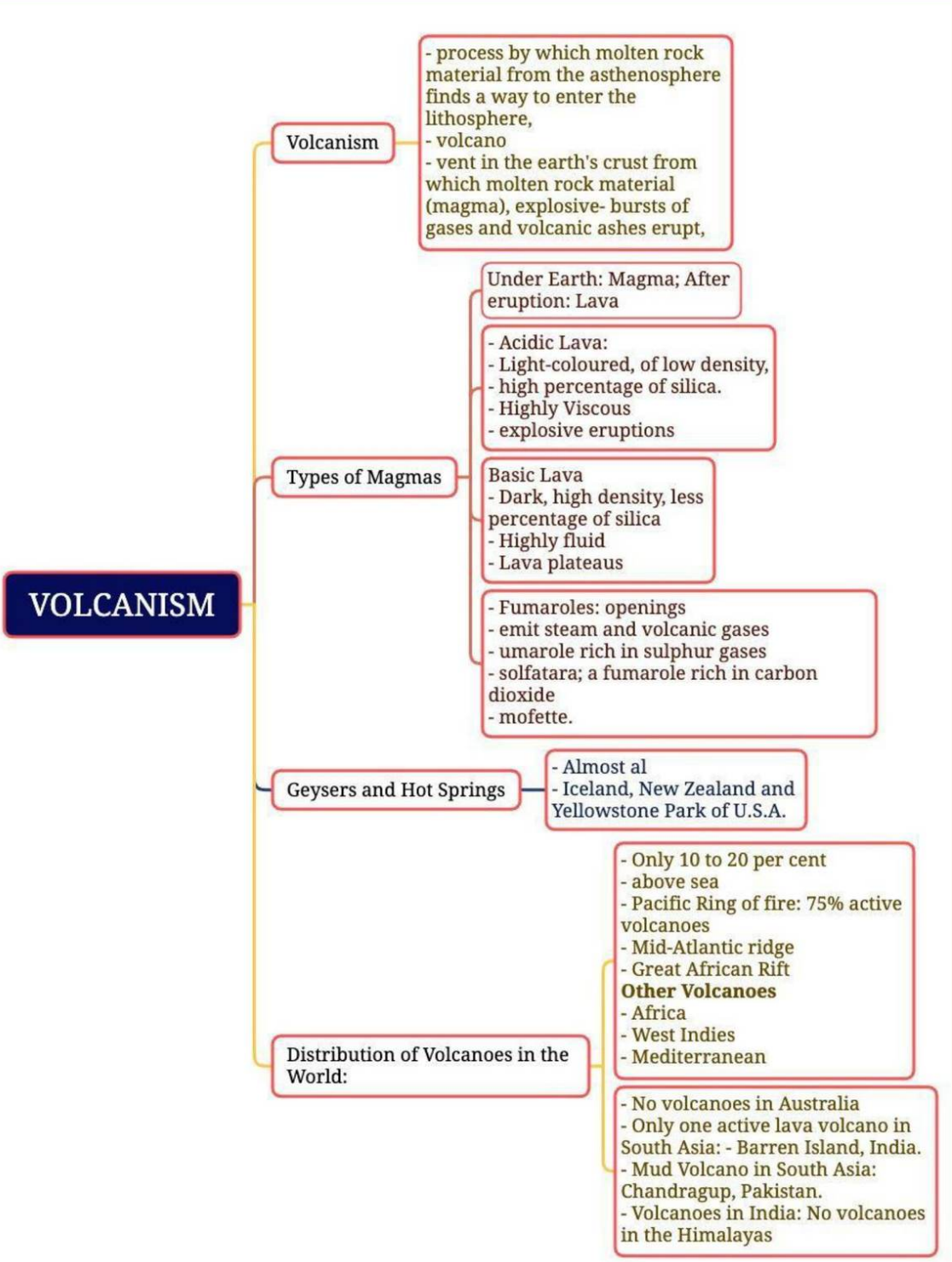


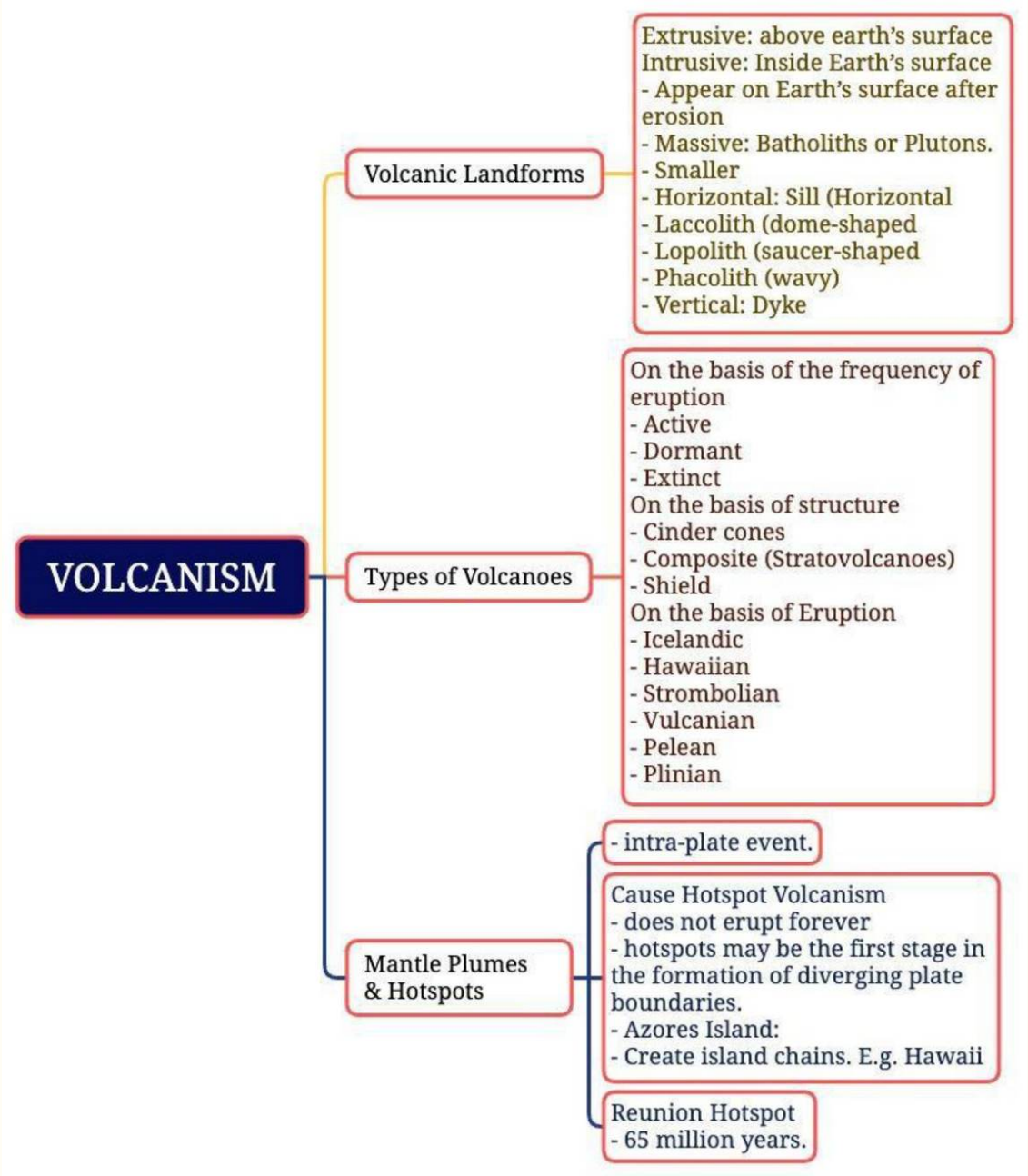
PLATE TECTONICS THEORY



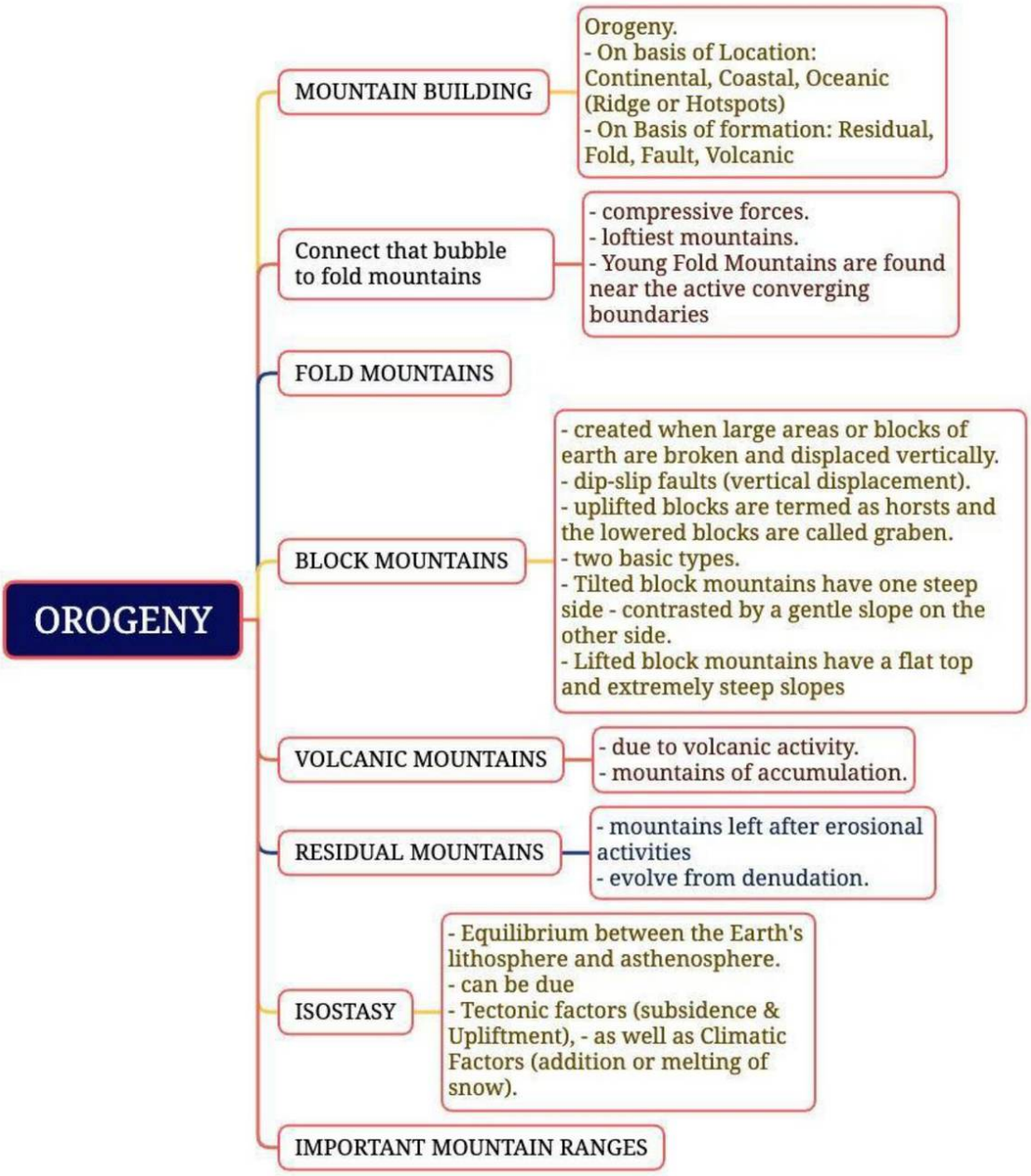

Plate Boundaries
Diverging/Constructive Boundaries
- plates move away
- spreading sites.
- crust formation (hence constructive) and volcanic landforms
- Earthquakes (shallow focus)
- Examples
Volcanoes, Block Mountains
olcanic Islands,
Converging/Destructive Boundaries
2 types
- C-C Convergence
- colliding boundary
- O-C Convergence
- Cordilleran convergence
- subducting boundary.
- folded mountains may emerge.
- formation of Fold mountains, and in case of O-C convergence, formation of Oceanic -Trenches and Volcanoes
Transforms/Conservative
- plates move past each other.
- no creation or destruction of landform but only deformation
- Generally perpendicular to oceanic ridges.
Indian Plate

VOLCANOES

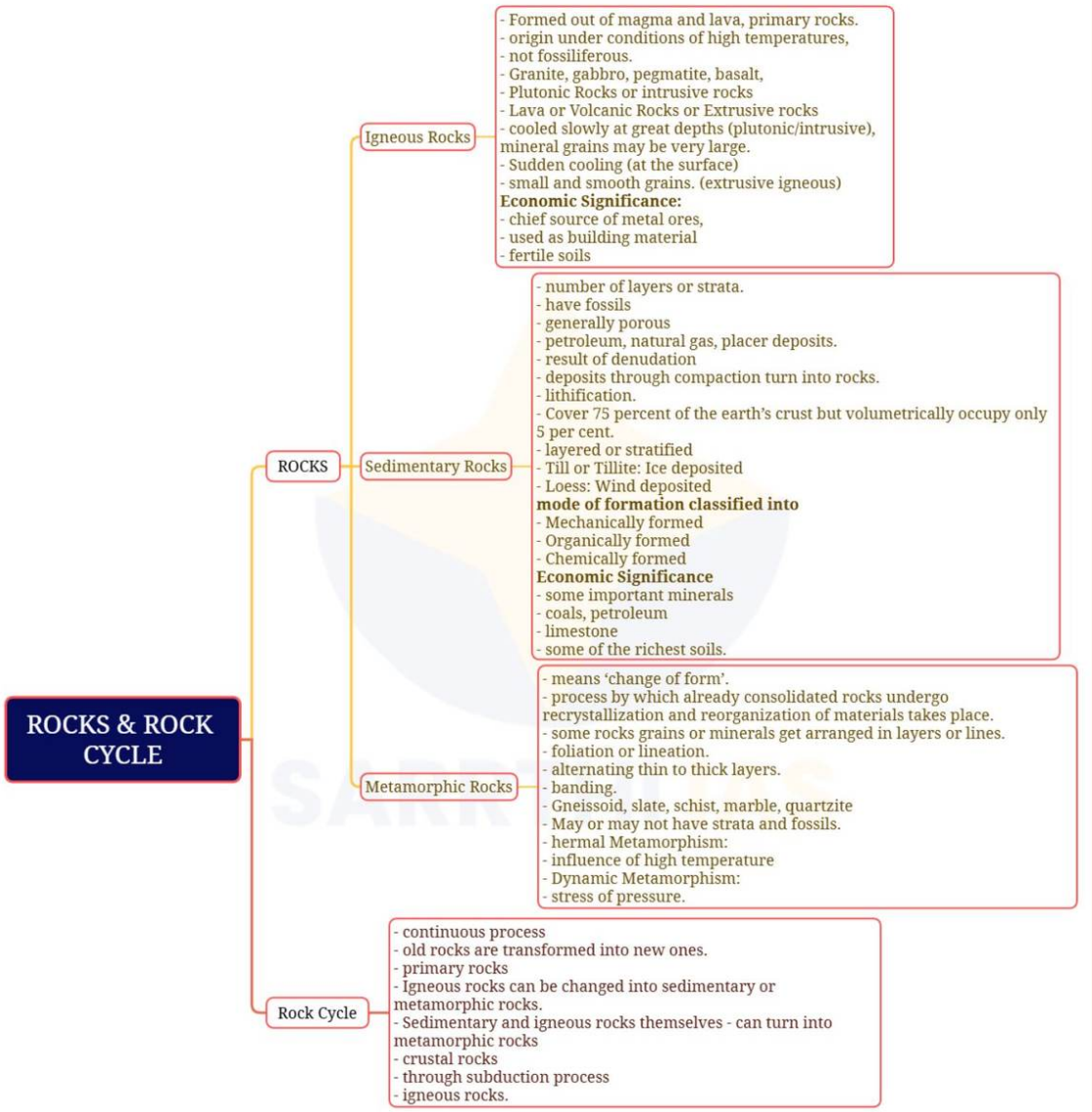

VOLCANISM
Volcanism
- process by which molten rock material from the asthenosphere finds a way to enter the lithosphere,
- volcano
- vent in the earth's crust from which molten rock material (magma), explosive- bursts of gases and volcanic ashes erupt,
Types of Magmas
Under Earth: Magma; After eruption: Lava
- Acidic Lava:
- Light-coloured, of low density,
- high percentage of silica.
- Highly Viscous
- explosive eruptions
Basic Lava
- Dark, high density, less percentage of silica
- Highly fluid
- Lava plateaus
- Fumaroles: openings
- emit steam and volcanic gases
- umarole rich in sulphur gases
- solfatara; a fumarole rich in carbon dioxide
- mofette.
Geysers and Hot Springs
- Almost al
- Iceland, New Zealand and Yellowstone Park of U.S.A.
Distribution of Volcanoes in the World:
- Only 10 to 20 per cent
- above sea
- Pacific Ring of fire: 75% active volcanoes
- Mid-Atlantic ridge
- Great African Rift
Other Volcanoes
- Africa
- West Indies
- Mediterranean
- No volcanoes in Australia
- Only one active lava volcano in South Asia: - Barren Island, India.
- Mud Volcano in South Asia: Chandragup, Pakistan.
- Volcanoes in India: No volcanoes in the Himalayas


VOLCANISM
Volcanic Landforms
Extrusive: above earth's surface
Intrusive: Inside Earth's surface
- Appear on Earth's surface after erosion
- Massive: Batholiths or Plutons.
- Smaller
- Horizontal: Sill (Horizontal
- Laccolith (dome-shaped
- Lopolith (saucer-shaped
- Phacolith (wavy)
- Vertical: Dyke
Types of Volcanoes
On the basis of the frequency of eruption
- Active
- Dormant
- Extinct
On the basis of structure
- Cinder cones
- Composite (Stratovolcanoes)
- Shield
On the basis of Eruption
- Icelandic
- Hawaiian
- Strombolian
- Vulcanian
- Pelean
- Plinian
Mantle Plumes & Hotspots
- intra-plate event.
Cause Hotspot Volcanism
- does not erupt forever
- hotspots may be the first stage in the formation of diverging plate boundaries.
- Azores Island:
- Create island chains. E.g. Hawaii
Reunion Hotspot
- 65 million years.

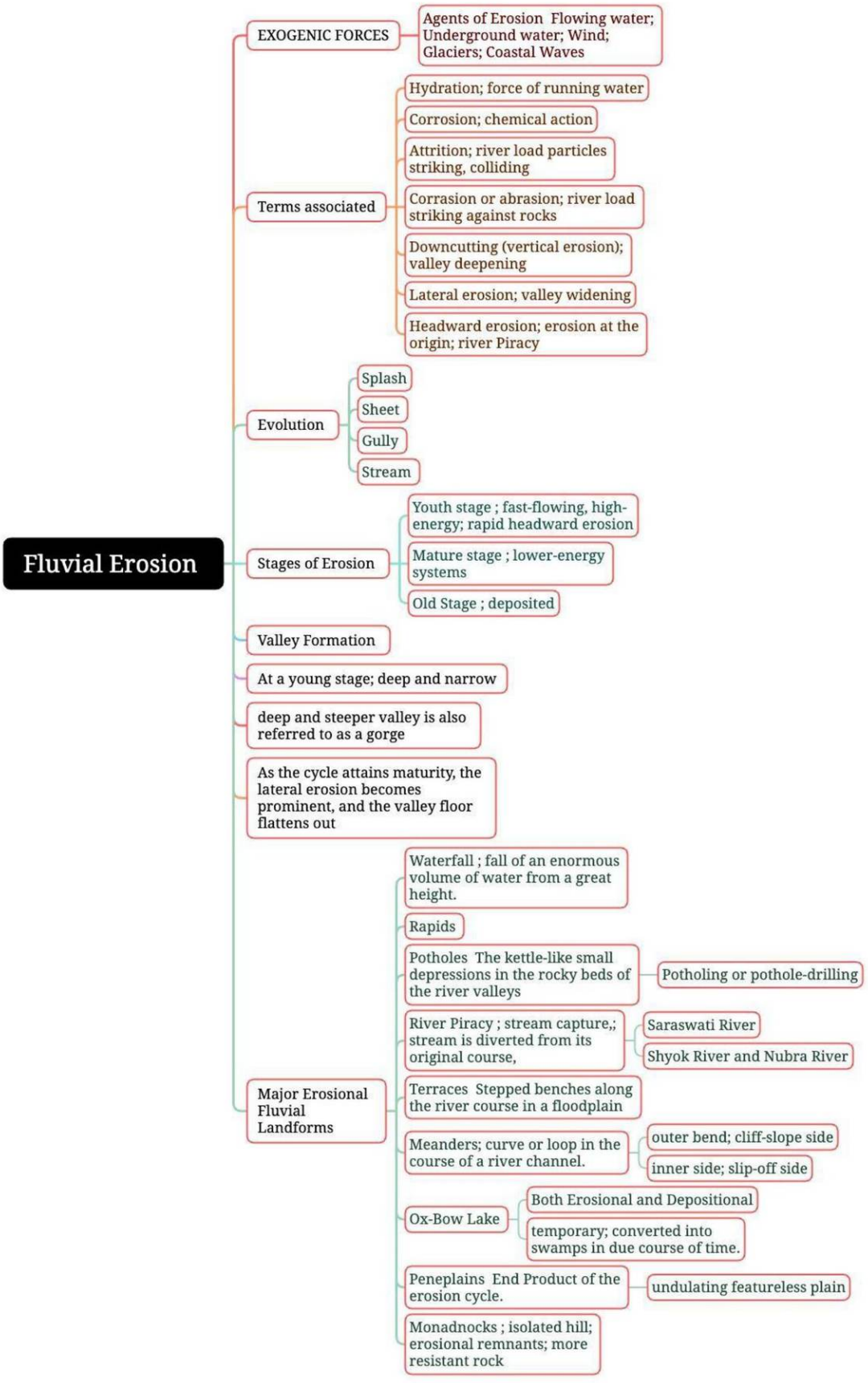
OROGENY

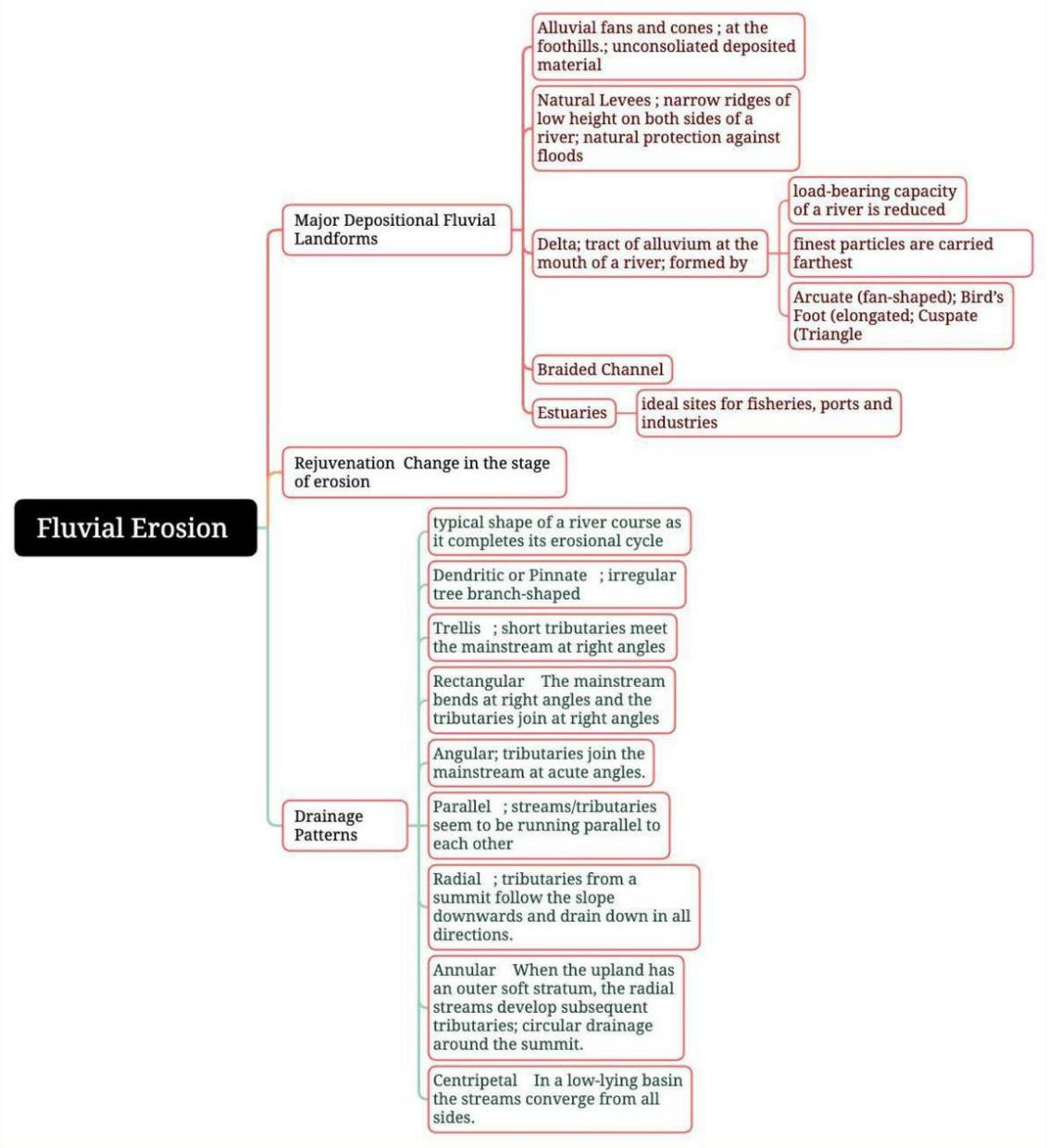

OROGENY
MOUNTAIN BUILDING
Orogeny.
- On basis of Location: Continental, Coastal, Oceanic (Ridge or Hotspots)
- On Basis of formation: Residual, Fold, Fault, Volcanic
Connect that bubble to fold mountains
- compressive forces.
- loftiest mountains.
- Young Fold Mountains are found near the active converging boundaries
FOLD MOUNTAINS
BLOCK MOUNTAINS
- created when large areas or blocks of earth are broken and displaced vertically.
- dip-slip faults (vertical displacement).
- uplifted blocks are termed as horsts and the lowered blocks are called graben.
- two basic types.
- Tilted block mountains have one steep side - contrasted by a gentle slope on the other side.
- Lifted block mountains have a flat top and extremely steep slopes
VOLCANIC MOUNTAINS
- due to volcanic activity.
- mountains of accumulation.
RESIDUAL MOUNTAINS
- mountains left after erosional activities
- evolve from denudation.
ISOSTASY
- Equilibrium between the Earth's lithosphere and asthenosphere.
- can be due
- Tectonic factors (subsidence & Upliftment), - as well as Climatic Factors (addition or melting of snow).
IMPORTANT MOUNTAIN RANGES

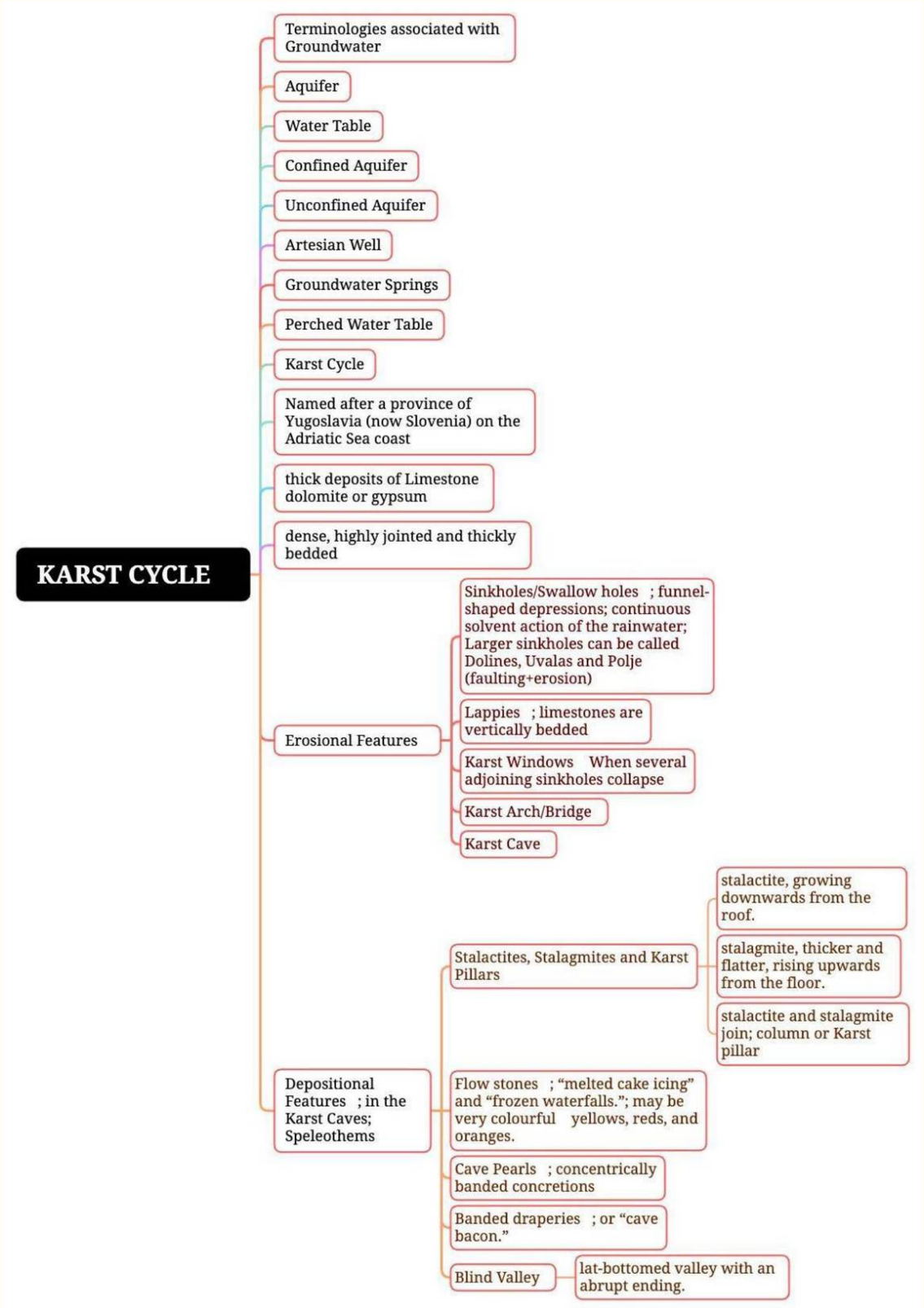
ROCKS & ROCK CYCLE

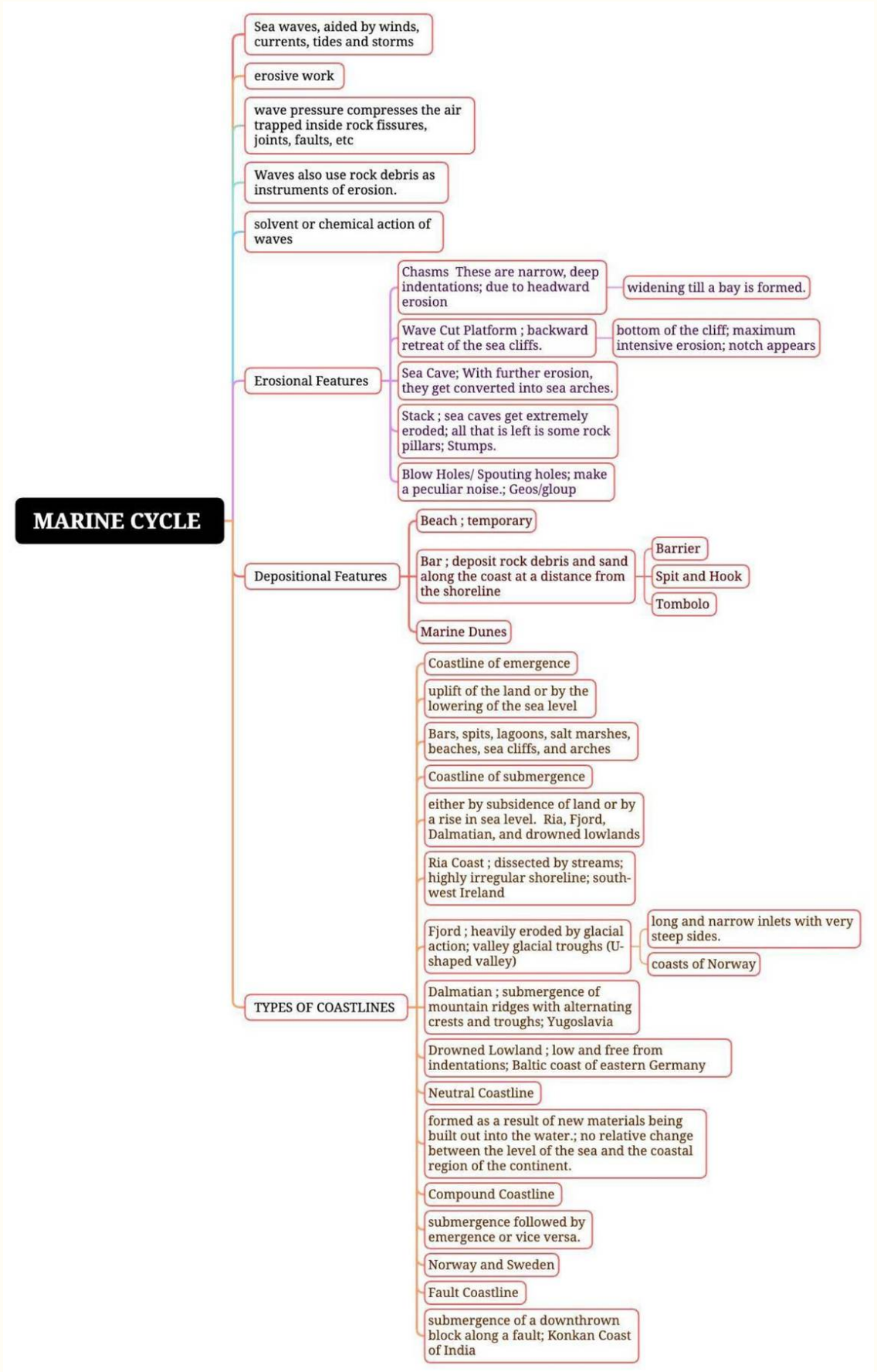


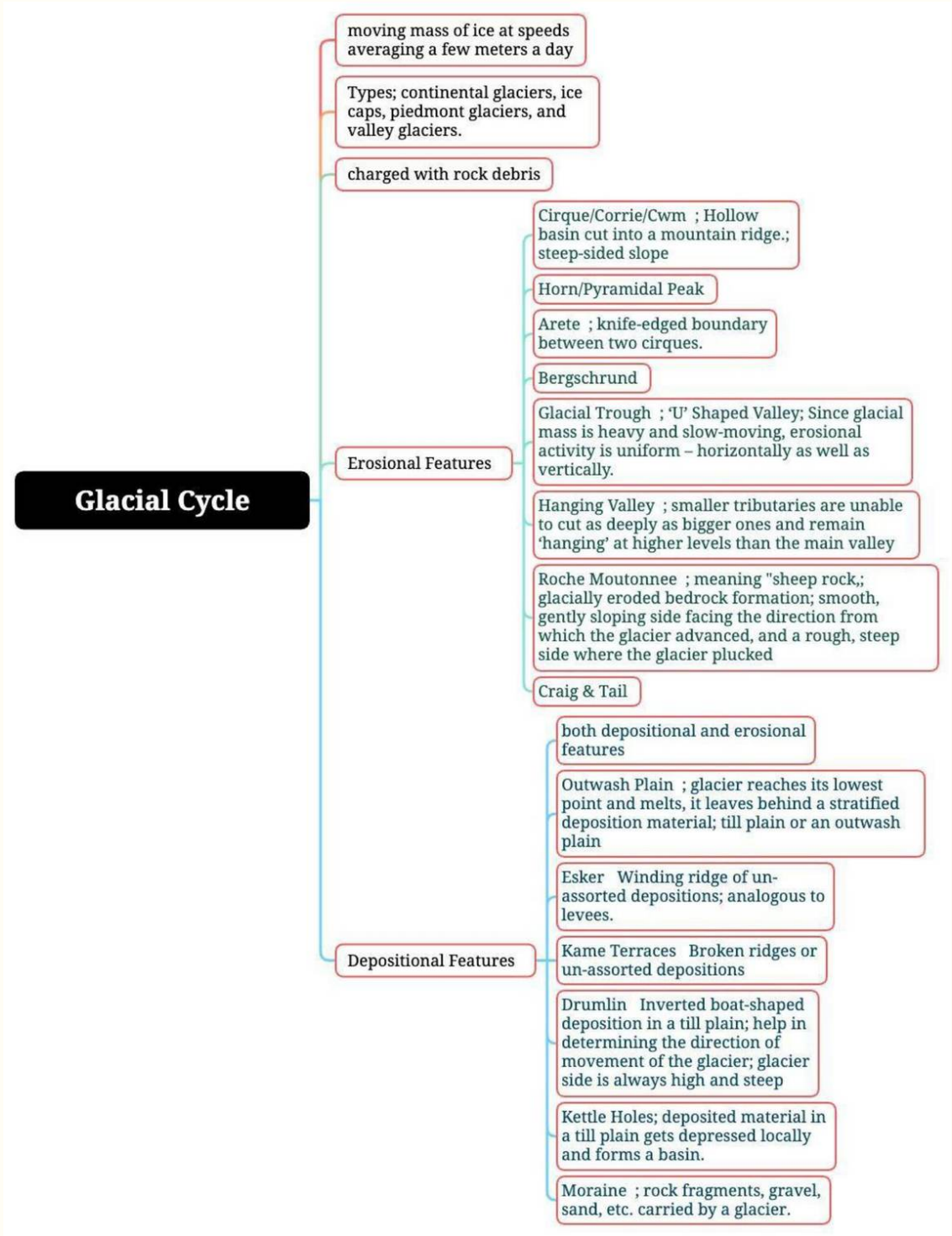
CYCLES OF EROSION

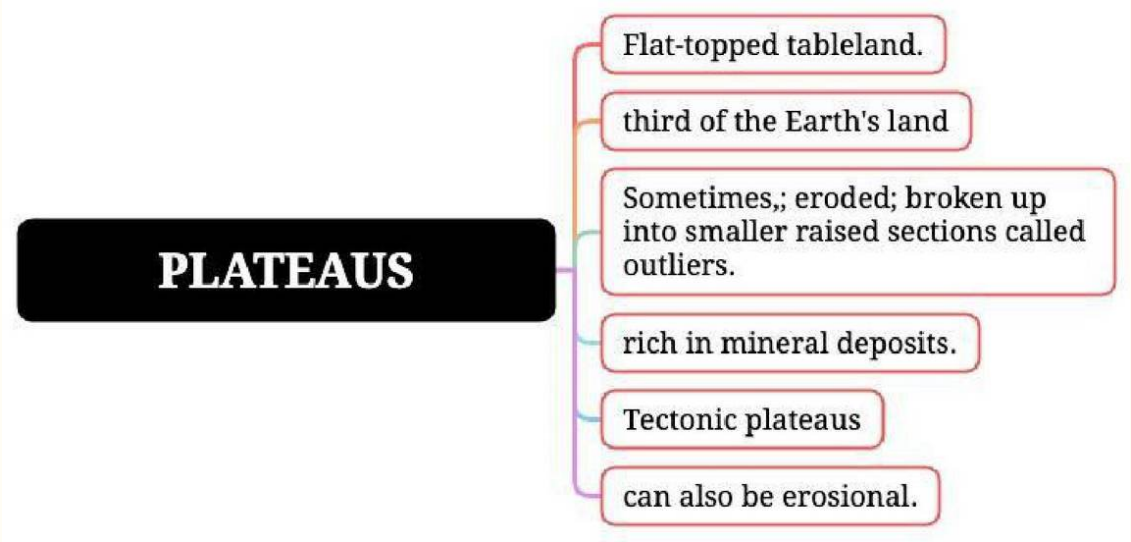


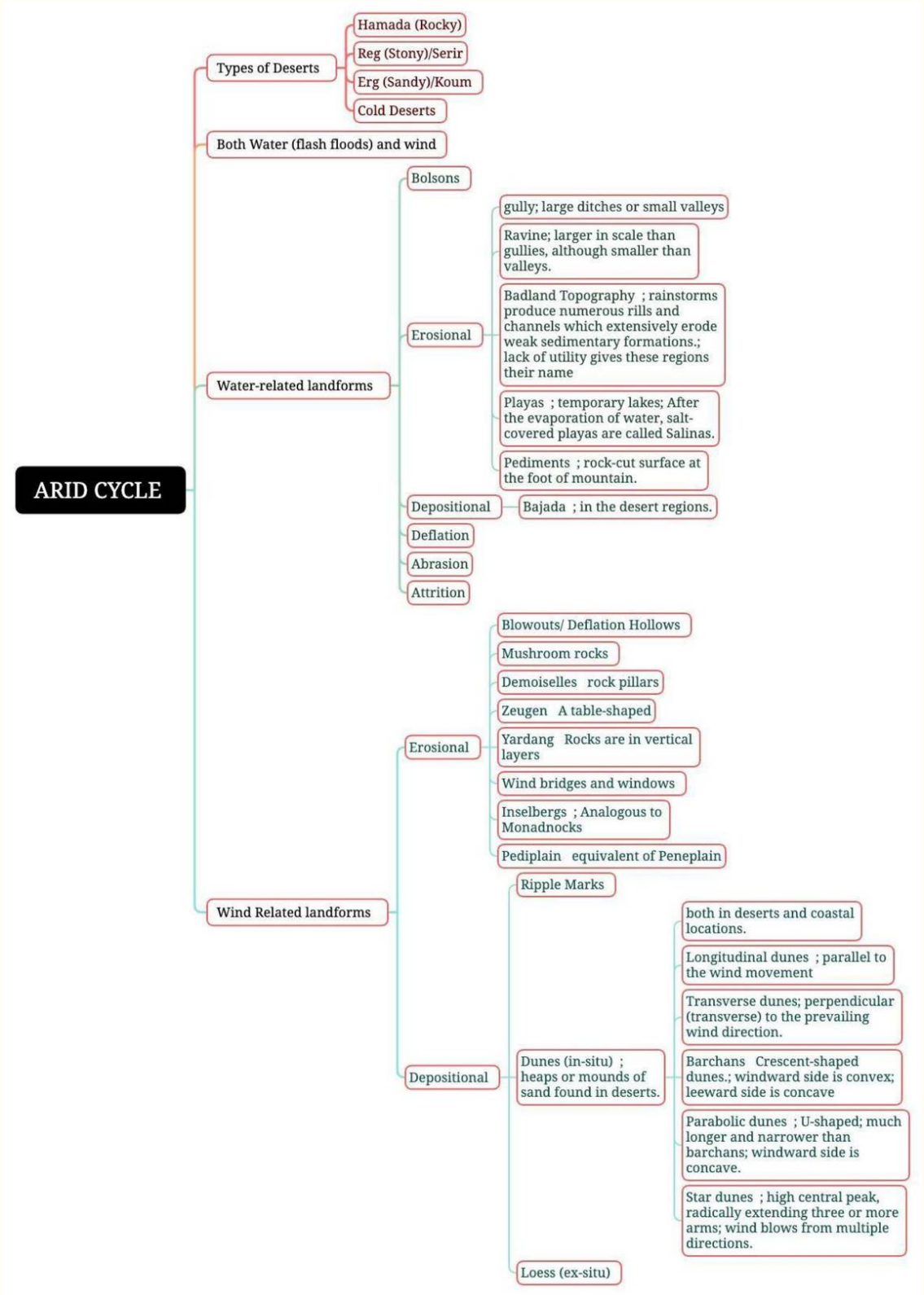

Fluvial Erosion
Major Depositional Fluvial Landforms
Alluvial fans and cones ; at the foothills.; unconsoliated deposited material
Natural Levees ; narrow ridges of low height on both sides of a river; natural protection against floods
Delta; tract of alluvium at the mouth of a river; formed by
load-bearing capacity of a river is reduced
finest particles are carried farthest
Arcuate (fan-shaped); Bird's Foot (elongated; Cuspate (Triangle
Braided Channel
Estuaries
ideal sites for fisheries, ports and industries
Rejuvenation Change in the stage of erosion
Drainage Patterns
typical shape of a river course as it completes its erosional cycle
Dendritic or Pinnate ; irregular tree branch-shaped
Trellis ; short tributaries meet the mainstream at right angles
Rectangular The mainstream bends at right angles and the tributaries join at right angles
Angular; tributaries join the mainstream at acute angles.
Parallel ; streams/tributaries seem to be running parallel to each other
Radial ; tributaries from a summit follow the slope downwards and drain down in all directions.
Annular When the upland has an outer soft stratum, the radial streams develop subsequent tributaries; circular drainage around the summit.
Centripetal In a low-lying basin the streams converge from all sides.

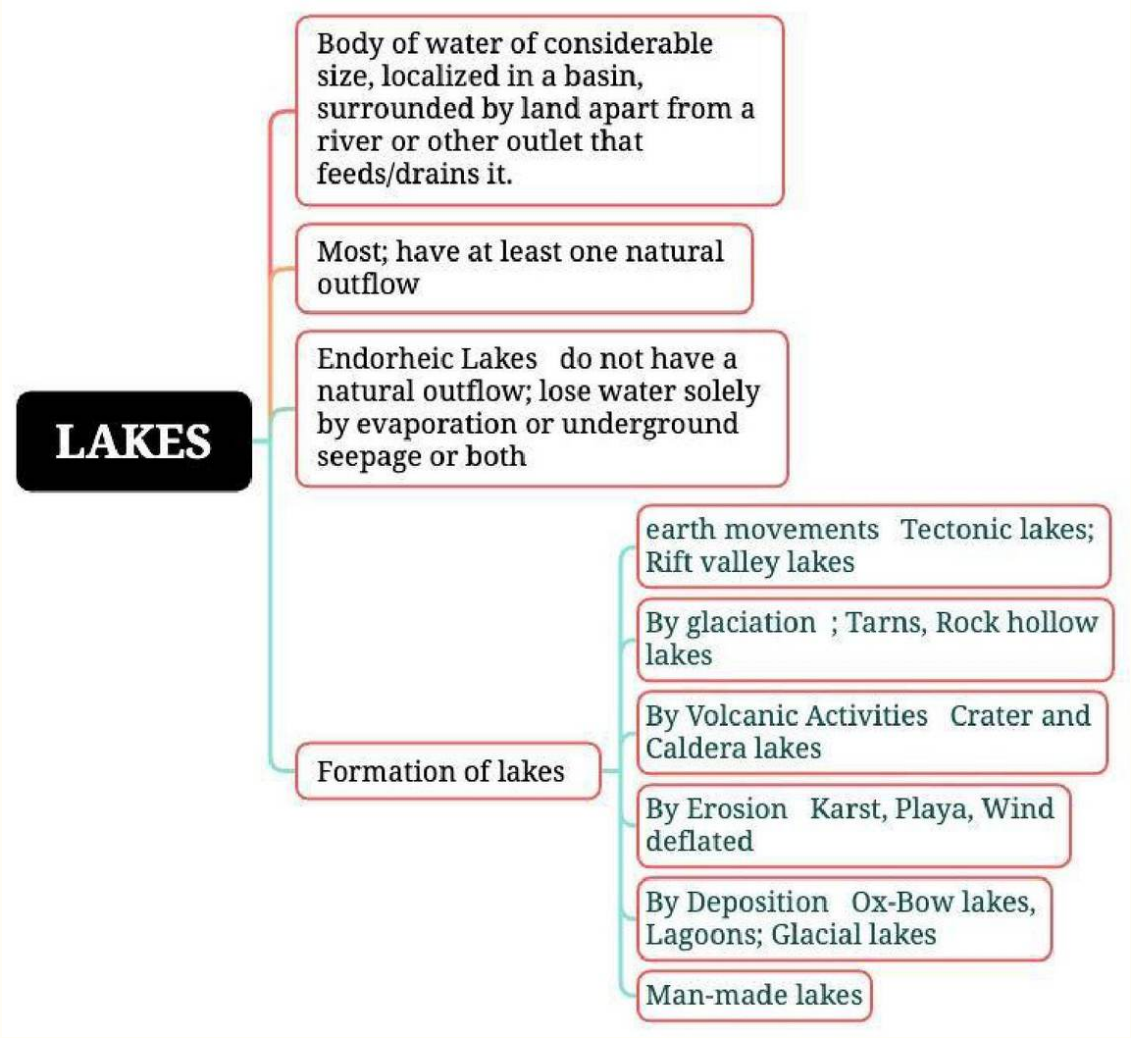

KARST CYCLE
Terminologies associated with Groundwater
Aquifer
Water Table
Confined Aquifer
Unconfined Aquifer
Artesian Well
Groundwater Springs
Perched Water Table
Karst Cycle
Named after a province of Yugoslavia (now Slovenia) on the Adriatic Sea coast
thick deposits of Limestone dolomite or gypsum
dense, highly jointed and thickly bedded
Erosional Features
Sinkholes/Swallow holes ; funnel-shaped depressions; continuous solvent action of the rainwater; Larger sinkholes can be called Dolines, Uvalas and Polje (faulting+erosion)
Lappies ; limestones are vertically bedded
Karst Windows When several adjoining sinkholes collapse
Karst Arch/Bridge
Karst Cave
Depositional Features ; in the Karst Caves; Speleothems
Stalactites, Stalagmites and Karst Pillars
stalactite, growing downwards from the roof.
stalagmite, thicker and flatter, rising upwards from the floor.
stalactite and stalagmite join; column or Karst pillar
Flow stones ; "melted cake icing" and "frozen waterfalls."; may be very colourful yellows, reds, and oranges.
Cave Pearls ; concentrically banded concretions
Banded draperies ; or "cave bacon."
Blind Valley
lat-bottomed valley with an abrupt ending.

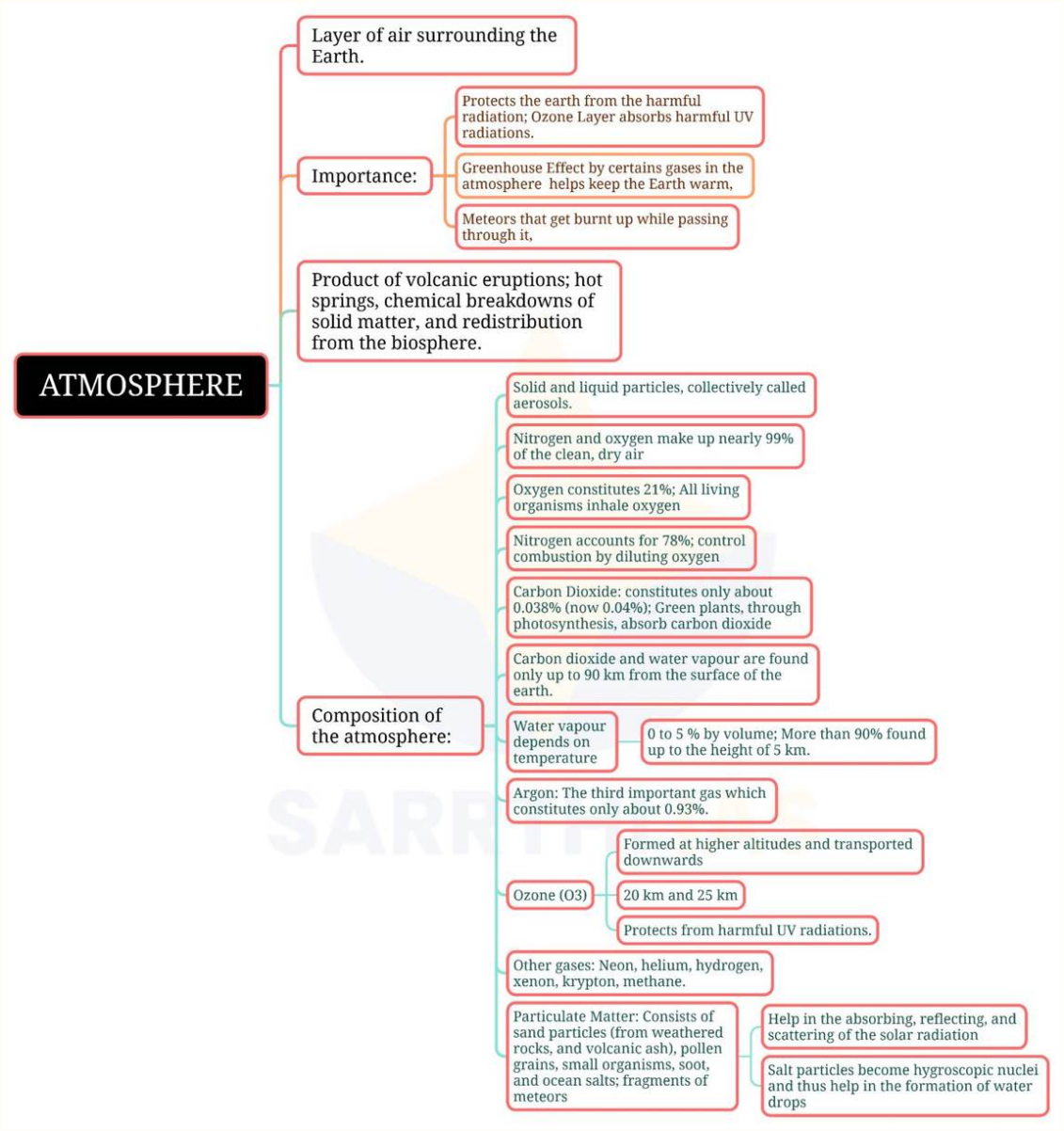

MARINE CYCLE
Sea waves, aided by winds, currents, tides and storms
erosive work
wave pressure compresses the air trapped inside rock fissures, joints, faults, etc
Waves also use rock debris as instruments of erosion.
solvent or chemical action of waves
Erosional Features
Chasms These are narrow, deep indentations; due to headward erosion
widening till a bay is formed.
Wave Cut Platform ; backward retreat of the sea cliffs.
bottom of the cliff; maximum intensive erosion; notch appears
Sea Cave; With further erosion, they get converted into sea arches.
Stack ; sea caves get extremely eroded; all that is left is some rock pillars; Stumps.
Blow Holes/ Spouting holes; make a peculiar noise.; Geos/gloup
Depositional Features
Beach ; temporary
Bar ; deposit rock debris and sand along the coast at a distance from the shoreline
Barrier
Spit and Hook
Tombolo
Marine Dunes
TYPES OF COASTLINES
Coastline of emergence
uplift of the land or by the lowering of the sea level
Bars, spits, lagoons, salt marshes, beaches, sea cliffs, and arches
Coastline of submergence
either by subsidence of land or by a rise in sea level. Ria, Fjord, Dalmatian, and drowned lowlands
Ria Coast ; dissected by streams; highly irregular shoreline; south-west Ireland
Fjord ; heavily eroded by glacial action; valley glacial troughs (U-shaped valley)
long and narrow inlets with very steep sides.
coasts of Norway
Dalmatian ; submergence of mountain ridges with alternating crests and troughs; Yugoslavia
Drowned Lowland ; low and free from indentations; Baltic coast of eastern Germany
Neutral Coastline
formed as a result of new materials being built out into the water.; no relative change between the level of the sea and the coastal region of the continent.
Compound Coastline
submergence followed by emergence or vice versa.
Norway and Sweden
Fault Coastline
submergence of a downthrown block along a fault; Konkan Coast of India

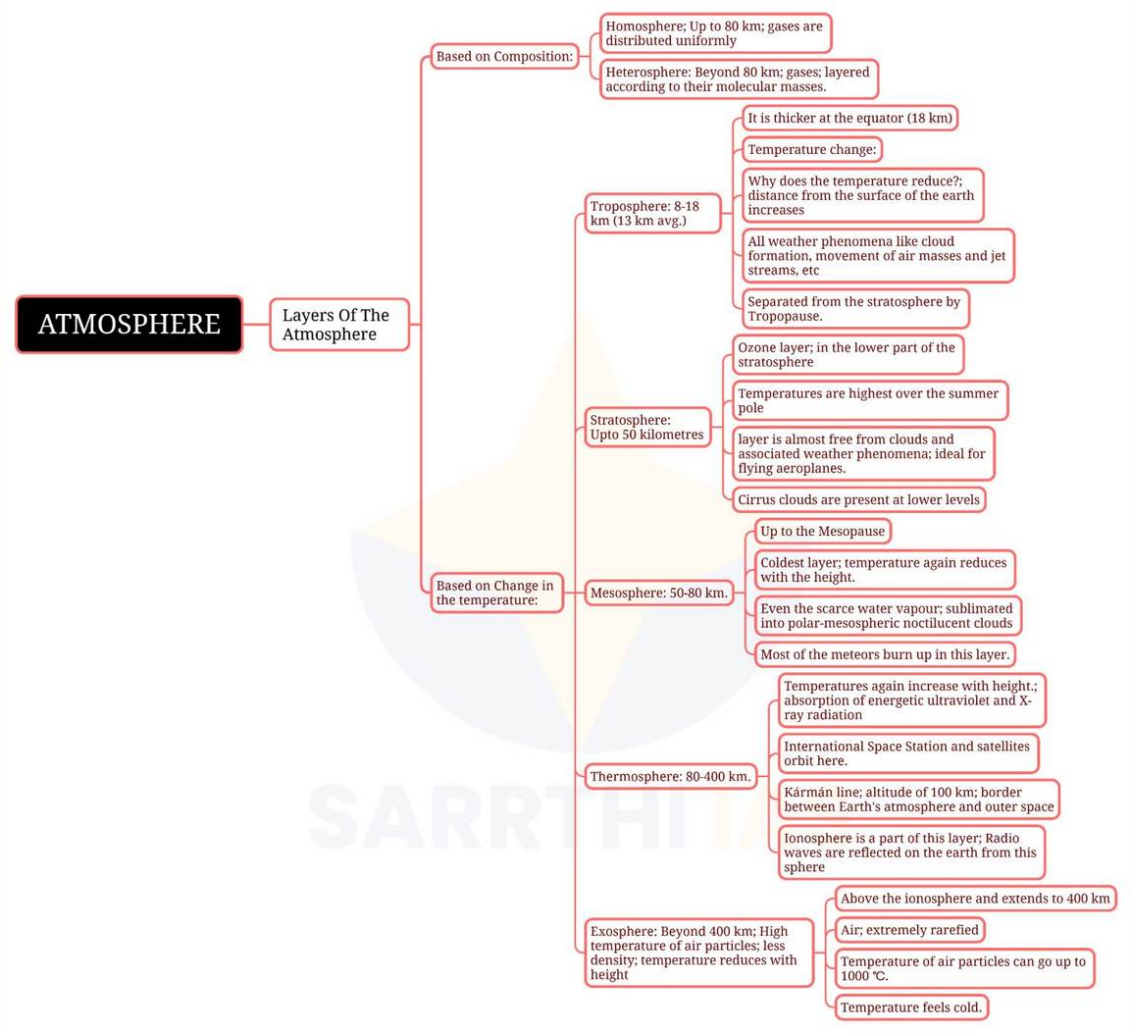

Glacial Cycle
moving mass of ice at speeds averaging a few meters a day
Types; continental glaciers, ice caps, piedmont glaciers, and valley glaciers.
charged with rock debris
Erosional Features
Cirque/Corrie/Cwm ; Hollow basin cut into a mountain ridge.; steep-sided slope
Horn/Pyramidal Peak
Arete ; knife-edged boundary between two cirques.
Bergschrund
Glacial Trough ; 'U' Shaped Valley; Since glacial mass is heavy and slow-moving, erosional activity is uniform – horizontally as well as vertically.
Hanging Valley ; smaller tributaries are unable to cut as deeply as bigger ones and remain 'hanging' at higher levels than the main valley
Roche Moutonnee ; meaning "sheep rock,; glacially eroded bedrock formation; smooth, gently sloping side facing the direction from which the glacier advanced, and a rough, steep side where the glacier plucked
Craig & Tail
Depositional Features
both depositional and erosional features
Outwash Plain ; glacier reaches its lowest point and melts, it leaves behind a stratified deposition material; till plain or an outwash plain
Esker Winding ridge of un-assorted depositions; analogous to levees.
Kame Terraces Broken ridges or un-assorted depositions
Drumlin Inverted boat-shaped deposition in a till plain; help in determining the direction of movement of the glacier; glacier side is always high and steep
Kettle Holes; deposited material in a till plain gets depressed locally and forms a basin.
Moraine ; rock fragments, gravel, sand, etc. carried by a glacier.

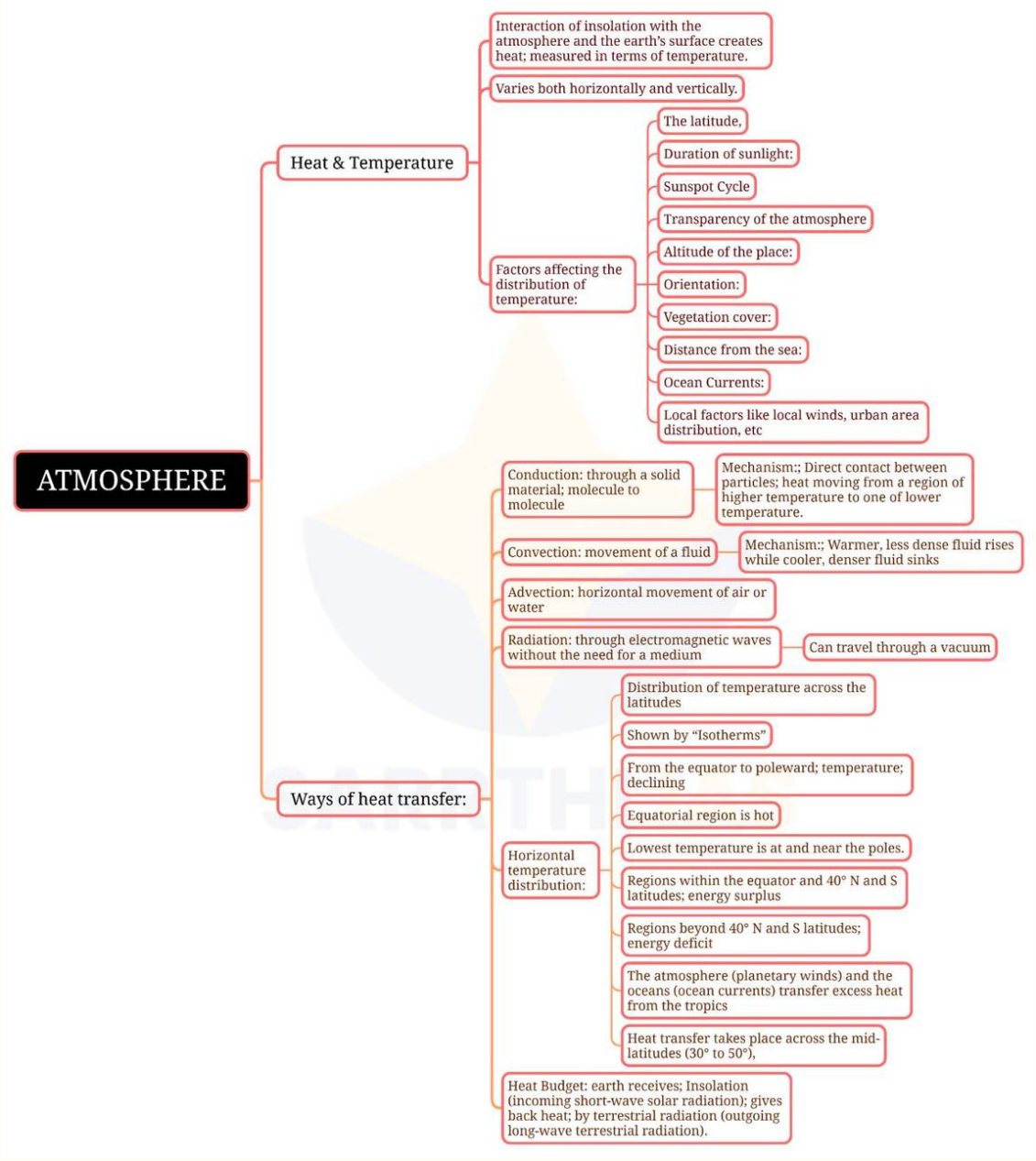

PLATEAUS
Flat-topped tableland.
third of the Earth's land
Sometimes,; eroded; broken up into smaller raised sections called outliers.
rich in mineral deposits.
Tectonic plateaus
can also be erosional.

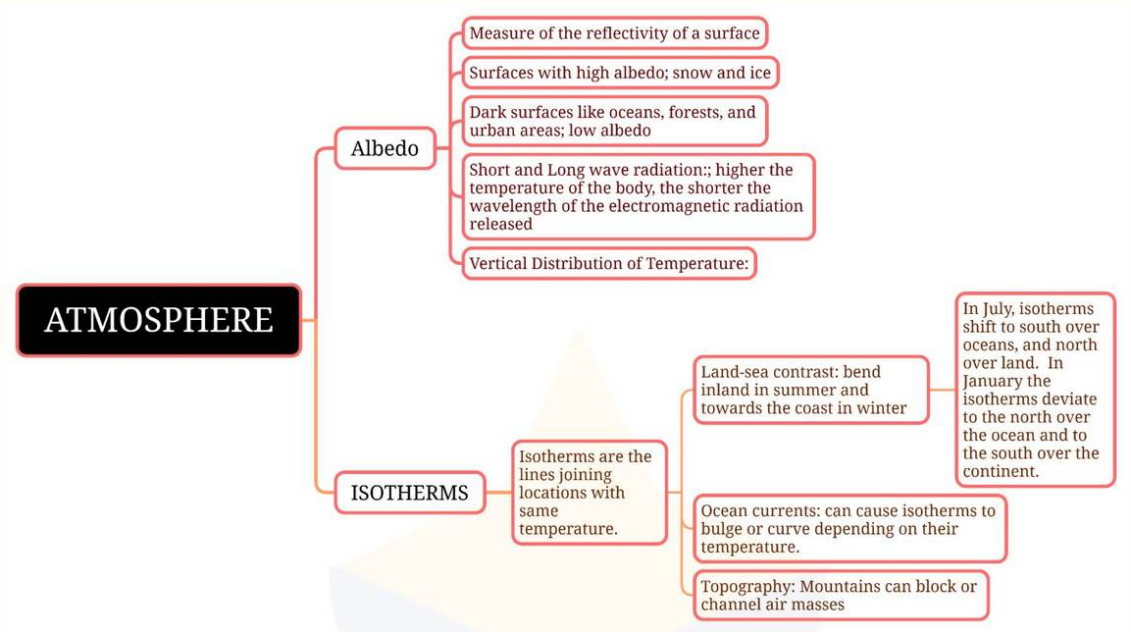

ARID CYCLE
Types of Deserts
Hamada (Rocky)
Reg (Stony)/Serir
Erg (Sandy)/Koum
Cold Deserts
Both Water (flash floods) and wind
Water-related landforms
Bolsons
Erosional
gully; large ditches or small valleys
Ravine; larger in scale than gullies, although smaller than valleys.
Badland Topography ; rainstorms produce numerous rills and channels which extensively erode weak sedimentary formations.; lack of utility gives these regions their name
Playas ; temporary lakes; After the evaporation of water, salt-covered playas are called Salinas.
Pediments ; rock-cut surface at the foot of mountain.
Depositional
Bajada ; in the desert regions.
Deflation
Abrasion
Attrition
Wind Related landforms
Erosional
Blowouts/ Deflation Hollows
Mushroom rocks
Demoiselles rock pillars
Zeugen A table-shaped
Yardang Rocks are in vertical layers
Wind bridges and windows
Inselbergs ; Analogous to Monadnocks
Pediplain equivalent of Peneplain
Depositional
Ripple Marks
Dunes (in-situ) ; heaps or mounds of sand found in deserts.
both in deserts and coastal locations.
Longitudinal dunes ; parallel to the wind movement
Transverse dunes; perpendicular (transverse) to the prevailing wind direction.
Barchans Crescent-shaped dunes.; windward side is convex; leeward side is concave
Parabolic dunes ; U-shaped; much longer and narrower than barchans; windward side is concave.
Star dunes ; high central peak, radically extending three or more arms; wind blows from multiple directions.
Loess (ex-situ)

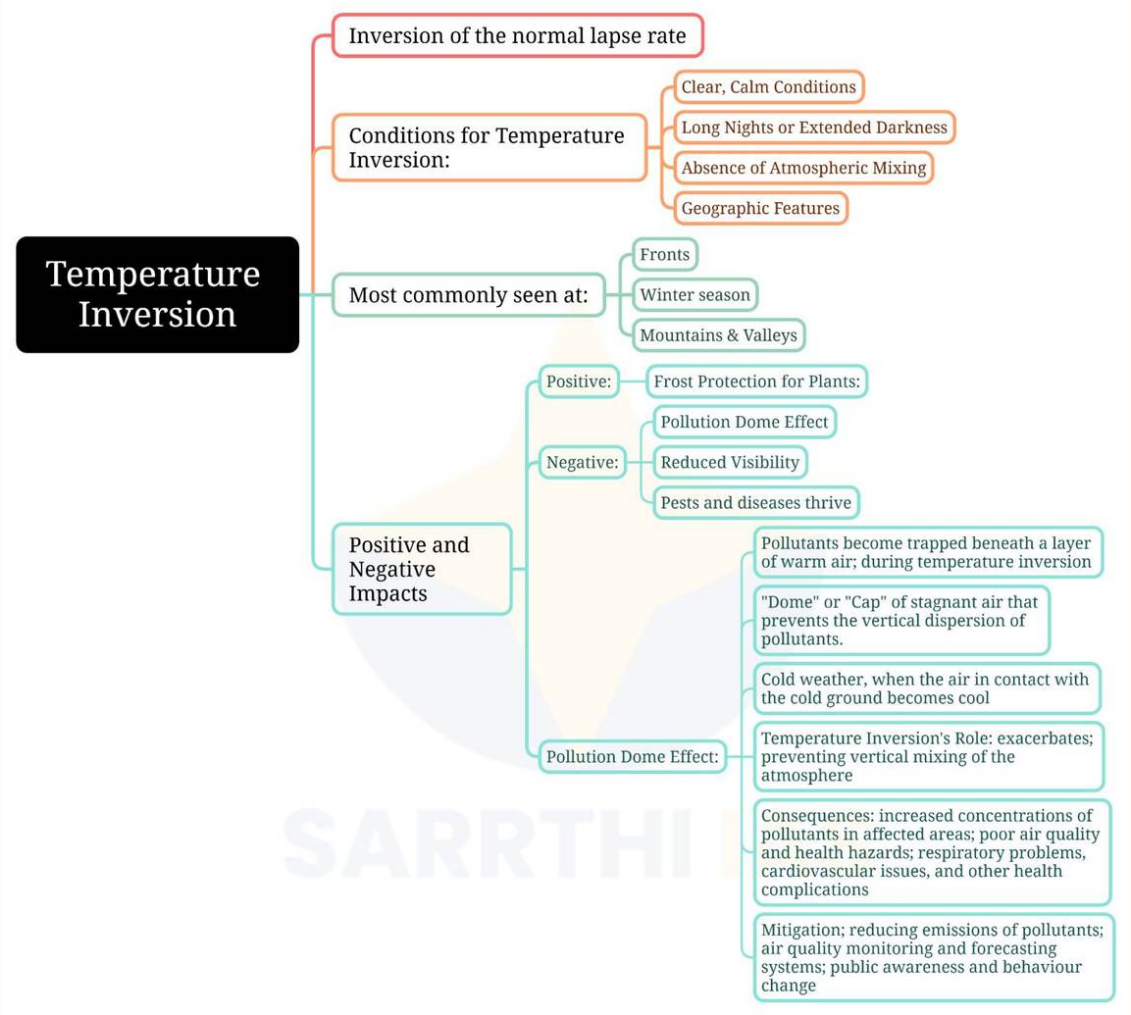

LAKES
Body of water of considerable size, localized in a basin, surrounded by land apart from a river or other outlet that feeds/drains it.
Most; have at least one natural outflow
Endorheic Lakes do not have a natural outflow; lose water solely by evaporation or underground seepage or both
Formation of lakes
earth movements Tectonic lakes; Rift valley lakes
By glaciation ; Tarns, Rock hollow lakes
By Volcanic Activities Crater and Caldera lakes
By Erosion Karst, Playa, Wind deflated
By Deposition Ox-Bow lakes, Lagoons; Glacial lakes
Man-made lakes

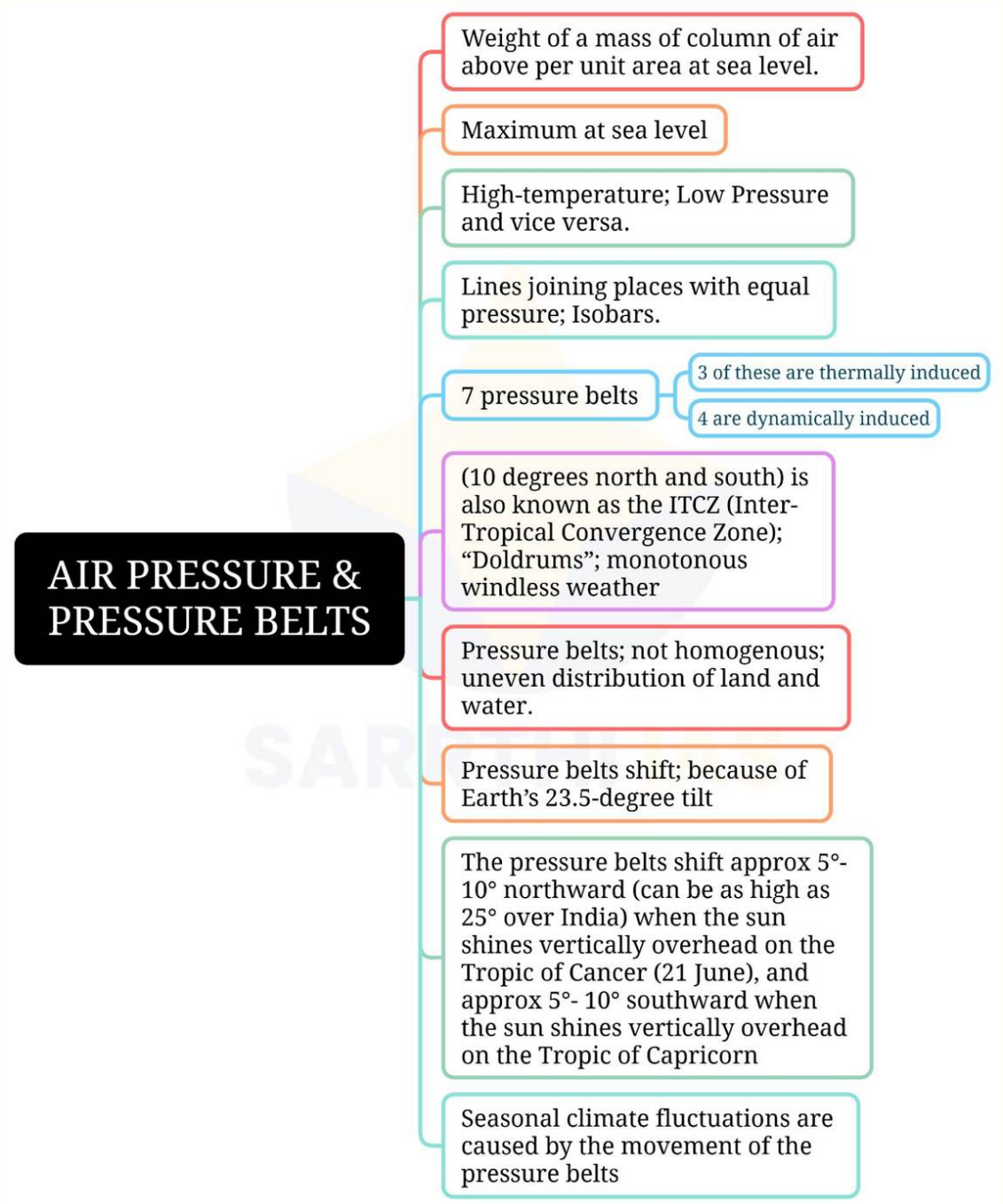

ATMOSPHERE
Layer of air surrounding the Earth.
Importance:
Protects the earth from the harmful radiation; Ozone Layer absorbs harmful UV radiations.
Greenhouse Effect by certains gases in the atmosphere helps keep the Earth warm,
Meteors that get burnt up while passing through it,
Product of volcanic eruptions; hot springs, chemical breakdowns of solid matter, and redistribution from the biosphere.
Composition of the atmosphere:
Solid and liquid particles, collectively called aerosols.
Nitrogen and oxygen make up nearly 99% of the clean, dry air
Oxygen constitutes 21%; All living organisms inhale oxygen
Nitrogen accounts for 78%; control combustion by diluting oxygen
Carbon Dioxide: constitutes only about 0.038% (now 0.04%); Green plants, through photosynthesis, absorb carbon dioxide
Carbon dioxide and water vapour are found only up to 90 km from the surface of the earth.
Water vapour depends on temperature
0 to 5 % by volume; More than 90% found up to the height of 5 km.
Argon: The third important gas which constitutes only about 0.93%.
Ozone (O3)
Formed at higher altitudes and transported downwards
20 km and 25 km
Protects from harmful UV radiations.
Other gases: Neon, helium, hydrogen, xenon, krypton, methane.
Particulate Matter: Consists of sand particles (from weathered rocks, and volcanic ash), pollen grains, small organisms, soot, and ocean salts; fragments of meteors
Help in the absorbing, reflecting, and scattering of the solar radiation
Salt particles become hygroscopic nuclei and thus help in the formation of water drops

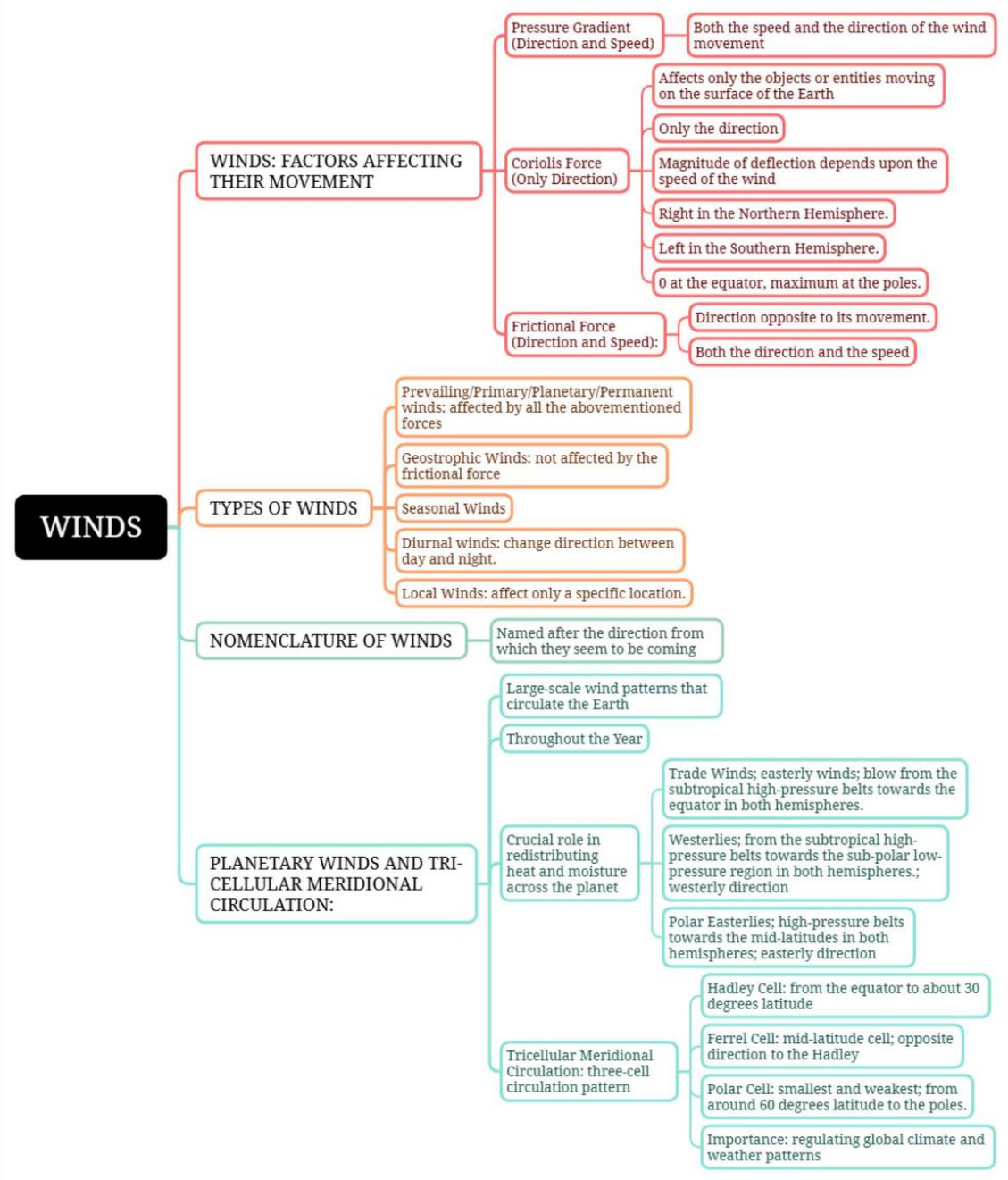

ATMOSPHERE
Layers Of The Atmosphere
Based on Composition:
Homosphere; Up to 80 km; gases are distributed uniformly
Heterosphere: Beyond 80 km; gases; layered according to their molecular masses.
Based on Change in the temperature:
Troposphere: 8-18 km (13 km avg.)
It is thicker at the equator (18 km)
Temperature change:
Why does the temperature reduce?; distance from the surface of the earth increases
All weather phenomena like cloud formation, movement of air masses and jet streams, etc
Separated from the stratosphere by Tropopause.
Stratosphere: Upto 50 kilometres
Ozone layer; in the lower part of the stratosphere
Temperatures are highest over the summer pole
layer is almost free from clouds and associated weather phenomena; ideal for flying aeroplanes.
Cirrus clouds are present at lower levels
Mesosphere: 50-80 km.
Up to the Mesopause
Coldest layer; temperature again reduces with the height.
Even the scarce water vapour; sublimated into polar-mesospheric noctilucent clouds
Most of the meteors burn up in this layer.
Thermosphere: 80-400 km.
Temperatures again increase with height.; absorption of energetic ultraviolet and X-ray radiation
International Space Station and satellites orbit here.
Kármán line; altitude of 100 km; border between Earth's atmosphere and outer space
Ionosphere is a part of this layer; Radio waves are reflected on the earth from this sphere
Exosphere: Beyond 400 km; High temperature of air particles; less density; temperature reduces with height
Above the ionosphere and extends to 400 km
Air; extremely rarefied
Temperature of air particles can go up to 1000 °C.
Temperature feels cold.

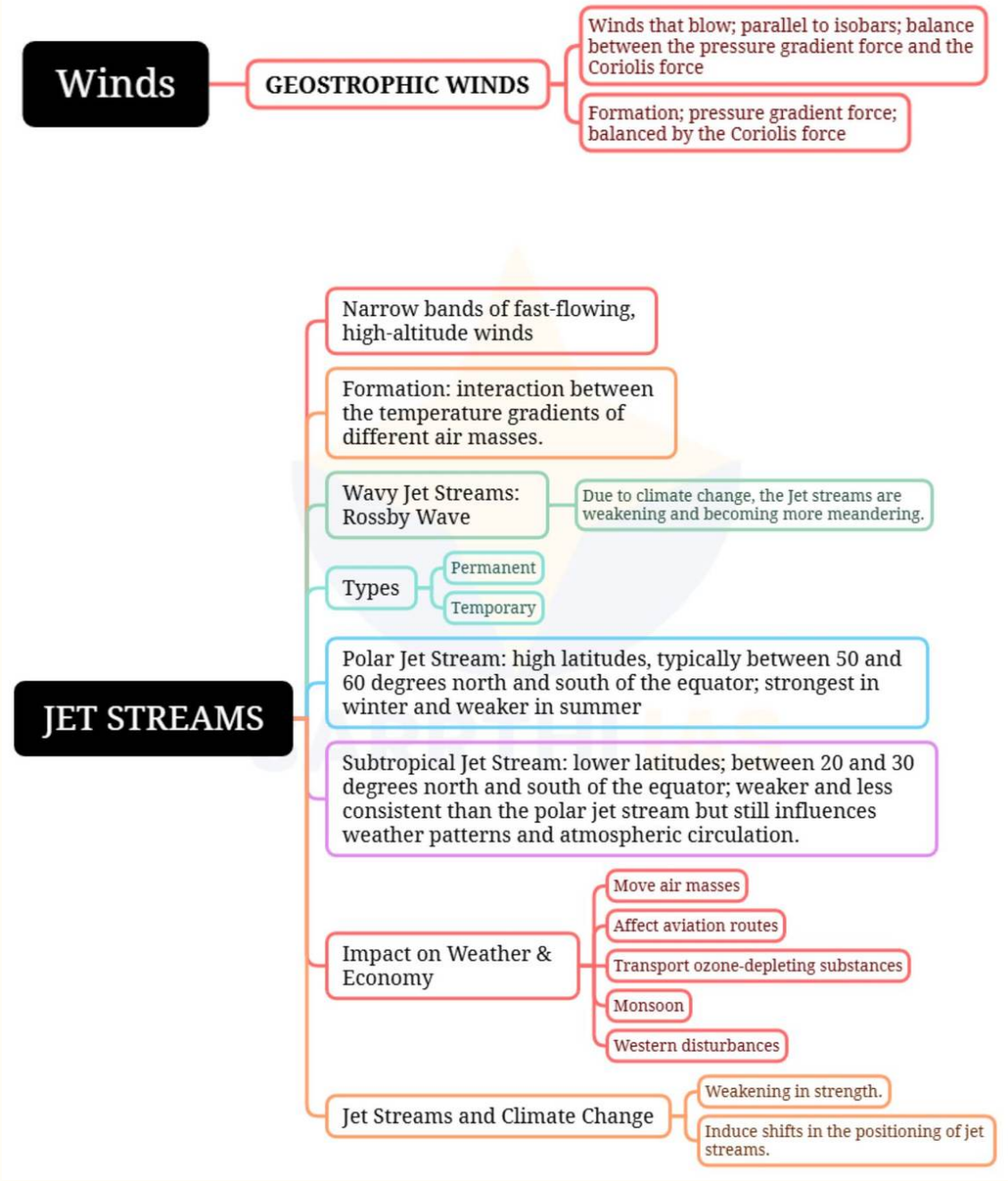

ATMOSPHERE
Heat & Temperature
Interaction of insolation with the atmosphere and the earth's surface creates heat; measured in terms of temperature.
Varies both horizontally and vertically.
Factors affecting the distribution of temperature:
The latitude,
Duration of sunlight:
Sunspot Cycle
Transparency of the atmosphere
Altitude of the place:
Orientation:
Vegetation cover:
Distance from the sea:
Ocean Currents:
Local factors like local winds, urban area distribution, etc
Ways of heat transfer:
Conduction: through a solid material; molecule to molecule
Mechanism:; Direct contact between particles; heat moving from a region of higher temperature to one of lower temperature.
Convection: movement of a fluid
Mechanism:; Warmer, less dense fluid rises while cooler, denser fluid sinks
Advection: horizontal movement of air or water
Radiation: through electromagnetic waves without the need for a medium
Can travel through a vacuum
Horizontal temperature distribution:
Distribution of temperature across the latitudes
Shown by "Isotherms"
From the equator to poleward; temperature; declining
Equatorial region is hot
Lowest temperature is at and near the poles.
Regions within the equator and 40° N and S latitudes; energy surplus
Regions beyond 40° N and S latitudes; energy deficit
The atmosphere (planetary winds) and the oceans (ocean currents) transfer excess heat from the tropics
Heat transfer takes place across the mid-latitudes (30° to 50°),
Heat Budget: earth receives; Insolation (incoming short-wave solar radiation); gives back heat; by terrestrial radiation (outgoing long-wave terrestrial radiation).

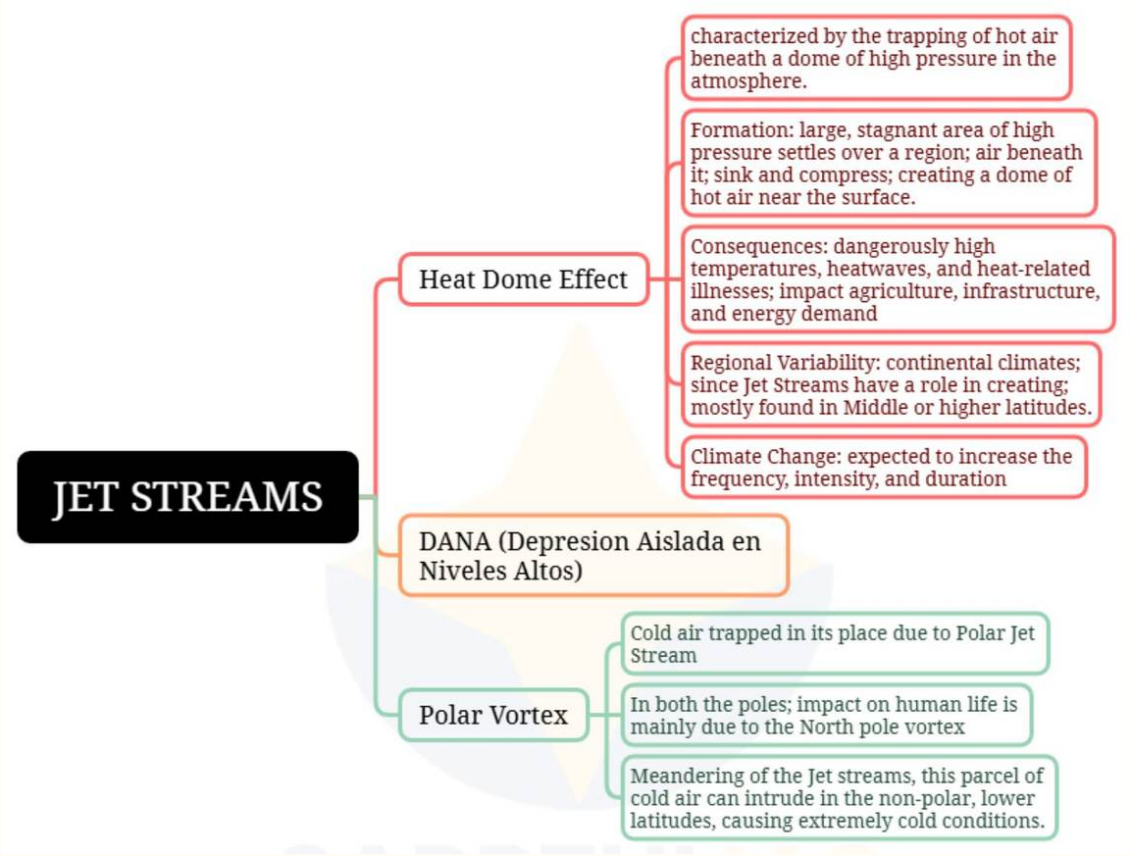

ATMOSPHERE
Albedo
Measure of the reflectivity of a surface
Surfaces with high albedo; snow and ice
Dark surfaces like oceans, forests, and urban areas; low albedo
Short and Long wave radiation:; higher the temperature of the body, the shorter the wavelength of the electromagnetic radiation released
Vertical Distribution of Temperature:
ISOTHERMS
Isotherms are the lines joining locations with same temperature.
Land-sea contrast: bend inland in summer and towards the coast in winter
In July, isotherms shift to south over oceans, and north over land. In January the isotherms deviate to the north over the ocean and to the south over the continent.
Ocean currents: can cause isotherms to bulge or curve depending on their temperature.
Topography: Mountains can block or channel air masses

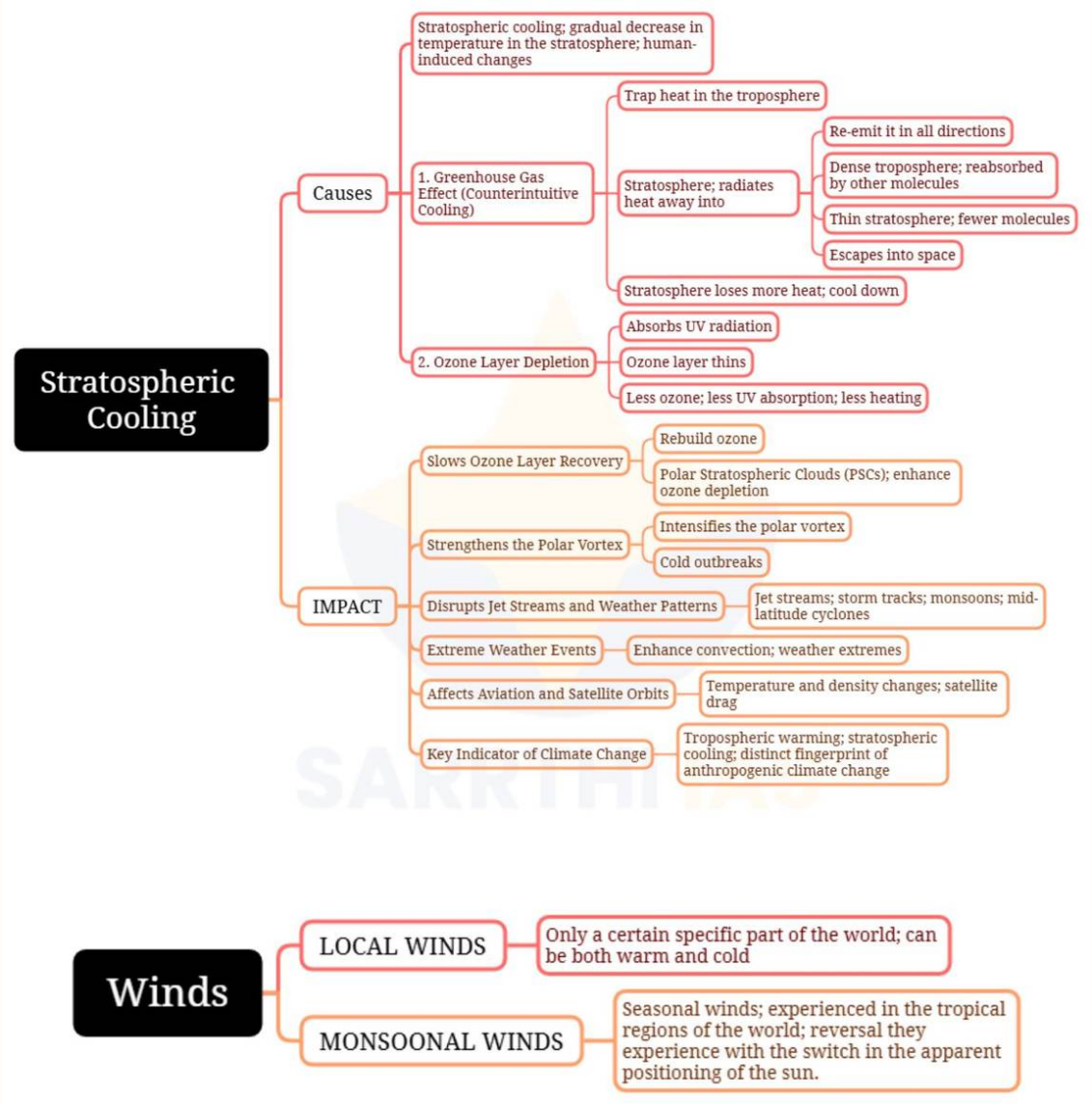

Temperature Inversion
Inversion of the normal lapse rate
Conditions for Temperature Inversion:
Clear, Calm Conditions
Long Nights or Extended Darkness
Absence of Atmospheric Mixing
Geographic Features
Most commonly seen at:
Fronts
Winter season
Mountains & Valleys
Positive and Negative Impacts
Positive:
Frost Protection for Plants:
Negative:
Pollution Dome Effect
Reduced Visibility
Pests and diseases thrive
Pollution Dome Effect:
Pollutants become trapped beneath a layer of warm air; during temperature inversion
"Dome" or "Cap" of stagnant air that prevents the vertical dispersion of pollutants.
Cold weather, when the air in contact with the cold ground becomes cool
Temperature Inversion's Role: exacerbates; preventing vertical mixing of the atmosphere
Consequences: increased concentrations of pollutants in affected areas; poor air quality and health hazards; respiratory problems, cardiovascular issues, and other health complications
Mitigation; reducing emissions of pollutants; air quality monitoring and forecasting systems; public awareness and behaviour change

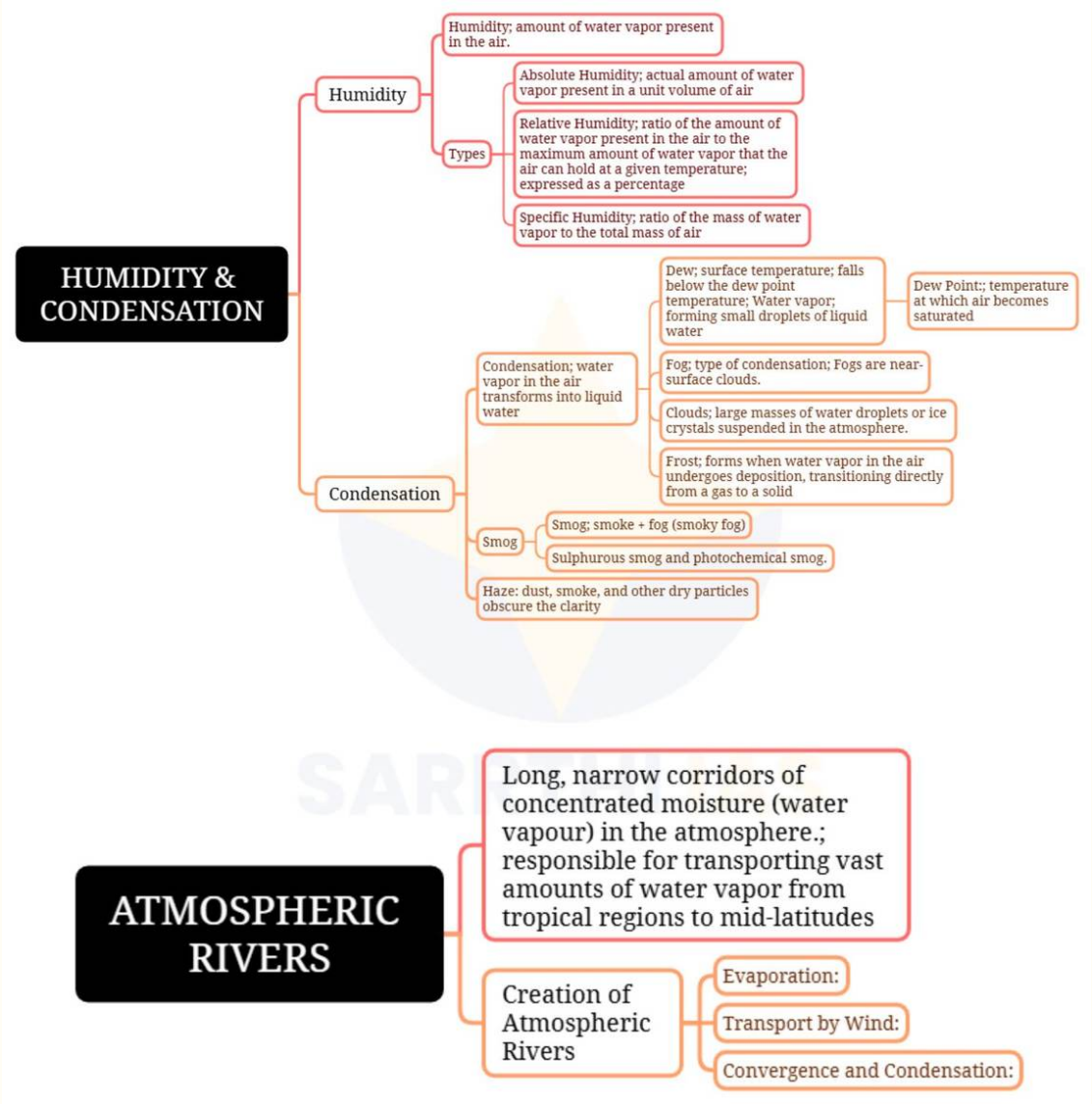

AIR PRESSURE & PRESSURE BELTS
Weight of a mass of column of air above per unit area at sea level.
Maximum at sea level
High-temperature; Low Pressure and vice versa.
Lines joining places with equal pressure; Isobars.
7 pressure belts
3 of these are thermally induced
4 are dynamically induced
(10 degrees north and south) is also known as the ITCZ (Inter-Tropical Convergence Zone); "Doldrums"; monotonous windless weather
Pressure belts; not homogenous; uneven distribution of land and water.
Pressure belts shift; because of Earth's 23.5-degree tilt
The pressure belts shift approx 5°-10° northward (can be as high as 25° over India) when the sun shines vertically overhead on the Tropic of Cancer (21 June), and approx 5°- 10° southward when the sun shines vertically overhead on the Tropic of Capricorn
Seasonal climate fluctuations are caused by the movement of the pressure belts

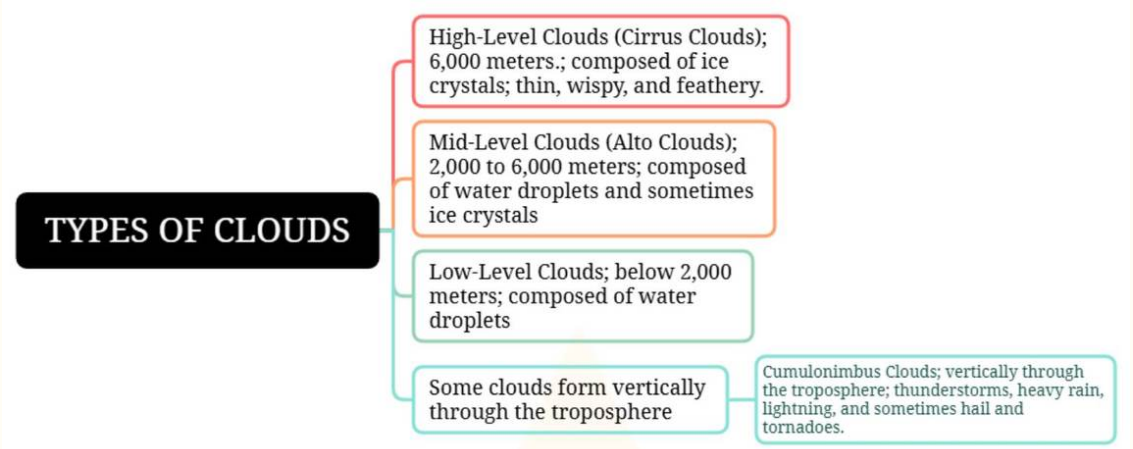

WINDS
WINDS: FACTORS AFFECTING THEIR MOVEMENT
Pressure Gradient (Direction and Speed)
Both the speed and the direction of the wind movement
Coriolis Force (Only Direction)
Affects only the objects or entities moving on the surface of the Earth
Only the direction
Magnitude of deflection depends upon the speed of the wind
Right in the Northern Hemisphere.
Left in the Southern Hemisphere.
0 at the equator, maximum at the poles.
Frictional Force (Direction and Speed):
Direction opposite to its movement.
Both the direction and the speed
TYPES OF WINDS
Prevailing/Primary/Planetary/Permanent winds: affected by all the abovementioned forces
Geostrophic Winds: not affected by the frictional force
Seasonal Winds
Diurnal winds: change direction between day and night.
Local Winds: affect only a specific location.
NOMENCLATURE OF WINDS
Named after the direction from which they seem to be coming
PLANETARY WINDS AND TRI-CELLULAR MERIDIONAL CIRCULATION:
Large-scale wind patterns that circulate the Earth
Throughout the Year
Crucial role in redistributing heat and moisture across the planet
Trade Winds; easterly winds; blow from the subtropical high-pressure belts towards the equator in both hemispheres.
Westerlies; from the subtropical high-pressure belts towards the sub-polar low-pressure region in both hemispheres.; westerly direction
Polar Easterlies; high-pressure belts towards the mid-latitudes in both hemispheres; easterly direction
Tricellular Meridional Circulation: three-cell circulation pattern
Hadley Cell: from the equator to about 30 degrees latitude
Ferrel Cell: mid-latitude cell; opposite direction to the Hadley
Polar Cell: smallest and weakest; from around 60 degrees latitude to the poles.
Importance: regulating global climate and weather patterns

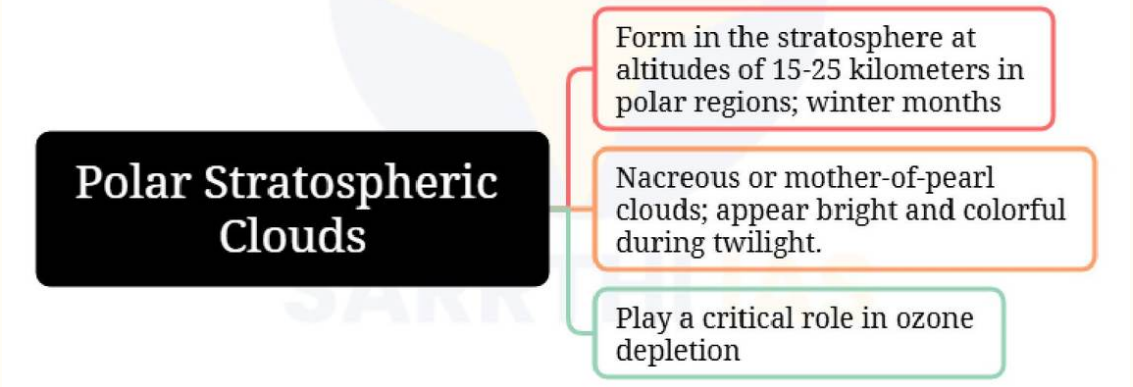

Winds
GEOSTROPHIC WINDS
Winds that blow; parallel to isobars; balance between the pressure gradient force and the Coriolis force
Formation; pressure gradient force; balanced by the Coriolis force
JET STREAMS
Narrow bands of fast-flowing, high-altitude winds
Formation: interaction between the temperature gradients of different air masses.
Wavy Jet Streams: Rossby Wave
Due to climate change, the Jet streams are weakening and becoming more meandering.
Types
Permanent
Temporary
Polar Jet Stream: high latitudes, typically between 50 and 60 degrees north and south of the equator; strongest in winter and weaker in summer
Subtropical Jet Stream: lower latitudes; between 20 and 30 degrees north and south of the equator; weaker and less consistent than the polar jet stream but still influences weather patterns and atmospheric circulation.
Impact on Weather & Economy
Move air masses
Affect aviation routes
Transport ozone-depleting substances
Monsoon
Western disturbances
Jet Streams and Climate Change
Weakening in strength.
Induce shifts in the positioning of jet streams.

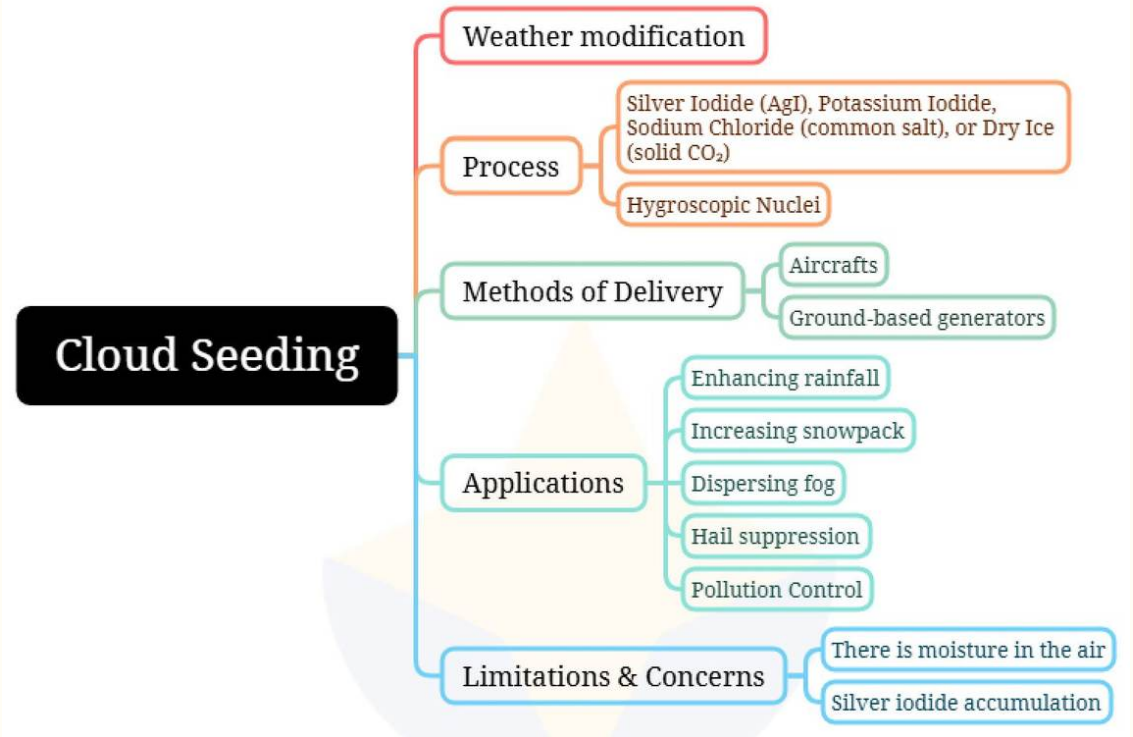

JET STREAMS
Heat Dome Effect
characterized by the trapping of hot air beneath a dome of high pressure in the atmosphere.
Formation: large, stagnant area of high pressure settles over a region; air beneath it; sink and compress; creating a dome of hot air near the surface.
Consequences: dangerously high temperatures, heatwaves, and heat-related illnesses; impact agriculture, infrastructure, and energy demand
Regional Variability: continental climates; since Jet Streams have a role in creating; mostly found in Middle or higher latitudes.
Climate Change: expected to increase the frequency, intensity, and duration
DANA (Depresion Aislada en Niveles Altos)
Polar Vortex
Cold air trapped in its place due to Polar Jet Stream
In both the poles; impact on human life is mainly due to the North pole vortex
Meandering of the Jet streams, this parcel of cold air can intrude in the non-polar, lower latitudes, causing extremely cold conditions.

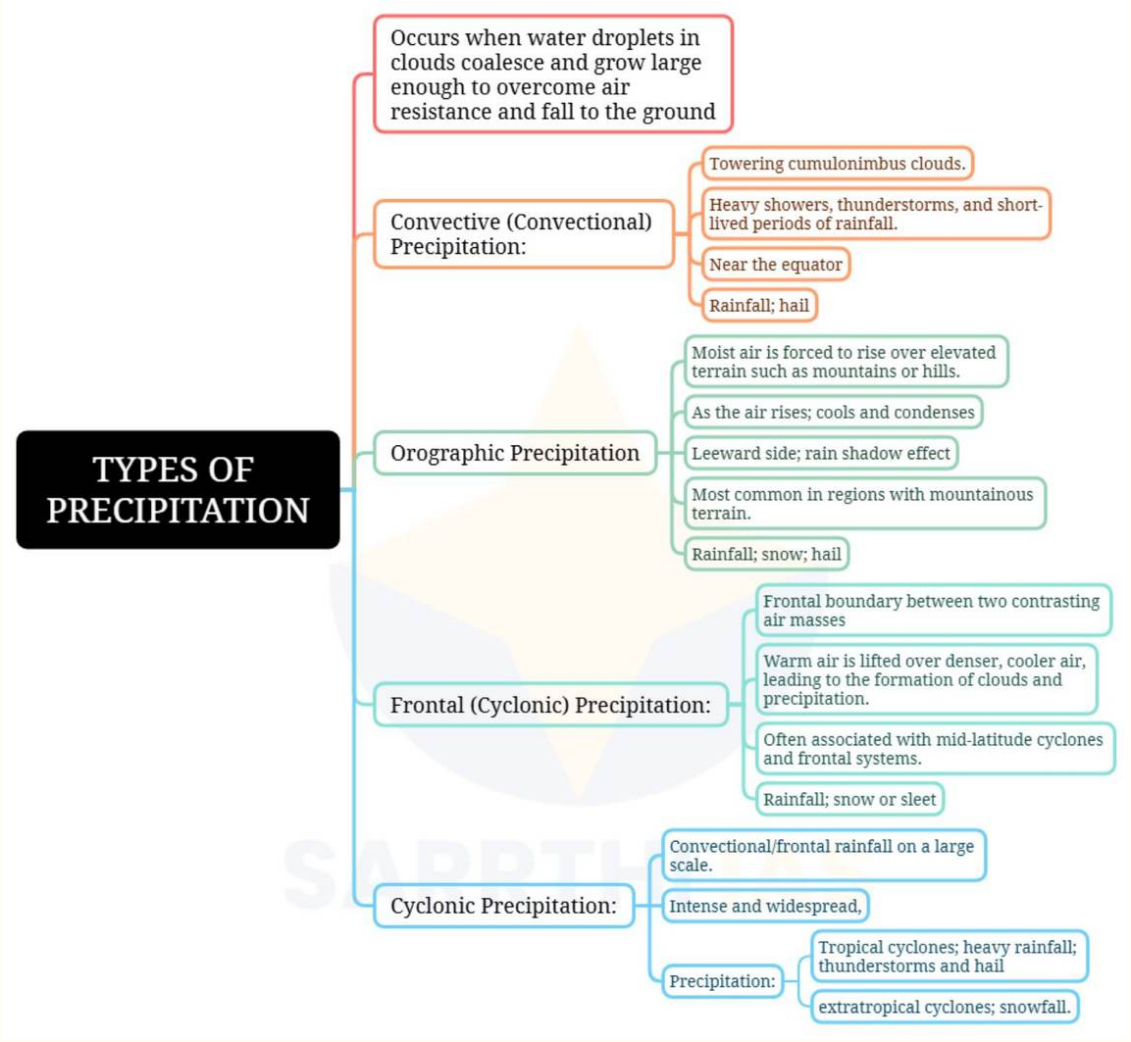

Stratospheric Cooling
Causes
Stratospheric cooling; gradual decrease in temperature in the stratosphere; human-induced changes
1. Greenhouse Gas Effect (Counterintuitive Cooling)
Trap heat in the troposphere
Stratosphere; radiates heat away into
Re-emit it in all directions
Dense troposphere; reabsorbed by other molecules
Thin stratosphere; fewer molecules
Escapes into space
Stratosphere loses more heat; cool down
2. Ozone Layer Depletion
Absorbs UV radiation
Ozone layer thins
Less ozone; less UV absorption; less heating
IMPACT
Slows Ozone Layer Recovery
Rebuild ozone
Polar Stratospheric Clouds (PSCs); enhance ozone depletion
Strengthens the Polar Vortex
Intensifies the polar vortex
Cold outbreaks
Disrupts Jet Streams and Weather Patterns
Jet streams; storm tracks; monsoons; mid-latitude cyclones
Extreme Weather Events
Enhance convection; weather extremes
Affects Aviation and Satellite Orbits
Temperature and density changes; satellite drag
Key Indicator of Climate Change
Tropospheric warming; stratospheric cooling; distinct fingerprint of anthropogenic climate change
Winds
LOCAL WINDS
Only a certain specific part of the world; can be both warm and cold
MONSOONAL WINDS
Seasonal winds; experienced in the tropical regions of the world; reversal they experience with the switch in the apparent positioning of the sun.

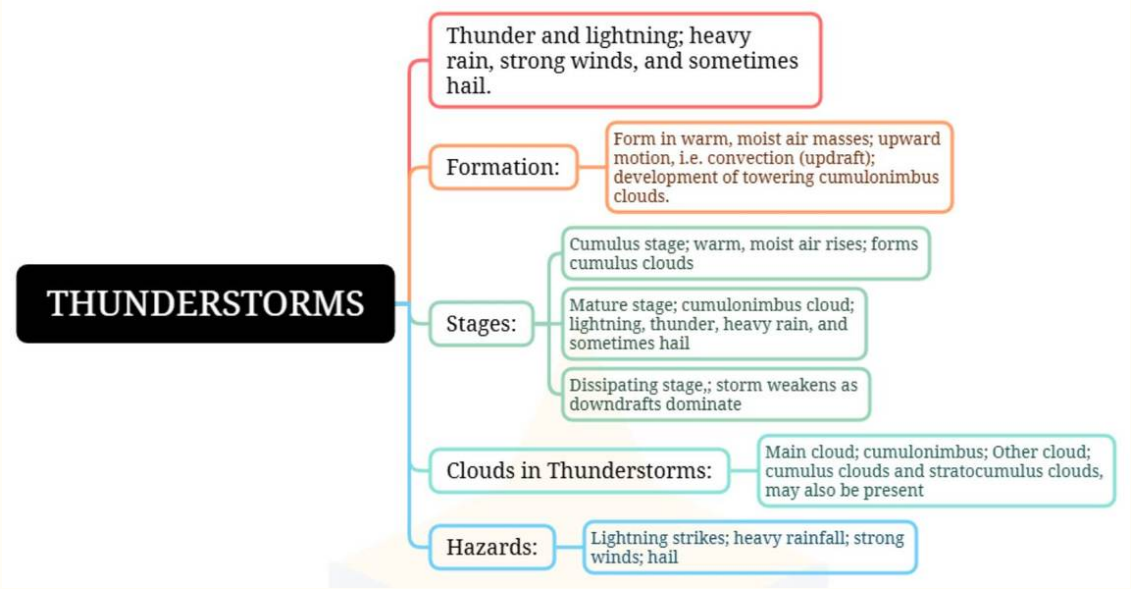

HUMIDITY & CONDENSATION
Humidity
Humidity; amount of water vapor present in the air.
Types
Absolute Humidity; actual amount of water vapor present in a unit volume of air
Relative Humidity; ratio of the amount of water vapor present in the air to the maximum amount of water vapor that the air can hold at a given temperature; expressed as a percentage
Specific Humidity; ratio of the mass of water vapor to the total mass of air
Condensation
Condensation; water vapor in the air transforms into liquid water
Dew; surface temperature; falls below the dew point temperature; Water vapor; forming small droplets of liquid water
Dew Point:; temperature at which air becomes saturated
Fog; type of condensation; Fogs are near-surface clouds.
Clouds; large masses of water droplets or ice crystals suspended in the atmosphere.
Frost; forms when water vapor in the air undergoes deposition, transitioning directly from a gas to a solid
Smog
Smog; smoke + fog (smoky fog)
Sulphurous smog and photochemical smog.
Haze: dust, smoke, and other dry particles obscure the clarity
ATMOSPHERIC RIVERS
Long, narrow corridors of concentrated moisture (water vapour) in the atmosphere.; responsible for transporting vast amounts of water vapor from tropical regions to mid-latitudes
Creation of Atmospheric Rivers
Evaporation:
Transport by Wind:
Convergence and Condensation:

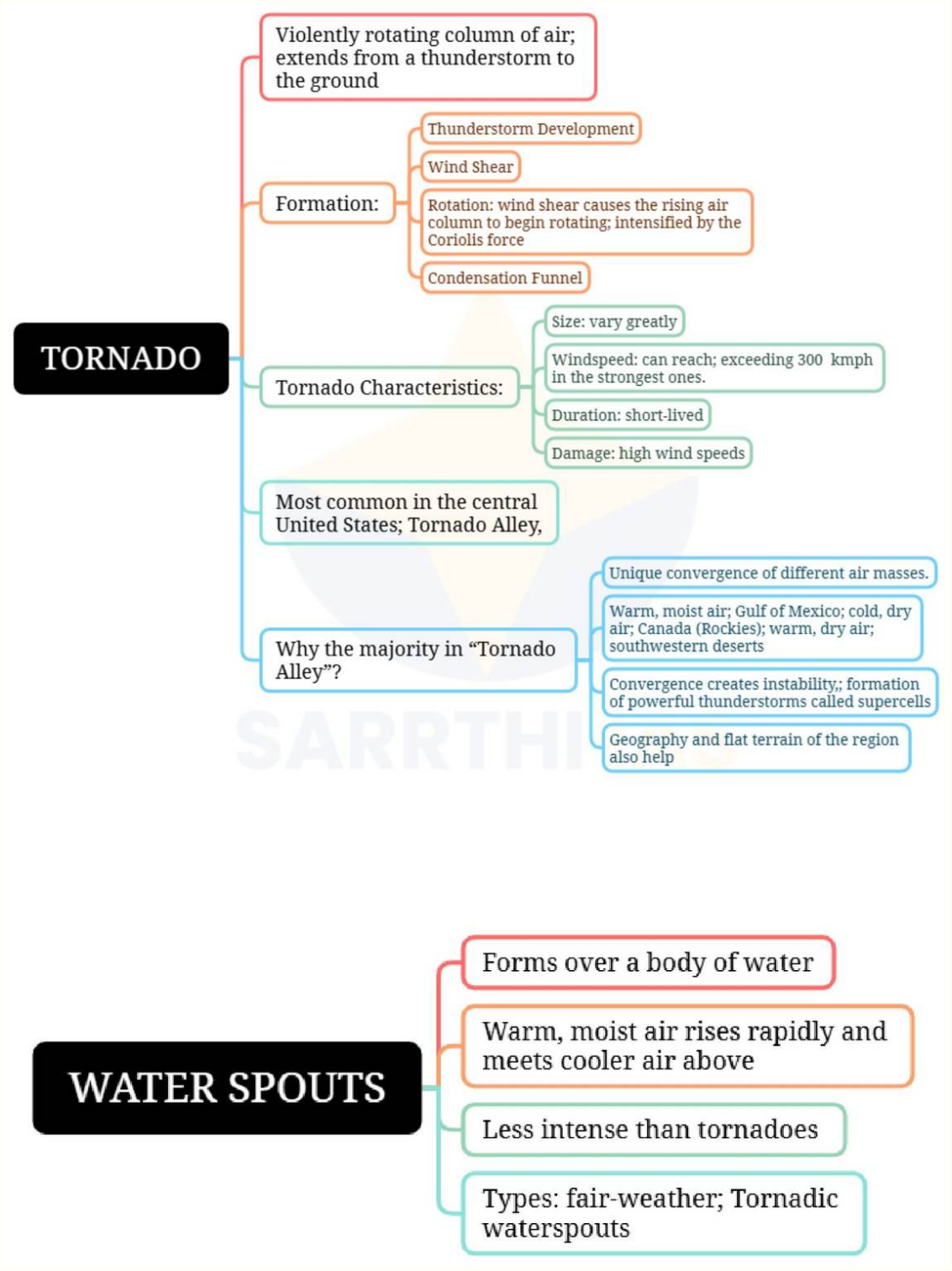

TYPES OF CLOUDS
High-Level Clouds (Cirrus Clouds); 6,000 meters.; composed of ice crystals; thin, wispy, and feathery.
Mid-Level Clouds (Alto Clouds); 2,000 to 6,000 meters; composed of water droplets and sometimes ice crystals
Low-Level Clouds; below 2,000 meters; composed of water droplets
Some clouds form vertically through the troposphere
Cumulonimbus Clouds; vertically through the troposphere; thunderstorms, heavy rain, lightning, and sometimes hail and tornadoes.

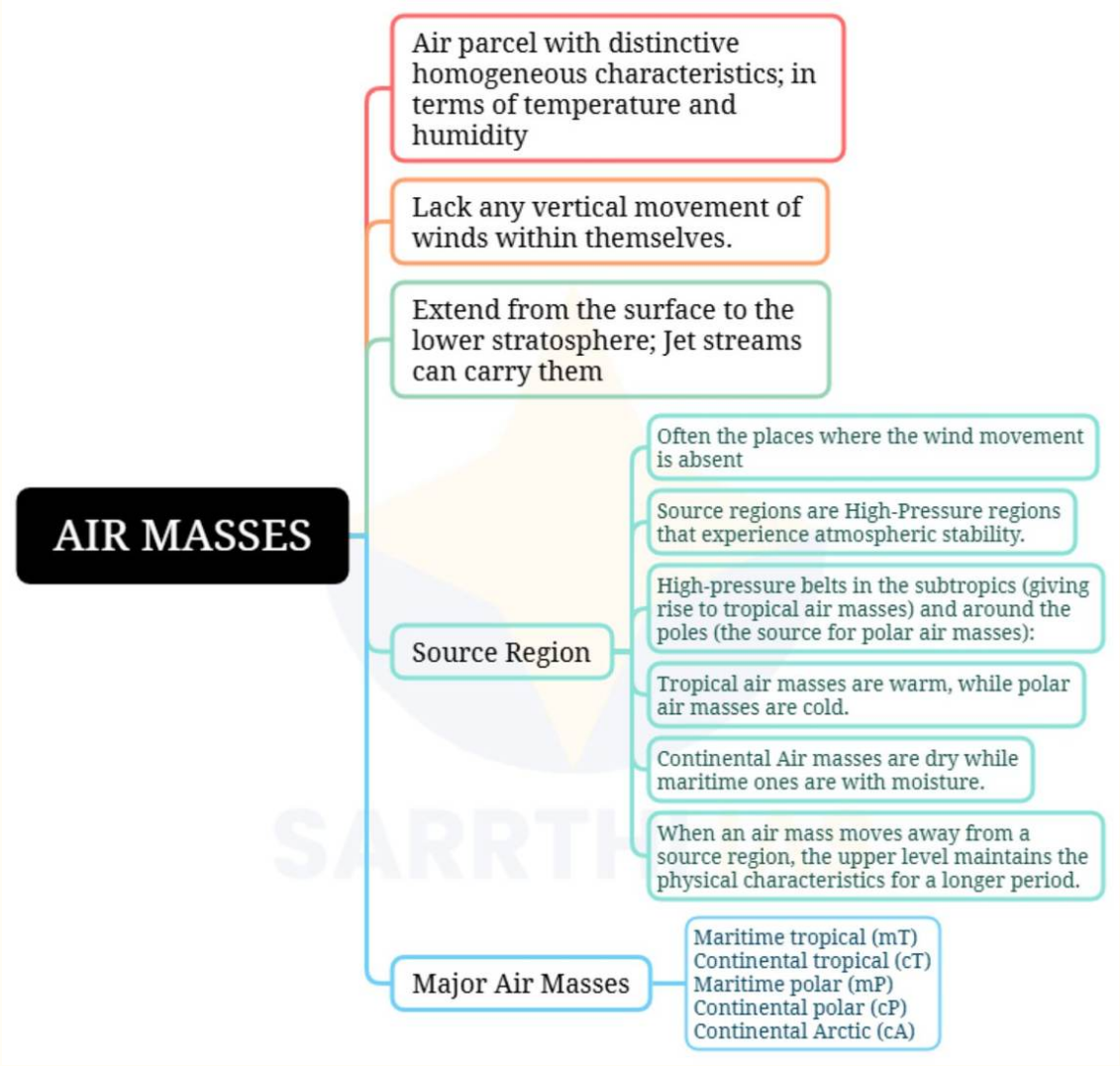

Polar Stratospheric Clouds
Form in the stratosphere at altitudes of 15-25 kilometers in polar regions; winter months
Nacreous or mother-of-pearl clouds; appear bright and colorful during twilight.
Play a critical role in ozone depletion

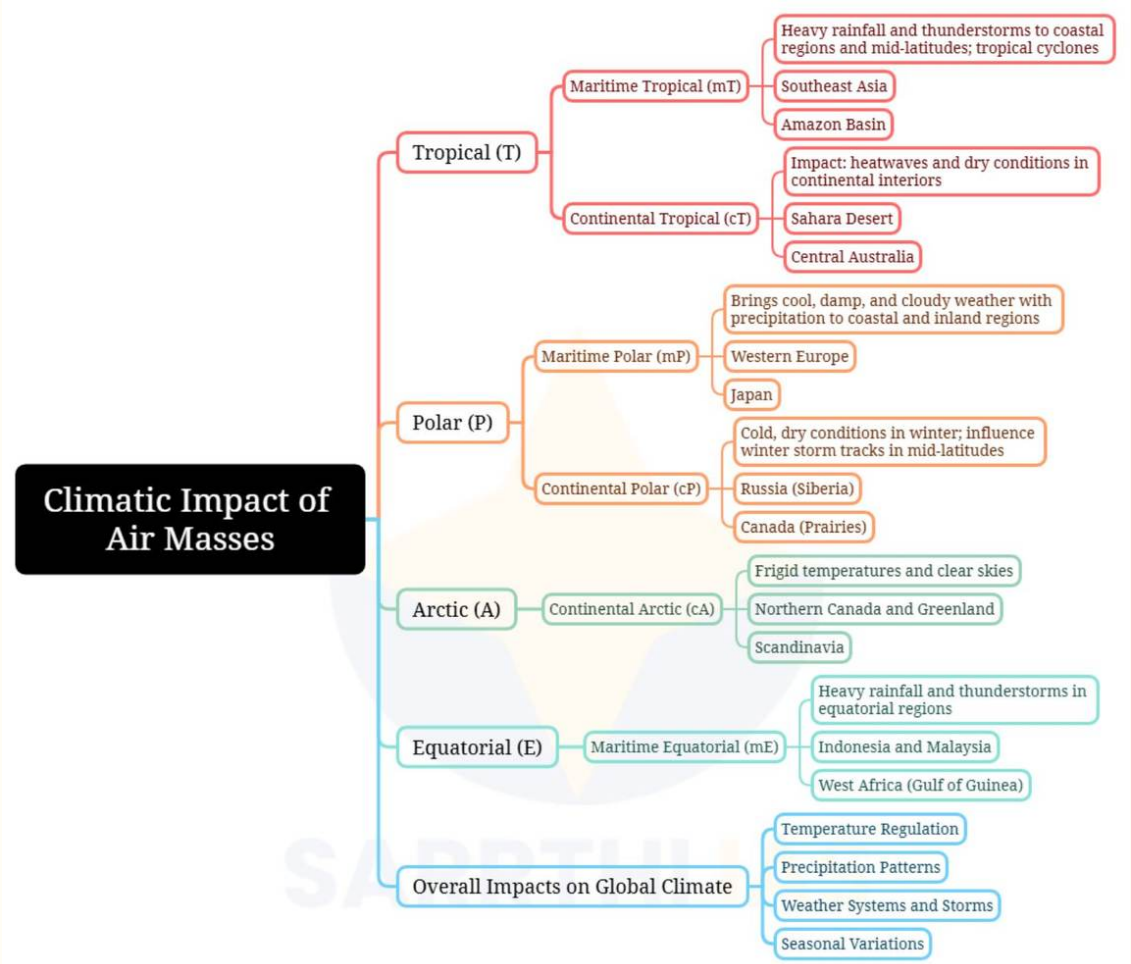

Cloud Seeding
Weather modification
Process
Silver Iodide (AgI), Potassium Iodide, Sodium Chloride (common salt), or Dry Ice (solid CO_2)
Hygroscopic Nuclei
Methods of Delivery
Aircrafts
Ground-based generators
Applications
Enhancing rainfall
Increasing snowpack
Dispersing fog
Hail suppression
Pollution Control
Limitations & Concerns
There is moisture in the air
Silver iodide accumulation

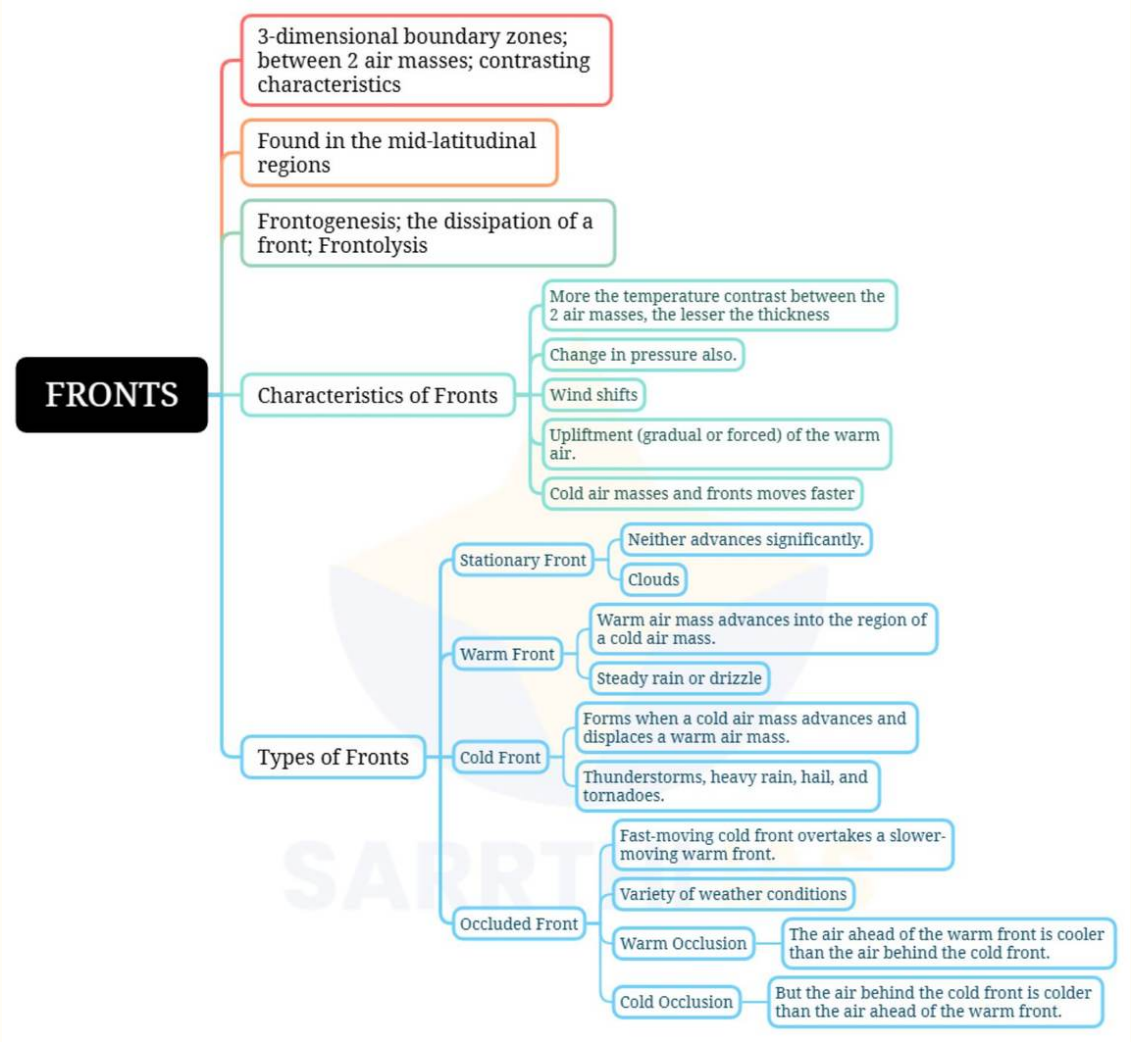

TYPES OF PRECIPITATION
Occurs when water droplets in clouds coalesce and grow large enough to overcome air resistance and fall to the ground
Convective (Convectional) Precipitation:
Towering cumulonimbus clouds.
Heavy showers, thunderstorms, and short-lived periods of rainfall.
Near the equator
Rainfall; hail
Orographic Precipitation
Moist air is forced to rise over elevated terrain such as mountains or hills.
As the air rises; cools and condenses
Leeward side; rain shadow effect
Most common in regions with mountainous terrain.
Rainfall; snow; hail
Frontal (Cyclonic) Precipitation:
Frontal boundary between two contrasting air masses
Warm air is lifted over denser, cooler air, leading to the formation of clouds and precipitation.
Often associated with mid-latitude cyclones and frontal systems.
Rainfall; snow or sleet
Cyclonic Precipitation:
Convectional/frontal rainfall on a large scale.
Intense and widespread,
Precipitation:
Tropical cyclones; heavy rainfall; thunderstorms and hail
extratropical cyclones; snowfall.

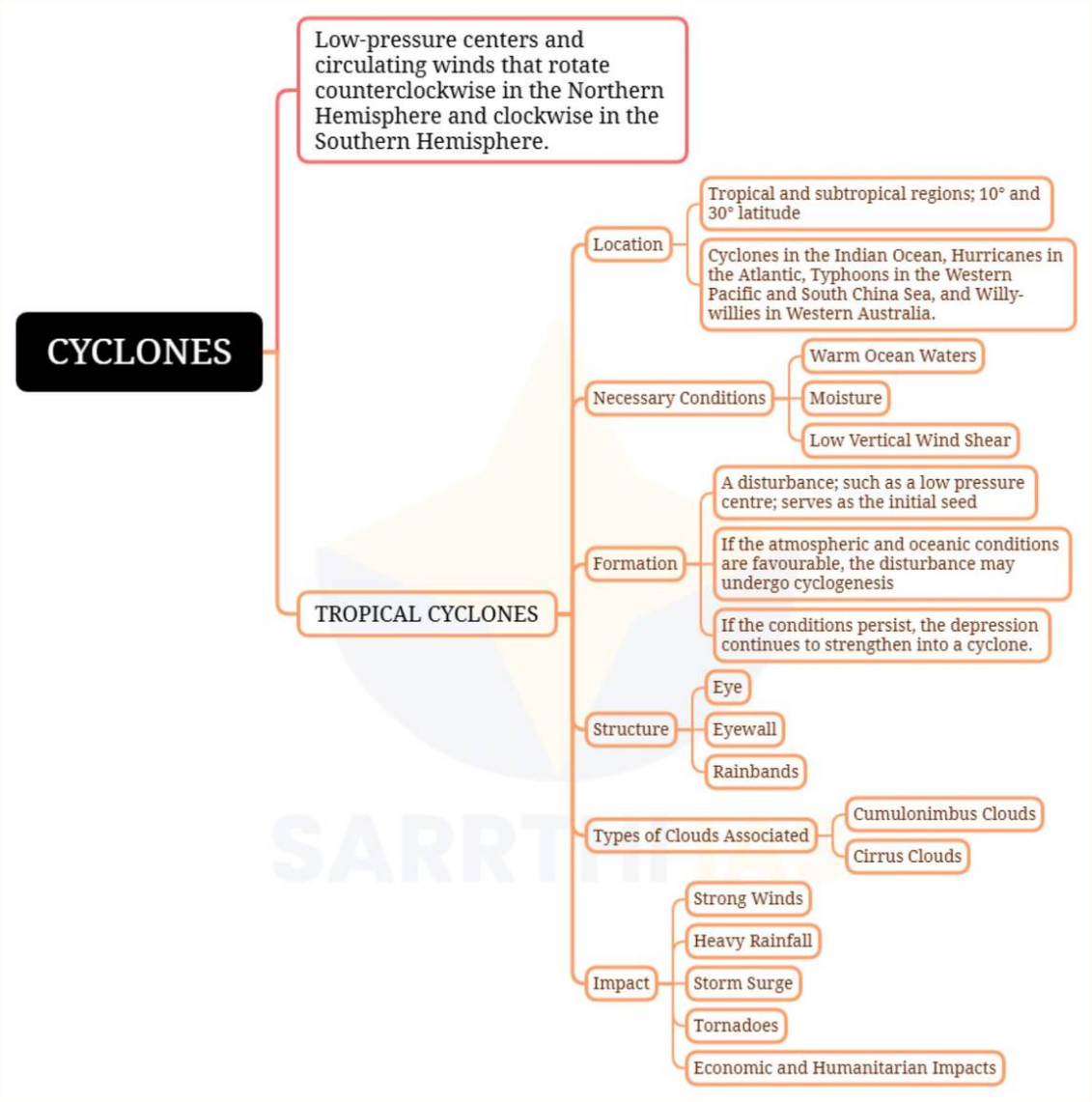

THUNDERSTORMS
Thunder and lightning; heavy rain, strong winds, and sometimes hail.
Formation:
Form in warm, moist air masses; upward motion, i.e. convection (updraft); development of towering cumulonimbus clouds.
Stages:
Cumulus stage; warm, moist air rises; forms cumulus clouds
Mature stage; cumulonimbus cloud; lightning, thunder, heavy rain, and sometimes hail
Dissipating stage,; storm weakens as downdrafts dominate
Clouds in Thunderstorms:
Main cloud; cumulonimbus; Other cloud; cumulus clouds and stratocumulus clouds, may also be present
Hazards:
Lightning strikes; heavy rainfall; strong winds; hail

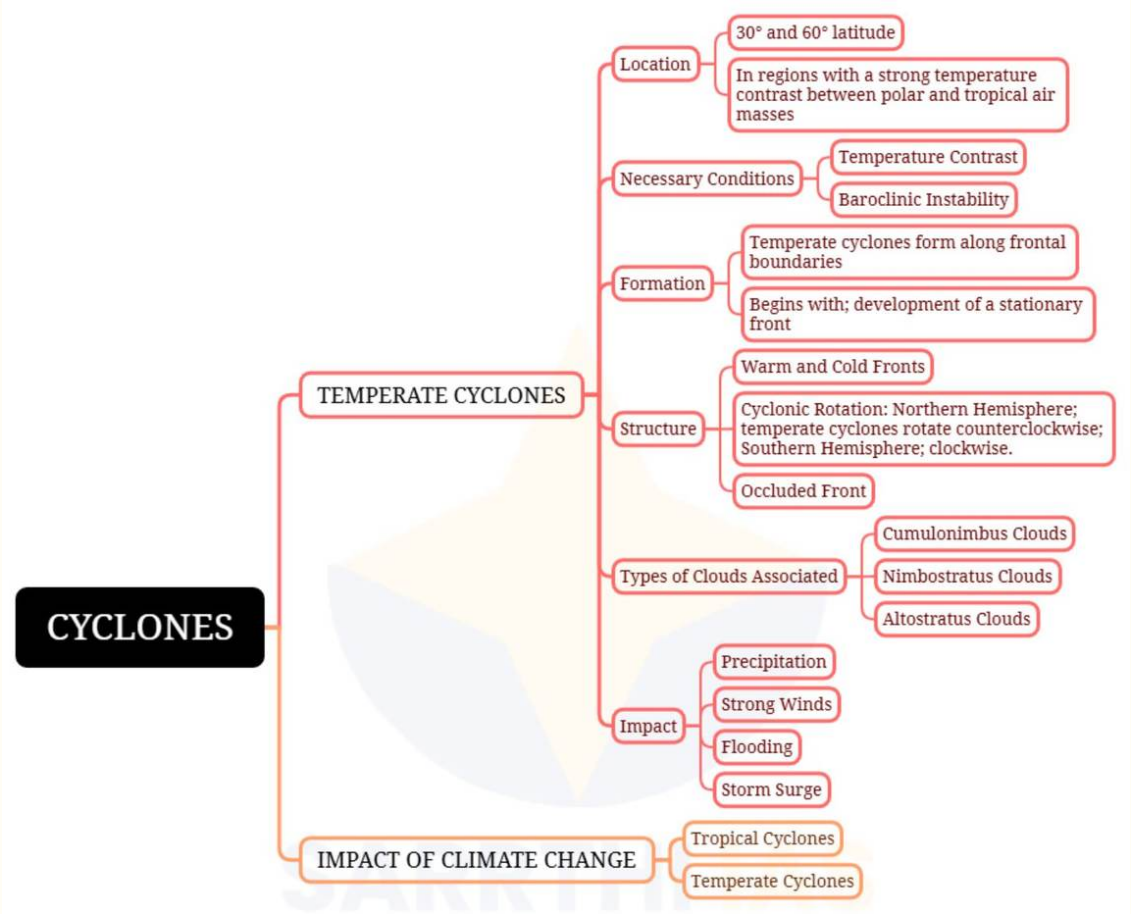

TORNADO
Violently rotating column of air; extends from a thunderstorm to the ground
Formation:
Thunderstorm Development
Wind Shear
Rotation: wind shear causes the rising air column to begin rotating; intensified by the Coriolis force
Condensation Funnel
Tornado Characteristics:
Size: vary greatly
Windspeed: can reach; exceeding 300 kmph in the strongest ones.
Duration: short-lived
Damage: high wind speeds
Most common in the central United States; Tornado Alley,
Why the majority in "Tornado Alley"?
Unique convergence of different air masses.
Warm, moist air; Gulf of Mexico; cold, dry air; Canada (Rockies); warm, dry air; southwestern deserts
Convergence creates instability,; formation of powerful thunderstorms called supercells
Geography and flat terrain of the region also help
WATER SPOUTS
Forms over a body of water
Warm, moist air rises rapidly and meets cooler air above
Less intense than tornadoes
Types: fair-weather; Tornadic waterspouts

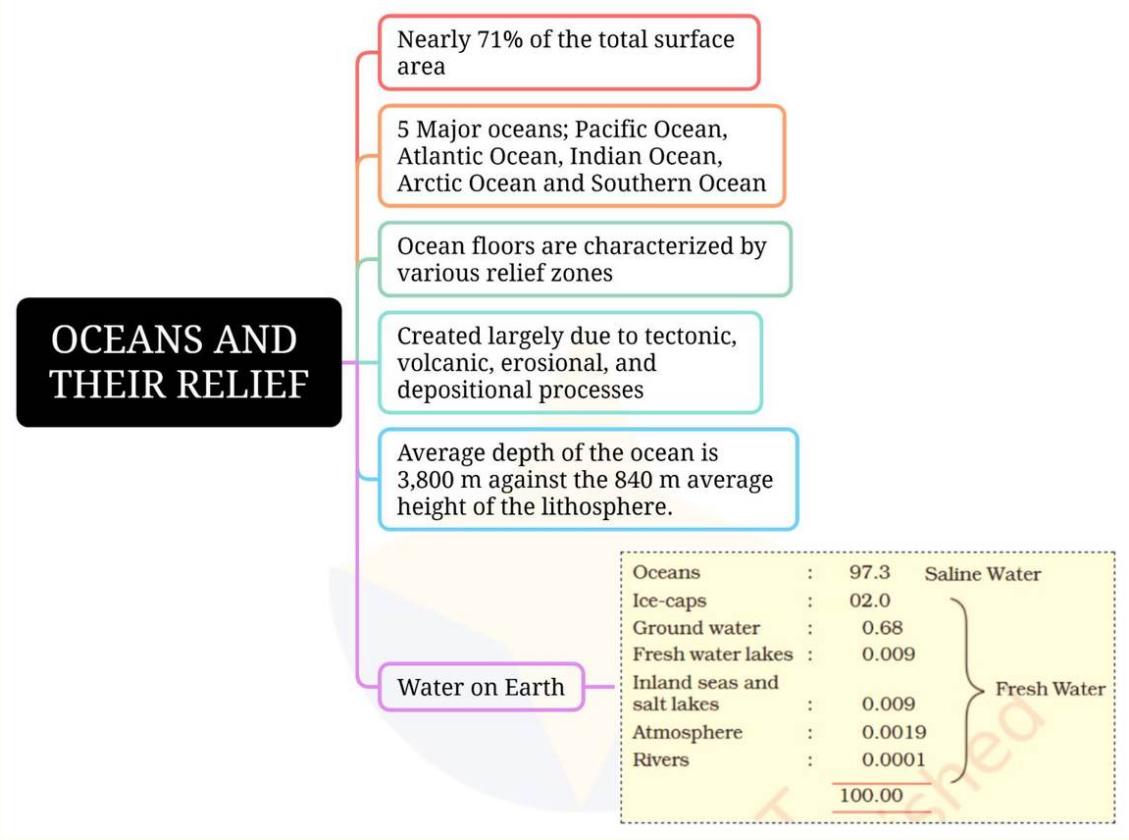

AIR MASSES
Air parcel with distinctive homogeneous characteristics; in terms of temperature and humidity
Lack any vertical movement of winds within themselves.
Extend from the surface to the lower stratosphere; Jet streams can carry them
Source Region
Often the places where the wind movement is absent
Source regions are High-Pressure regions that experience atmospheric stability.
High-pressure belts in the subtropics (giving rise to tropical air masses) and around the poles (the source for polar air masses):
Tropical air masses are warm, while polar air masses are cold.
Continental Air masses are dry while maritime ones are with moisture.
When an air mass moves away from a source region, the upper level maintains the physical characteristics for a longer period.
Major Air Masses
Maritime tropical (mT)
Continental tropical (cT)
Maritime polar (mP)
Continental polar (cP)
Continental Arctic (cA)

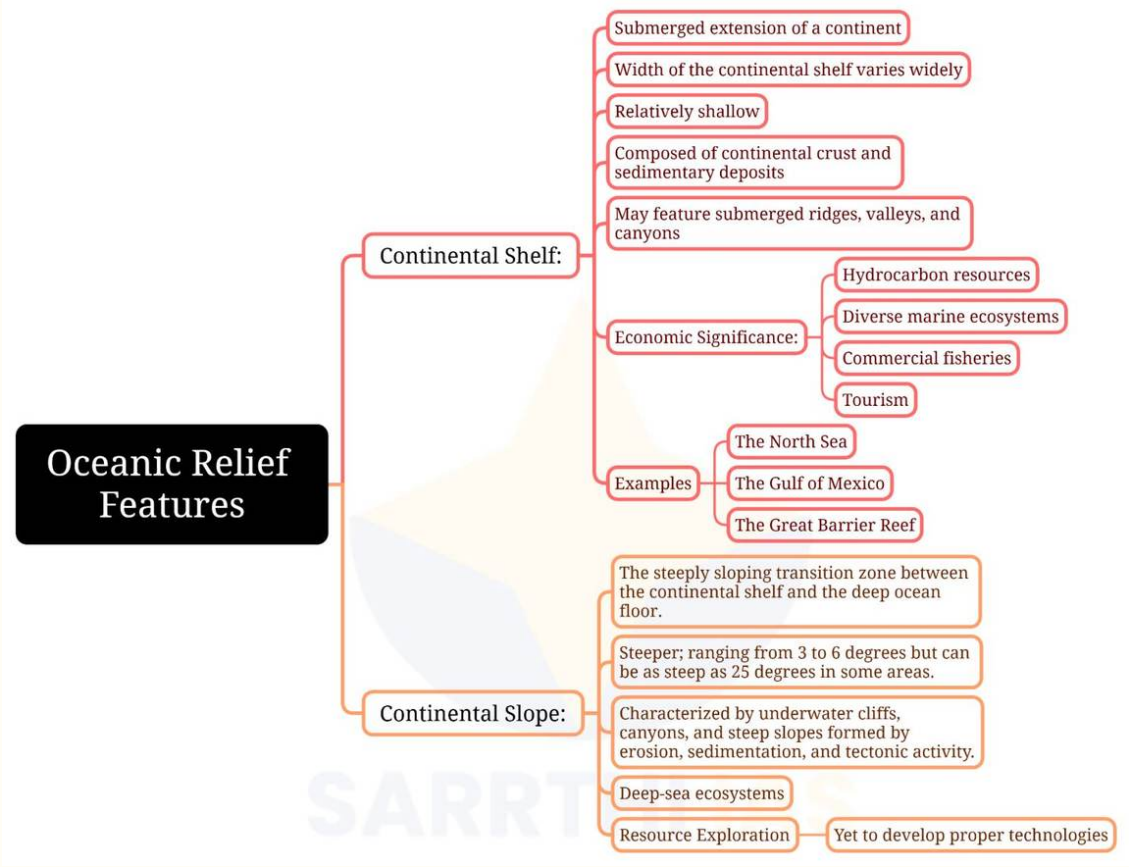

Climatic Impact of Air Masses
Tropical (T)
Maritime Tropical (mT)
Heavy rainfall and thunderstorms to coastal regions and mid-latitudes; tropical cyclones
Southeast Asia
Amazon Basin
Continental Tropical (cT)
Impact: heatwaves and dry conditions in continental interiors
Sahara Desert
Central Australia
Polar (P)
Maritime Polar (mP)
Brings cool, damp, and cloudy weather with precipitation to coastal and inland regions
Western Europe
Japan
Continental Polar (cP)
Cold, dry conditions in winter; influence winter storm tracks in mid-latitudes
Russia (Siberia)
Canada (Prairies)
Arctic (A)
Continental Arctic (cA)
Frigid temperatures and clear skies
Northern Canada and Greenland
Scandinavia
Equatorial (E)
Maritime Equatorial (mE)
Heavy rainfall and thunderstorms in equatorial regions
Indonesia and Malaysia
West Africa (Gulf of Guinea)
Overall Impacts on Global Climate
Temperature Regulation
Precipitation Patterns
Weather Systems and Storms
Seasonal Variations

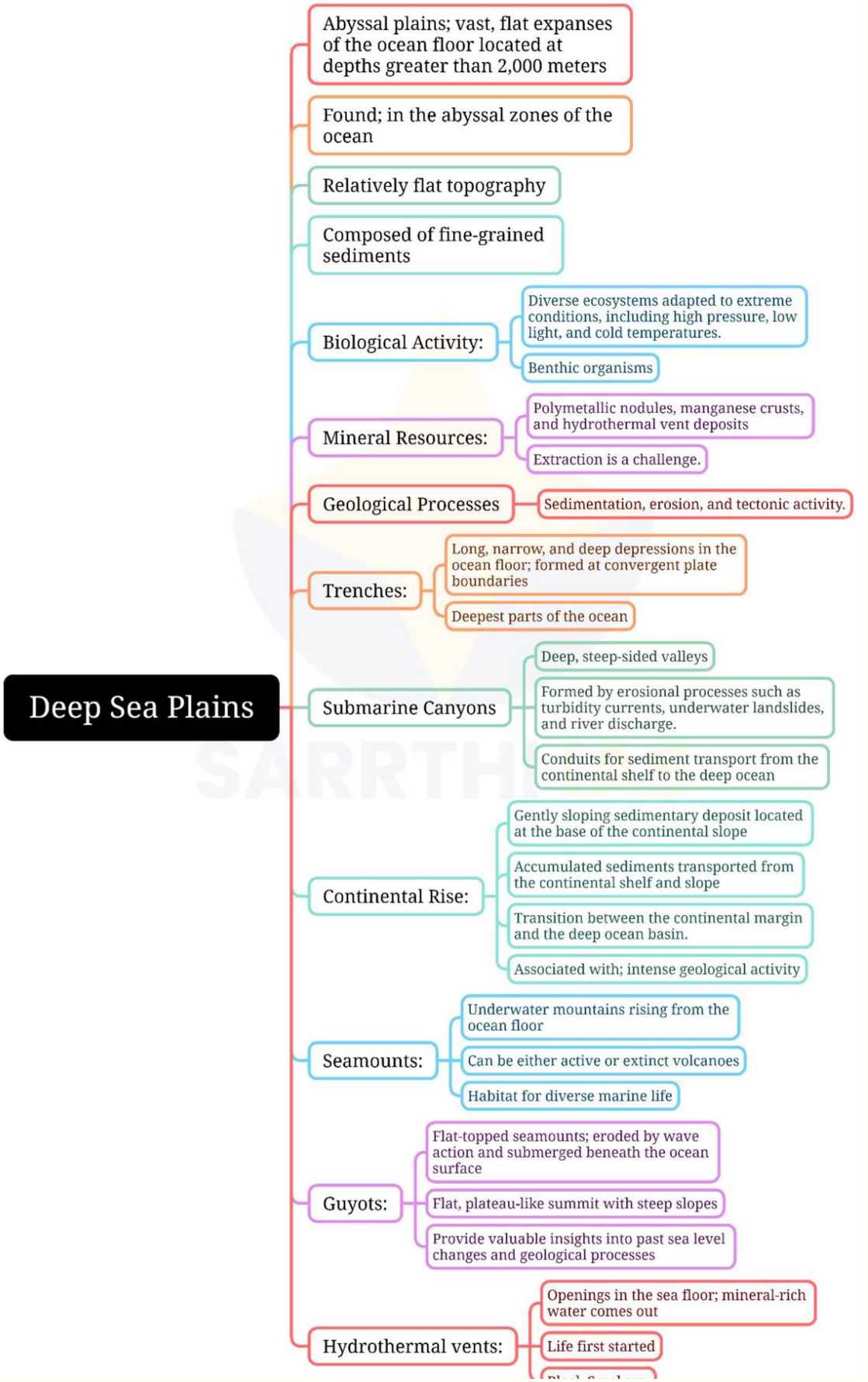

FRONTS
3-dimensional boundary zones; between 2 air masses; contrasting characteristics
Found in the mid-latitudinal regions
Frontogenesis; the dissipation of a front; Frontolysis
Characteristics of Fronts
More the temperature contrast between the 2 air masses, the lesser the thickness
Change in pressure also.
Wind shifts
Upliftment (gradual or forced) of the warm air.
Cold air masses and fronts moves faster
Types of Fronts
Stationary Front
Neither advances significantly.
Clouds
Warm Front
Warm air mass advances into the region of a cold air mass.
Steady rain or drizzle
Cold Front
Forms when a cold air mass advances and displaces a warm air mass.
Thunderstorms, heavy rain, hail, and tornadoes.
Occluded Front
Fast-moving cold front overtakes a slower-moving warm front.
Variety of weather conditions
Warm Occlusion
The air ahead of the warm front is cooler than the air behind the cold front.
Cold Occlusion
But the air behind the cold front is colder than the air ahead of the warm front.

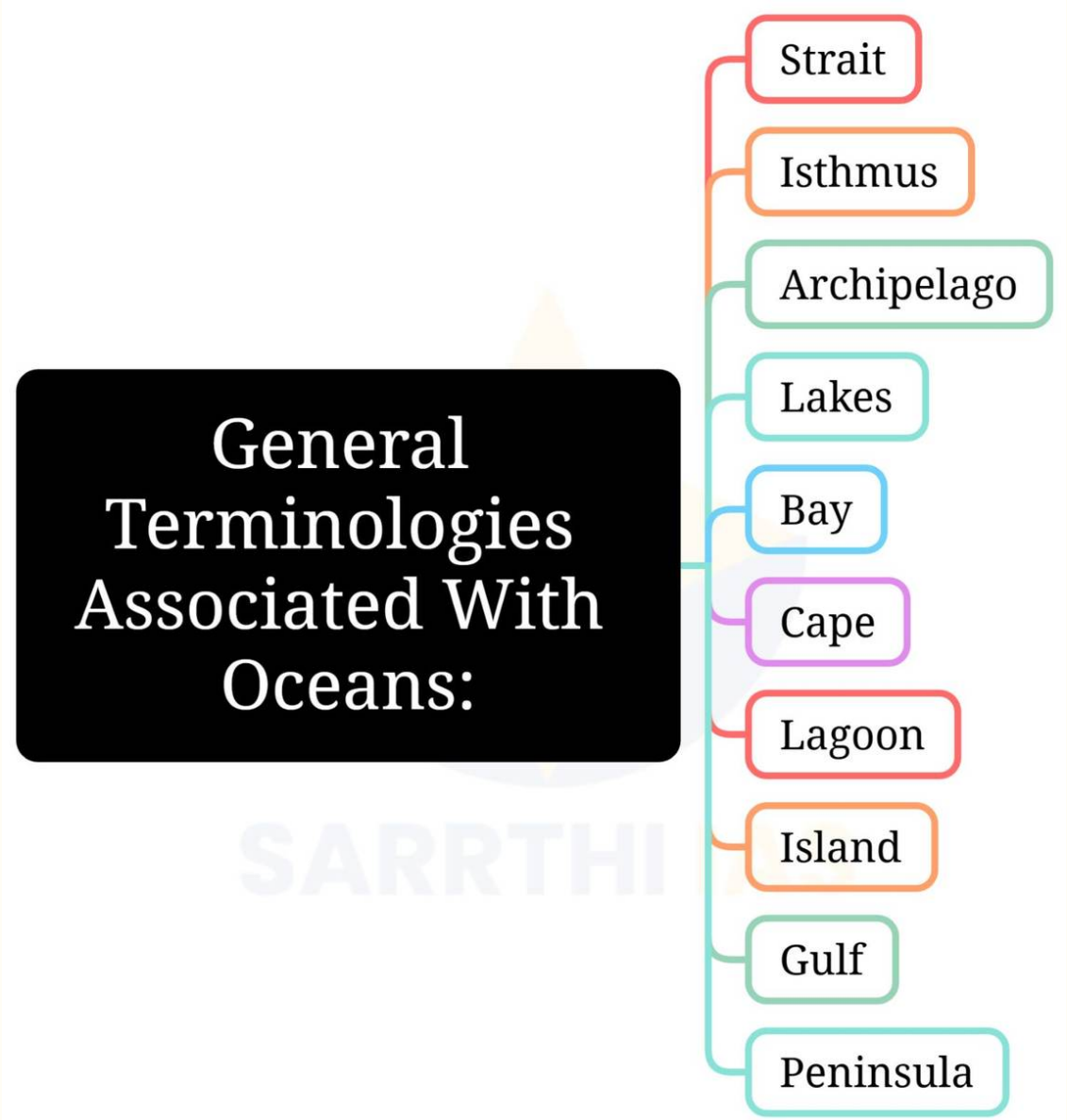

CYCLONES
Low-pressure centers and circulating winds that rotate counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere.
TROPICAL CYCLONES
Location
Tropical and subtropical regions; 10° and 30° latitude
Cyclones in the Indian Ocean, Hurricanes in the Atlantic, Typhoons in the Western Pacific and South China Sea, and Willy-willies in Western Australia.
Necessary Conditions
Warm Ocean Waters
Moisture
Low Vertical Wind Shear
Formation
A disturbance; such as a low pressure centre; serves as the initial seed
If the atmospheric and oceanic conditions are favourable, the disturbance may undergo cyclogenesis
If the conditions persist, the depression continues to strengthen into a cyclone.
Structure
Eye
Eyewall
Rainbands
Types of Clouds Associated
Cumulonimbus Clouds
Cirrus Clouds
Impact
Strong Winds
Heavy Rainfall
Storm Surge
Tornadoes
Economic and Humanitarian Impacts

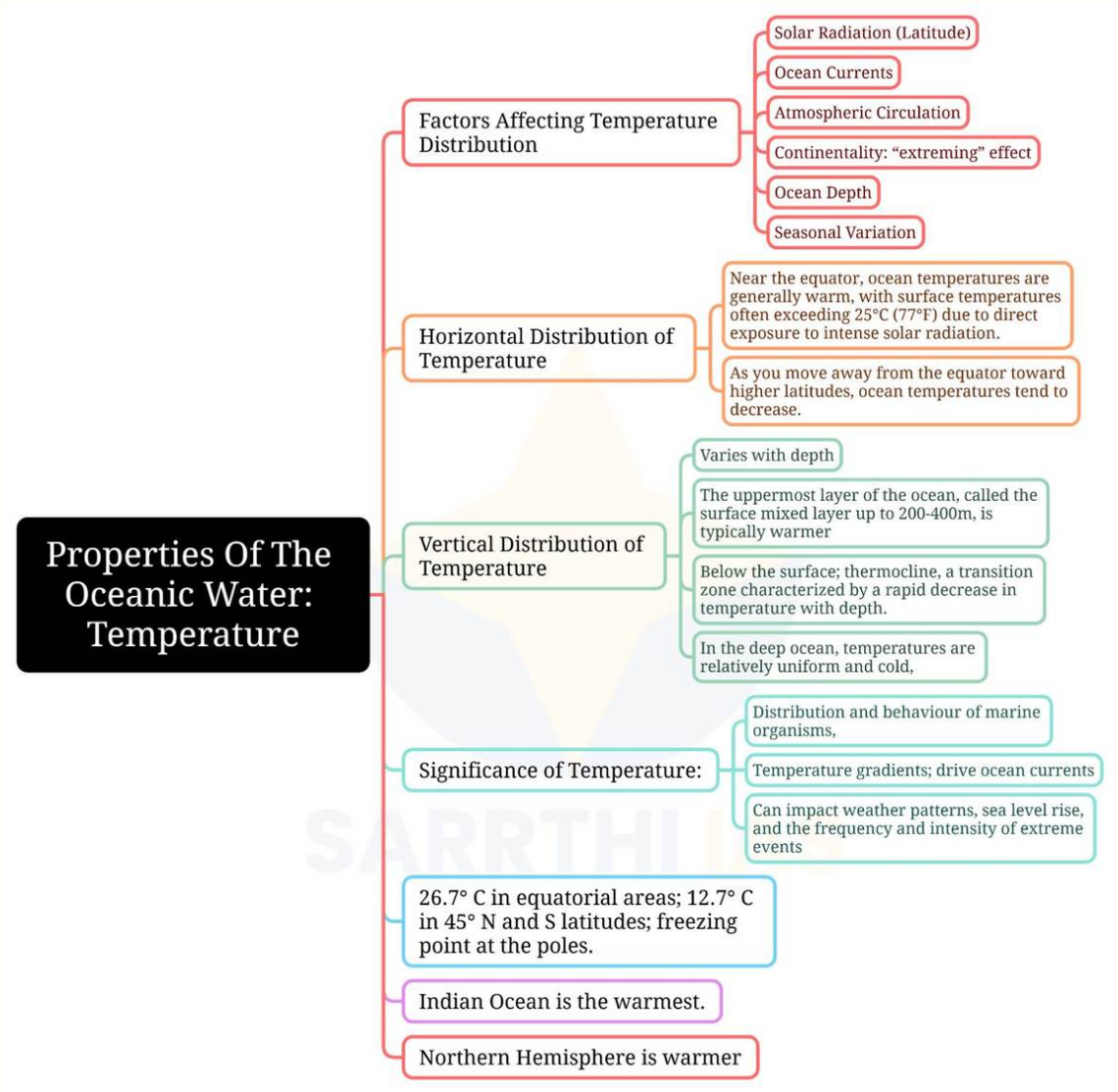

CYCLONES
TEMPERATE CYCLONES
Location
30° and 60° latitude
In regions with a strong temperature contrast between polar and tropical air masses
Necessary Conditions
Temperature Contrast
Baroclinic Instability
Formation
Temperate cyclones form along frontal boundaries
Begins with; development of a stationary front
Structure
Warm and Cold Fronts
Cyclonic Rotation: Northern Hemisphere; temperate cyclones rotate counterclockwise; Southern Hemisphere; clockwise.
Occluded Front
Types of Clouds Associated
Cumulonimbus Clouds
Nimbostratus Clouds
Altostratus Clouds
Impact
Precipitation
Strong Winds
Flooding
Storm Surge
IMPACT OF CLIMATE CHANGE
Tropical Cyclones
Temperate Cyclones

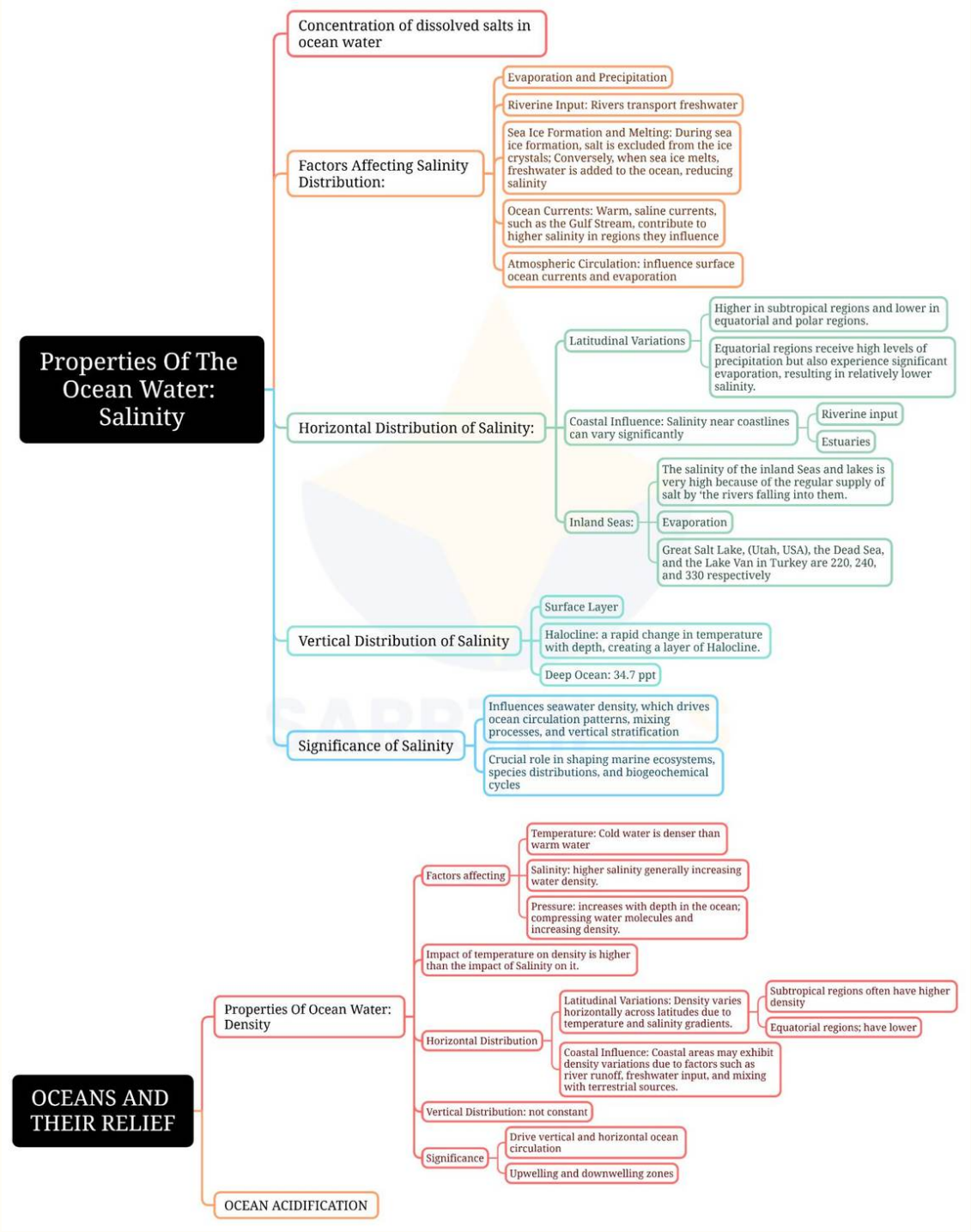

OCEANS AND THEIR RELIEF
Nearly 71% of the total surface area
5 Major oceans; Pacific Ocean, Atlantic Ocean, Indian Ocean, Arctic Ocean and Southern Ocean
Ocean floors are characterized by various relief zones
Created largely due to tectonic, volcanic, erosional, and depositional processes
Average depth of the ocean is 3,800 m against the 840 m average height of the lithosphere.
Water on Earth
Oceans : 97.3 Saline Water
Ice-caps : 02.0
Ground water : 0.68
Fresh water lakes : 0.009
Inland seas and salt lakes : 0.009
Atmosphere : 0.0019
Rivers : 0.0001
Fresh Water
100.00

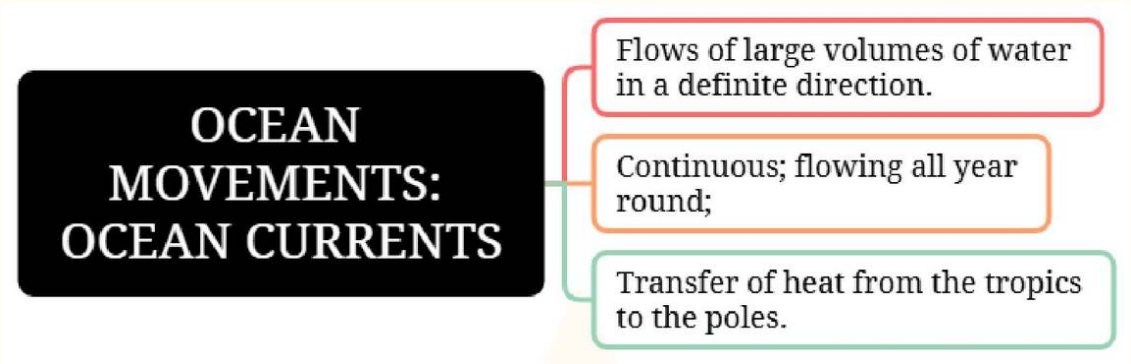

Oceanic Relief Features
Continental Shelf:
Submerged extension of a continent
Width of the continental shelf varies widely
Relatively shallow
Composed of continental crust and sedimentary deposits
May feature submerged ridges, valleys, and canyons
Economic Significance:
Hydrocarbon resources
Diverse marine ecosystems
Commercial fisheries
Tourism
Examples
The North Sea
The Gulf of Mexico
The Great Barrier Reef
Continental Slope:
The steeply sloping transition zone between the continental shelf and the deep ocean floor.
Steeper; ranging from 3 to 6 degrees but can be as steep as 25 degrees in some areas.
Characterized by underwater cliffs, canyons, and steep slopes formed by erosion, sedimentation, and tectonic activity.
Deep-sea ecosystems
Resource Exploration
Yet to develop proper technologies

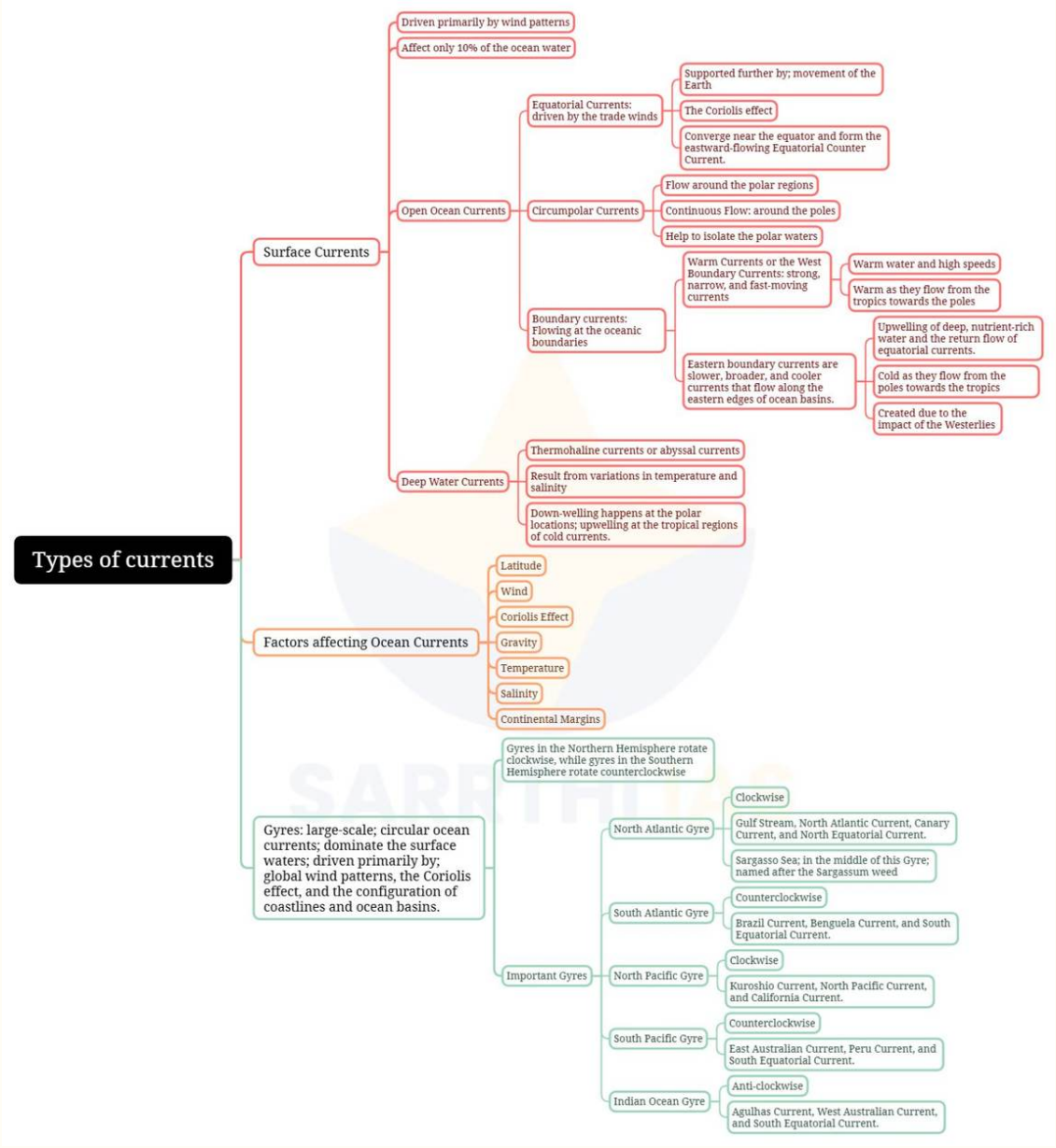

Deep Sea Plains
Abyssal plains; vast, flat expanses of the ocean floor located at depths greater than 2,000 meters
Found; in the abyssal zones of the ocean
Relatively flat topography
Composed of fine-grained sediments
Biological Activity:
Diverse ecosystems adapted to extreme conditions, including high pressure, low light, and cold temperatures.
Benthic organisms
Mineral Resources:
Polymetallic nodules, manganese crusts, and hydrothermal vent deposits
Extraction is a challenge.
Geological Processes
Sedimentation, erosion, and tectonic activity.
Trenches:
Long, narrow, and deep depressions in the ocean floor; formed at convergent plate boundaries
Deepest parts of the ocean
Submarine Canyons
Deep, steep-sided valleys
Formed by erosional processes such as turbidity currents, underwater landslides, and river discharge.
Conduits for sediment transport from the continental shelf to the deep ocean
Continental Rise:
Gently sloping sedimentary deposit located at the base of the continental slope
Accumulated sediments transported from the continental shelf and slope
Transition between the continental margin and the deep ocean basin.
Associated with; intense geological activity
Seamounts:
Underwater mountains rising from the ocean floor
Can be either active or extinct volcanoes
Habitat for diverse marine life
Guyots:
Flat-topped seamounts; eroded by wave action and submerged beneath the ocean surface
Flat, plateau-like summit with steep slopes
Provide valuable insights into past sea level changes and geological processes
Hydrothermal vents:
Openings in the sea floor; mineral-rich water comes out
Life first started

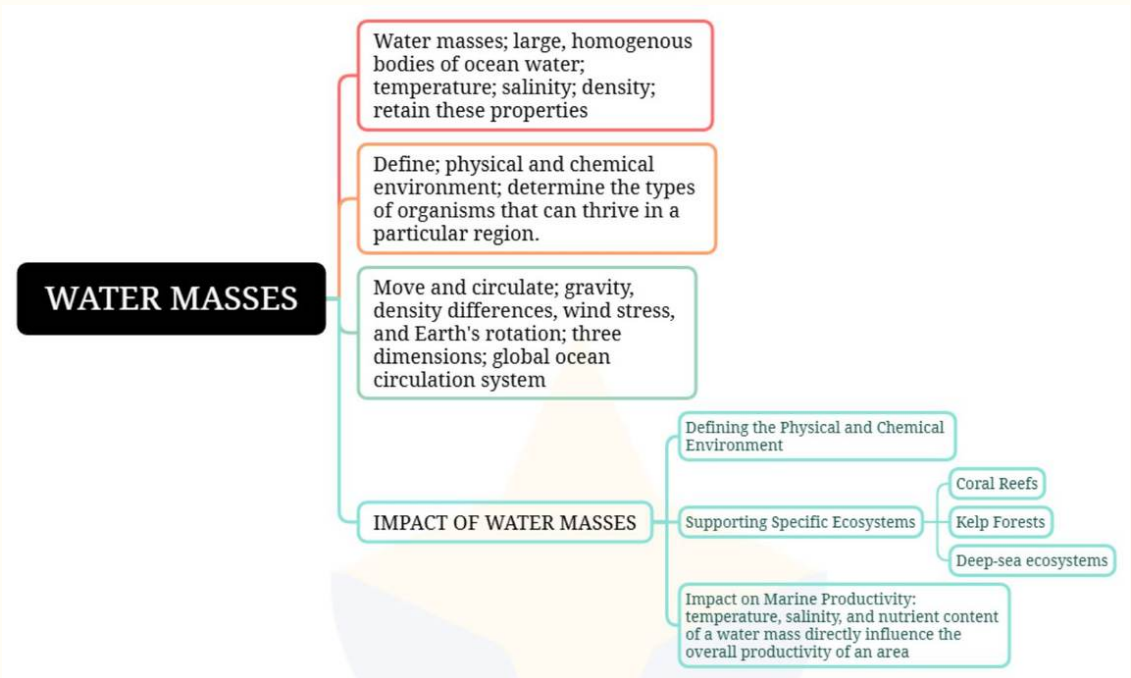

General Terminologies Associated With Oceans:
Strait
Isthmus
Archipelago
Lakes
Bay
Cape
Lagoon
Island
Gulf
Peninsula

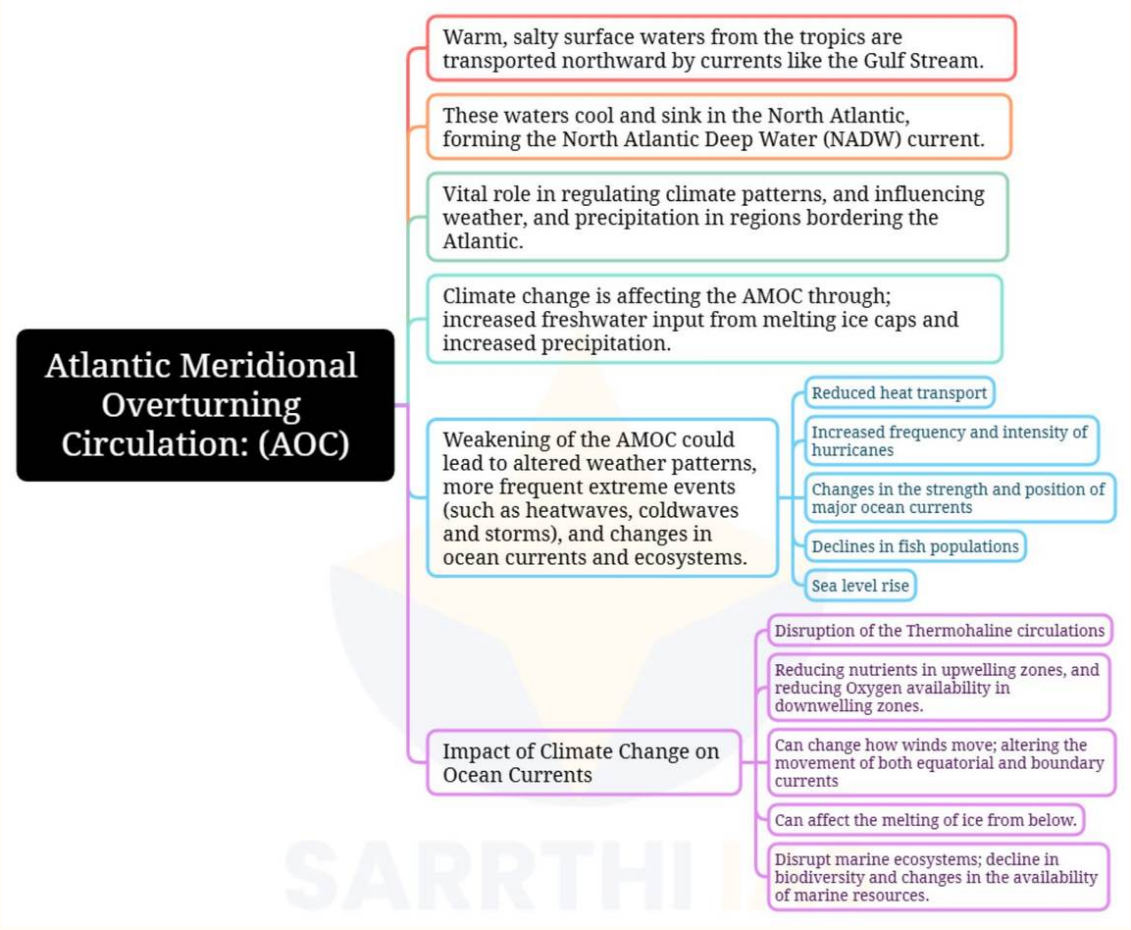

Properties Of The Oceanic Water: Temperature
Factors Affecting Temperature Distribution
Solar Radiation (Latitude)
Ocean Currents
Atmospheric Circulation
Continentality: "extreming" effect
Ocean Depth
Seasonal Variation
Horizontal Distribution of Temperature
Near the equator, ocean temperatures are generally warm, with surface temperatures often exceeding 25°C (77°F) due to direct exposure to intense solar radiation.
As you move away from the equator toward higher latitudes, ocean temperatures tend to decrease.
Vertical Distribution of Temperature
Varies with depth
The uppermost layer of the ocean, called the surface mixed layer up to 200-400m, is typically warmer
Below the surface; thermocline, a transition zone characterized by a rapid decrease in temperature with depth.
In the deep ocean, temperatures are relatively uniform and cold,
Significance of Temperature:
Distribution and behaviour of marine organisms,
Temperature gradients; drive ocean currents
Can impact weather patterns, sea level rise, and the frequency and intensity of extreme events
26.7° C in equatorial areas; 12.7° C in 45° N and S latitudes; freezing point at the poles.
Indian Ocean is the warmest.
Northern Hemisphere is warmer

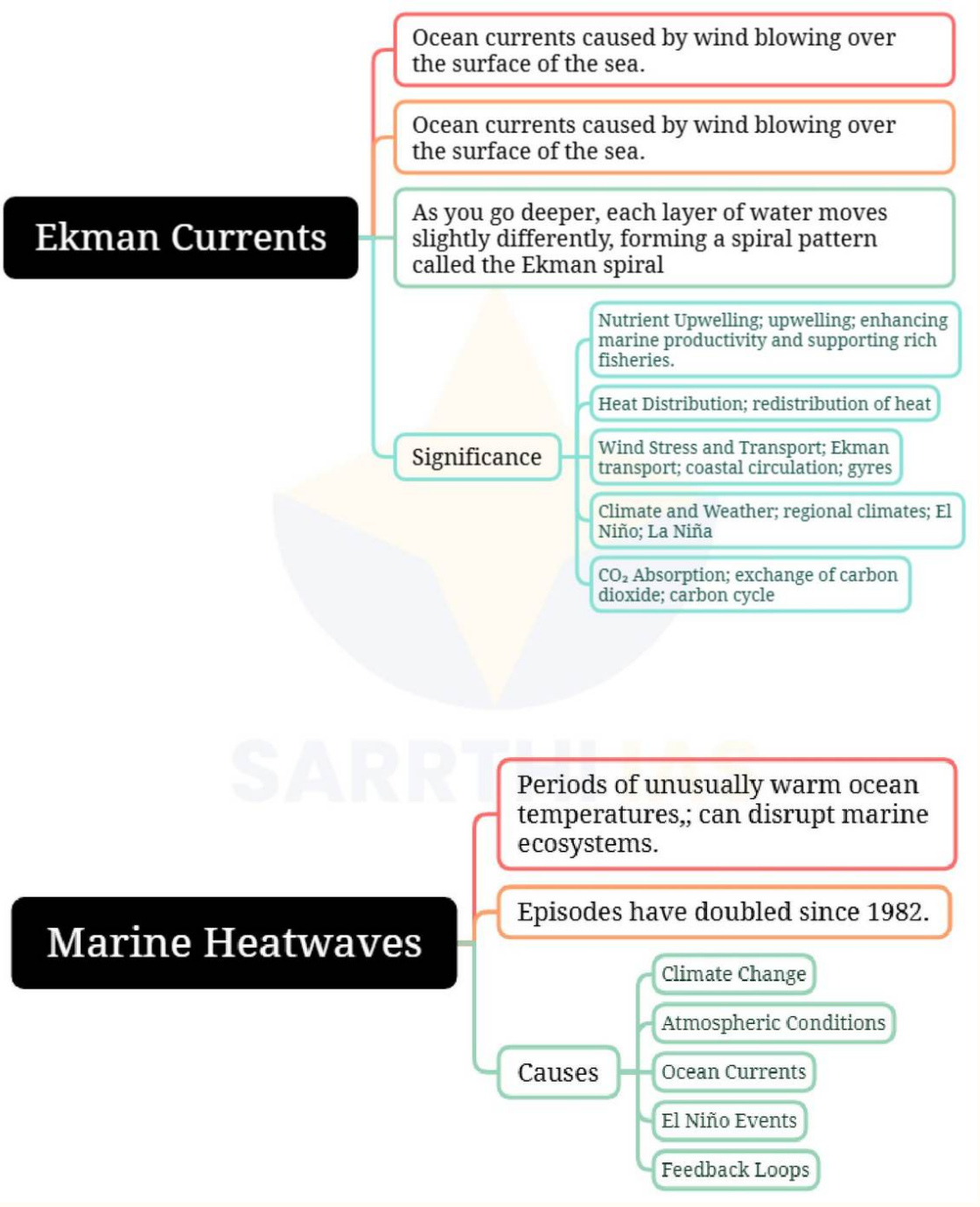

Properties Of The Ocean Water: Salinity
Concentration of dissolved salts in ocean water
Factors Affecting Salinity Distribution:
Evaporation and Precipitation
Riverine Input: Rivers transport freshwater
Sea Ice Formation and Melting: During sea ice formation, salt is excluded from the ice crystals; Conversely, when sea ice melts, freshwater is added to the ocean, reducing salinity
Ocean Currents: Warm, saline currents, such as the Gulf Stream, contribute to higher salinity in regions they influence
Atmospheric Circulation: influence surface ocean currents and evaporation
Horizontal Distribution of Salinity:
Latitudinal Variations
Higher in subtropical regions and lower in equatorial and polar regions.
Equatorial regions receive high levels of precipitation but also experience significant evaporation, resulting in relatively lower salinity.
Coastal Influence: Salinity near coastlines can vary significantly
Riverine input
Estuaries
Inland Seas:
The salinity of the inland Seas and lakes is very high because of the regular supply of salt by 'the rivers falling into them.
Evaporation
Great Salt Lake, (Utah, USA), the Dead Sea, and the Lake Van in Turkey are 220, 240, and 330 respectively
Vertical Distribution of Salinity
Surface Layer
Halocline: a rapid change in temperature with depth, creating a layer of Halocline.
Deep Ocean: 34.7 ppt
Significance of Salinity
Influences seawater density, which drives ocean circulation patterns, mixing processes, and vertical stratification
Crucial role in shaping marine ecosystems, species distributions, and biogeochemical cycles
OCEANS AND THEIR RELIEF
Properties Of Ocean Water: Density
Factors affecting
Temperature: Cold water is denser than warm water
Salinity: higher salinity generally increasing water density.
Pressure: increases with depth in the ocean; compressing water molecules and increasing density.
Impact of temperature on density is higher than the impact of Salinity on it.
Horizontal Distribution
Latitudinal Variations: Density varies horizontally across latitudes due to temperature and salinity gradients.
Subtropical regions often have higher density
Equatorial regions; have lower
Coastal Influence: Coastal areas may exhibit density variations due to factors such as river runoff, freshwater input, and mixing with terrestrial sources.
Vertical Distribution: not constant
Significance
Drive vertical and horizontal ocean circulation
Upwelling and downwelling zones
OCEAN ACIDIFICATION

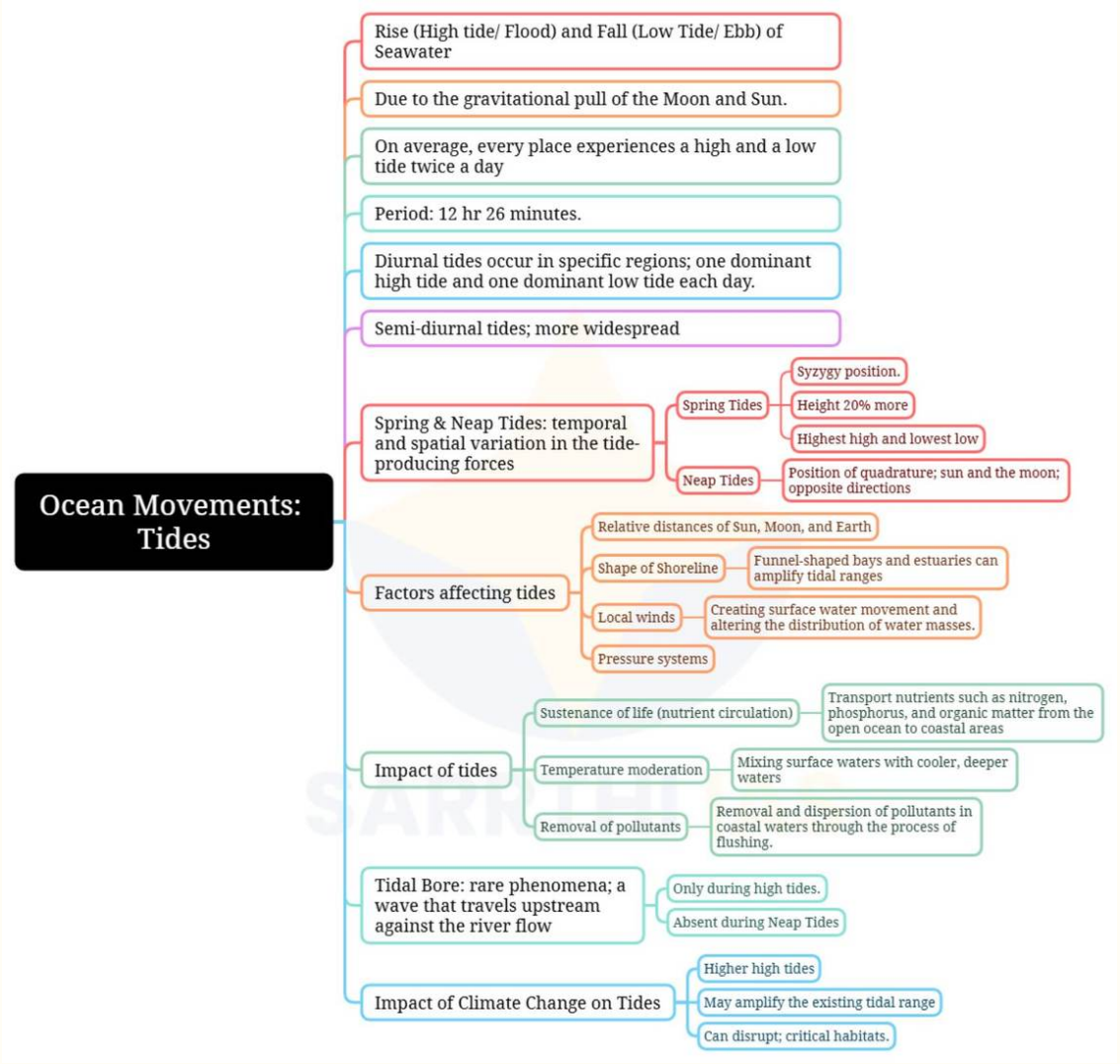

OCEAN MOVEMENTS: OCEAN CURRENTS
Flows of large volumes of water in a definite direction.
Continuous; flowing all year round;
Transfer of heat from the tropics to the poles.

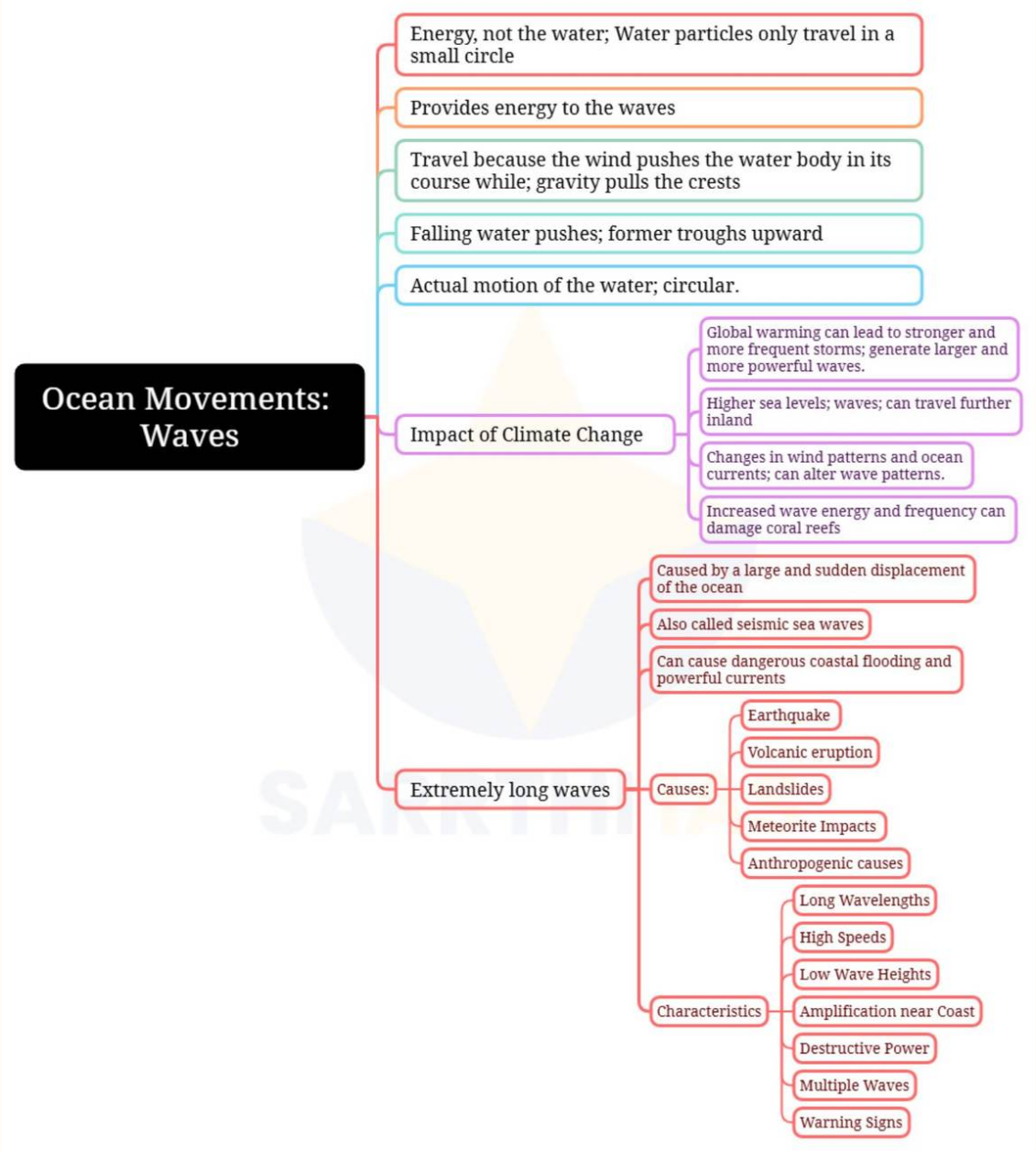

Types of currents
Surface Currents
Driven primarily by wind patterns
Affect only 10% of the ocean water
Open Ocean Currents
Equatorial Currents: driven by the trade winds
Supported further by; movement of the Earth
The Coriolis effect
Converge near the equator and form the eastward-flowing Equatorial Counter Current.
Circumpolar Currents
Flow around the polar regions
Continuous Flow: around the poles
Help to isolate the polar waters
Boundary currents: Flowing at the oceanic boundaries
Warm Currents or the West Boundary Currents: strong, narrow, and fast-moving currents
Warm water and high speeds
Warm as they flow from the tropics towards the poles
Eastern boundary currents are slower, broader, and cooler currents that flow along the eastern edges of ocean basins.
Upwelling of deep, nutrient-rich water and the return flow of equatorial currents.
Cold as they flow from the poles towards the tropics
Created due to the impact of the Westerlies
Deep Water Currents
Thermohaline currents or abyssal currents
Result from variations in temperature and salinity
Down-welling happens at the polar locations; upwelling at the tropical regions of cold currents.
Factors affecting Ocean Currents
Latitude
Wind
Coriolis Effect
Gravity
Temperature
Salinity
Continental Margins
Gyres: large-scale; circular ocean currents; dominate the surface waters; driven primarily by; global wind patterns, the Coriolis effect, and the configuration of coastlines and ocean basins.
Gyres in the Northern Hemisphere rotate clockwise, while gyres in the Southern Hemisphere rotate counterclockwise
Important Gyres
North Atlantic Gyre
Clockwise
Gulf Stream, North Atlantic Current, Canary Current, and North Equatorial Current.
Sargasso Sea; in the middle of this Gyre; named after the Sargassum weed
South Atlantic Gyre
Counterclockwise
Brazil Current, Benguela Current, and South Equatorial Current.
North Pacific Gyre
Clockwise
Kuroshio Current, North Pacific Current, and California Current.
South Pacific Gyre
Counterclockwise
East Australian Current, Peru Current, and South Equatorial Current.
Indian Ocean Gyre
Anti-clockwise
Agulhas Current, West Australian Current, and South Equatorial Current.

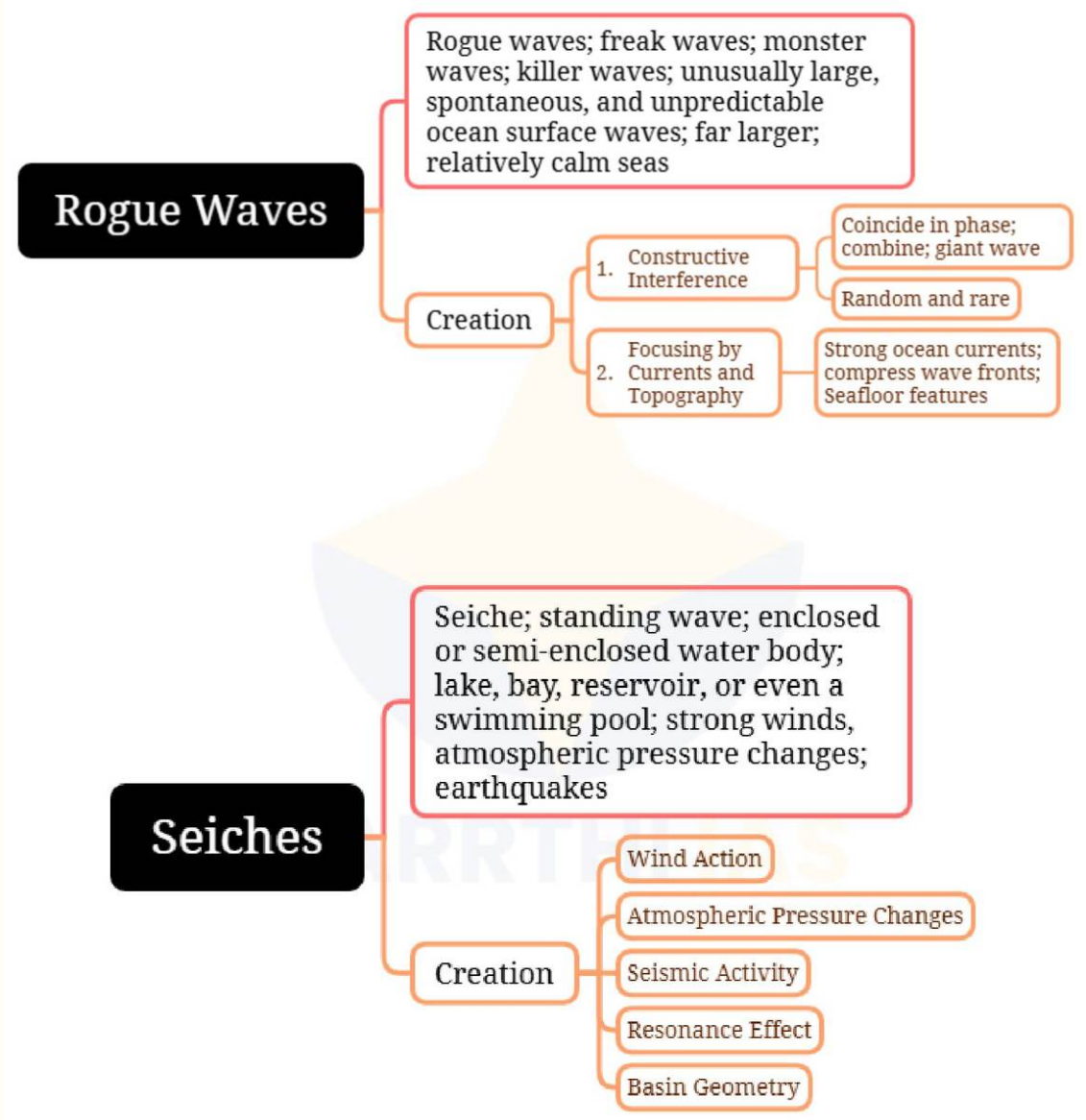

WATER MASSES
Water masses; large, homogenous bodies of ocean water; temperature; salinity; density; retain these properties
Define; physical and chemical environment; determine the types of organisms that can thrive in a particular region.
Move and circulate; gravity, density differences, wind stress, and Earth's rotation; three dimensions; global ocean circulation system
IMPACT OF WATER MASSES
Defining the Physical and Chemical Environment
Supporting Specific Ecosystems
Coral Reefs
Kelp Forests
Deep-sea ecosystems
Impact on Marine Productivity: temperature, salinity, and nutrient content of a water mass directly influence the overall productivity of an area

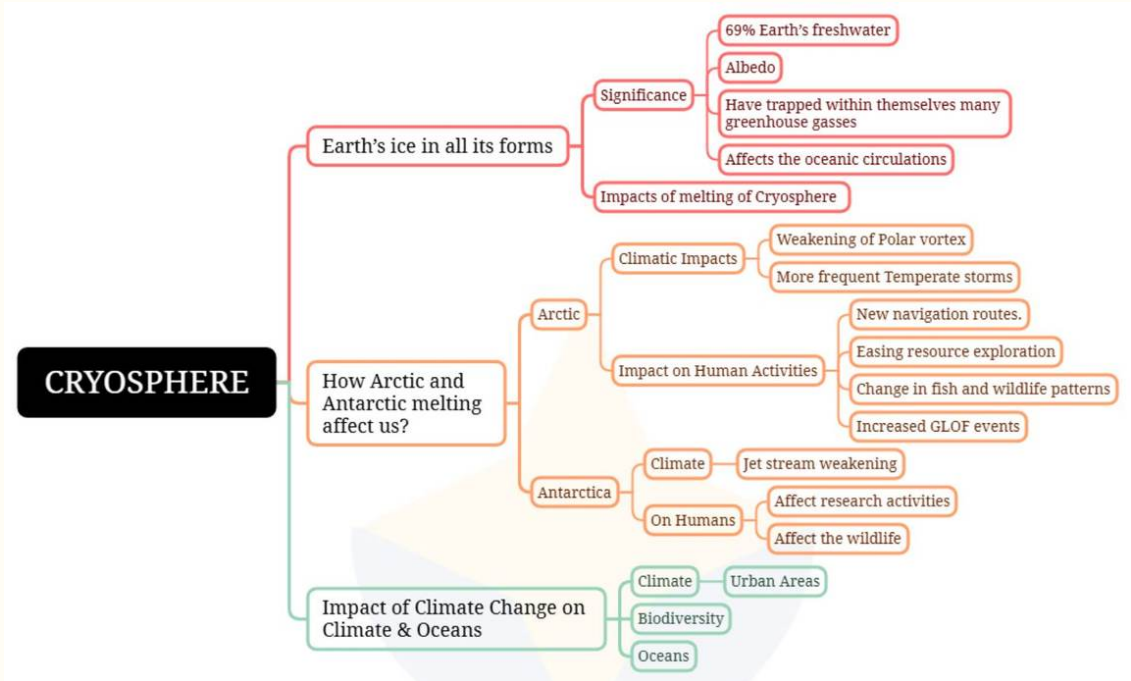

Atlantic Meridional Overturning Circulation: (AOC)
Warm, salty surface waters from the tropics are transported northward by currents like the Gulf Stream.
These waters cool and sink in the North Atlantic, forming the North Atlantic Deep Water (NADW) current.
Vital role in regulating climate patterns, and influencing weather, and precipitation in regions bordering the Atlantic.
Climate change is affecting the AMOC through; increased freshwater input from melting ice caps and increased precipitation.
Weakening of the AMOC could lead to altered weather patterns, more frequent extreme events (such as heatwaves, coldwaves and storms), and changes in ocean currents and ecosystems.
Reduced heat transport
Increased frequency and intensity of hurricanes
Changes in the strength and position of major ocean currents
Declines in fish populations
Sea level rise
Impact of Climate Change on Ocean Currents
Disruption of the Thermohaline circulations
Reducing nutrients in upwelling zones, and reducing Oxygen availability in downwelling zones.
Can change how winds move; altering the movement of both equatorial and boundary currents
Can affect the melting of ice from below.
Disrupt marine ecosystems; decline in biodiversity and changes in the availability of marine resources.

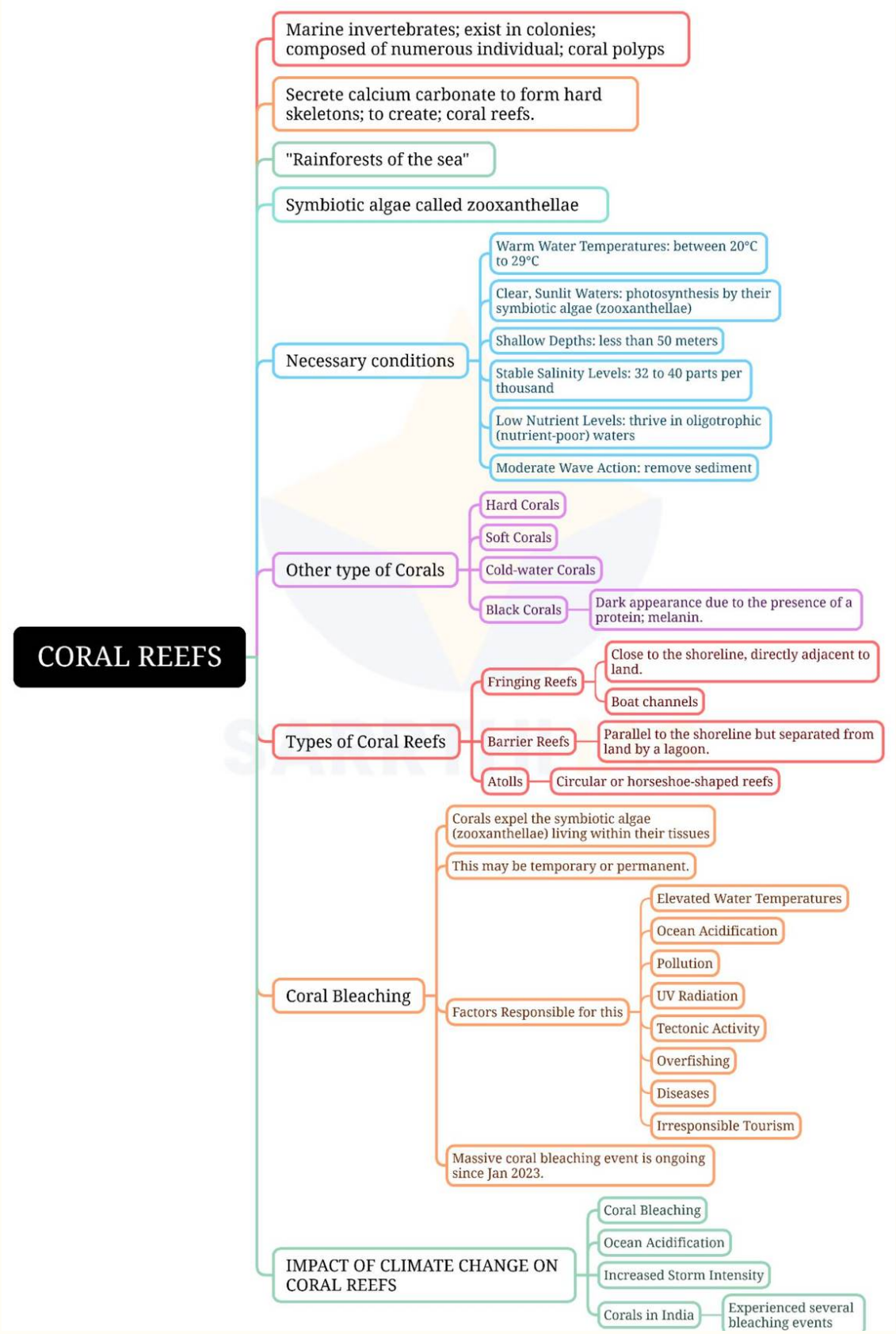

Ekman Currents
Ocean currents caused by wind blowing over the surface of the sea.
Ocean currents caused by wind blowing over the surface of the sea.
As you go deeper, each layer of water moves slightly differently, forming a spiral pattern called the Ekman spiral
Significance
Nutrient Upwelling; upwelling; enhancing marine productivity and supporting rich fisheries.
Heat Distribution; redistribution of heat
Wind Stress and Transport; Ekman transport; coastal circulation; gyres
Climate and Weather; regional climates; El Niño; La Niña
CO_2 Absorption; exchange of carbon dioxide; carbon cycle
Marine Heatwaves
Periods of unusually warm ocean temperatures,; can disrupt marine ecosystems.
Episodes have doubled since 1982.
Causes
Climate Change
Atmospheric Conditions
Ocean Currents
El Niño Events
Feedback Loops

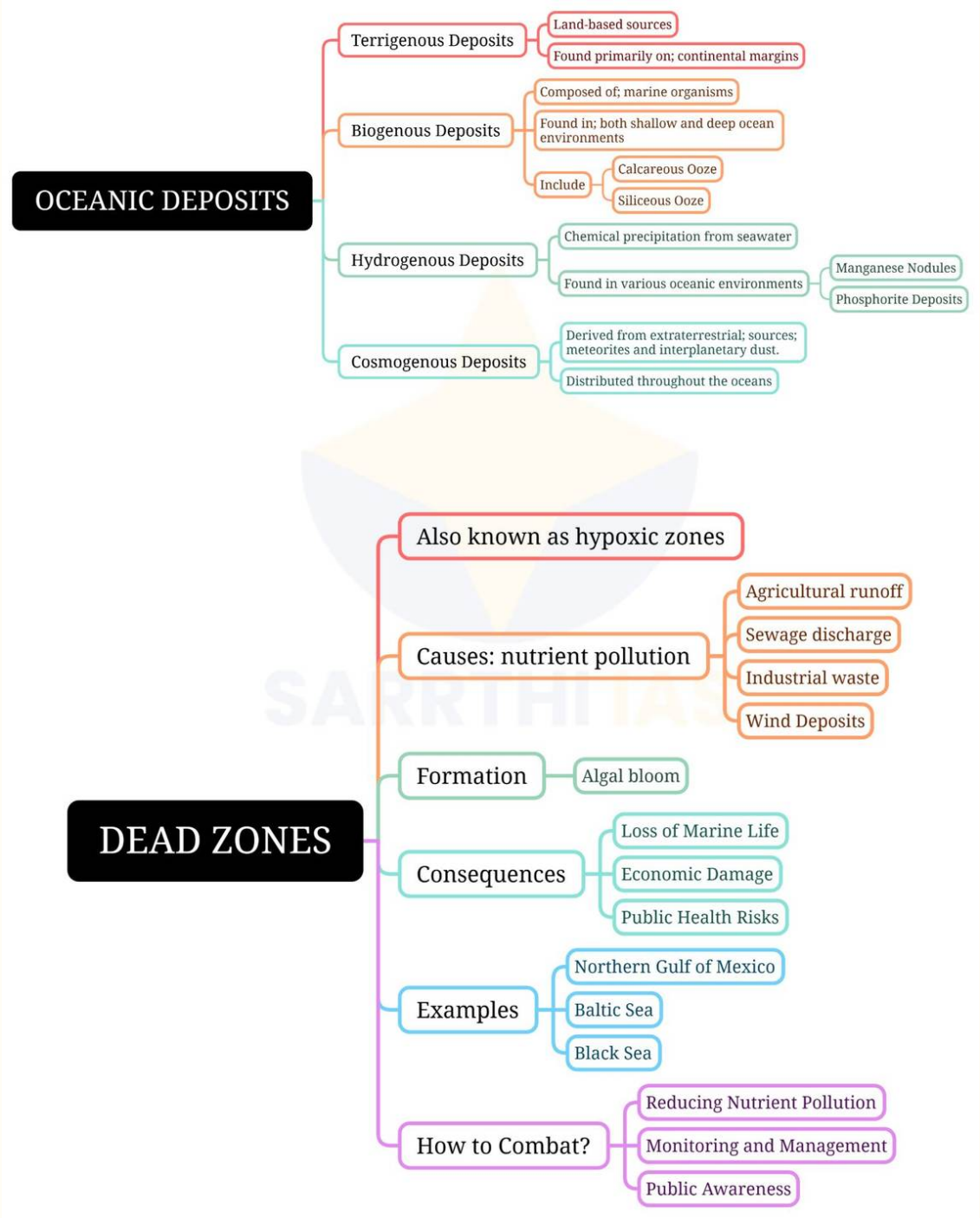

Ocean Movements: Tides
Rise (High tide/ Flood) and Fall (Low Tide/ Ebb) of Seawater
Due to the gravitational pull of the Moon and Sun.
On average, every place experiences a high and a low tide twice a day
Period: 12 hr 26 minutes.
Diurnal tides occur in specific regions; one dominant high tide and one dominant low tide each day.
Semi-diurnal tides; more widespread
Spring & Neap Tides: temporal and spatial variation in the tide-producing forces
Spring Tides
Syzygy position.
Height 20% more
Highest high and lowest low
Neap Tides
Position of quadrature; sun and the moon; opposite directions
Factors affecting tides
Relative distances of Sun, Moon, and Earth
Shape of Shoreline
Funnel-shaped bays and estuaries can amplify tidal ranges
Local winds
Creating surface water movement and altering the distribution of water masses.
Pressure systems
Impact of tides
Sustenance of life (nutrient circulation)
Transport nutrients such as nitrogen, phosphorus, and organic matter from the open ocean to coastal areas
Temperature moderation
Mixing surface waters with cooler, deeper waters
Removal of pollutants
Removal and dispersion of pollutants in coastal waters through the process of flushing.
Tidal Bore: rare phenomena; a wave that travels upstream against the river flow
Only during high tides.
Absent during Neap Tides
Impact of Climate Change on Tides
Higher high tides
May amplify the existing tidal range
Can disrupt; critical habitats.

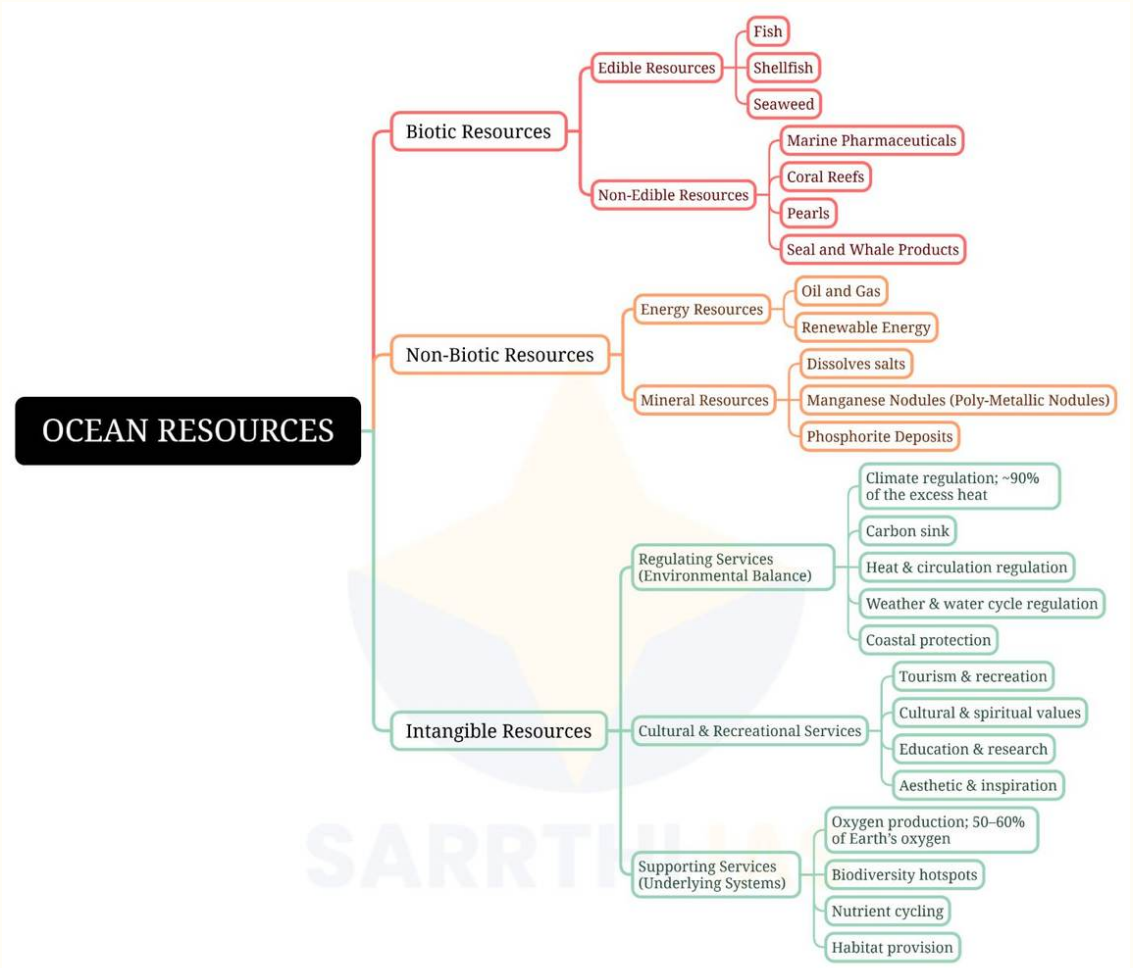

Ocean Movements: Waves
Energy, not the water; Water particles only travel in a small circle
Provides energy to the waves
Travel because the wind pushes the water body in its course while; gravity pulls the crests
Falling water pushes; former troughs upward
Actual motion of the water; circular.
Impact of Climate Change
Global warming can lead to stronger and more frequent storms; generate larger and more powerful waves.
Higher sea levels; waves; can travel further inland
Changes in wind patterns and ocean currents; can alter wave patterns.
Increased wave energy and frequency can damage coral reefs
Extremely long waves
Caused by a large and sudden displacement of the ocean
Also called seismic sea waves
Can cause dangerous coastal flooding and powerful currents
Causes:
Earthquake
Volcanic eruption
Landslides
Meteorite Impacts
Anthropogenic causes
Characteristics
Long Wavelengths
High Speeds
Low Wave Heights
Amplification near Coast
Destructive Power
Multiple Waves
Warning Signs

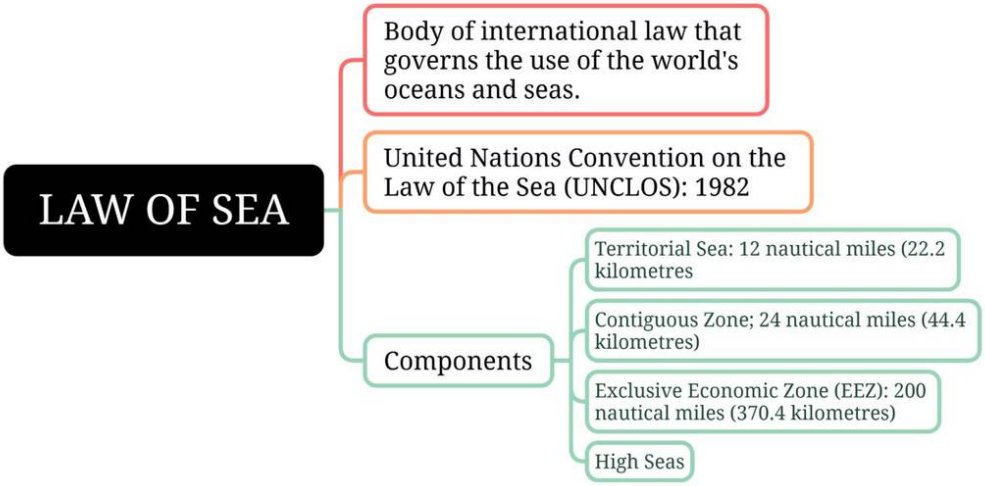

Rogue Waves
Rogue waves; freak waves; monster waves; killer waves; unusually large, spontaneous, and unpredictable ocean surface waves; far larger; relatively calm seas
Creation
1. Constructive Interference
Coincide in phase; combine; giant wave
Random and rare
2. Focusing by Currents and Topography
Strong ocean currents; compress wave fronts; Seafloor features
Seiches
Seiche; standing wave; enclosed or semi-enclosed water body; lake, bay, reservoir, or even a swimming pool; strong winds, atmospheric pressure changes; earthquakes
Creation
Wind Action
Atmospheric Pressure Changes
Seismic Activity
Resonance Effect
Basin Geometry


CRYOSPHERE
Earth's ice in all its forms
Significance
69% Earth's freshwater
Albedo
Have trapped within themselves many greenhouse gasses
Affects the oceanic circulations
Impacts of melting of Cryosphere
How Arctic and Antarctic melting affect us?
Arctic
Climatic Impacts
Weakening of Polar vortex
More frequent Temperate storms
Impact on Human Activities
New navigation routes.
Easing resource exploration
Change in fish and wildlife patterns
Increased GLOF events
Antarctica
Climate
Jet stream weakening
On Humans
Affect research activities
Affect the wildlife
Impact of Climate Change on Climate & Oceans
Climate
Urban Areas
Biodiversity
Oceans

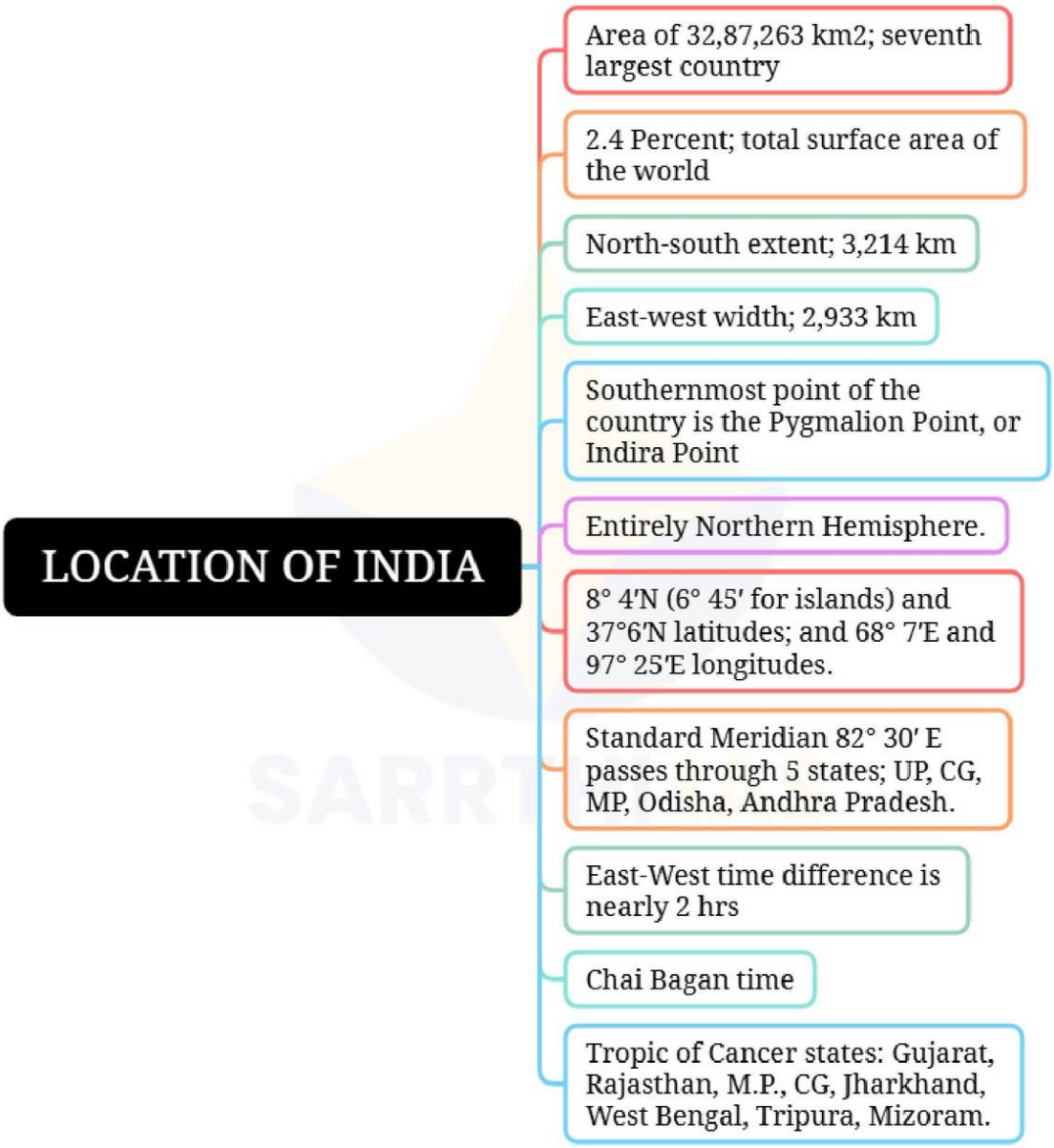

CORAL REEFS
Marine invertebrates; exist in colonies; composed of numerous individual; coral polyps
Secrete calcium carbonate to form hard skeletons; to create; coral reefs.
"Rainforests of the sea"
Symbiotic algae called zooxanthellae
Necessary conditions
Warm Water Temperatures: between 20°C to 29°C
Clear, Sunlit Waters: photosynthesis by their symbiotic algae (zooxanthellae)
Shallow Depths: less than 50 meters
Stable Salinity Levels: 32 to 40 parts per thousand
Low Nutrient Levels: thrive in oligotrophic (nutrient-poor) waters
Moderate Wave Action: remove sediment
Other type of Corals
Hard Corals
Soft Corals
Cold-water Corals
Black Corals
Dark appearance due to the presence of a protein; melanin.
Types of Coral Reefs
Fringing Reefs
Close to the shoreline, directly adjacent to land.
Boat channels
Barrier Reefs
Parallel to the shoreline but separated from land by a lagoon.
Atolls
Circular or horseshoe-shaped reefs
Coral Bleaching
Corals expel the symbiotic algae (zooxanthellae) living within their tissues
This may be temporary or permanent.
Factors Responsible for this
Elevated Water Temperatures
Ocean Acidification
Pollution
UV Radiation
Tectonic Activity
Overfishing
Diseases
Irresponsible Tourism
Massive coral bleaching event is ongoing since Jan 2023.
IMPACT OF CLIMATE CHANGE ON CORAL REEFS
Coral Bleaching
Ocean Acidification
Increased Storm Intensity
Corals in India
Experienced several bleaching events

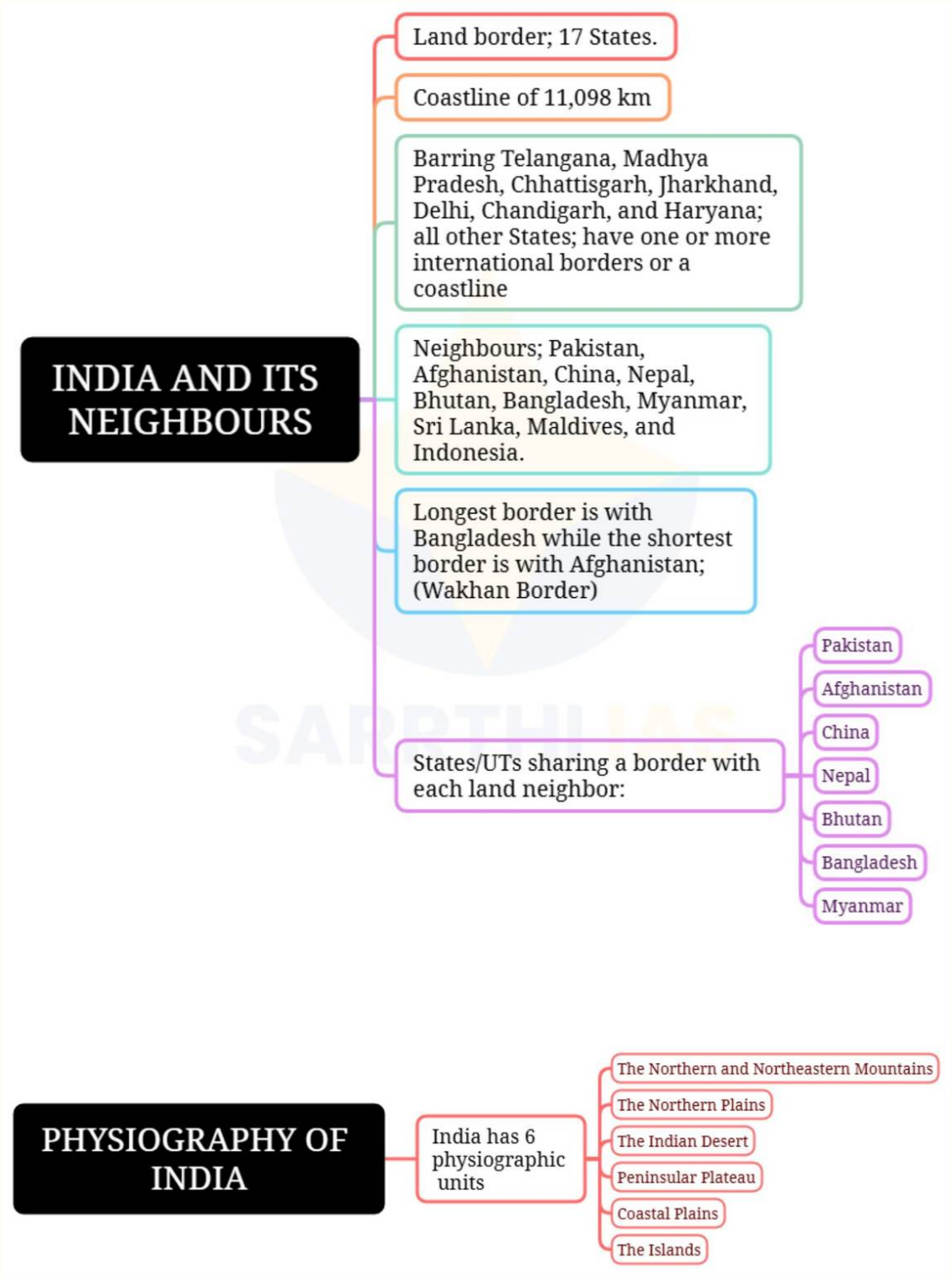

OCEANIC DEPOSITS
Terrigenous Deposits
Land-based sources
Found primarily on; continental margins
Biogenous Deposits
Composed of; marine organisms
Found in; both shallow and deep ocean environments
Include
Calcareous Ooze
Siliceous Ooze
Hydrogenous Deposits
Chemical precipitation from seawater
Found in various oceanic environments
Manganese Nodules
Phosphorite Deposits
Cosmogenous Deposits
Derived from extraterrestrial; sources; meteorites and interplanetary dust.
Distributed throughout the oceans
DEAD ZONES
Also known as hypoxic zones
Causes: nutrient pollution
Agricultural runoff
Sewage discharge
Industrial waste
Wind Deposits
Formation
Algal bloom
Consequences
Loss of Marine Life
Economic Damage
Public Health Risks
Examples
Northern Gulf of Mexico
Baltic Sea
Black Sea
How to Combat?
Reducing Nutrient Pollution
Monitoring and Management
Public Awareness

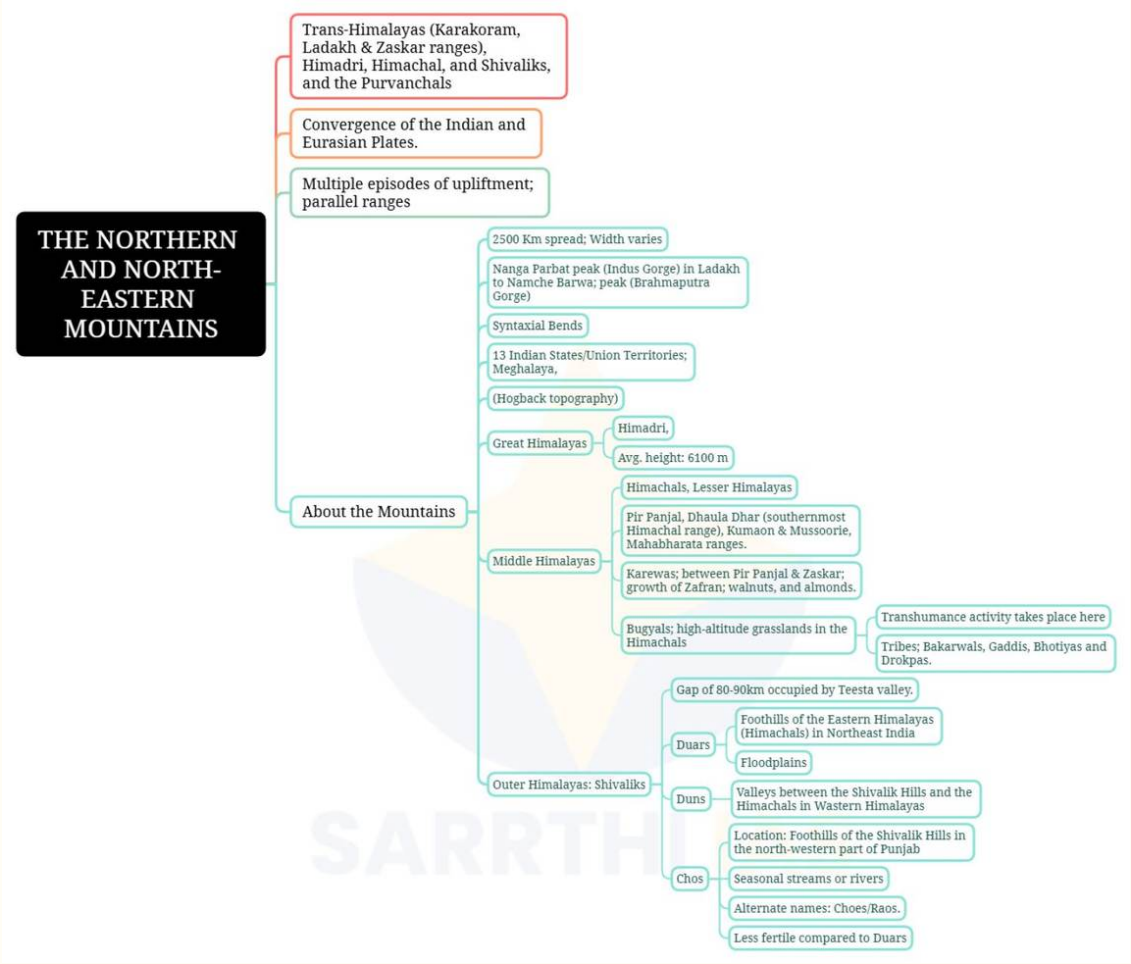

OCEAN RESOURCES
Biotic Resources
Edible Resources
Fish
Shellfish
Seaweed
Non-Edible Resources
Marine Pharmaceuticals
Coral Reefs
Pearls
Seal and Whale Products
Non-Biotic Resources
Energy Resources
Oil and Gas
Renewable Energy
Mineral Resources
Dissolves salts
Manganese Nodules (Poly-Metallic Nodules)
Phosphorite Deposits
Intangible Resources
Regulating Services (Environmental Balance)
Climate regulation; ~90% of the excess heat
Carbon sink
Heat & circulation regulation
Weather & water cycle regulation
Coastal protection
Cultural & Recreational Services
Tourism & recreation
Cultural & spiritual values
Education & research
Aesthetic & inspiration
Supporting Services (Underlying Systems)
Oxygen production; 50–60% of Earth's oxygen
Biodiversity hotspots
Nutrient cycling
Habitat provision

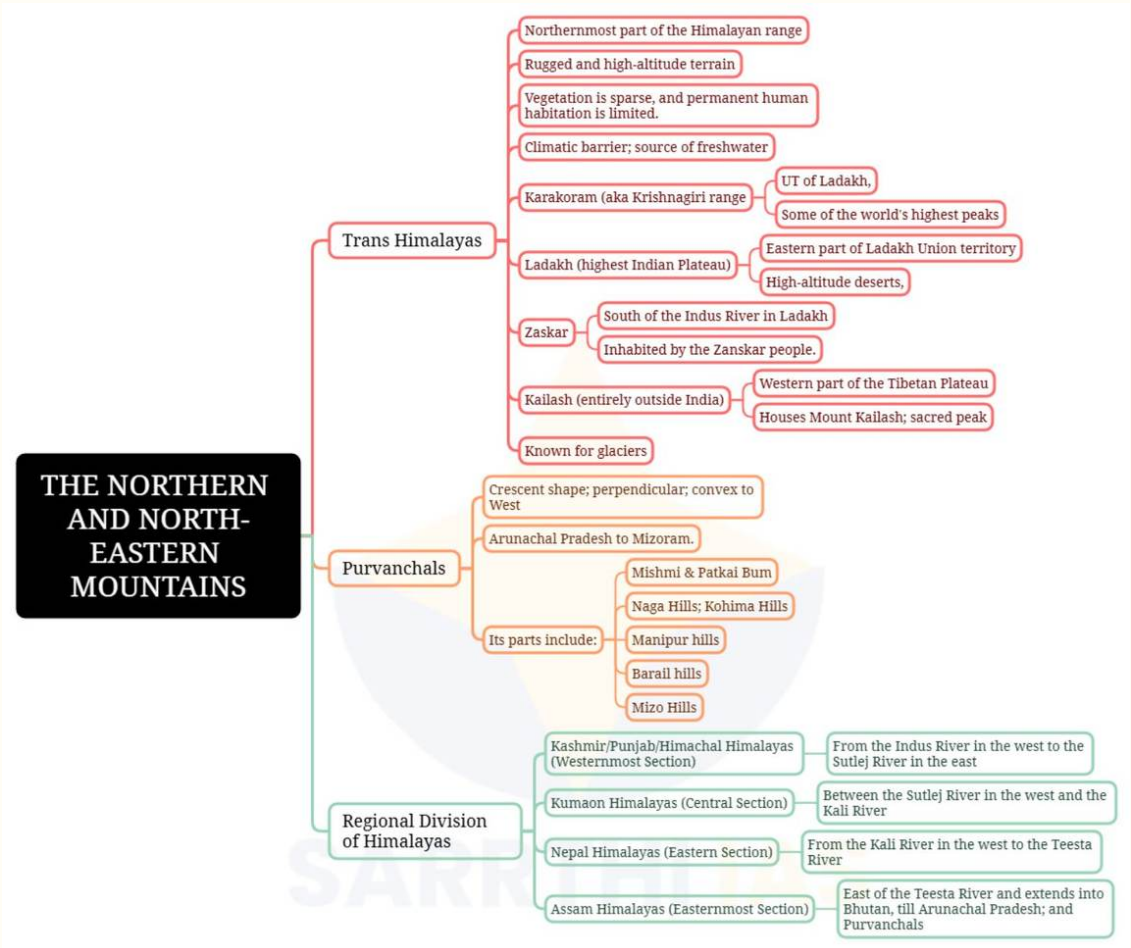

LAW OF SEA
Body of international law that governs the use of the world's oceans and seas.
United Nations Convention on the Law of the Sea (UNCLOS): 1982
Components
Territorial Sea: 12 nautical miles (22.2 kilometres
Contiguous Zone; 24 nautical miles (44.4 kilometres)
Exclusive Economic Zone (EEZ): 200 nautical miles (370.4 kilometres)
High Seas

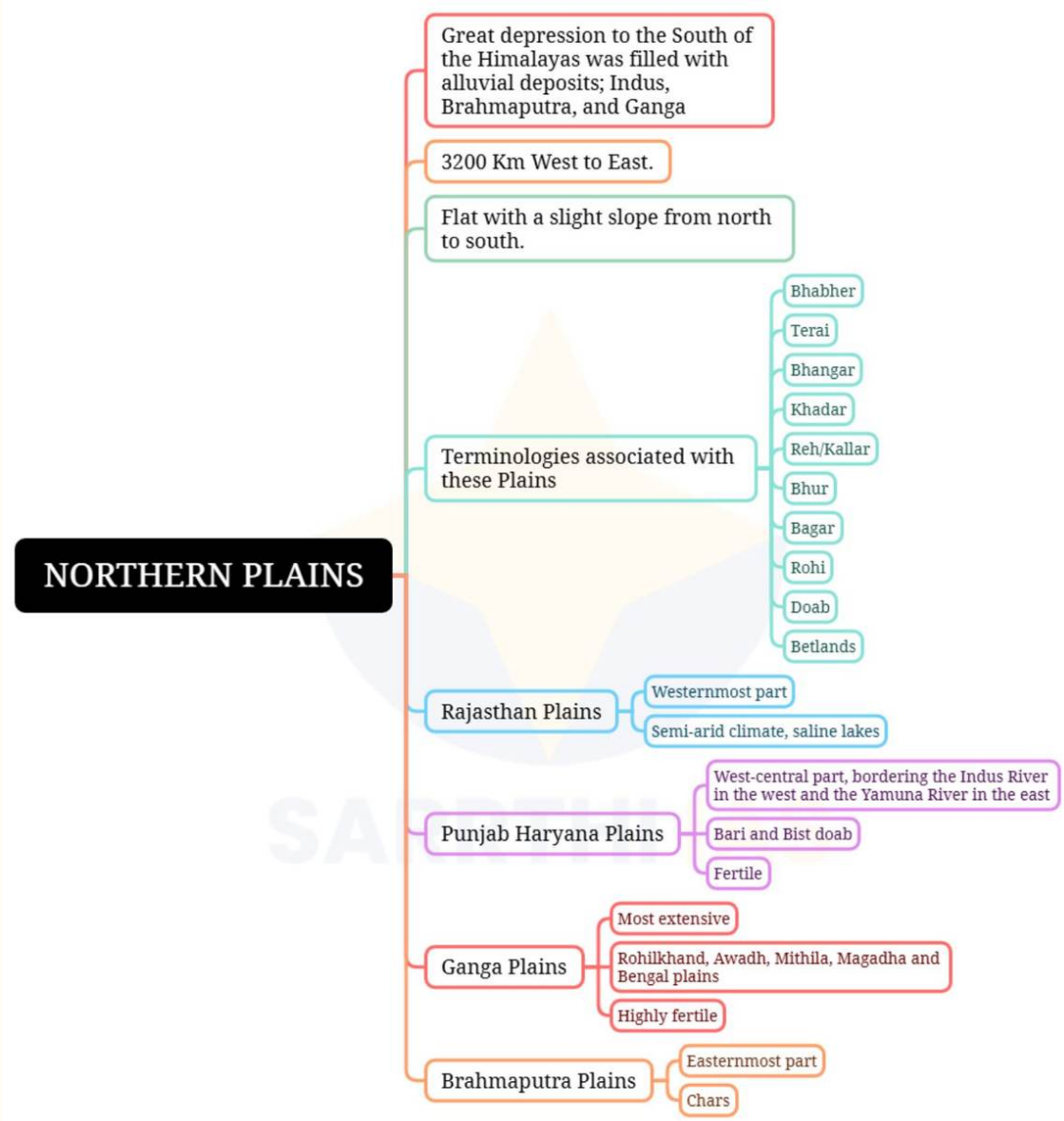
Location & Physiography of India

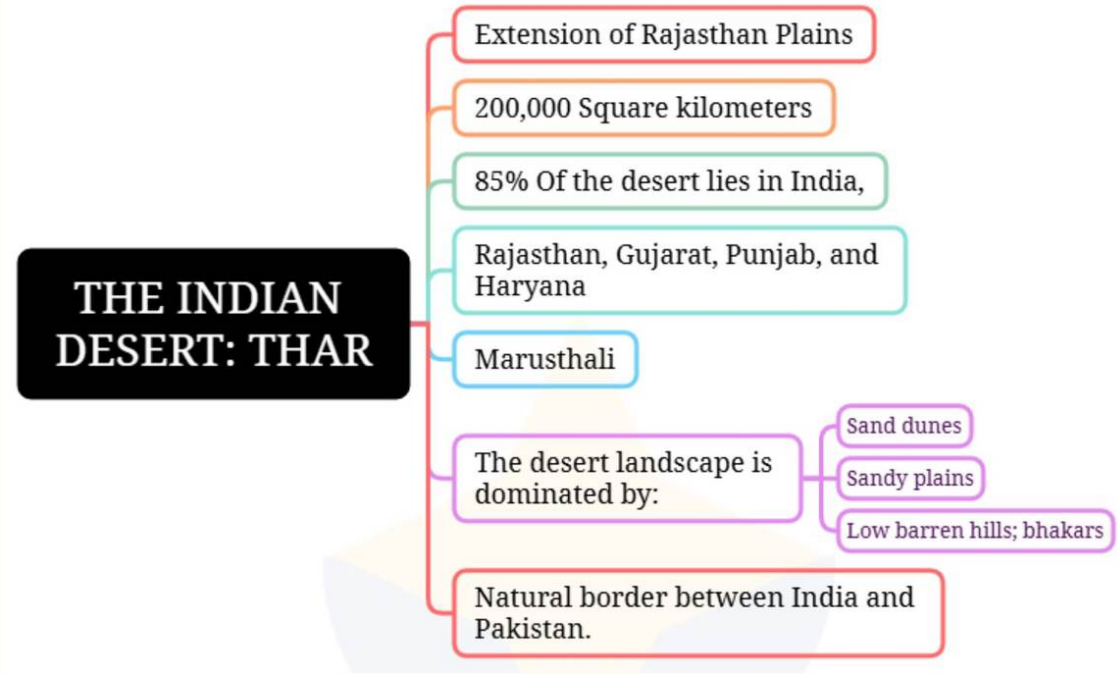


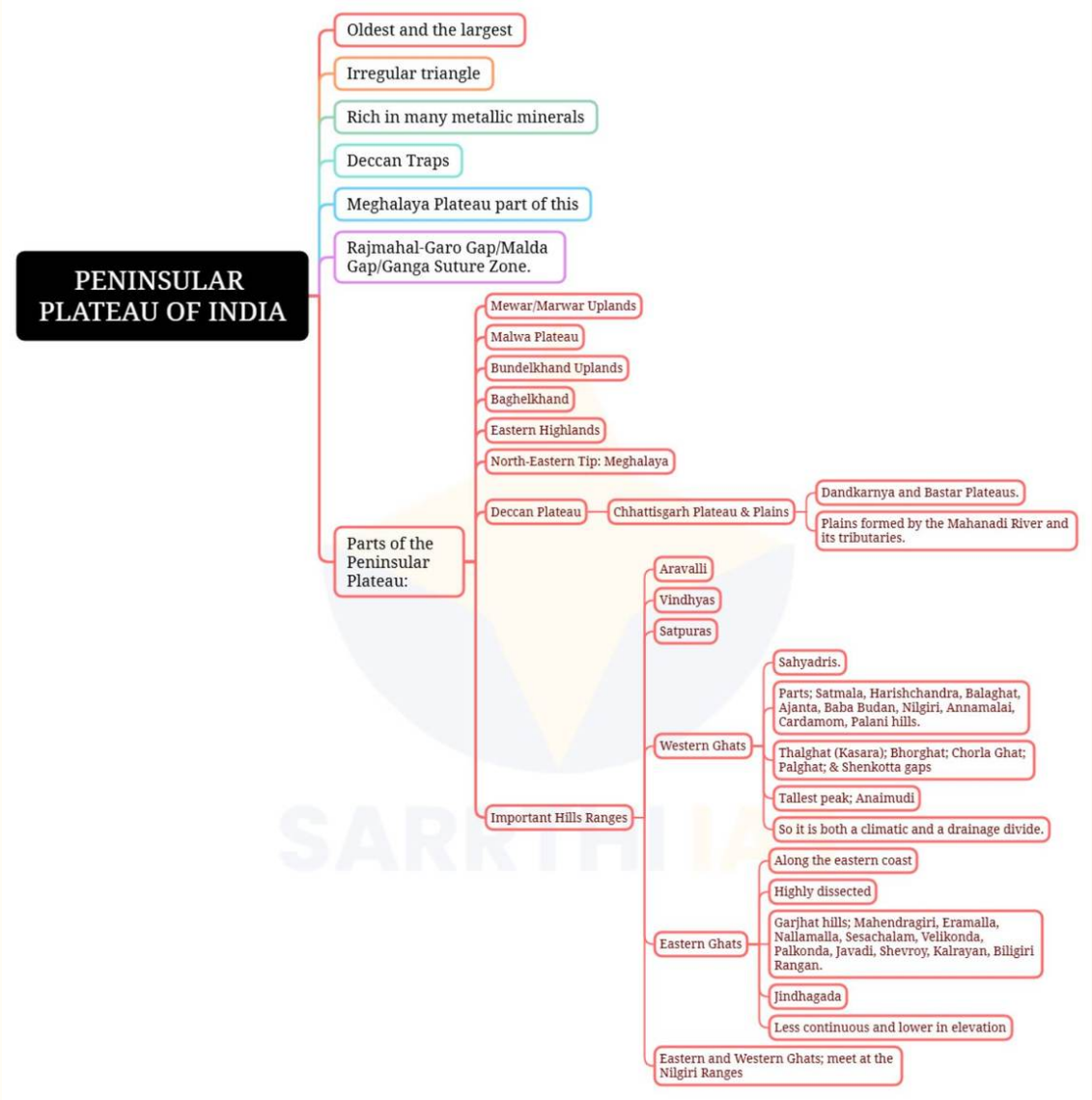

INDIA AND ITS NEIGHBOURS
Land border; 17 States.
Coastline of 11,098 km
Barring Telangana, Madhya Pradesh, Chhattisgarh, Jharkhand, Delhi, Chandigarh, and Haryana; all other States; have one or more international borders or a coastline
Neighbours; Pakistan, Afghanistan, China, Nepal, Bhutan, Bangladesh, Myanmar, Sri Lanka, Maldives, and Indonesia.
Longest border is with Bangladesh while the shortest border is with Afghanistan; (Wakhan Border)
States/UTs sharing a border with each land neighbor:
Pakistan
Afghanistan
China
Nepal
Bhutan
Bangladesh
Myanmar
PHYSIOGRAPHY OF INDIA
India has 6 physiographic units
The Northern and Northeastern Mountains
The Northern Plains
The Indian Desert
Peninsular Plateau
Coastal Plains
The Islands

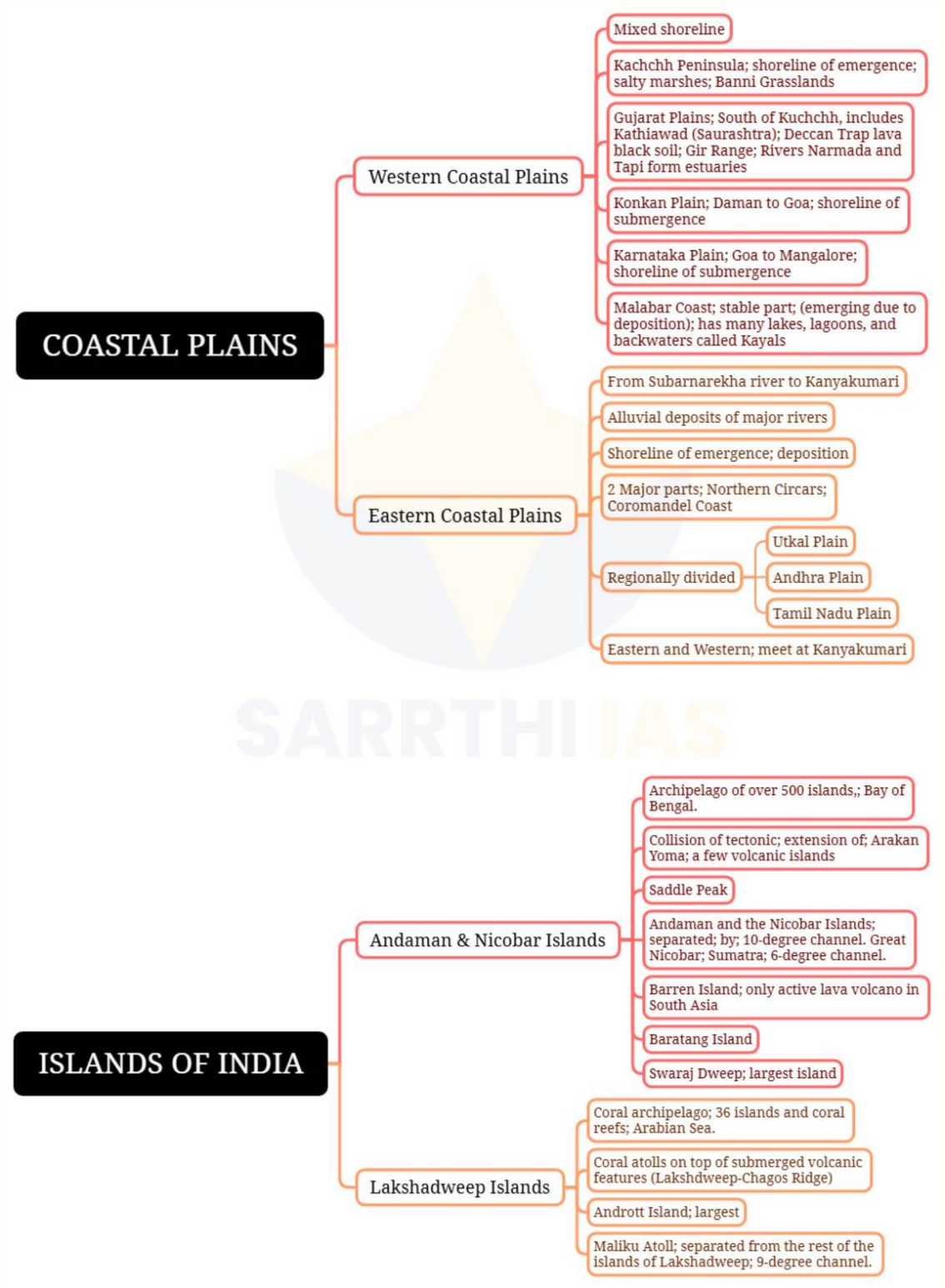

THE NORTHERN AND NORTH-EASTERN MOUNTAINS
Trans-Himalayas (Karakoram, Ladakh & Zaskar ranges), Himadri, Himachal, and Shivaliks, and the Purvanchals
Convergence of the Indian and Eurasian Plates.
Multiple episodes of upliftment; parallel ranges
About the Mountains
2500 Km spread; Width varies
Nanga Parbat peak (Indus Gorge) in Ladakh to Namche Barwa; peak (Brahmaputra Gorge)
Syntaxial Bends
13 Indian States/Union Territories; Meghalaya,
(Hogback topography)
Great Himalayas
Himadri,
Avg. height: 6100 m
Middle Himalayas
Himachals, Lesser Himalayas
Pir Panjal, Dhaula Dhar (southernmost Himachal range), Kumaon & Mussoorie, Mahabharata ranges.
Karewas; between Pir Panjal & Zaskar; growth of Zafran; walnuts, and almonds.
Bugyals; high-altitude grasslands in the Himachals
Transhumance activity takes place here
Tribes; Bakarwals, Gaddis, Bhotiyas and Drokpas.
Outer Himalayas: Shivaliks
Gap of 80-90km occupied by Teesta valley.
Duars
Foothills of the Eastern Himalayas (Himachals) in Northeast India
Floodplains
Duns
Valleys between the Shivalik Hills and the Himachals in Western Himalayas
Chos
Location: Foothills of the Shivalik Hills in the north-western part of Punjab
Seasonal streams or rivers
Alternate names: Choes/Raos.
Less fertile compared to Duars


THE NORTHERN AND NORTH-EASTERN MOUNTAINS
Trans Himalayas
Northernmost part of the Himalayan range
Rugged and high-altitude terrain
Vegetation is sparse, and permanent human habitation is limited.
Climatic barrier; source of freshwater
Karakoram (aka Krishnagiri range
UT of Ladakh,
Some of the world's highest peaks
Ladakh (highest Indian Plateau)
Eastern part of Ladakh Union territory
High-altitude deserts,
Zaskar
South of the Indus River in Ladakh
Inhabited by the Zanskar people.
Kailash (entirely outside India)
Western part of the Tibetan Plateau
Houses Mount Kailash; sacred peak
Known for glaciers
Purvanchals
Crescent shape; perpendicular; convex to West
Arunachal Pradesh to Mizoram.
Its parts include:
Mishmi & Patkai Bum
Naga Hills; Kohima Hills
Manipur hills
Barail hills
Mizo Hills
Regional Division of Himalayas
Kashmir/Punjab/Himachal Himalayas (Westernmost Section)
From the Indus River in the west to the Sutlej River in the east
Kumaon Himalayas (Central Section)
Between the Sutlej River in the west and the Kali River
Nepal Himalayas (Eastern Section)
From the Kali River in the west to the Teesta River
Assam Himalayas (Easternmost Section)
East of the Teesta River and extends into Bhutan, till Arunachal Pradesh; and Purvanchals

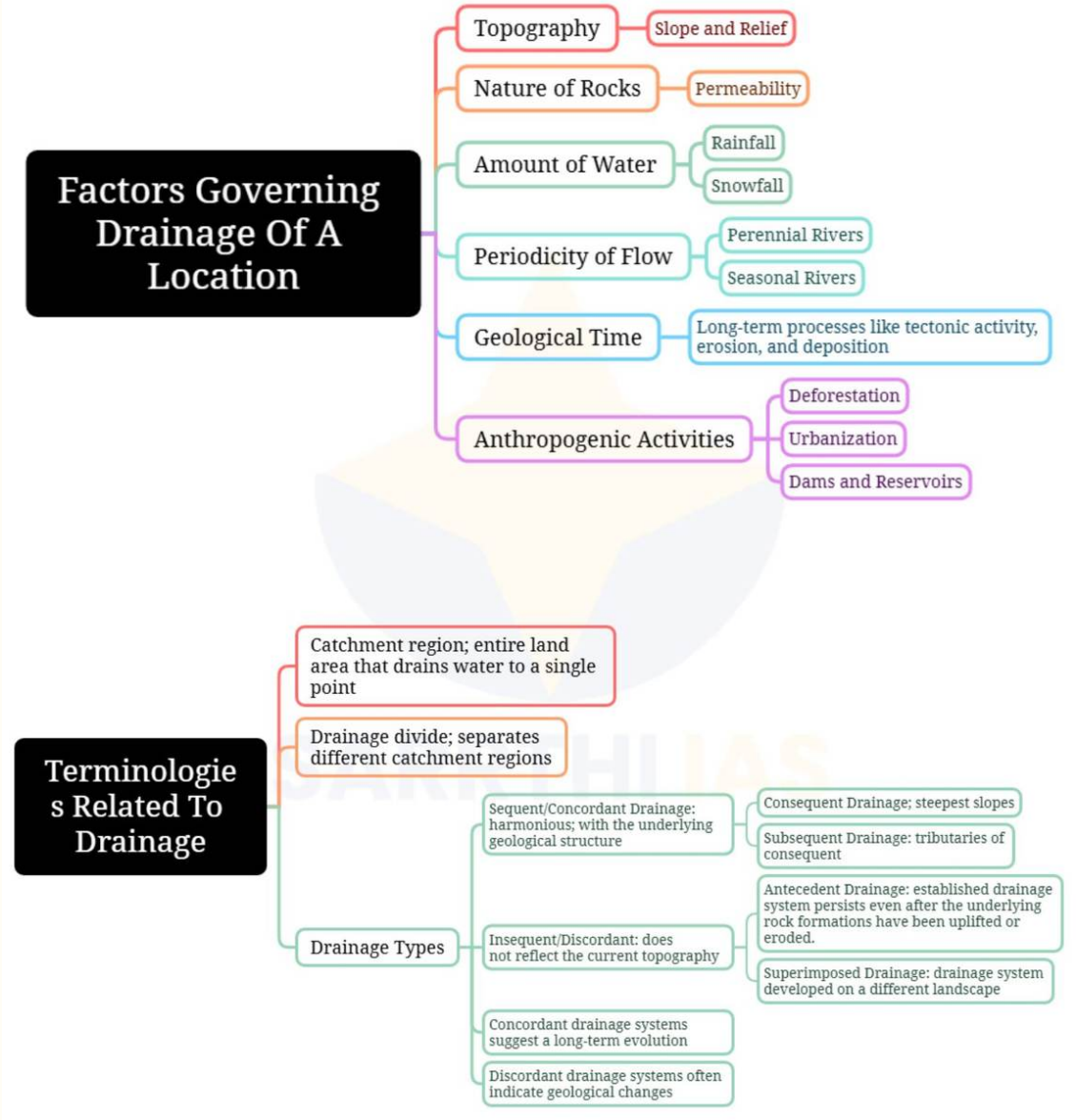

NORTHERN PLAINS
Great depression to the South of the Himalayas was filled with alluvial deposits; Indus, Brahmaputra, and Ganga
3200 Km West to East.
Flat with a slight slope from north to south.
Terminologies associated with these Plains
Bhabher
Terai
Bhangar
Khadar
Reh/Kallar
Bhur
Bagar
Rohi
Doab
Betlands
Rajasthan Plains
Westernmost part
Semi-arid climate, saline lakes
Punjab Haryana Plains
West-central part, bordering the Indus River in the west and the Yamuna River in the east
Bari and Bist doab
Fertile
Ganga Plains
Most extensive
Rohilkhand, Awadh, Mithila, Magadha and Bengal plains
Highly fertile
Brahmaputra Plains
Easternmost part
Chars

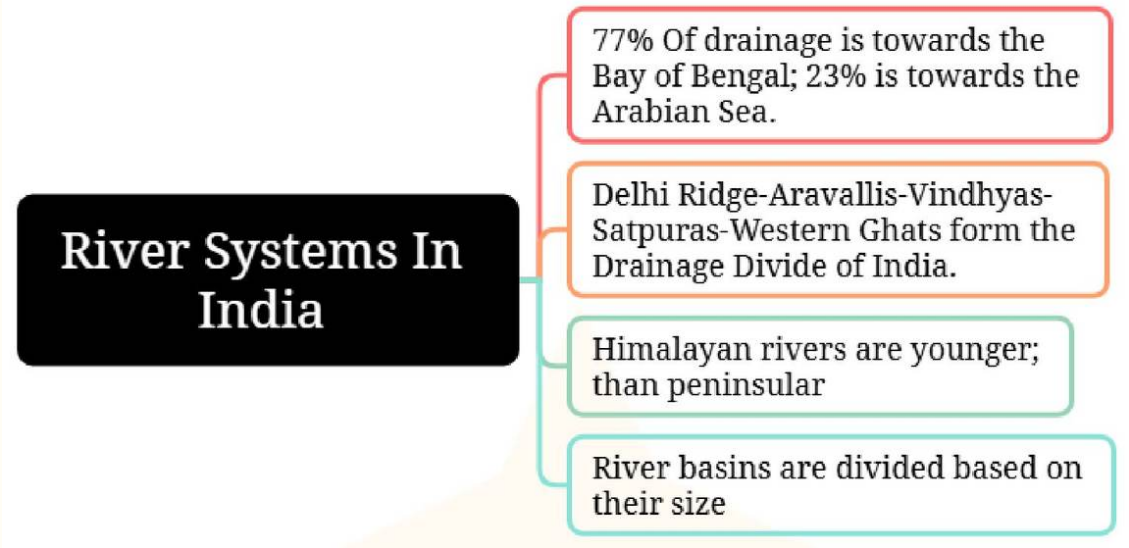

THE INDIAN DESERT: THAR
Extension of Rajasthan Plains
200,000 Square kilometers
85% Of the desert lies in India,
Rajasthan, Gujarat, Punjab, and Haryana
Marusthali
The desert landscape is dominated by:
Sand dunes
Sandy plains
Low barren hills; bhakars
Natural border between India and Pakistan.

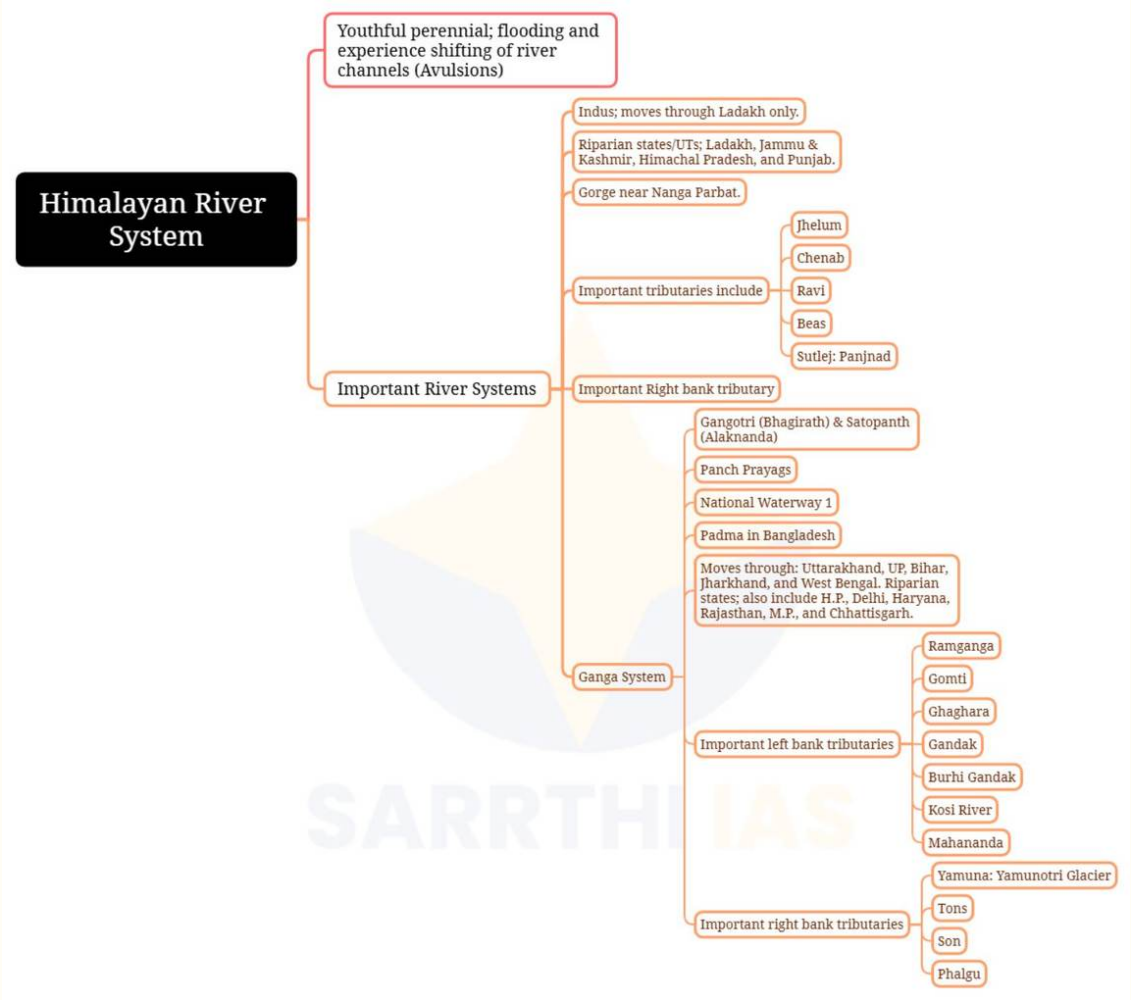

PENINSULAR PLATEAU OF INDIA
Oldest and the largest
Irregular triangle
Rich in many metallic minerals
Deccan Traps
Meghalaya Plateau part of this
Rajmahal-Garo Gap/Malda Gap/Ganga Suture Zone.
Parts of the Peninsular Plateau:
Mewar/Marwar Uplands
Malwa Plateau
Bundelkhand Uplands
Baghelkhand
Eastern Highlands
North-Eastern Tip: Meghalaya
Deccan Plateau
Chhattisgarh Plateau & Plains
Dandkarnya and Bastar Plateaus.
Plains formed by the Mahanadi River and its tributaries.
Important Hills Ranges
Aravalli
Vindhyas
Satpuras
Western Ghats
Sahyadris.
Parts; Satmala, Harishchandra, Balaghat, Ajanta, Baba Budan, Nilgiri, Annamalai, Cardamom, Palani hills.
Thalghat (Kasara); Bhorghat; Chorla Ghat; Palghat; & Shenkotta gaps
Tallest peak; Anaimudi
So it is both a climatic and a drainage divide.
Eastern Ghats
Along the eastern coast
Highly dissected
Garjhat hills; Mahendragiri, Eramalla, Nallamalla, Sesachalam, Velikonda, Palkonda, Javadi, Shevroy, Kalrayan, Biligiri Rangan.
Jindhagada
Less continuous and lower in elevation
Eastern and Western Ghats; meet at the Nilgiri Ranges

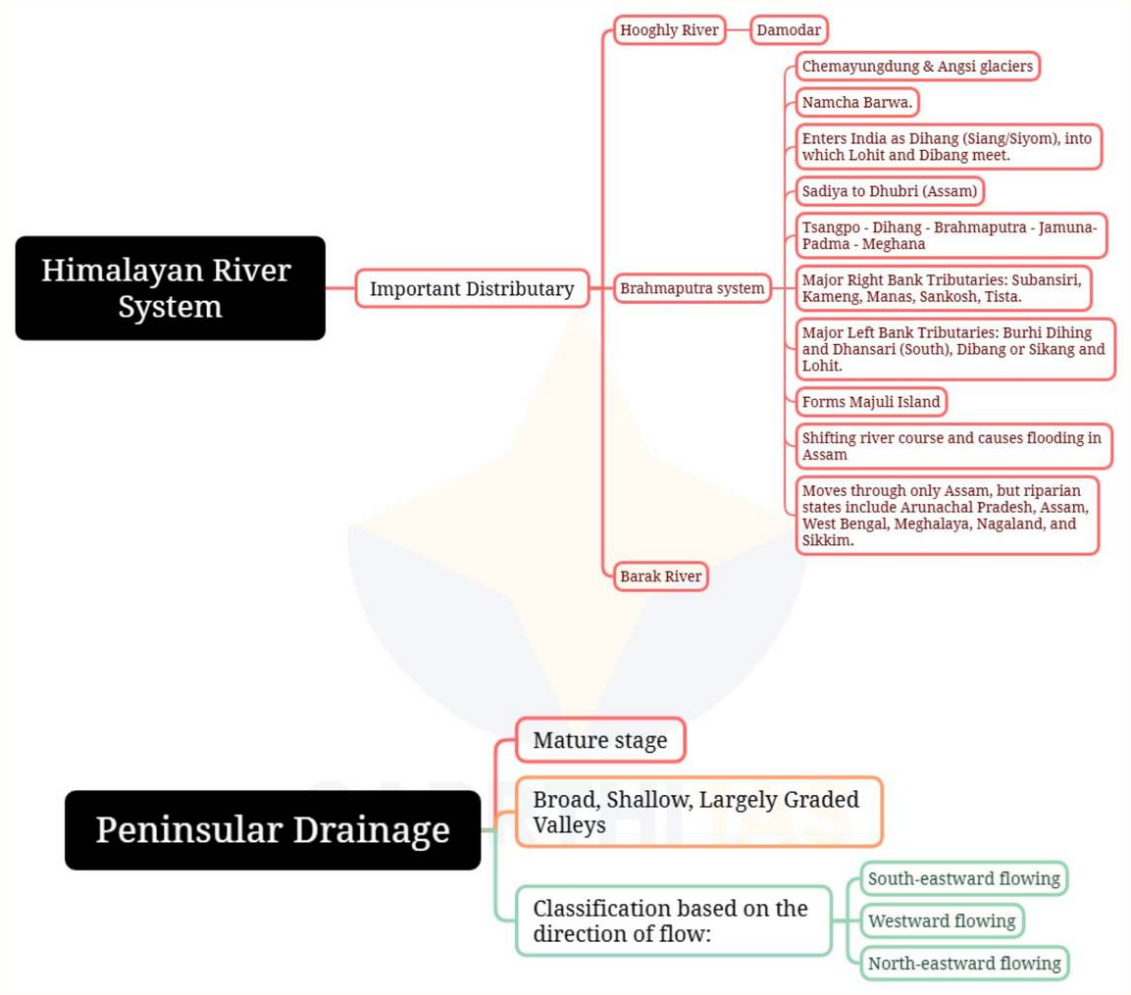

COASTAL PLAINS
Western Coastal Plains
Mixed shoreline
Kachchh Peninsula; shoreline of emergence; salty marshes; Banni Grasslands
Gujarat Plains; South of Kuchchh, includes Kathiawad (Saurashtra); Deccan Trap lava black soil; Gir Range; Rivers Narmada and Tapi form estuaries
Konkan Plain; Daman to Goa; shoreline of submergence
Karnataka Plain; Goa to Mangalore; shoreline of submergence
Malabar Coast; stable part; (emerging due to deposition); has many lakes, lagoons, and backwaters called Kayals
Eastern Coastal Plains
From Subarnarekha river to Kanyakumari
Alluvial deposits of major rivers
Shoreline of emergence; deposition
2 Major parts; Northern Circars; Coromandel Coast
Regionally divided
Utkal Plain
Andhra Plain
Tamil Nadu Plain
Eastern and Western; meet at Kanyakumari
ISLANDS OF INDIA
Andaman & Nicobar Islands
Archipelago of over 500 islands,; Bay of Bengal.
Collision of tectonic; extension of; Arakan Yoma; a few volcanic islands
Saddle Peak
Andaman and the Nicobar Islands; separated; by; 10-degree channel. Great Nicobar; Sumatra; 6-degree channel.
Barren Island; only active lava volcano in South Asia
Baratang Island
Swaraj Dweep; largest island
Lakshadweep Islands
Coral archipelago; 36 islands and coral reefs; Arabian Sea.
Coral atolls on top of submerged volcanic features (Lakshdweep-Chagos Ridge)
Andrott Island; largest
Maliku Atoll; separated from the rest of the islands of Lakshadweep; 9-degree channel.

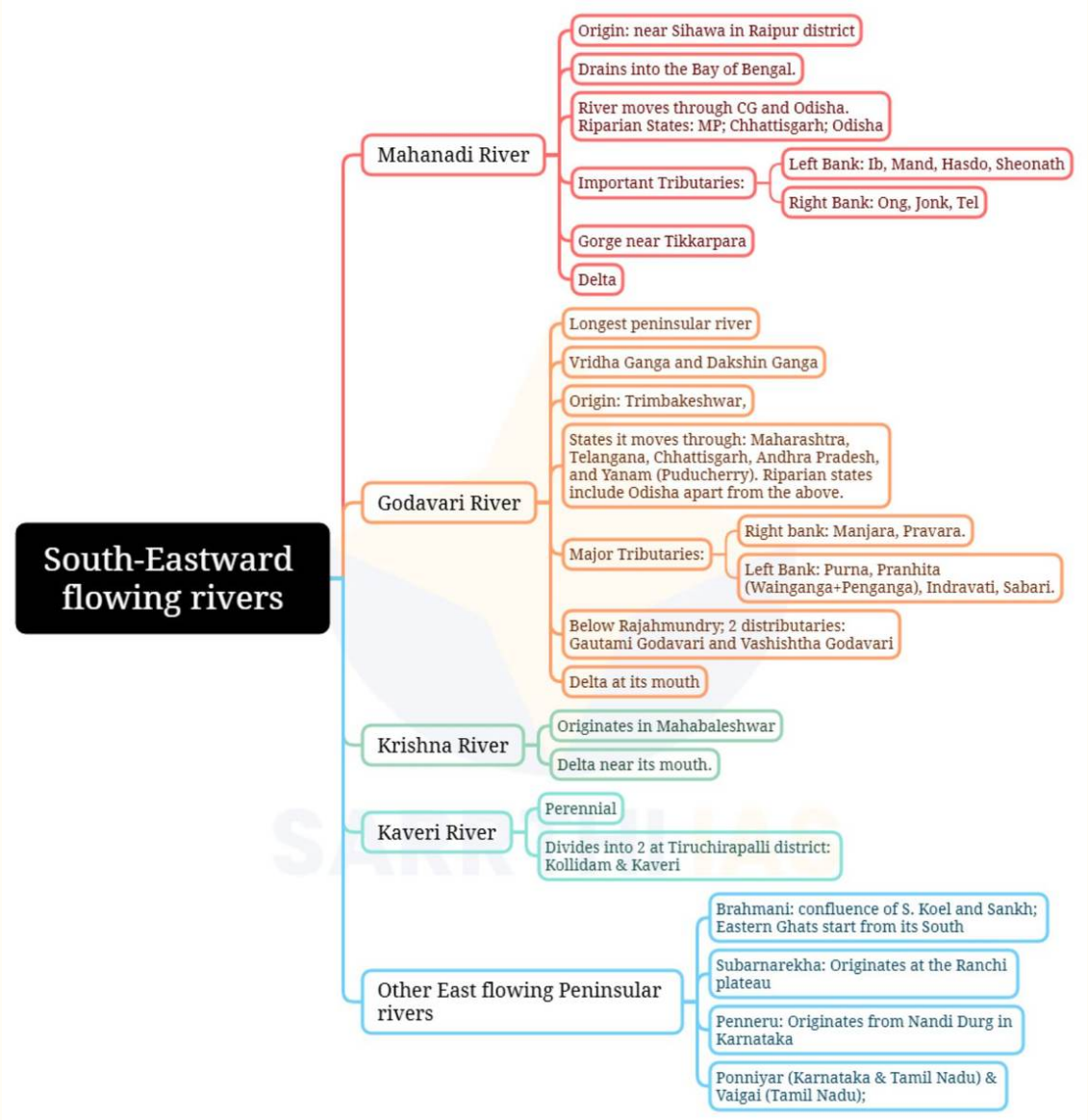
Indian Drainage system

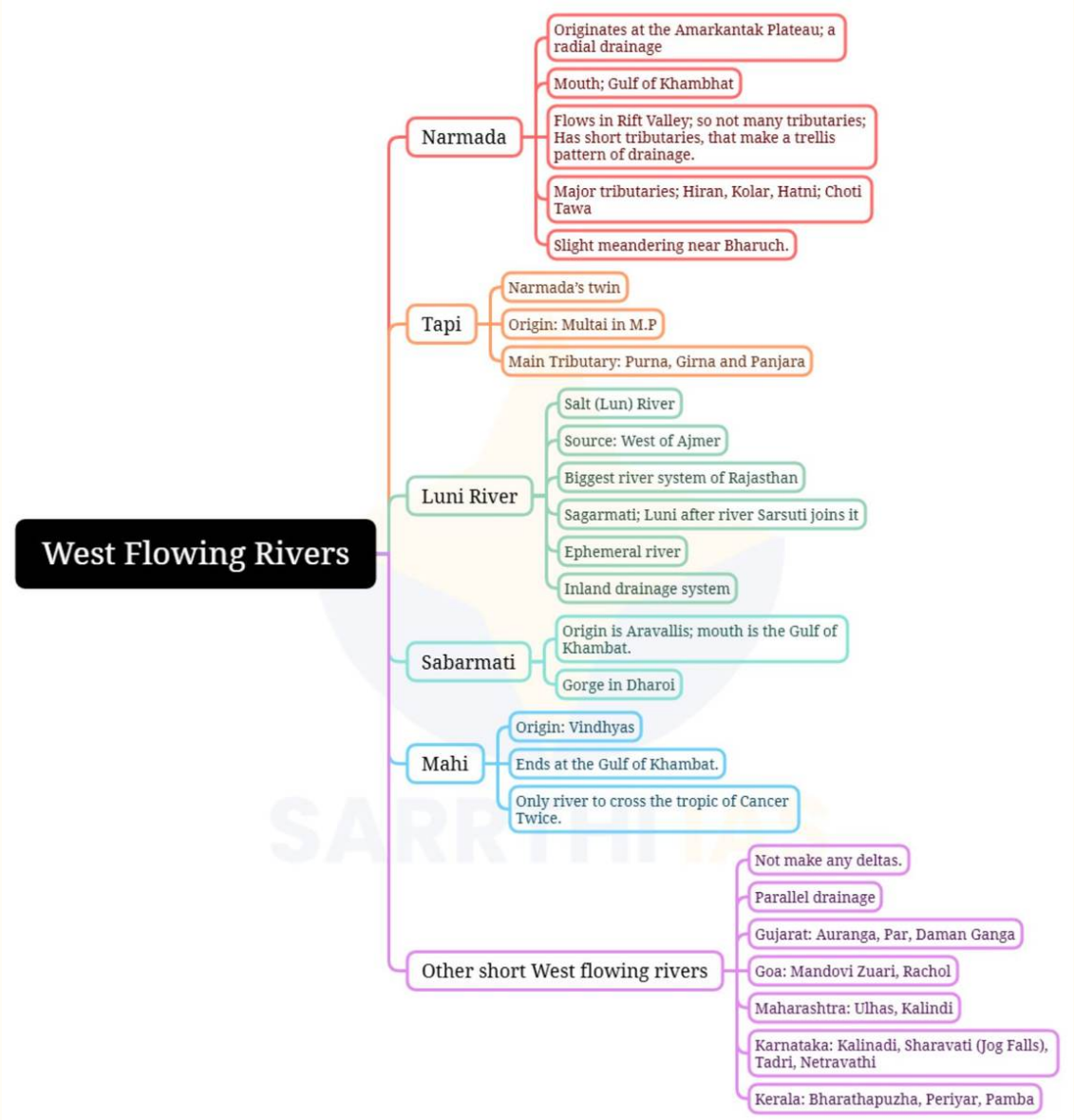


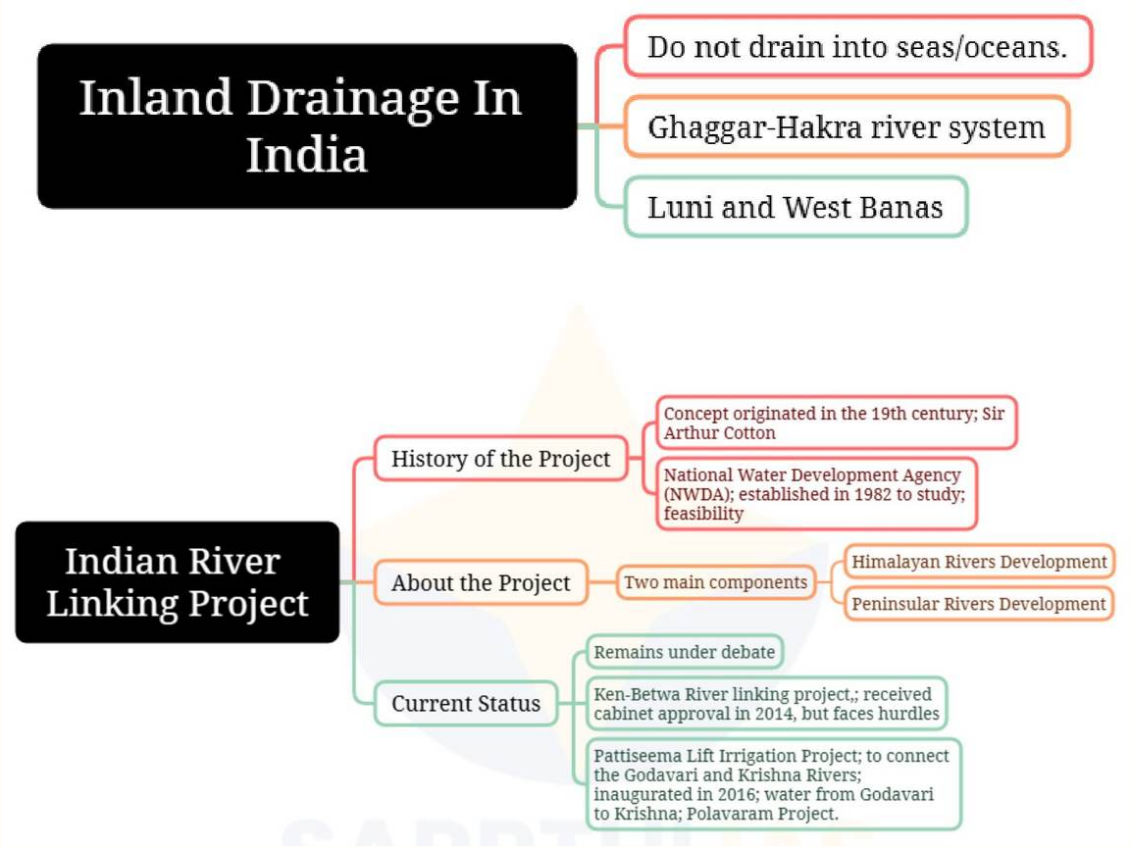



Himalayan River System
Youthful perennial; flooding and experience shifting of river channels (Avulsions)
Important River Systems
Indus; moves through Ladakh only.
Riparian states/UTs; Ladakh, Jammu & Kashmir, Himachal Pradesh, and Punjab.
Gorge near Nanga Parbat.
Important tributaries include
Jhelum
Chenab
Ravi
Beas
Sutlej: Panjnad
Important Right bank tributary
Ganga System
Gangotri (Bhagirath) & Satopanth (Alaknanda)
Panch Prayags
National Waterway 1
Padma in Bangladesh
Moves through: Uttarakhand, UP, Bihar, Jharkhand, and West Bengal. Riparian states; also include H.P., Delhi, Haryana, Rajasthan, M.P., and Chhattisgarh.
Important left bank tributaries
Ramganga
Gomti
Ghaghara
Gandak
Burhi Gandak
Kosi River
Mahananda
Important right bank tributaries
Yamuna: Yamunotri Glacier
Tons
Son
Phalgu

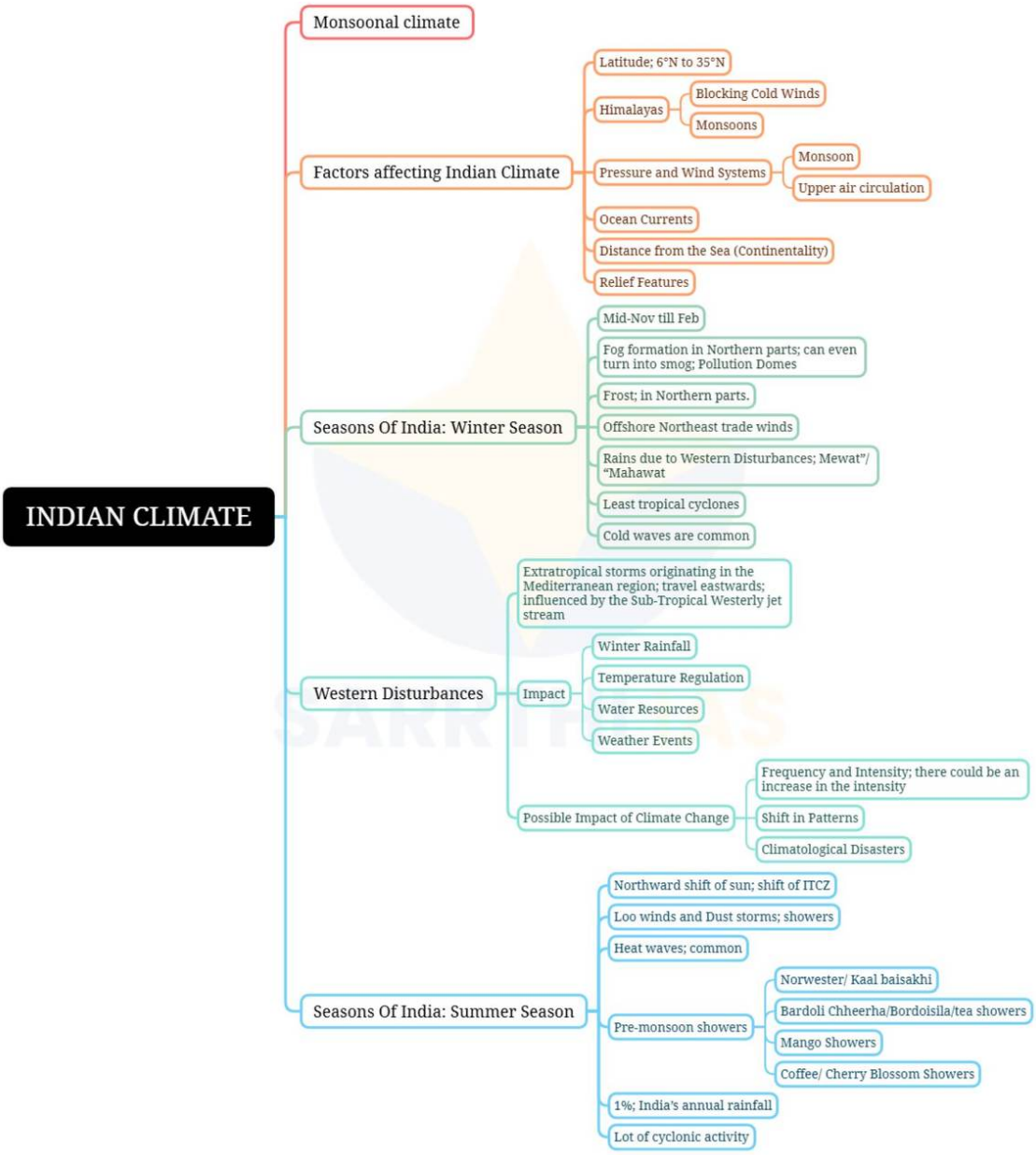

Himalayan River System
Important Distributary
Hooghly River
Damodar
Brahmaputra system
Chemayungdung & Angsi glaciers
Namcha Barwa.
Enters India as Dihang (Siang/Siyom), into which Lohit and Dibang meet.
Sadiya to Dhubri (Assam)
Tsangpo - Dihang - Brahmaputra - Jamuna-Padma - Meghana
Major Right Bank Tributaries: Subansiri, Kameng, Manas, Sankosh, Tista.
Major Left Bank Tributaries: Burhi Dihing and Dhansari (South), Dibang or Sikang and Lohit.
Forms Majuli Island
Shifting river course and causes flooding in Assam
Moves through only Assam, but riparian states include Arunachal Pradesh, Assam, West Bengal, Meghalaya, Nagaland, and Sikkim.
Barak River
Peninsular Drainage
Mature stage
Broad, Shallow, Largely Graded Valleys
Classification based on the direction of flow:
South-eastward flowing
Westward flowing
North-eastward flowing

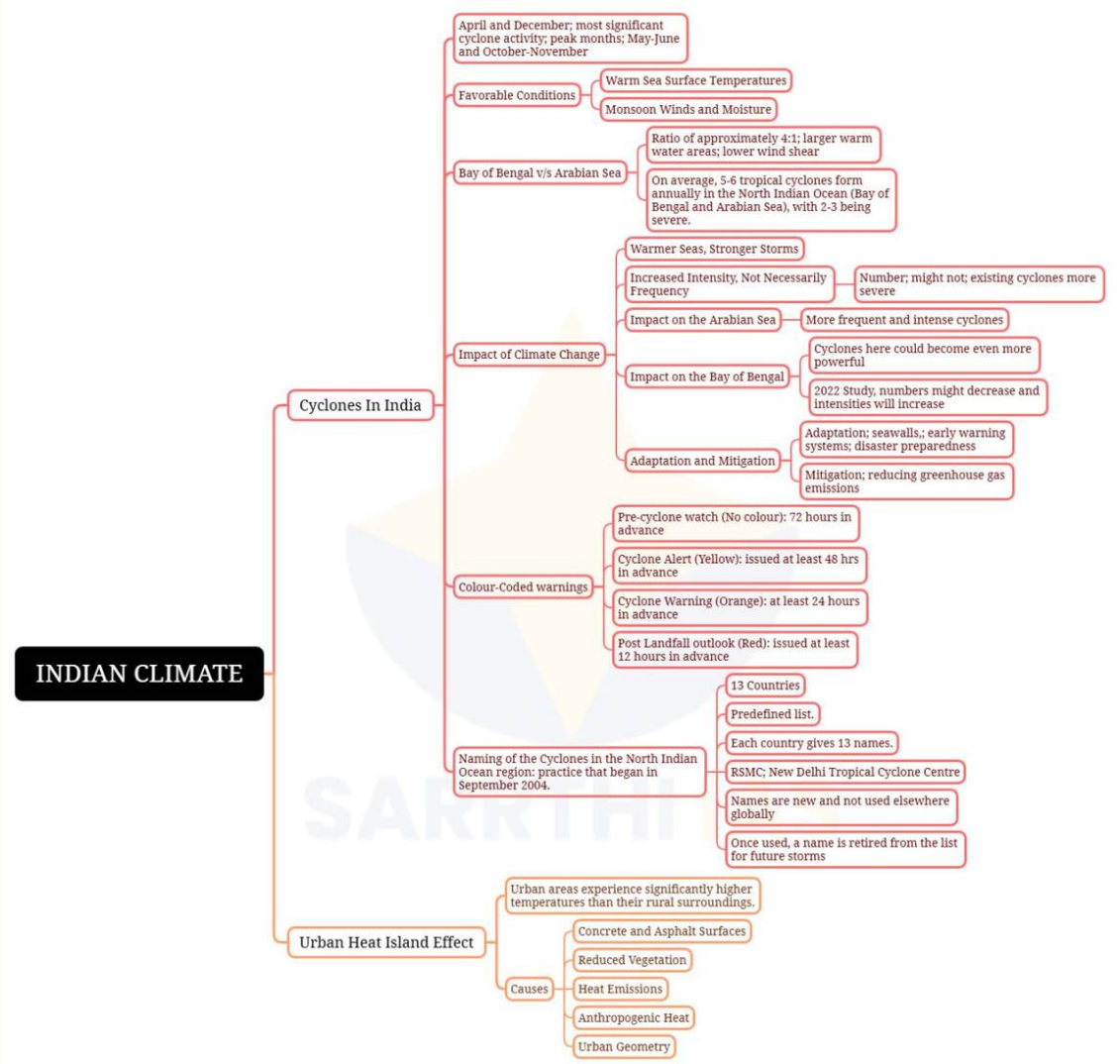

South-Eastward flowing rivers
Mahanadi River
Origin: near Sihawa in Raipur district
Drains into the Bay of Bengal.
River moves through CG and Odisha.
Riparian States: MP; Chhattisgarh; Odisha
Important Tributaries:
Left Bank: Ib, Mand, Hasdo, Sheonath
Right Bank: Ong, Jonk, Tel
Gorge near Tikkarpara
Delta
Godavari River
Longest peninsular river
Vridha Ganga and Dakshin Ganga
Origin: Trimbakeshwar,
States it moves through: Maharashtra, Telangana, Chhattisgarh, Andhra Pradesh, and Yanam (Puducherry). Riparian states include Odisha apart from the above.
Major Tributaries:
Right bank: Manjara, Pravara.
Left Bank: Purna, Pranhita (Wainganga+Penganga), Indravati, Sabari.
Below Rajahmundry; 2 distributaries: Gautami Godavari and Vashishtha Godavari
Delta at its mouth
Krishna River
Originates in Mahabaleshwar
Delta near its mouth.
Kaveri River
Perennial
Divides into 2 at Tiruchirapalli district: Kollidam & Kaveri
Other East flowing Peninsular rivers
Brahmani: confluence of S. Koel and Sankh; Eastern Ghats start from its South
Subarnarekha: Originates at the Ranchi plateau
Penneru: Originates from Nandi Durg in Karnataka
Ponniyar (Karnataka & Tamil Nadu) & Vaigai (Tamil Nadu);

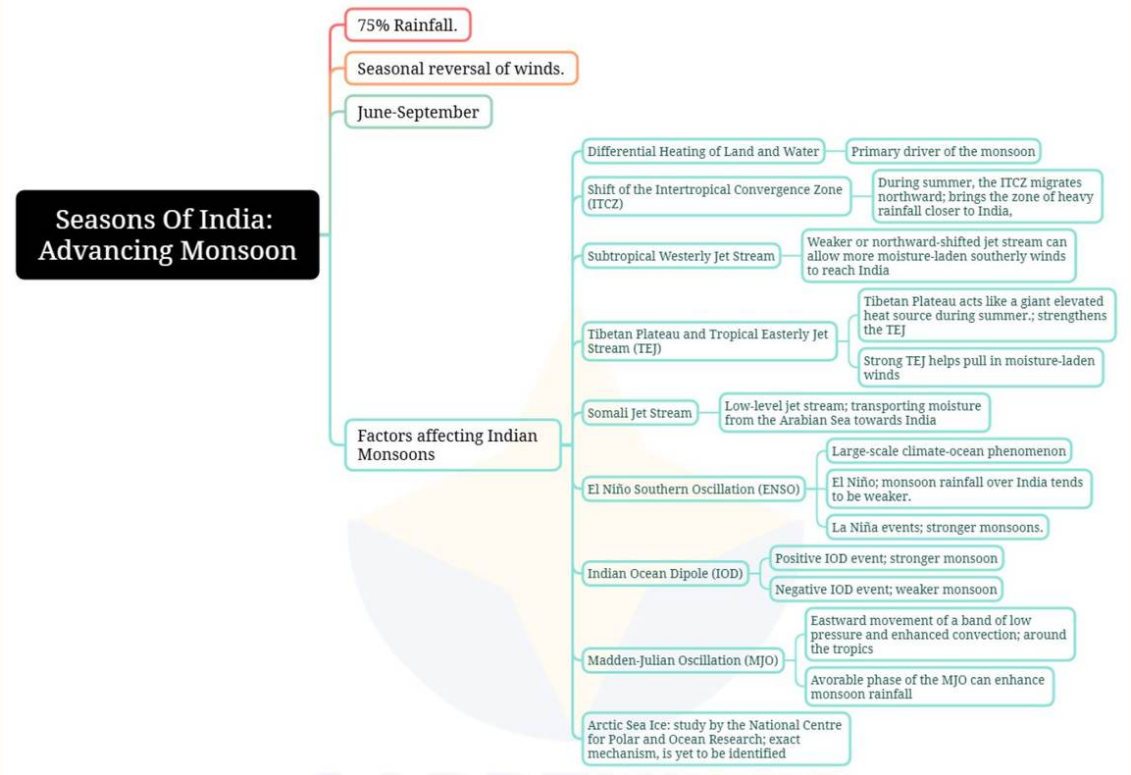

West Flowing Rivers
Narmada
Originates at the Amarkantak Plateau; a radial drainage
Mouth; Gulf of Khambhat
Flows in Rift Valley; so not many tributaries; Has short tributaries, that make a trellis pattern of drainage.
Major tributaries; Hiran, Kolar, Hatni; Choti Tawa
Slight meandering near Bharuch.
Tapi
Narmada's twin
Origin: Multai in M.P
Main Tributary: Purna, Girna and Panjara
Luni River
Salt (Lun) River
Source: West of Ajmer
Biggest river system of Rajasthan
Sagarmati; Luni after river Sarsuti joins it
Ephemeral river
Inland drainage system
Sabarmati
Origin is Aravallis; mouth is the Gulf of Khambat.
Gorge in Dharoi
Mahi
Origin: Vindhyas
Ends at the Gulf of Khambat.
Only river to cross the tropic of Cancer Twice.
Other short West flowing rivers
Not make any deltas.
Parallel drainage
Gujarat: Auranga, Par, Daman Ganga
Goa: Mandovi Zuari, Rachol
Maharashtra: Ulhas, Kalindi
Karnataka: Kalinadi, Sharavati (Jog Falls), Tadri, Netravathi
Kerala: Bharathapuzha, Periyar, Pamba

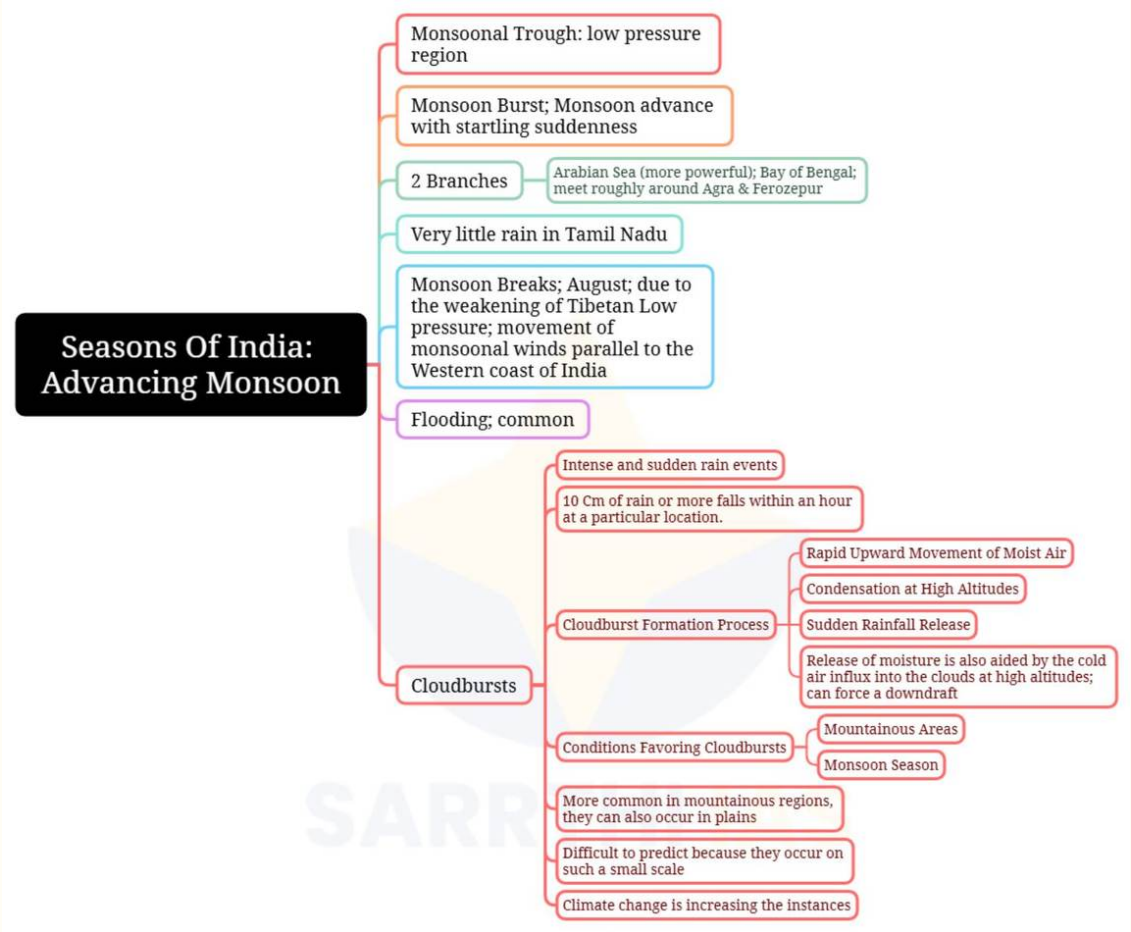

Inland Drainage In India
Do not drain into seas/oceans.
Ghaggar-Hakra river system
Luni and West Banas
Indian River Linking Project
History of the Project
Concept originated in the 19th century; Sir Arthur Cotton
National Water Development Agency (NWDA); established in 1982 to study; feasibility
About the Project
Two main components
Himalayan Rivers Development
Peninsular Rivers Development
Current Status
Remains under debate
Ken-Betwa River linking project,; received cabinet approval in 2014, but faces hurdles
Pattiseema Lift Irrigation Project; to connect the Godavari and Krishna Rivers; inaugurated in 2016; water from Godavari to Krishna; Polavaram Project.

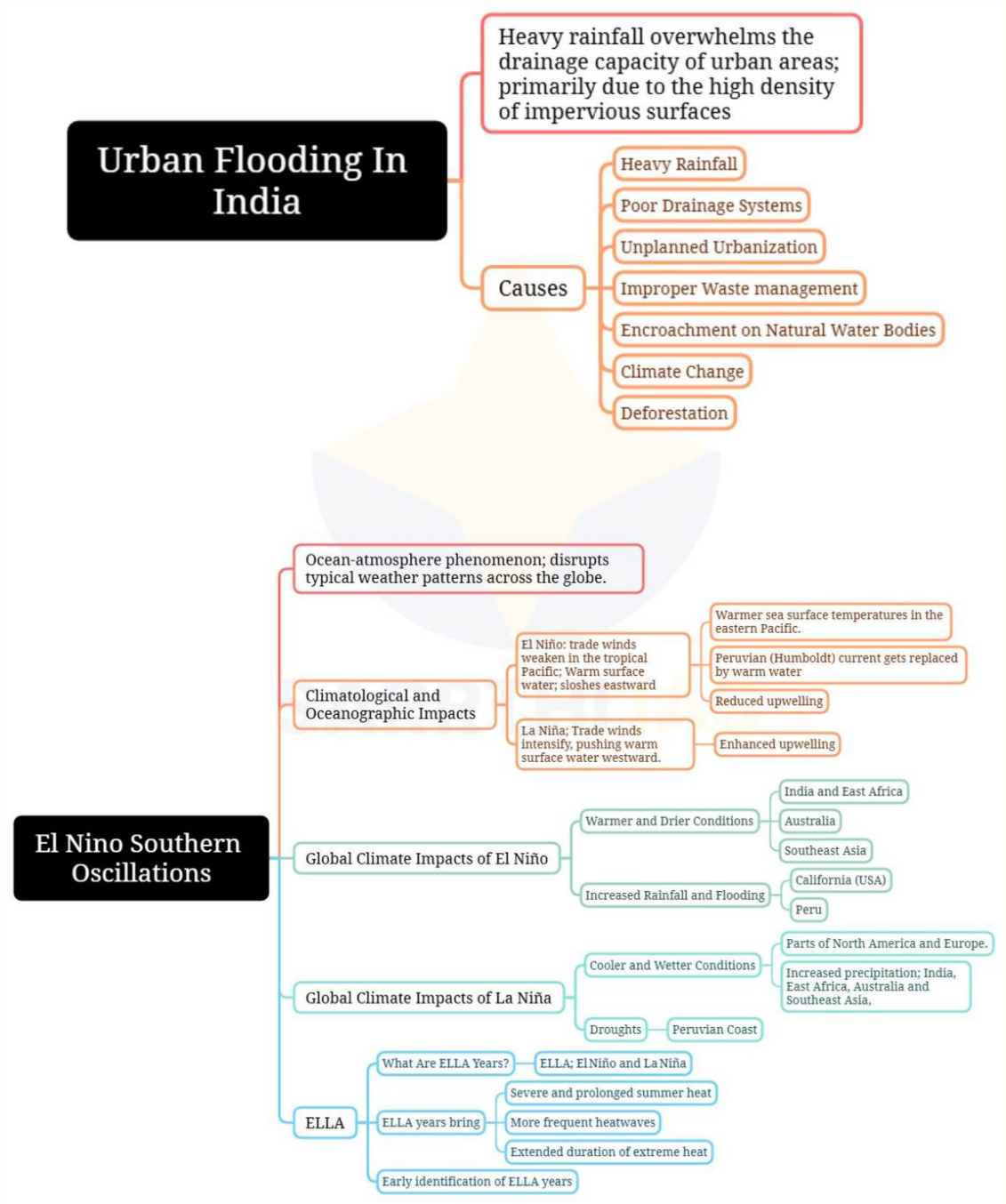
Indian Climate

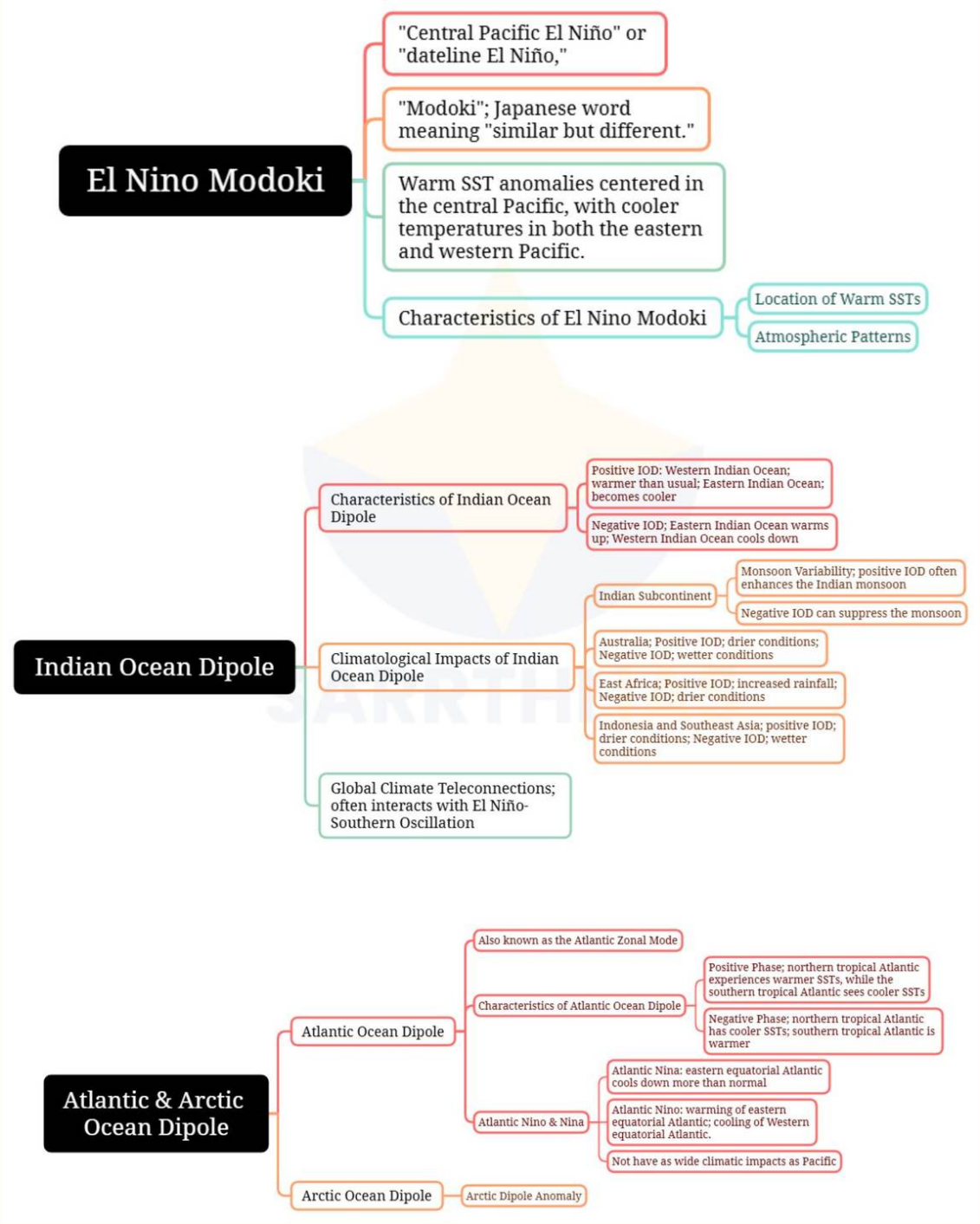


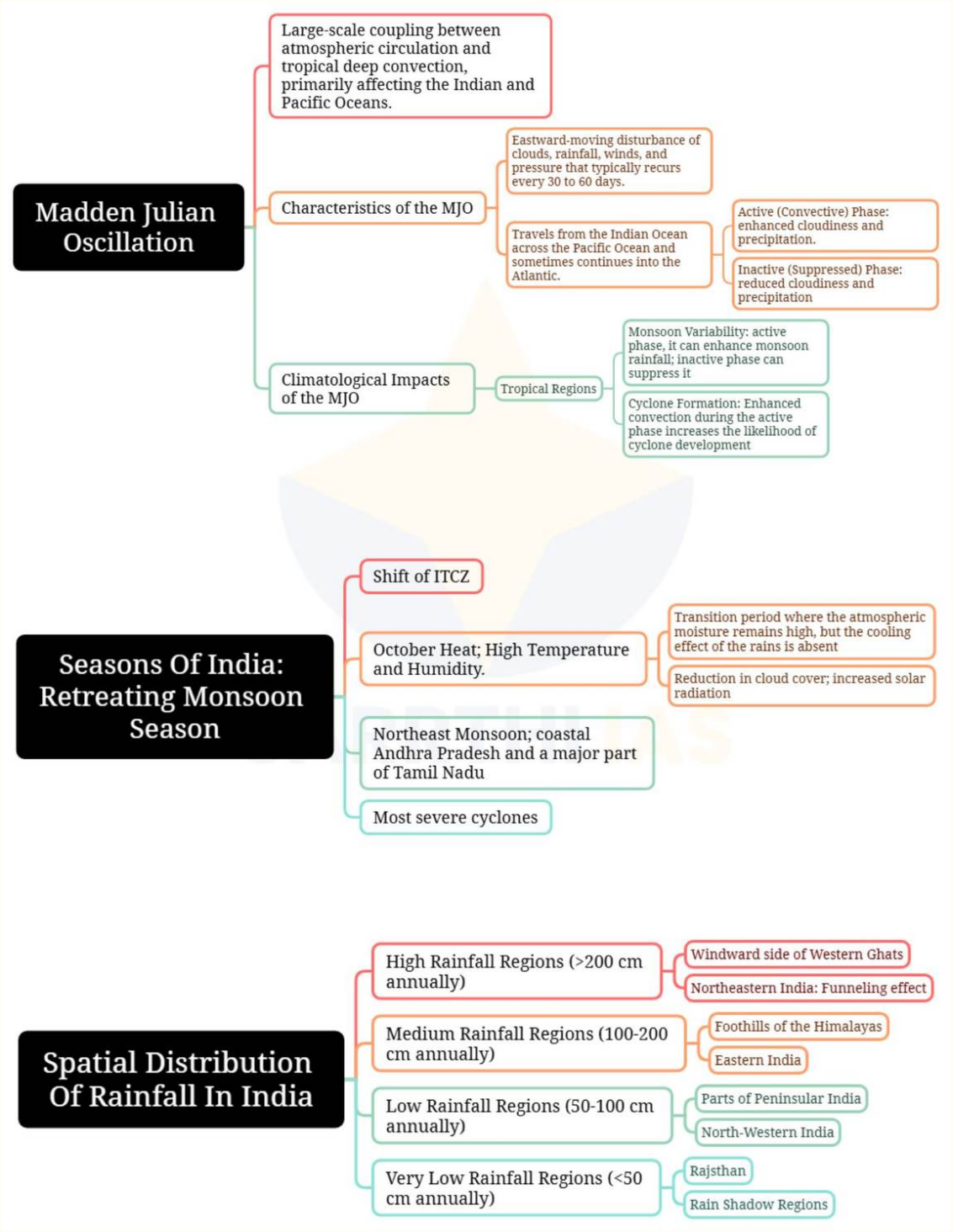

INDIAN CLIMATE
Cyclones In India
April and December; most significant cyclone activity; peak months; May-June and October-November
Favorable Conditions
Warm Sea Surface Temperatures
Monsoon Winds and Moisture
Bay of Bengal v/s Arabian Sea
Ratio of approximately 4:1; larger warm water areas; lower wind shear
On average, 5-6 tropical cyclones form annually in the North Indian Ocean (Bay of Bengal and Arabian Sea), with 2-3 being severe.
Impact of Climate Change
Warmer Seas, Stronger Storms
Increased Intensity, Not Necessarily Frequency
Number; might not; existing cyclones more severe
Impact on the Arabian Sea
More frequent and intense cyclones
Impact on the Bay of Bengal
Cyclones here could become even more powerful
2022 Study, numbers might decrease and intensities will increase
Adaptation and Mitigation
Adaptation; seawalls,; early warning systems; disaster preparedness
Mitigation; reducing greenhouse gas emissions
Colour-Coded warnings
Pre-cyclone watch (No colour): 72 hours in advance
Cyclone Alert (Yellow): issued at least 48 hrs in advance
Cyclone Warning (Orange): at least 24 hours in advance
Post Landfall outlook (Red): issued at least 12 hours in advance
Naming of the Cyclones in the North Indian Ocean region: practice that began in September 2004.
13 Countries
Predefined list.
Each country gives 13 names.
RSMC; New Delhi Tropical Cyclone Centre
Names are new and not used elsewhere globally
Once used, a name is retired from the list for future storms
Urban Heat Island Effect
Urban areas experience significantly higher temperatures than their rural surroundings.
Causes
Concrete and Asphalt Surfaces
Reduced Vegetation
Heat Emissions
Anthropogenic Heat
Urban Geometry


Seasons Of India: Advancing Monsoon
75% Rainfall.
Seasonal reversal of winds.
June-September
Factors affecting Indian Monsoons
Differential Heating of Land and Water
Primary driver of the monsoon
Shift of the Intertropical Convergence Zone (ITCZ)
During summer, the ITCZ migrates northward; brings the zone of heavy rainfall closer to India,
Subtropical Westerly Jet Stream
Weaker or northward-shifted jet stream can allow more moisture-laden southerly winds to reach India
Tibetan Plateau and Tropical Easterly Jet Stream (TEJ)
Tibetan Plateau acts like a giant elevated heat source during summer.; strengthens the TEJ
Strong TEJ helps pull in moisture-laden winds
Somali Jet Stream
Low-level jet stream; transporting moisture from the Arabian Sea towards India
El Niño Southern Oscillation (ENSO)
Large-scale climate-ocean phenomenon
El Niño; monsoon rainfall over India tends to be weaker.
La Niña events; stronger monsoons.
Indian Ocean Dipole (IOD)
Positive IOD event; stronger monsoon
Negative IOD event; weaker monsoon
Madden-Julian Oscillation (MJO)
Eastward movement of a band of low pressure and enhanced convection; around the tropics
Avorable phase of the MJO can enhance monsoon rainfall
Arctic Sea Ice: study by the National Centre for Polar and Ocean Research; exact mechanism, is yet to be identified

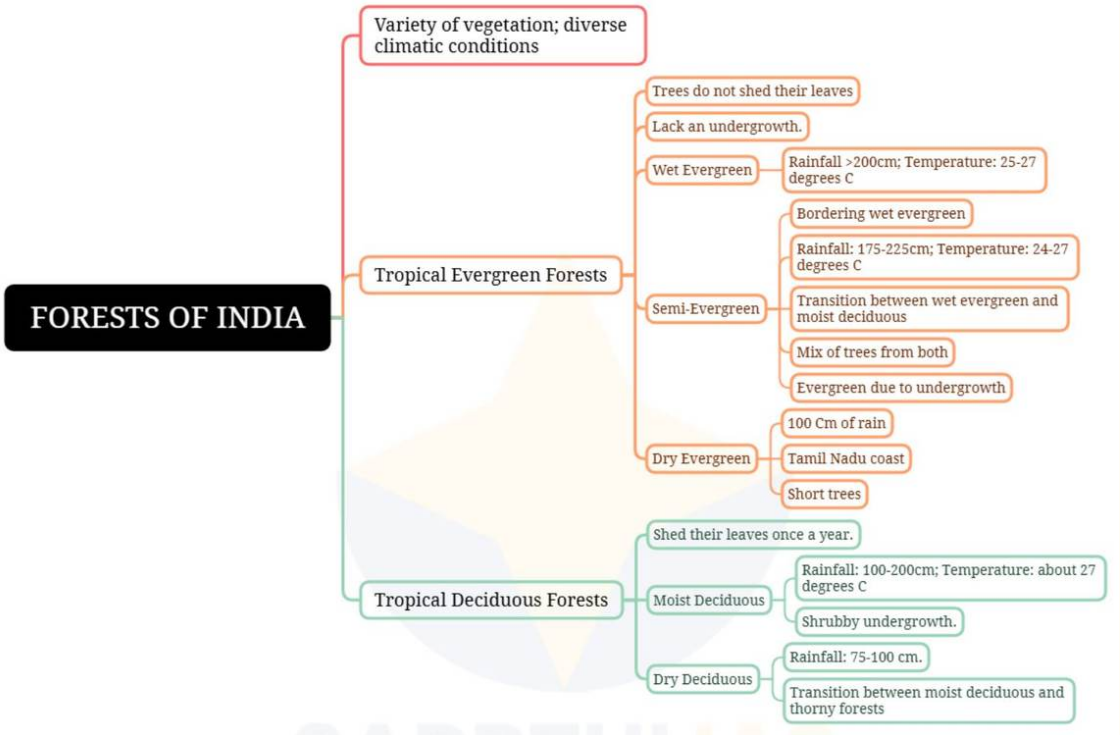

Seasons Of India: Advancing Monsoon
Monsoonal Trough: low pressure region
Monsoon Burst; Monsoon advance with startling suddenness
2 Branches
Arabian Sea (more powerful); Bay of Bengal; meet roughly around Agra & Ferozepur
Very little rain in Tamil Nadu
Monsoon Breaks; August; due to the weakening of Tibetan Low pressure; movement of monsoonal winds parallel to the Western coast of India
Flooding; common
Cloudbursts
Intense and sudden rain events
10 Cm of rain or more falls within an hour at a particular location.
Cloudburst Formation Process
Rapid Upward Movement of Moist Air
Condensation at High Altitudes
Sudden Rainfall Release
Release of moisture is also aided by the cold air influx into the clouds at high altitudes; can force a downdraft
Conditions Favoring Cloudbursts
Mountainous Areas
Monsoon Season
More common in mountainous regions, they can also occur in plains
Difficult to predict because they occur on such a small scale
Climate change is increasing the instances

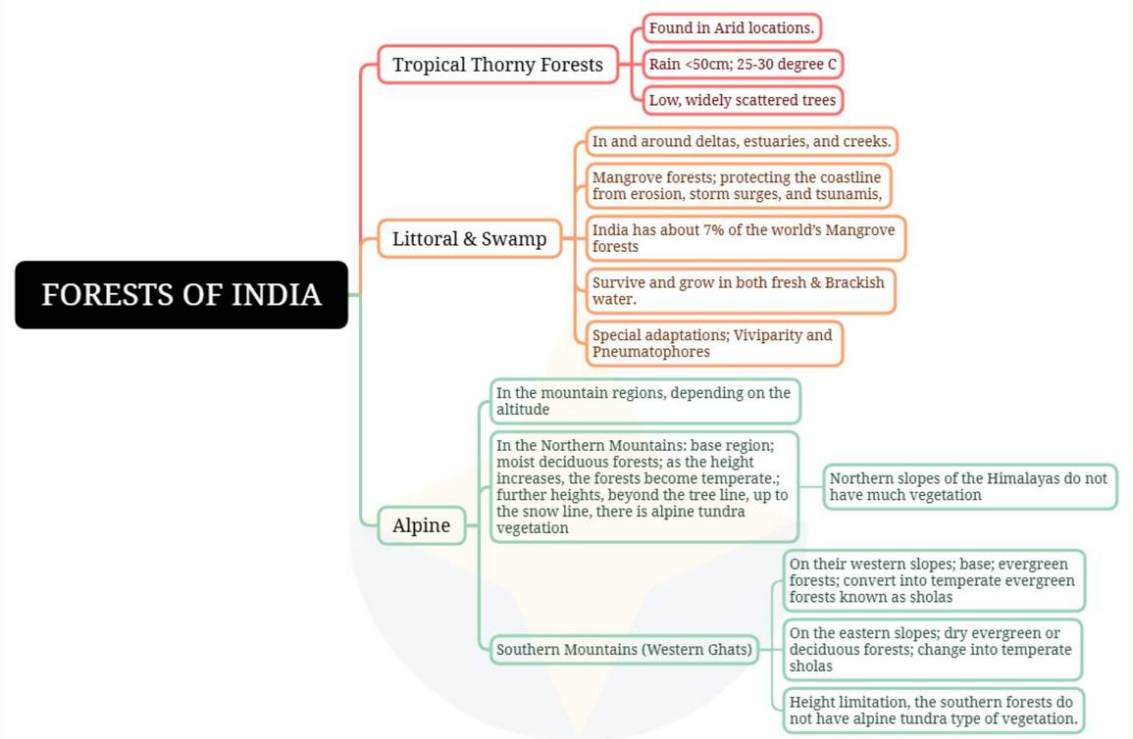

Urban Flooding In India
Heavy rainfall overwhelms the drainage capacity of urban areas; primarily due to the high density of impervious surfaces
Causes
Heavy Rainfall
Poor Drainage Systems
Unplanned Urbanization
Improper Waste management
Encroachment on Natural Water Bodies
Climate Change
Deforestation
El Nino Southern Oscillations
Ocean-atmosphere phenomenon; disrupts typical weather patterns across the globe.
Climatological and Oceanographic Impacts
El Niño: trade winds weaken in the tropical Pacific; Warm surface water; sloshes eastward
Warmer sea surface temperatures in the eastern Pacific.
Peruvian (Humboldt) current gets replaced by warm water
Reduced upwelling
La Niña; Trade winds intensify, pushing warm surface water westward.
Enhanced upwelling
Global Climate Impacts of El Niño
Warmer and Drier Conditions
India and East Africa
Australia
Southeast Asia
Increased Rainfall and Flooding
California (USA)
Peru
Global Climate Impacts of La Niña
Cooler and Wetter Conditions
Parts of North America and Europe.
Increased precipitation; India, East Africa, Australia and Southeast Asia,
Droughts
Peruvian Coast
ELLA
What Are ELLA Years?
ELLA; El Niño and La Niña
ELLA years bring
Severe and prolonged summer heat
More frequent heatwaves
Extended duration of extreme heat
Early identification of ELLA years

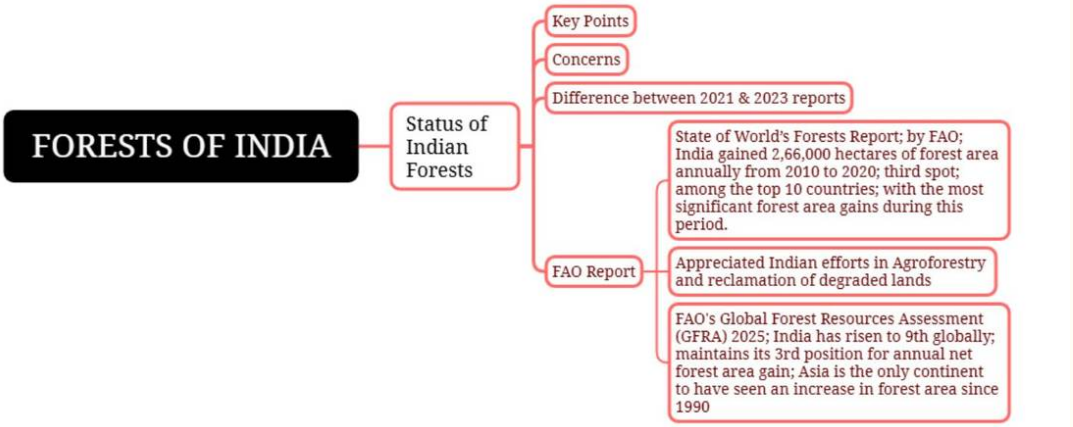

El Nino Modoki
"Central Pacific El Niño" or "dateline El Niño,"
"Modoki"; Japanese word meaning "similar but different."
Warm SST anomalies centered in the central Pacific, with cooler temperatures in both the eastern and western Pacific.
Characteristics of El Nino Modoki
Location of Warm SSTs
Atmospheric Patterns
Indian Ocean Dipole
Characteristics of Indian Ocean Dipole
Positive IOD: Western Indian Ocean; warmer than usual; Eastern Indian Ocean; becomes cooler
Negative IOD; Eastern Indian Ocean warms up; Western Indian Ocean cools down
Climatological Impacts of Indian Ocean Dipole
Indian Subcontinent
Monsoon Variability; positive IOD often enhances the Indian monsoon
Negative IOD can suppress the monsoon
Australia; Positive IOD; drier conditions; Negative IOD; wetter conditions
East Africa; Positive IOD; increased rainfall; Negative IOD; drier conditions
Indonesia and Southeast Asia; positive IOD; drier conditions; Negative IOD; wetter conditions
Global Climate Teleconnections; often interacts with El Niño-Southern Oscillation
Atlantic & Arctic Ocean Dipole
Atlantic Ocean Dipole
Also known as the Atlantic Zonal Mode
Characteristics of Atlantic Ocean Dipole
Positive Phase; northern tropical Atlantic experiences warmer SSTs, while the southern tropical Atlantic sees cooler SSTs
Negative Phase; northern tropical Atlantic has cooler SSTs; southern tropical Atlantic is warmer
Atlantic Nino & Nina
Atlantic Nina: eastern equatorial Atlantic cools down more than normal
Atlantic Nino: warming of eastern equatorial Atlantic; cooling of Western equatorial Atlantic.
Not have as wide climatic impacts as Pacific
Arctic Ocean Dipole
Arctic Dipole Anomaly

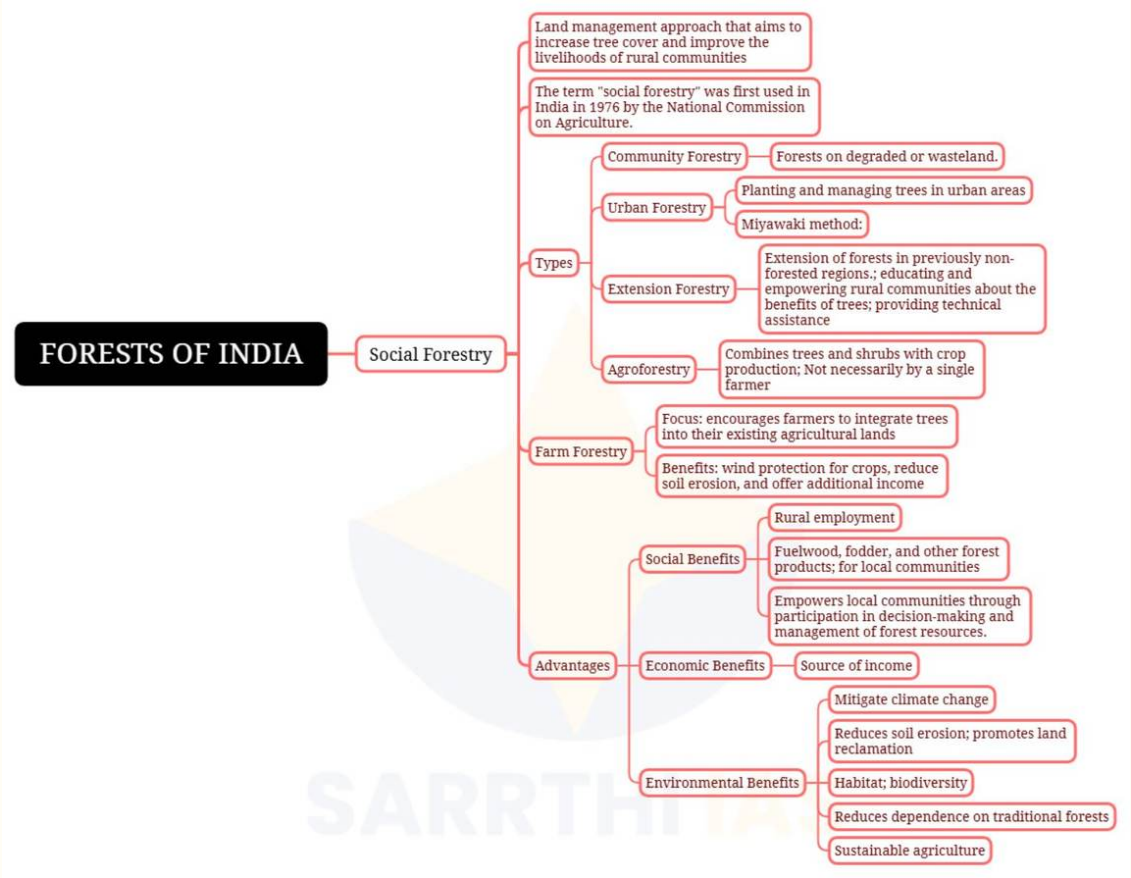

Madden Julian Oscillation
Large-scale coupling between atmospheric circulation and tropical deep convection, primarily affecting the Indian and Pacific Oceans.
Characteristics of the MJO
Eastward-moving disturbance of clouds, rainfall, winds, and pressure that typically recurs every 30 to 60 days.
Travels from the Indian Ocean across the Pacific Ocean and sometimes continues into the Atlantic.
Active (Convective) Phase: enhanced cloudiness and precipitation.
Inactive (Suppressed) Phase: reduced cloudiness and precipitation
Climatological Impacts of the MJO
Tropical Regions
Monsoon Variability: active phase, it can enhance monsoon rainfall; inactive phase can suppress it
Cyclone Formation: Enhanced convection during the active phase increases the likelihood of cyclone development
Seasons Of India: Retreating Monsoon Season
Shift of ITCZ
October Heat; High Temperature and Humidity.
Transition period where the atmospheric moisture remains high, but the cooling effect of the rains is absent
Reduction in cloud cover; increased solar radiation
Northeast Monsoon; coastal Andhra Pradesh and a major part of Tamil Nadu
Most severe cyclones
Spatial Distribution Of Rainfall In India
High Rainfall Regions (>200 cm annually)
Windward side of Western Ghats
Northeastern India: Funneling effect
Medium Rainfall Regions (100-200 cm annually)
Foothills of the Himalayas
Eastern India
Low Rainfall Regions (50-100 cm annually)
Parts of Peninsular India
North-Western India
Very Low Rainfall Regions (<50 cm annually)
Rajsthan
Rain Shadow Regions

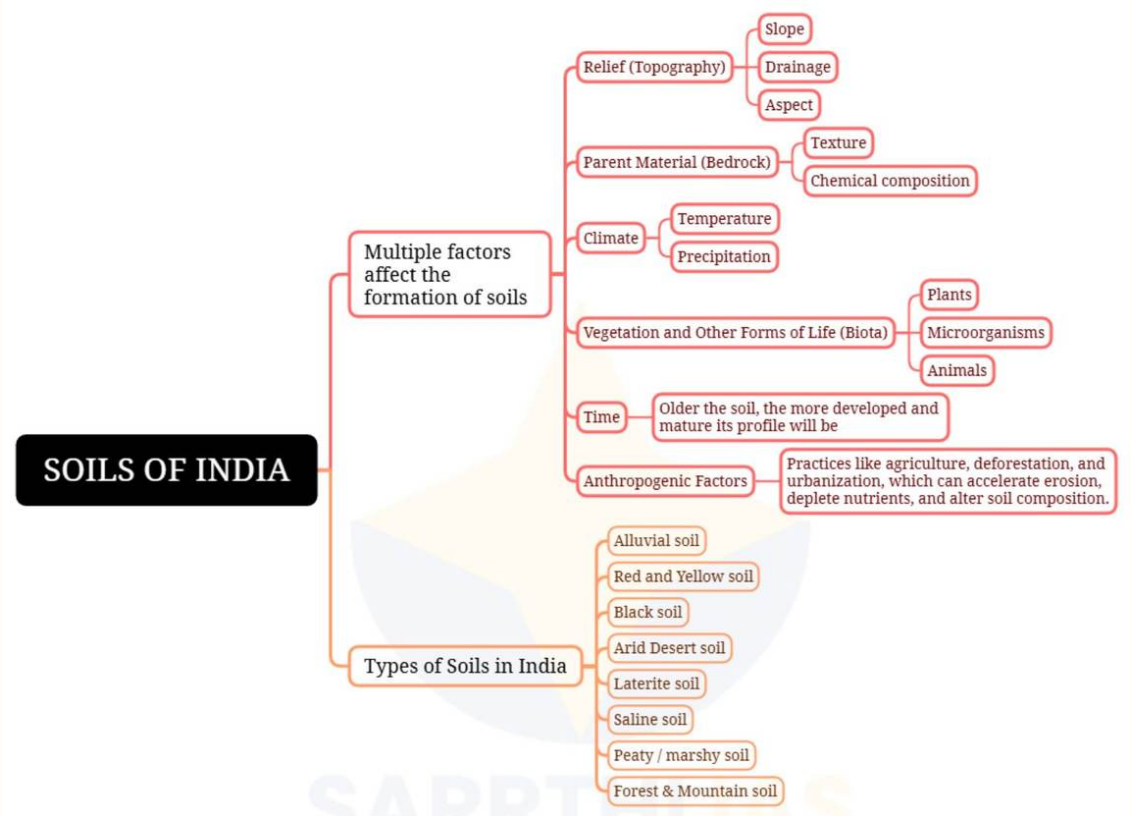
Indian Forest and Soils

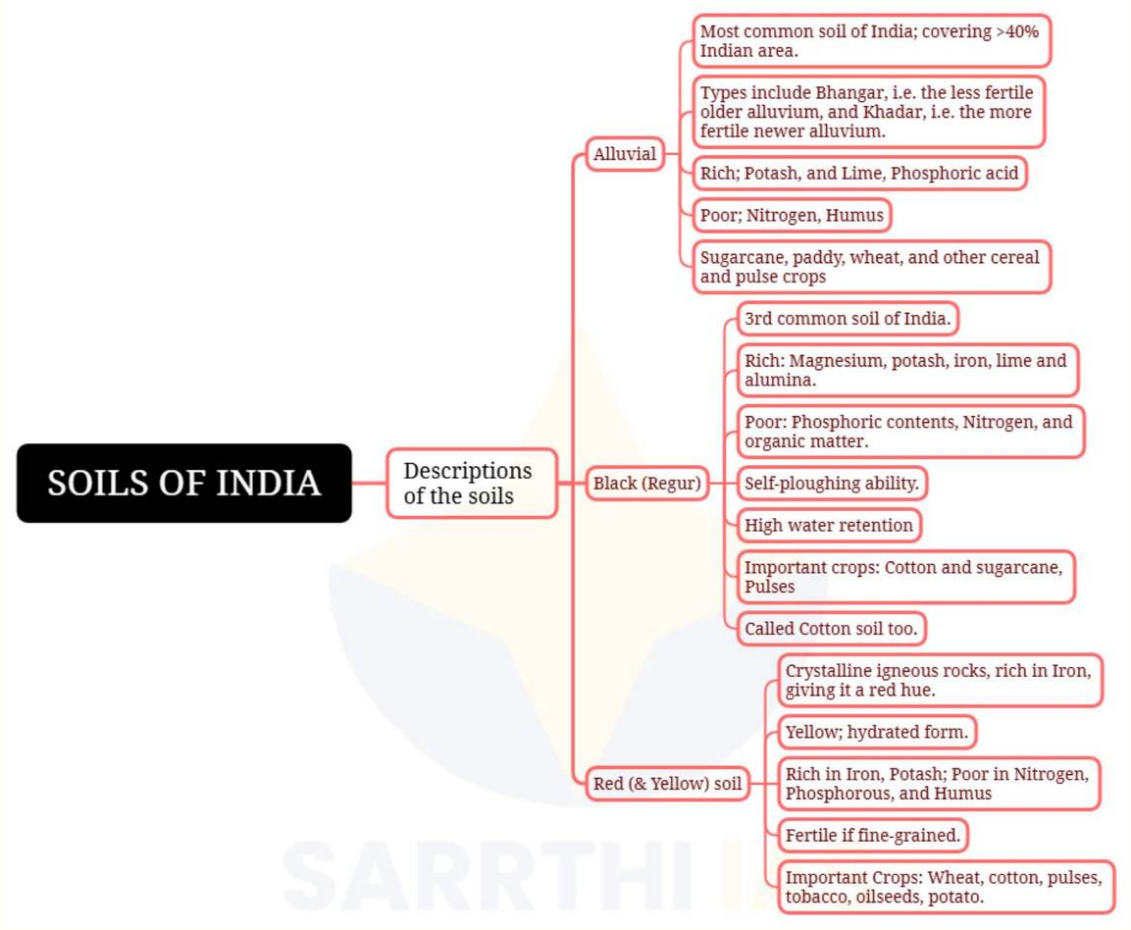



FORESTS OF INDIA
Tropical Thorny Forests
Found in Arid locations.
Rain <50cm; 25-30 degree C
Low, widely scattered trees
Littoral & Swamp
In and around deltas, estuaries, and creeks.
Mangrove forests; protecting the coastline from erosion, storm surges, and tsunamis,
India has about 7% of the world's Mangrove forests
Survive and grow in both fresh & Brackish water.
Special adaptations; Viviparity and Pneumatophores
Alpine
In the mountain regions, depending on the altitude
In the Northern Mountains: base region; moist deciduous forests; as the height increases, the forests become temperate.; further heights, beyond the tree line, up to the snow line, there is alpine tundra vegetation
Northern slopes of the Himalayas do not have much vegetation
Southern Mountains (Western Ghats)
On their western slopes; base; evergreen forests; convert into temperate evergreen forests known as sholas
On the eastern slopes; dry evergreen or deciduous forests; change into temperate sholas
Height limitation, the southern forests do not have alpine tundra type of vegetation.

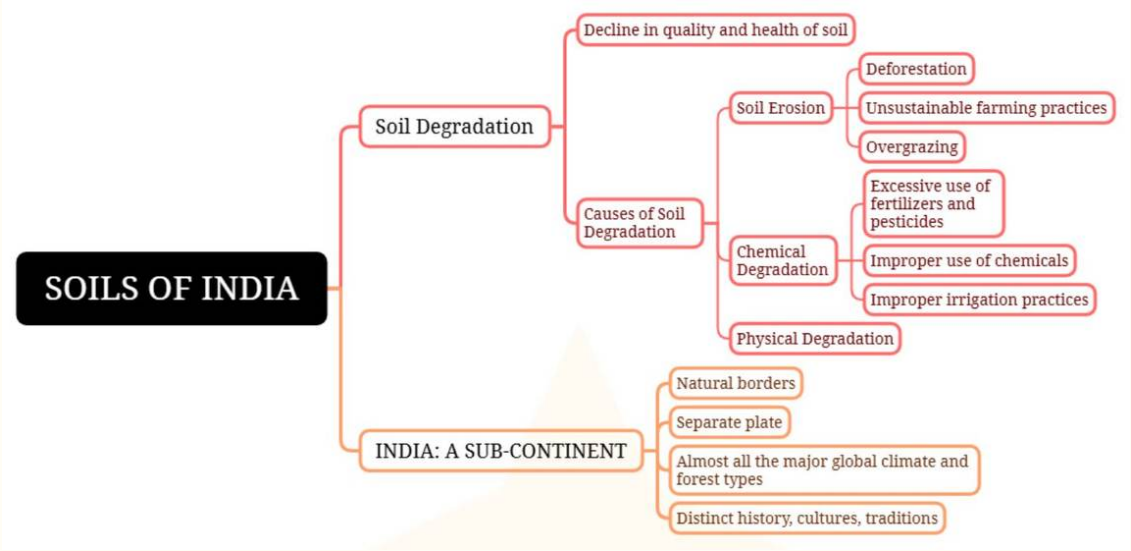

FORESTS OF INDIA
Status of Indian Forests
Key Points
Concerns
Difference between 2021 & 2023 reports
FAO Report
State of World's Forests Report; by FAO; India gained 2,66,000 hectares of forest area annually from 2010 to 2020; third spot; among the top 10 countries; with the most significant forest area gains during this period.
Appreciated Indian efforts in Agroforestry and reclamation of degraded lands
FAO's Global Forest Resources Assessment (GFRA) 2025; India has risen to 9th globally; maintains its 3rd position for annual net forest area gain; Asia is the only continent to have seen an increase in forest area since 1990


FORESTS OF INDIA
Social Forestry
Land management approach that aims to increase tree cover and improve the livelihoods of rural communities
The term "social forestry" was first used in India in 1976 by the National Commission on Agriculture.
Types
Community Forestry
Forests on degraded or wasteland.
Urban Forestry
Planting and managing trees in urban areas
Miyawaki method:
Extension Forestry
Extension of forests in previously non-forested regions.; educating and empowering rural communities about the benefits of trees; providing technical assistance
Agroforestry
Combines trees and shrubs with crop production; Not necessarily by a single farmer
Farm Forestry
Focus: encourages farmers to integrate trees into their existing agricultural lands
Benefits: wind protection for crops, reduce soil erosion, and offer additional income
Advantages
Social Benefits
Rural employment
Fuelwood, fodder, and other forest products; for local communities
Empowers local communities through participation in decision-making and management of forest resources.
Economic Benefits
Source of income
Environmental Benefits
Mitigate climate change
Reduces soil erosion; promotes land reclamation
Habitat; biodiversity
Reduces dependence on traditional forests
Sustainable agriculture


SOILS OF INDIA
Multiple factors affect the formation of soils
Relief (Topography)
Slope
Drainage
Aspect
Parent Material (Bedrock)
Texture
Chemical composition
Climate
Temperature
Precipitation
Vegetation and Other Forms of Life (Biota)
Plants
Microorganisms
Animals
Time
Older the soil, the more developed and mature its profile will be
Anthropogenic Factors
Practices like agriculture, deforestation, and urbanization, which can accelerate erosion, deplete nutrients, and alter soil composition.
Types of Soils in India
Alluvial soil
Red and Yellow soil
Black soil
Arid Desert soil
Laterite soil
Saline soil
Peaty / marshy soil
Forest & Mountain soil

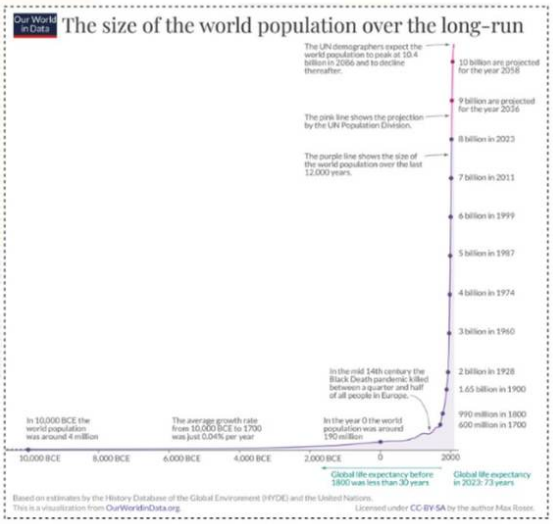

SOILS OF INDIA
Descriptions of the soils
Alluvial
Most common soil of India; covering >40% Indian area.
Types include Bhangar, i.e. the less fertile older alluvium, and Khadar, i.e. the more fertile newer alluvium.
Rich; Potash, and Lime, Phosphoric acid
Poor; Nitrogen, Humus
Sugarcane, paddy, wheat, and other cereal and pulse crops
Black (Regur)
3rd common soil of India.
Rich: Magnesium, potash, iron, lime and alumina.
Poor: Phosphoric contents, Nitrogen, and organic matter.
Self-ploughing ability.
High water retention
Important crops: Cotton and sugarcane, Pulses
Called Cotton soil too.
Red (& Yellow) soil
Crystalline igneous rocks, rich in Iron, giving it a red hue.
Yellow; hydrated form.
Rich in Iron, Potash; Poor in Nitrogen, Phosphorous, and Humus
Fertile if fine-grained.
Important Crops: Wheat, cotton, pulses, tobacco, oilseeds, potato.

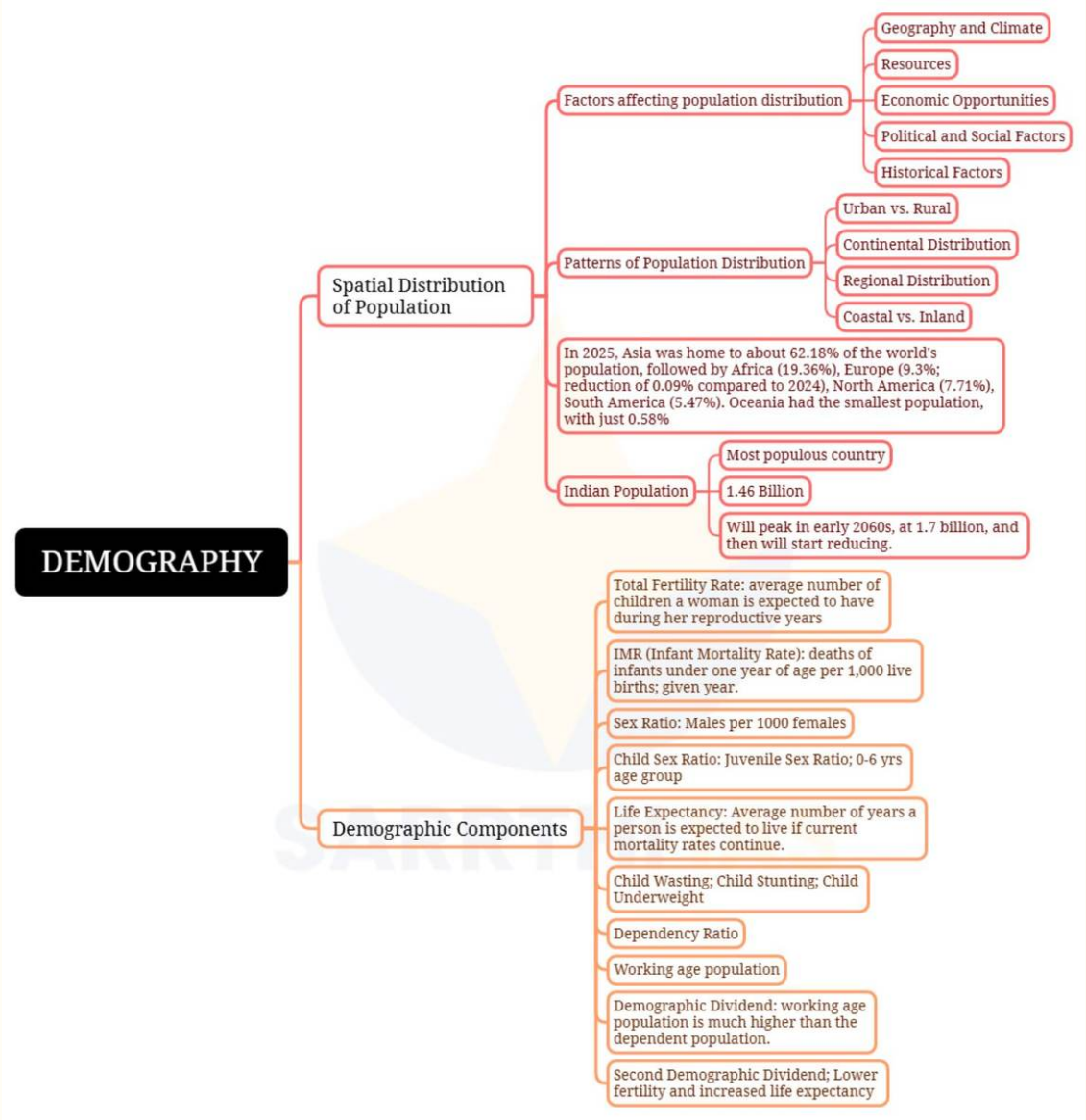

SOILS OF INDIA
Descriptions of the soils
Laterite Soil
In the wetter margins, the red soil converts into Laterite soil.
Acidic
Poor: organic matter, nitrogen, phosphate, and calcium.
Rich: iron oxide and potash are in excess.
Tea, coffee, Cashew
Arid Soils
Sandy & Saline
Lack of moisture, Humus, and Nitrogen; Phosphorous is adequate.
Include Calcium Kankar nodules
Can be cultivated if irrigation is available
Millets, Fodder crops, some beans (Guar Beans), wheat if irrigated
Saline Soils
Usara soil.
Infertile, with a large proportion of salts
In some parts, coconuts, and salt resistant fodder crops; White Revolution Impact
Peaty/Marshy
Areas of heavy rainfall and high humidity.
Rich in humus; organic content may go even up to 40-50 percent
If properly drained and fertilized; crops like rice
Forest & Mountain Soil
Forest areas, with adequate rainfall
Vary in their structure and texture depending upon the mountain environment where they are formed.
Valley sides- loamy and silty
Upper slopes- coarse-grained

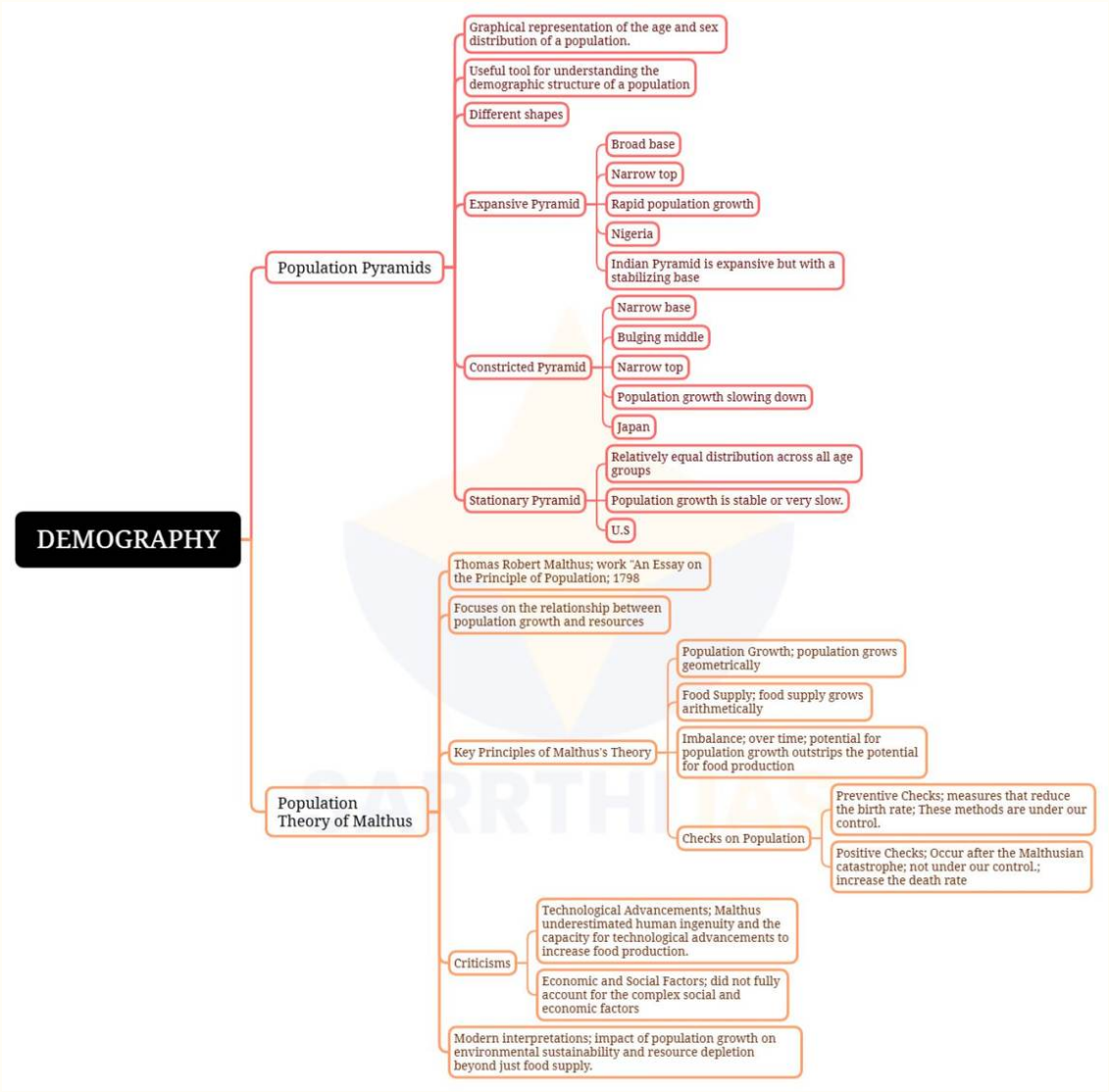

SOILS OF INDIA
Soil Degradation
Decline in quality and health of soil
Causes of Soil Degradation
Soil Erosion
Deforestation
Unsustainable farming practices
Overgrazing
Chemical Degradation
Excessive use of fertilizers and pesticides
Improper use of chemicals
Improper irrigation practices
Physical Degradation
INDIA: A SUB-CONTINENT
Natural borders
Separate plate
Almost all the major global climate and forest types
Distinct history, cultures, traditions

DEMOGRAPHY

- Demography is the study of populations, including their size, structure, distribution, and dynamics.
- Factors Affecting Population Growth
  - Birth Rate: live births per thousand people in a given population over a specific period
  - Death Rate: number of deaths per thousand people in a given population over a specific period
  - Migration: movement of people from one place to another; Immigration (inward migration); emigration (outward migration)
  - Fertility Rate: average number of children born to women of childbearing age in a population.
  - Age Structure; distribution of individuals in different age groups within a population
  - Economic and Social Factors
- Temporal Distribution of Population




DEMOGRAPHY
Spatial Distribution of Population
Factors affecting population distribution
Geography and Climate
Resources
Economic Opportunities
Political and Social Factors
Historical Factors
Patterns of Population Distribution
Urban vs. Rural
Continental Distribution
Regional Distribution
Coastal vs. Inland
In 2025, Asia was home to about 62.18% of the world's population, followed by Africa (19.36%), Europe (9.3%; reduction of 0.09% compared to 2024), North America (7.71%), South America (5.47%). Oceania had the smallest population, with just 0.58%
Indian Population
Most populous country
1.46 Billion
Will peak in early 2060s, at 1.7 billion, and then will start reducing.
Demographic Components
Total Fertility Rate: average number of children a woman is expected to have during her reproductive years
IMR (Infant Mortality Rate): deaths of infants under one year of age per 1,000 live births; given year.
Sex Ratio: Males per 1000 females
Child Sex Ratio: Juvenile Sex Ratio; 0-6 yrs age group
Life Expectancy: Average number of years a person is expected to live if current mortality rates continue.
Child Wasting; Child Stunting; Child Underweight
Dependency Ratio
Working age population
Demographic Dividend: working age population is much higher than the dependent population.
Second Demographic Dividend; Lower fertility and increased life expectancy


DEMOGRAPHY
Population Pyramids
Graphical representation of the age and sex distribution of a population.
Useful tool for understanding the demographic structure of a population
Different shapes
Expansive Pyramid
Broad base
Narrow top
Rapid population growth
Nigeria
Indian Pyramid is expansive but with a stabilizing base
Constricted Pyramid
Narrow base
Bulging middle
Narrow top
Population growth slowing down
Japan
Stationary Pyramid
Relatively equal distribution across all age groups
Population growth is stable or very slow.
U.S
Population Theory of Malthus
Thomas Robert Malthus; work "An Essay on the Principle of Population; 1798
Focuses on the relationship between population growth and resources
Key Principles of Malthus's Theory
Population Growth; population grows geometrically
Food Supply; food supply grows arithmetically
Imbalance; over time; potential for population growth outstrips the potential for food production
Checks on Population
Preventive Checks; measures that reduce the birth rate; These methods are under our control.
Positive Checks; Occur after the Malthusian catastrophe; not under our control.; increase the death rate
Criticisms
Technological Advancements; Malthus underestimated human ingenuity and the capacity for technological advancements to increase food production.
Economic and Social Factors; did not fully account for the complex social and economic factors
Modern interpretations; impact of population growth on environmental sustainability and resource depletion beyond just food supply.

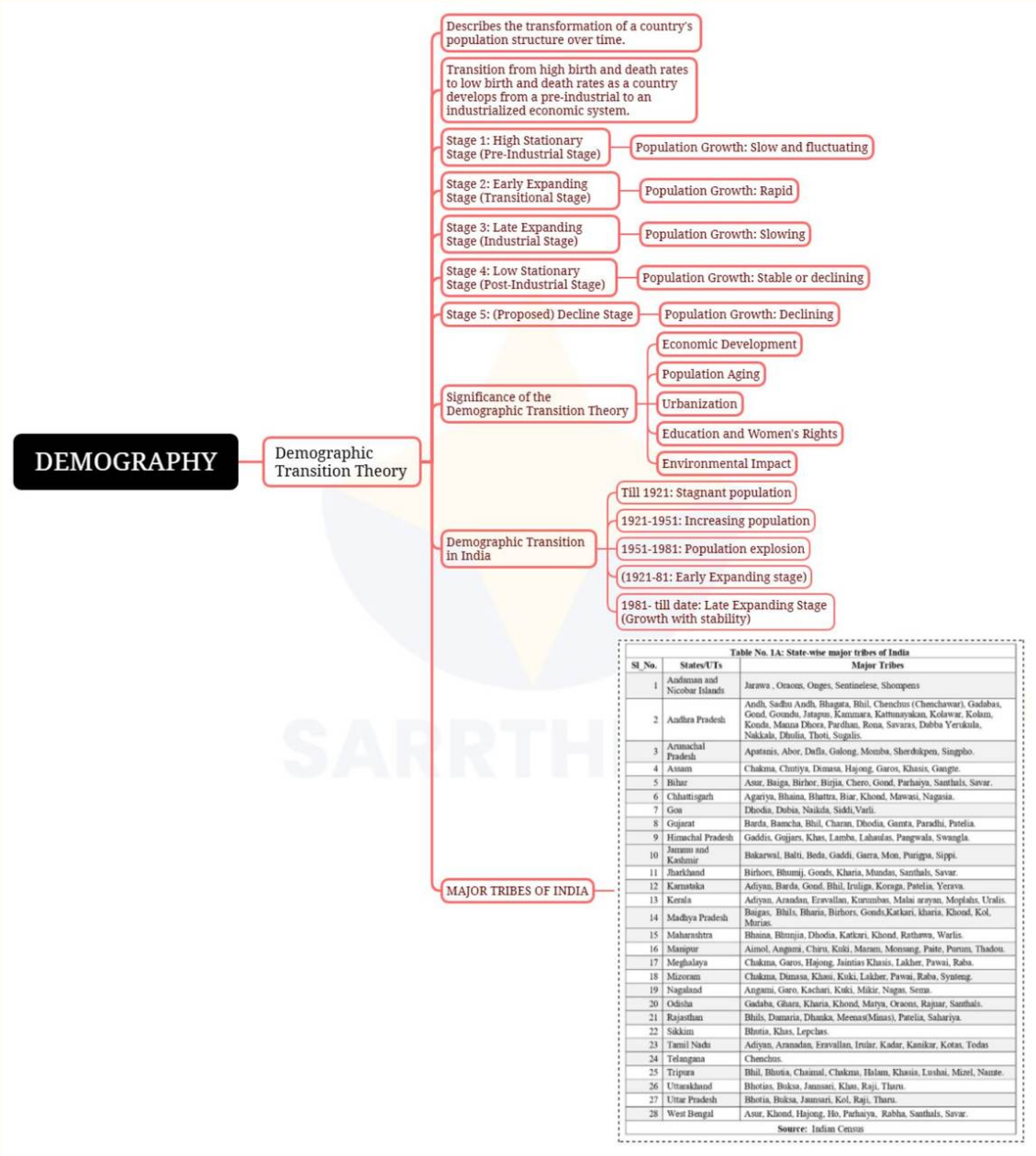
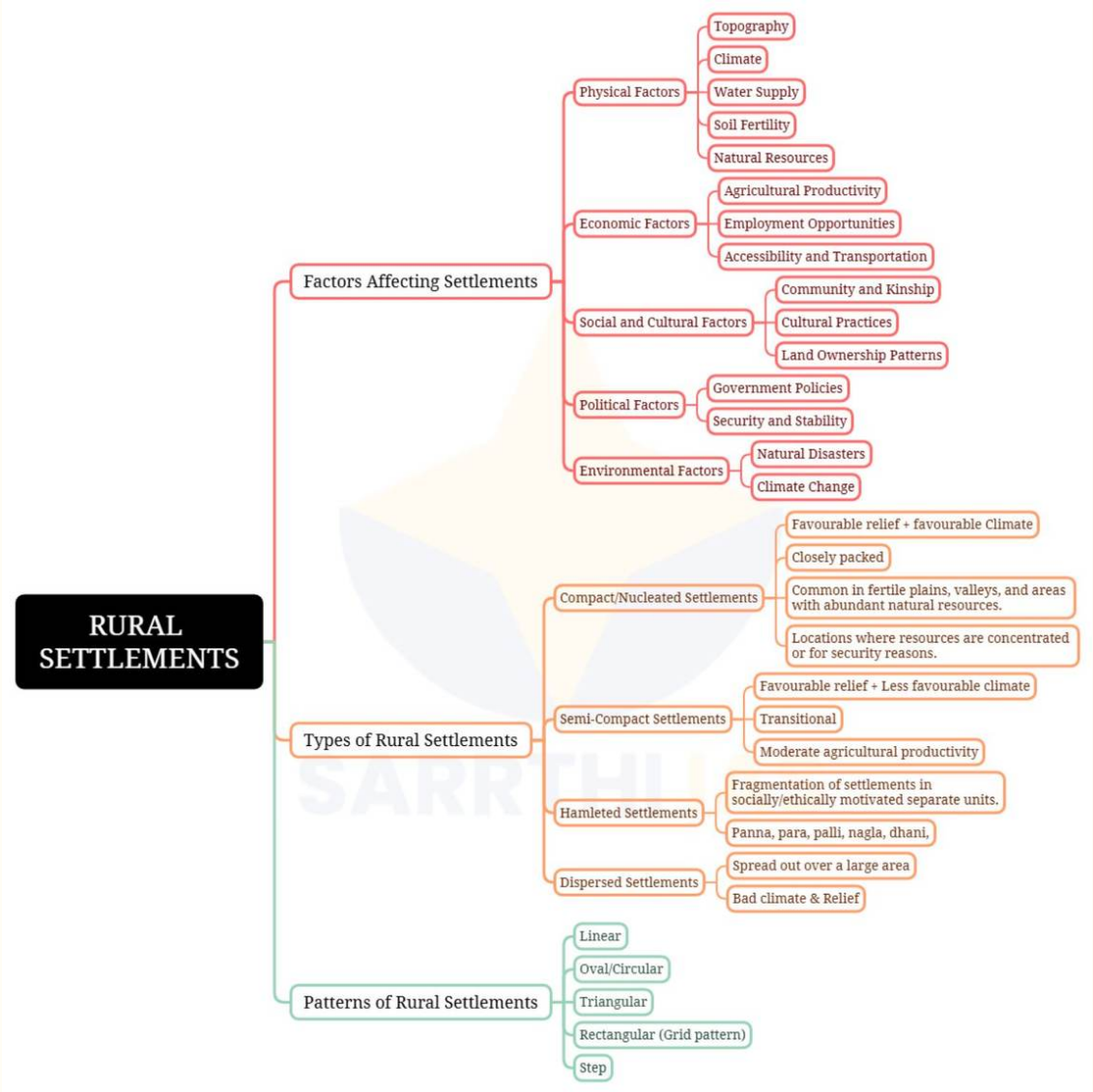
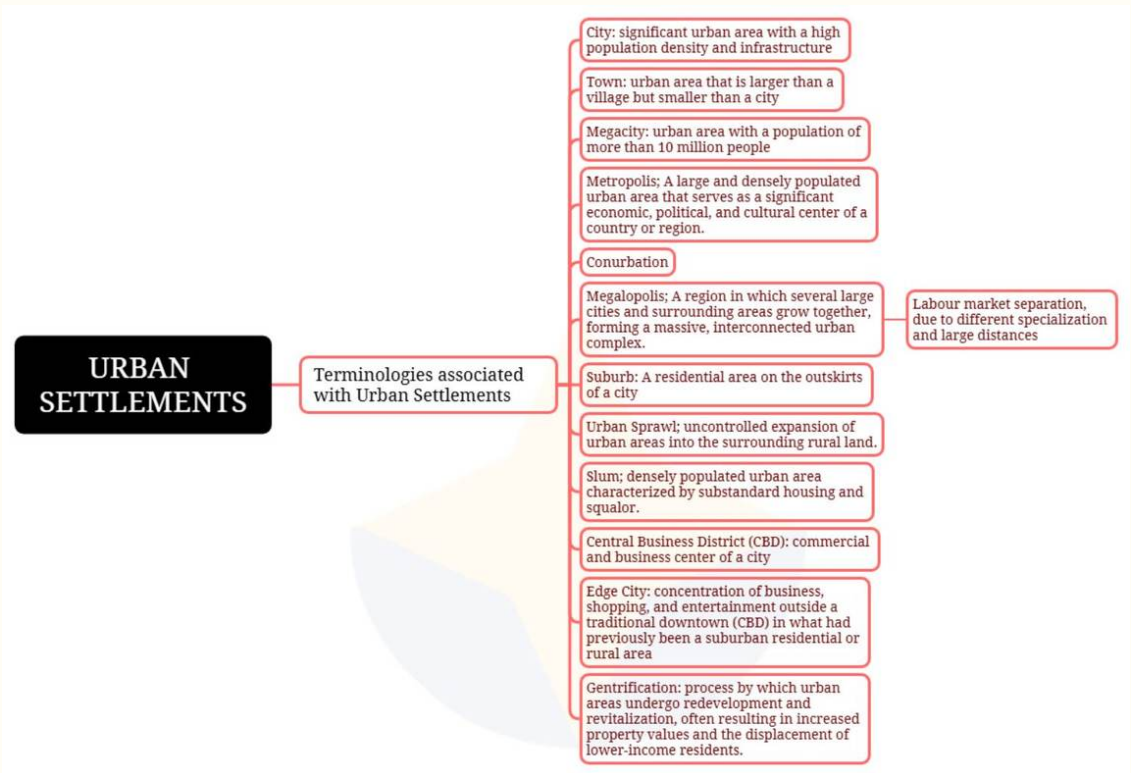


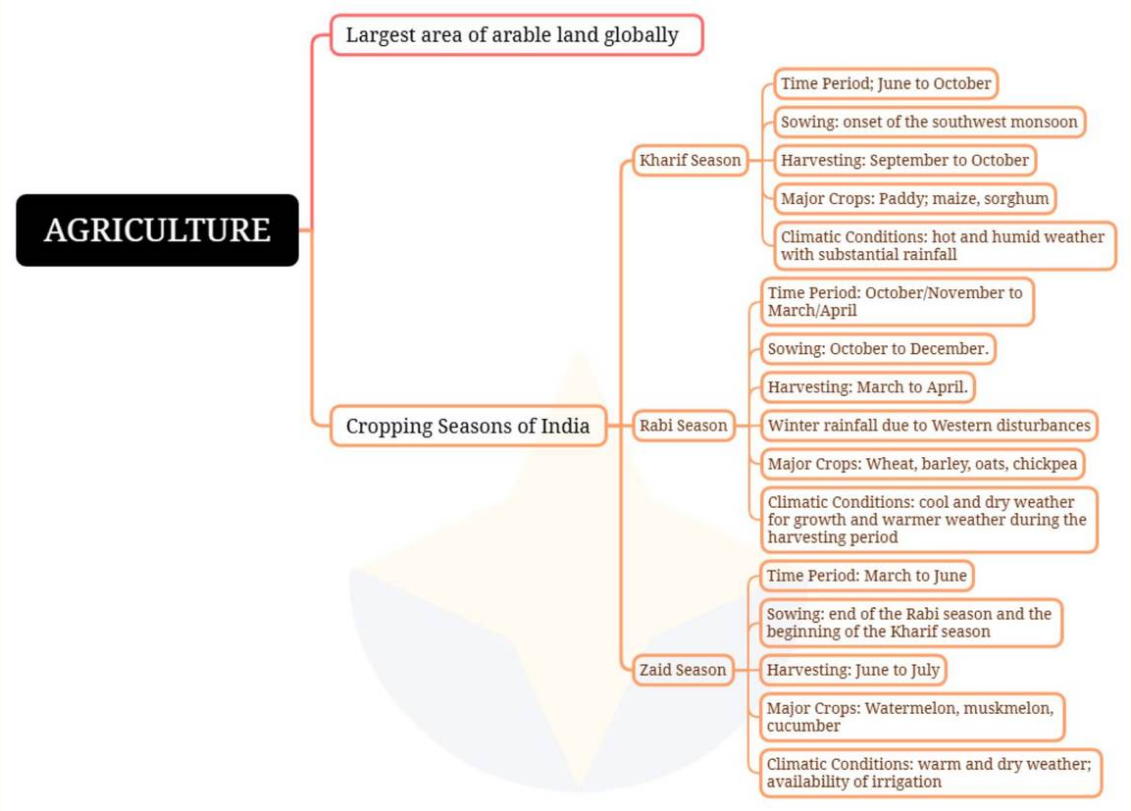
Table No. 1A: State-wise major tribes of India

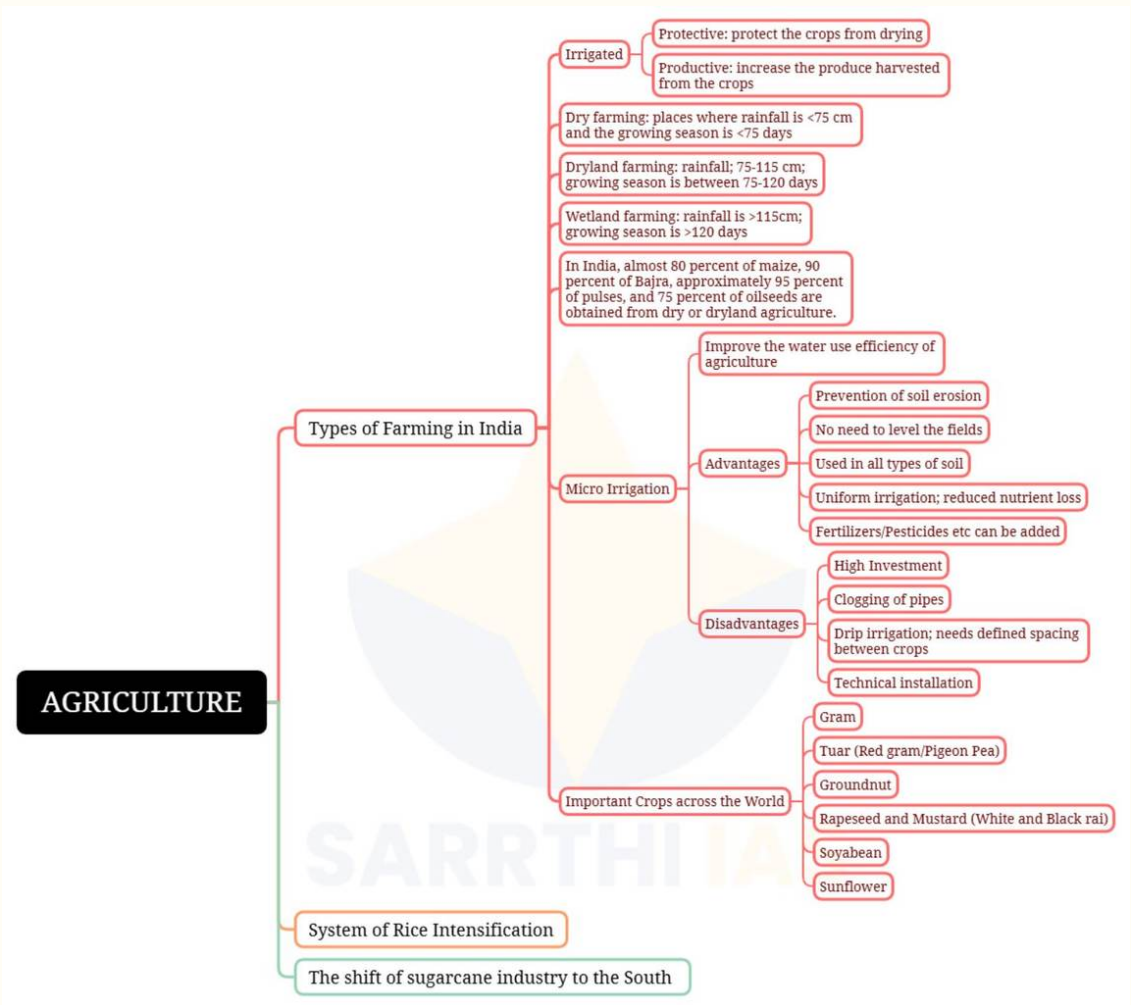
Sl. No.	States/UTs	Major Tribes
1	Andaman and Nicobar Islands	Jarawa , Oraons, Onges, Sentinelese, Shompens
2	Andhra Pradesh	Andh, Sadhu Andh, Bhagata, Bhil, Chenchus (Chenchawar), Gadabas, Gond, Goundu, Jatapus, Kammara, Kattunayakan, Kolawar, Kolam, Konda, Manna Dhora, Pardhan, Rona, Savaras, Dubba Yerukula, Nakkala, Dhulia, Thoti, Sugalis.
3	Arunachal Pradesh	Apatanis, Abor, Dafla, Galong, Monba, Sherdukpen, Singpho.
4	Assam	Chakma, Chutiya, Dimasa, Hajong, Garos, Khasis, Gangte.
5	Bihar	Asur, Baiga, Birhor, Birjia, Chero, Gond, Parhaiya, Santhals, Savar.
6	Chhattisgarh	Agariya, Bhaina, Bhattra, Biar, Khond, Mawasi, Nagasia.
7	Goa	Dhodia, Dubia, Naikda, Siddi,Varli.
8	Gujarat	Barda, Bamcha, Bhil, Charan, Dhodia, Gamta, Paradhi, Patelia.
9	Himachal Pradesh	Gaddis, Gujars, Khas, Lamba, Lahaulas, Pangwala, Swangla.
10	Jammu and Kashmir	Bakarwal, Balti, Beda, Gaddi, Garra, Mon, Purigpa, Sippi.
11	Jharkhand	Birhors, Bhumij, Gonds, Kharia, Mundas, Santhals, Savar.
12	Karnataka	Adiyan, Barda, Gond, Bhil, Iruliga, Koraga, Patelia, Yerava.
13	Kerala	Adiyan, Arandan, Eravallan, Kurumbas, Malai arayan, Moplahs, Uralis.
14	Madhya Pradesh	Baigas, Bhils, Bharia, Birhors, Gonds,Katkari, kharia, Khond, Kol, Murias.
15	Maharashtra	Bhaina, Bhunjia, Dhodia, Katkari, Khond, Rathawa, Warlis.
16	Manipur	Aimol, Angami, Chiru, Kuki, Maram, Monsang, Paite, Purum, Thadou.
17	Meghalaya	Chakma, Garos, Hajong, Jaintias Khasis, Lakher, Pawai, Raba.
18	Mizoram	Chakma, Dimasa, Khasi, Kuki, Lakher, Pawai, Raba, Synteng.
19	Nagaland	Angami, Garo, Kachari, Kuki, Mikir, Nagas, Sema.
20	Odisha	Gadaba, Ghara, Kharia, Khond, Marya, Oraons, Rajuar, Santhals.
21	Rajasthan	Bhils, Damaria, Dhanka, Meenas(Minas), Patelia, Sahariya.
22	Sikkim	Bhutia, Khas, Lepchas.
23	Tamil Nadu	Adiyan, Aranadan, Eravallan, Irular, Kadar, Kanikar, Kotas, Todas
24	Telangana	Chenchus.
25	Tripura	Bhil, Bhutia, Chaimal, Chakma, Halam, Khasia, Lushai, Mizel, Naute.
26	Uttarakhand	Bhotias, Buksa, Jaunsari, Khas, Raji, Tharu.
27	Uttar Pradesh	Bhotia, Buksa, Jaunsari, Kol, Raji, Tharu.
28	West Bengal	Asur, Khond, Hajong, Ho, Parhaiya, Rabha, Santhals, Savar.

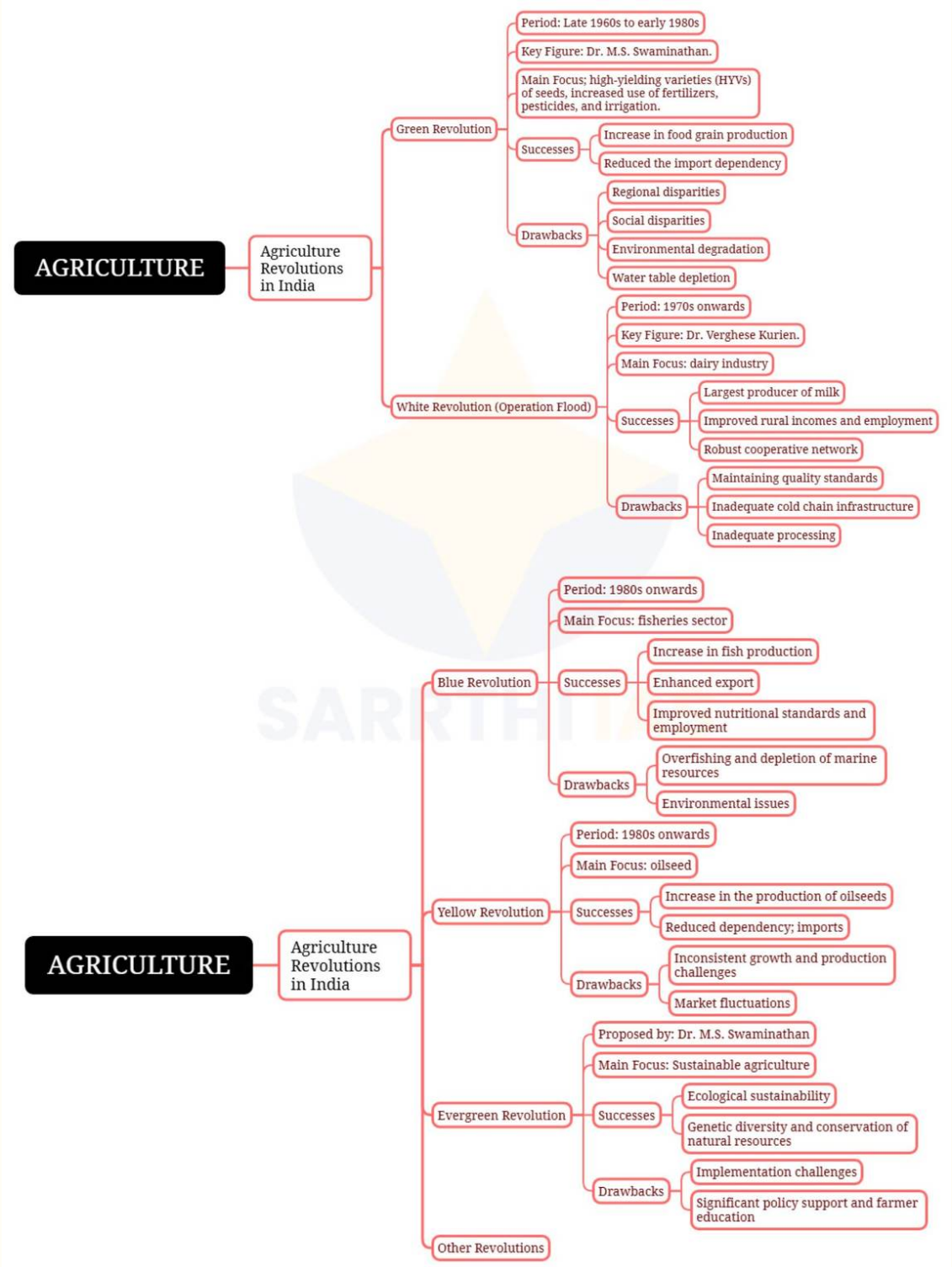
Source: Indian Census

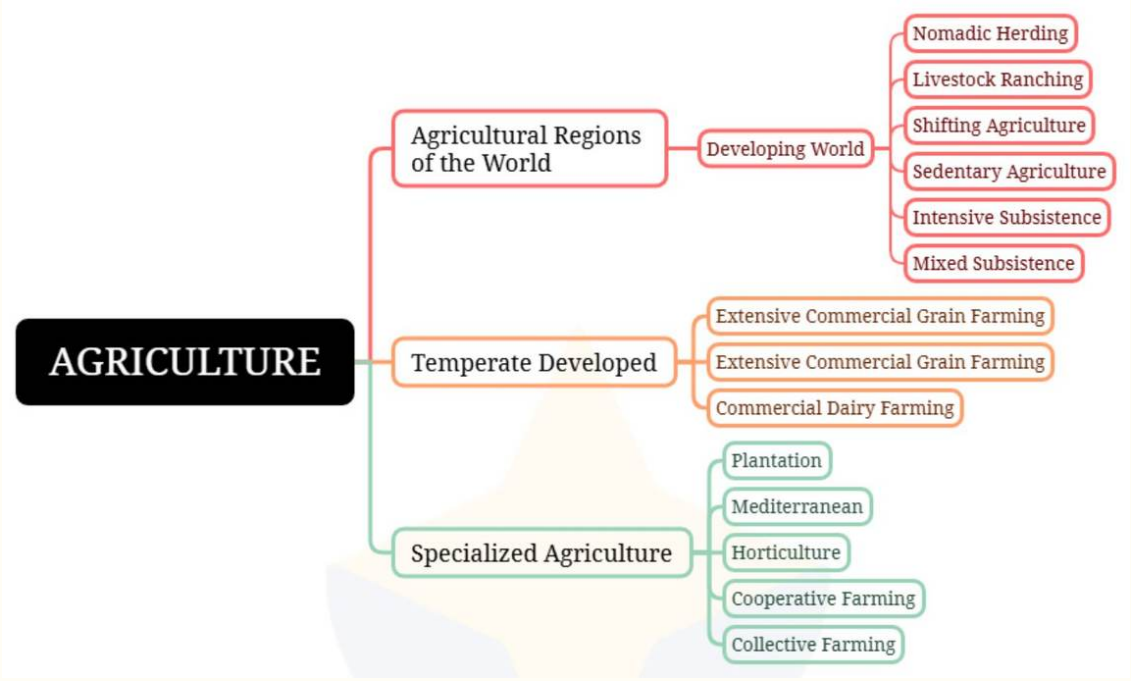

RURAL SETTLEMENTS
Factors Affecting Settlements
Physical Factors
Topography
Climate
Water Supply
Soil Fertility
Natural Resources
Economic Factors
Agricultural Productivity
Employment Opportunities
Accessibility and Transportation
Social and Cultural Factors
Community and Kinship
Cultural Practices
Land Ownership Patterns
Political Factors
Government Policies
Security and Stability
Environmental Factors
Natural Disasters
Climate Change
Types of Rural Settlements
Compact/Nucleated Settlements
Favourable relief + favourable Climate
Closely packed
Common in fertile plains, valleys, and areas with abundant natural resources.
Locations where resources are concentrated or for security reasons.
Semi-Compact Settlements
Favourable relief + Less favourable climate
Transitional
Moderate agricultural productivity
Hamleted Settlements
Fragmentation of settlements in socially/ethically motivated separate units.
Panna, para, palli, nagla, dhani,
Dispersed Settlements
Spread out over a large area
Bad climate & Relief
Patterns of Rural Settlements
Linear
Oval/Circular
Triangular
Rectangular (Grid pattern)
Step

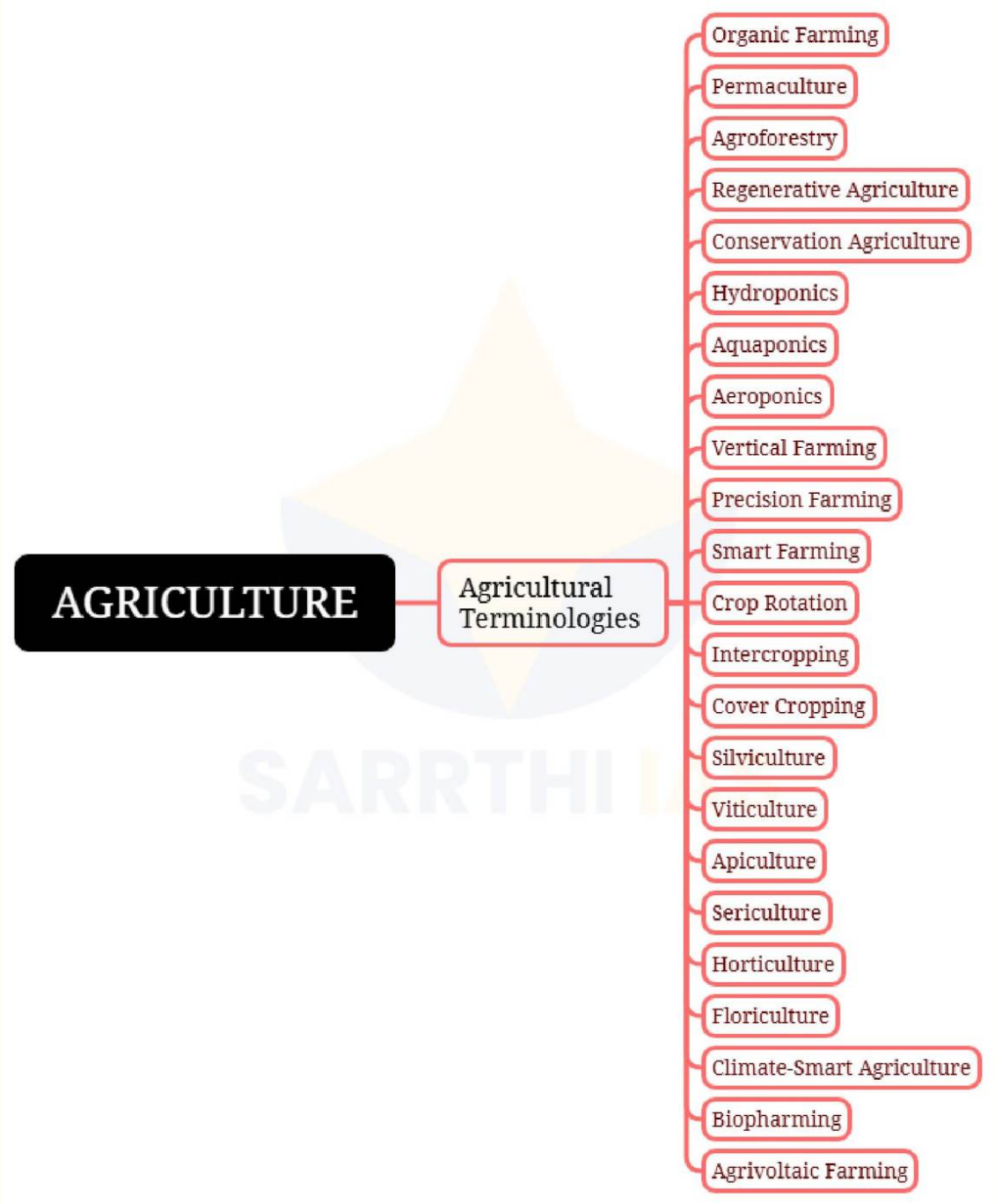

URBAN SETTLEMENTS
Terminologies associated with Urban Settlements
City: significant urban area with a high population density and infrastructure
Town: urban area that is larger than a village but smaller than a city
Megacity: urban area with a population of more than 10 million people
Metropolis; A large and densely populated urban area that serves as a significant economic, political, and cultural center of a country or region.
Conurbation
Megalopolis; A region in which several large cities and surrounding areas grow together, forming a massive, interconnected urban complex.
Labour market separation, due to different specialization and large distances
Suburb: A residential area on the outskirts of a city
Urban Sprawl; uncontrolled expansion of urban areas into the surrounding rural land.
Slum; densely populated urban area characterized by substandard housing and squalor.
Central Business District (CBD): commercial and business center of a city
Edge City: concentration of business, shopping, and entertainment outside a traditional downtown (CBD) in what had previously been a suburban residential or rural area
Gentrification: process by which urban areas undergo redevelopment and revitalization, often resulting in increased property values and the displacement of lower-income residents.

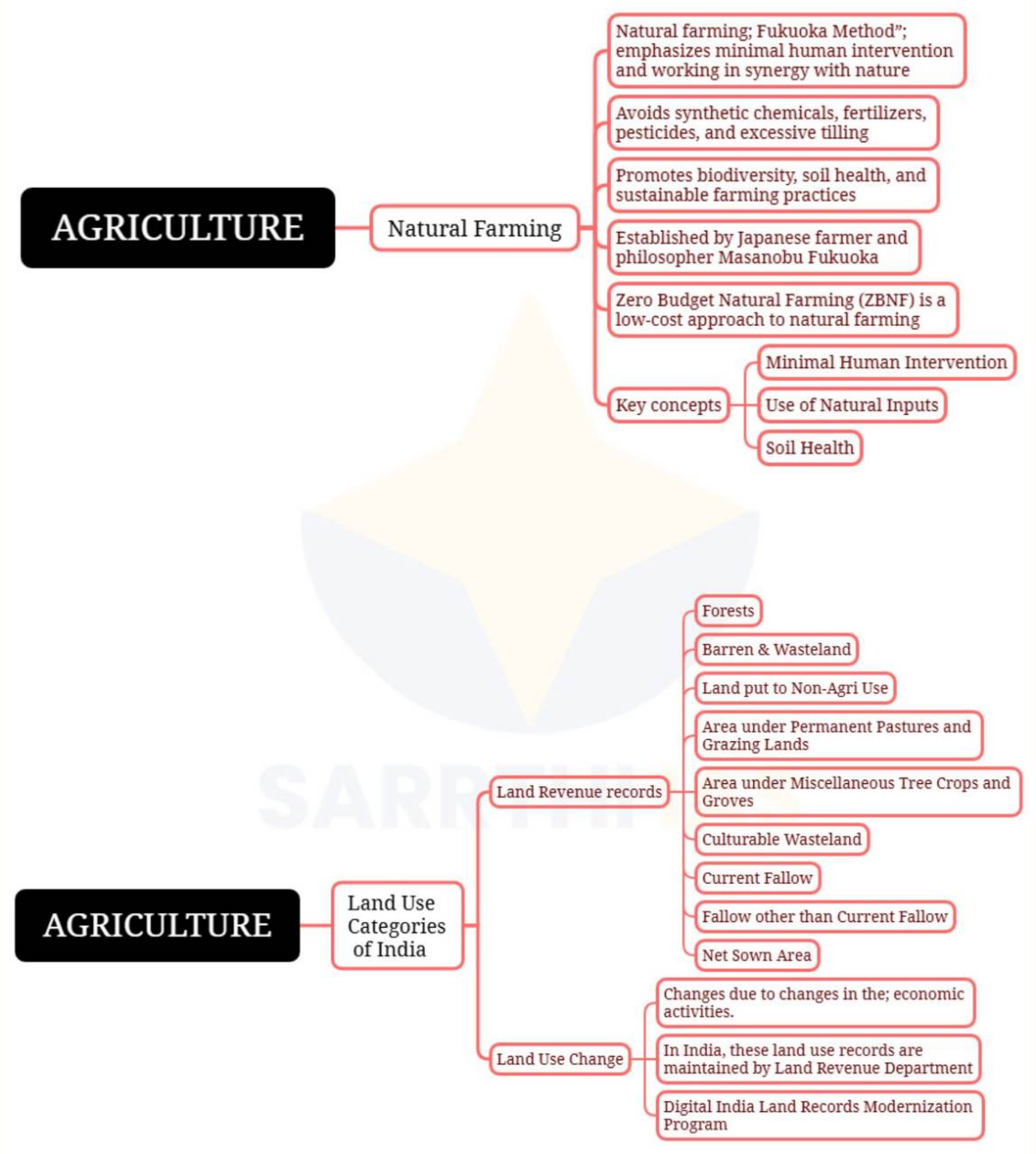

AGRICULTURE
Largest area of arable land globally
Cropping Seasons of India
Kharif Season
Time Period; June to October
Sowing: onset of the southwest monsoon
Harvesting: September to October
Major Crops: Paddy; maize, sorghum
Climatic Conditions: hot and humid weather with substantial rainfall
Rabi Season
Time Period: October/November to March/April
Sowing: October to December.
Harvesting: March to April.
Winter rainfall due to Western disturbances
Major Crops: Wheat, barley, oats, chickpea
Climatic Conditions: cool and dry weather for growth and warmer weather during the harvesting period
Zaid Season
Time Period: March to June
Sowing: end of the Rabi season and the beginning of the Kharif season
Harvesting: June to July
Major Crops: Watermelon, muskmelon, cucumber
Climatic Conditions: warm and dry weather; availability of irrigation

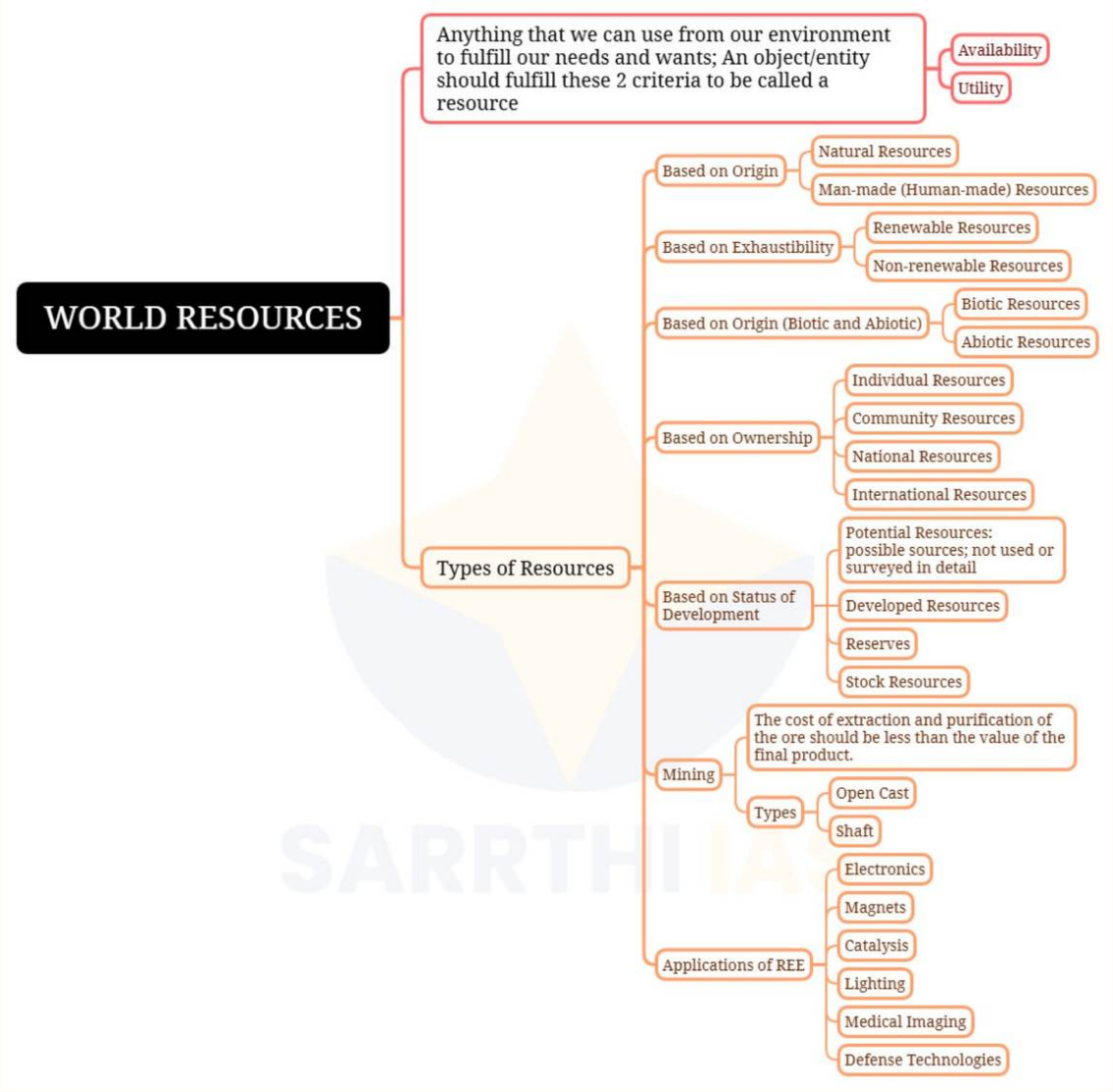

AGRICULTURE
Types of Farming in India
Irrigated
Protective: protect the crops from drying
Productive: increase the produce harvested from the crops
Dry farming: places where rainfall is <75 cm and the growing season is <75 days
Dryland farming: rainfall; 75-115 cm; growing season is between 75-120 days
Wetland farming: rainfall is >115cm; growing season is >120 days
In India, almost 80 percent of maize, 90 percent of Bajra, approximately 95 percent of pulses, and 75 percent of oilseeds are obtained from dry or dryland agriculture.
Micro Irrigation
Improve the water use efficiency of agriculture
Advantages
Prevention of soil erosion
No need to level the fields
Used in all types of soil
Uniform irrigation; reduced nutrient loss
Fertilizers/Pesticides etc can be added
Disadvantages
High Investment
Clogging of pipes
Drip irrigation; needs defined spacing between crops
Technical installation
Important Crops across the World
Gram
Tuar (Red gram/Pigeon Pea)
Groundnut
Rapeseed and Mustard (White and Black rai)
Soyabean
Sunflower
System of Rice Intensification
The shift of sugarcane industry to the South

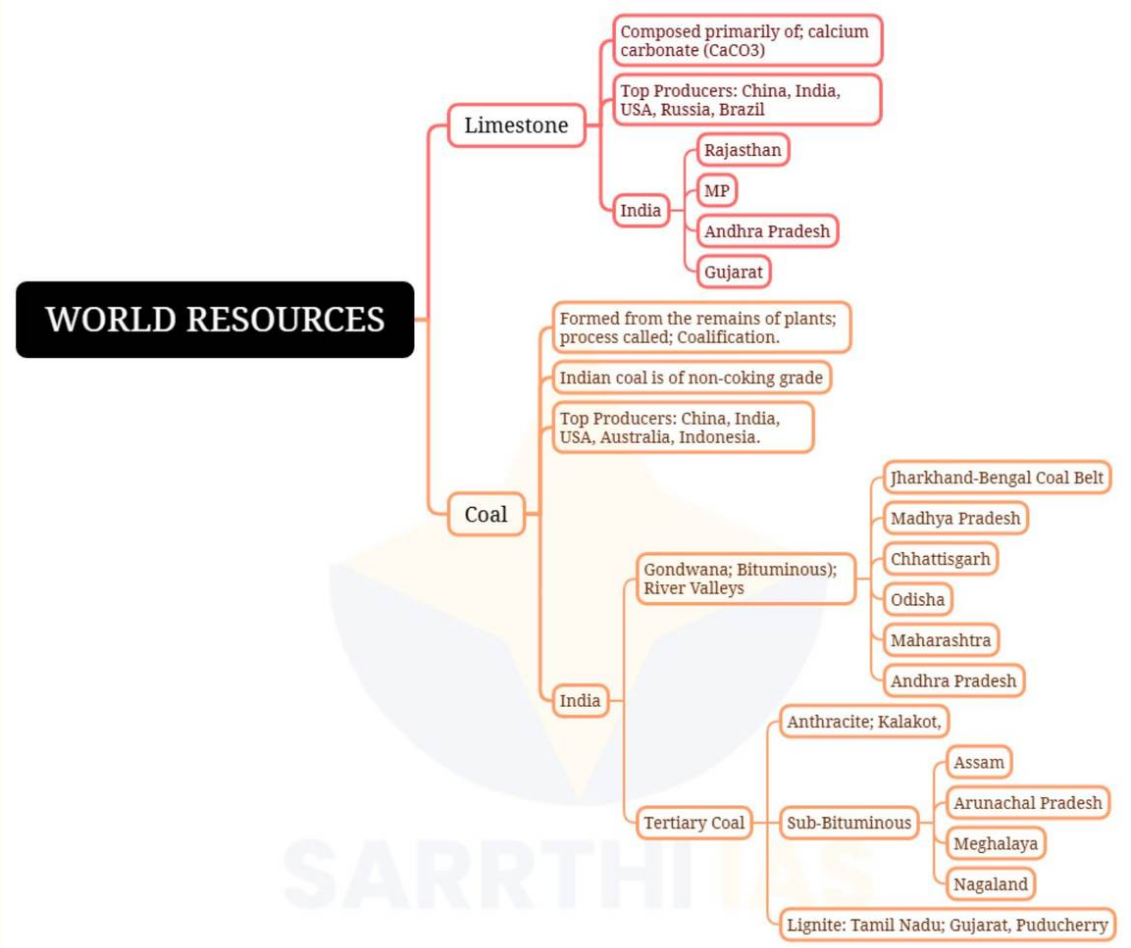

AGRICULTURE
Agriculture Revolutions in India
Green Revolution
Period: Late 1960s to early 1980s
Key Figure: Dr. M.S. Swaminathan.
Main Focus; high-yielding varieties (HYVs) of seeds, increased use of fertilizers, pesticides, and irrigation.
Successes
Increase in food grain production
Reduced the import dependency
Drawbacks
Regional disparities
Social disparities
Environmental degradation
Water table depletion
White Revolution (Operation Flood)
Period: 1970s onwards
Key Figure: Dr. Verghese Kurien.
Main Focus: dairy industry
Successes
Largest producer of milk
Improved rural incomes and employment
Robust cooperative network
Drawbacks
Maintaining quality standards
Inadequate cold chain infrastructure
Inadequate processing
AGRICULTURE
Agriculture Revolutions in India
Blue Revolution
Period: 1980s onwards
Main Focus: fisheries sector
Successes
Increase in fish production
Enhanced export
Improved nutritional standards and employment
Drawbacks
Overfishing and depletion of marine resources
Environmental issues
Yellow Revolution
Period: 1980s onwards
Main Focus: oilseed
Successes
Increase in the production of oilseeds
Reduced dependency; imports
Drawbacks
Inconsistent growth and production challenges
Market fluctuations
Evergreen Revolution
Proposed by: Dr. M.S. Swaminathan
Main Focus: Sustainable agriculture
Successes
Ecological sustainability
Genetic diversity and conservation of natural resources
Drawbacks
Implementation challenges
Significant policy support and farmer education
Other Revolutions

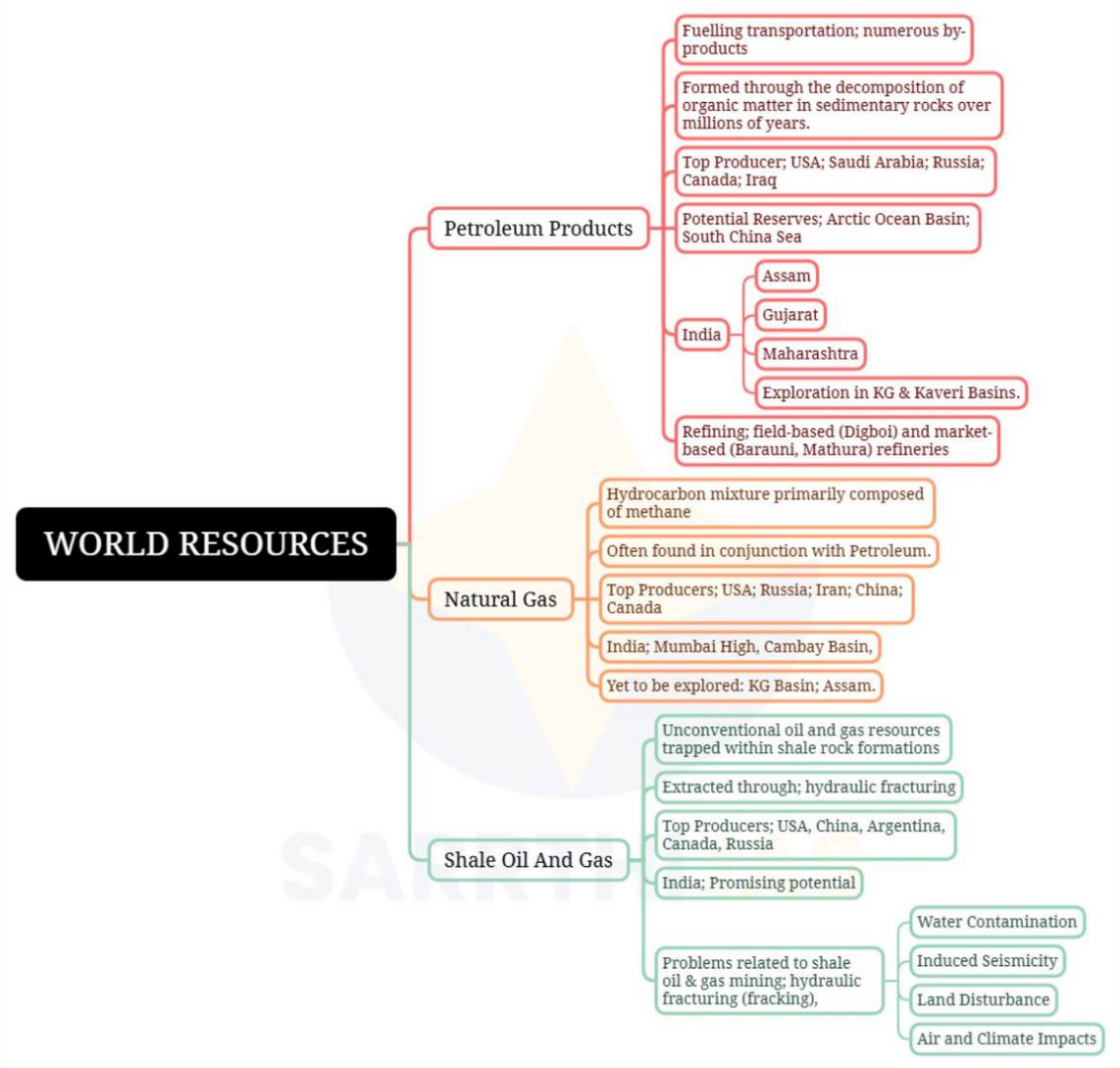

AGRICULTURE
Agricultural Regions of the World
Developing World
Nomadic Herding
Livestock Ranching
Shifting Agriculture
Sedentary Agriculture
Intensive Subsistence
Mixed Subsistence
Temperate Developed
Extensive Commercial Grain Farming
Extensive Commercial Grain Farming
Commercial Dairy Farming
Specialized Agriculture
Plantation
Mediterranean
Horticulture
Cooperative Farming
Collective Farming

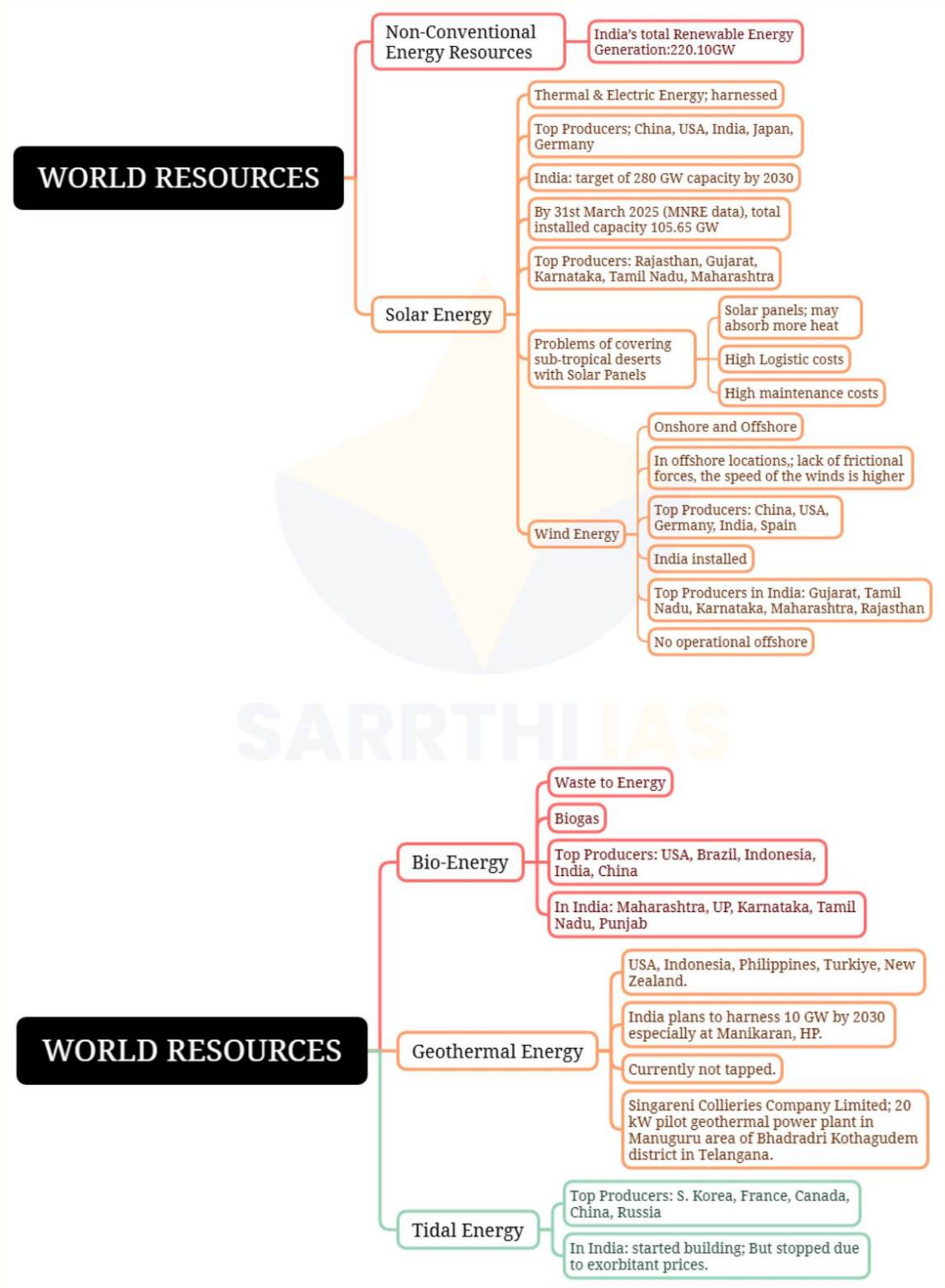

AGRICULTURE
Agricultural Terminologies
Organic Farming
Permaculture
Agroforestry
Regenerative Agriculture
Conservation Agriculture
Hydroponics
Aquaponics
Aeroponics
Vertical Farming
Precision Farming
Smart Farming
Crop Rotation
Intercropping
Cover Cropping
Silviculture
Viticulture
Apiculture
Sericulture
Horticulture
Floriculture
Climate-Smart Agriculture
Biopharming
Agrivoltaic Farming

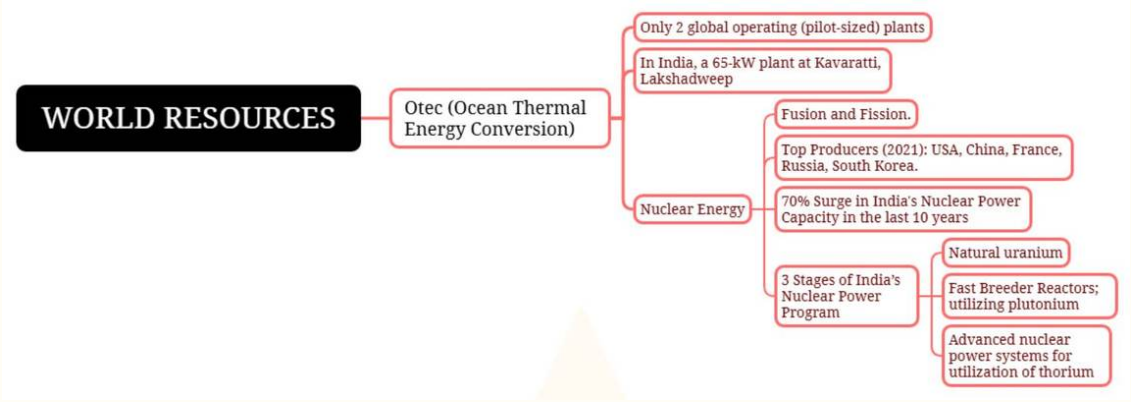

AGRICULTURE
Natural Farming
Natural farming; Fukuoka Method"; emphasizes minimal human intervention and working in synergy with nature
Avoids synthetic chemicals, fertilizers, pesticides, and excessive tilling
Promotes biodiversity, soil health, and sustainable farming practices
Established by Japanese farmer and philosopher Masanobu Fukuoka
Zero Budget Natural Farming (ZBNF) is a low-cost approach to natural farming
Key concepts
Minimal Human Intervention
Use of Natural Inputs
Soil Health
AGRICULTURE
Land Use Categories of India
Land Revenue records
Forests
Barren & Wasteland
Land put to Non-Agri Use
Area under Permanent Pastures and Grazing Lands
Area under Miscellaneous Tree Crops and Groves
Culturable Wasteland
Current Fallow
Fallow other than Current Fallow
Net Sown Area
Land Use Change
Changes due to changes in the; economic activities.
In India, these land use records are maintained by Land Revenue Department
Digital India Land Records Modernization Program

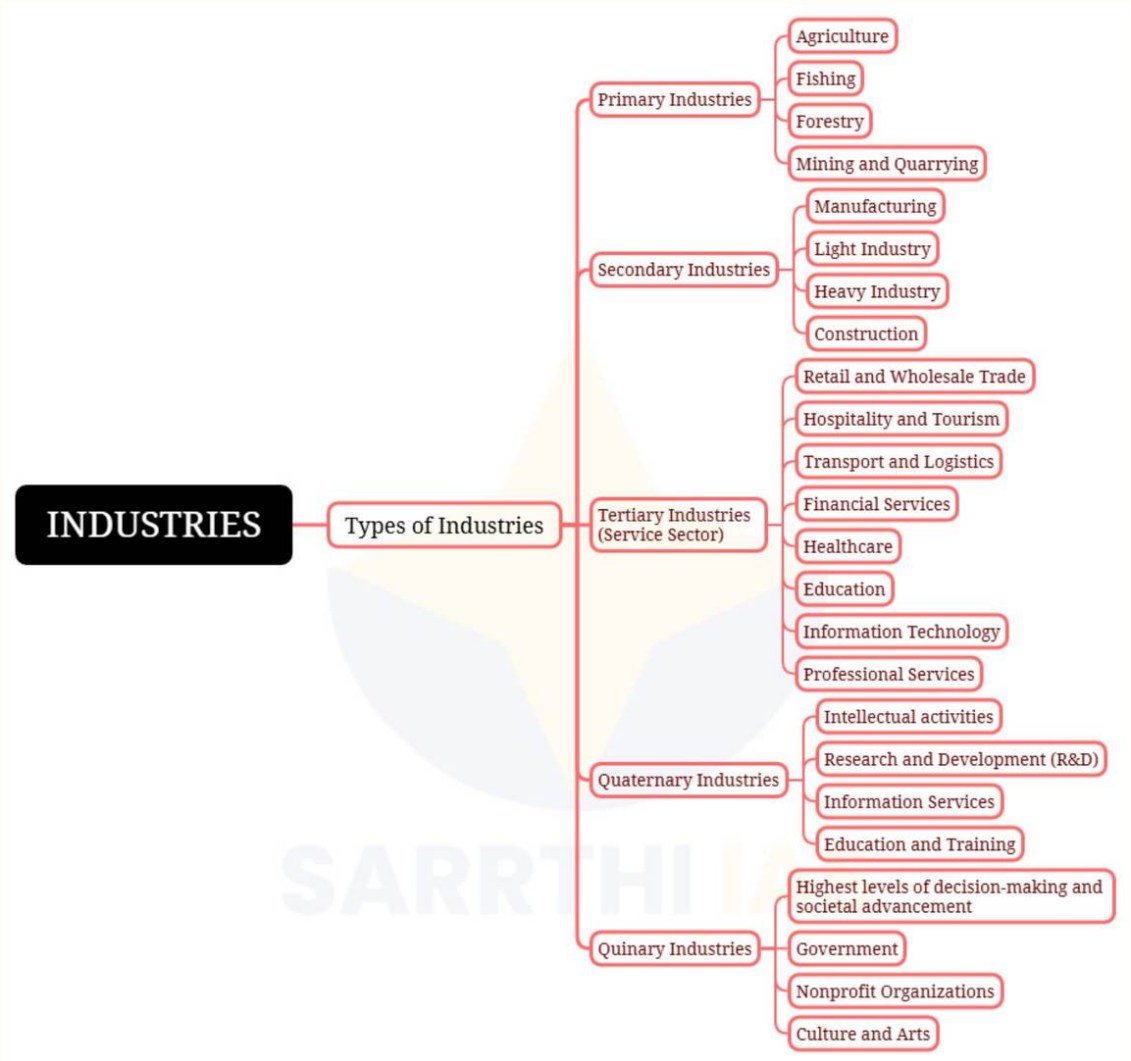

WORLD RESOURCES
Anything that we can use from our environment to fulfill our needs and wants; An object/entity should fulfill these 2 criteria to be called a resource
Availability
Utility
Types of Resources
Based on Origin
Natural Resources
Man-made (Human-made) Resources
Based on Exhaustibility
Renewable Resources
Non-renewable Resources
Based on Origin (Biotic and Abiotic)
Biotic Resources
Abiotic Resources
Based on Ownership
Individual Resources
Community Resources
National Resources
International Resources
Based on Status of Development
Potential Resources: possible sources; not used or surveyed in detail
Developed Resources
Reserves
Stock Resources
Mining
The cost of extraction and purification of the ore should be less than the value of the final product.
Types
Open Cast
Shaft
Applications of REE
Electronics
Magnets
Catalysis
Lighting
Medical Imaging
Defense Technologies

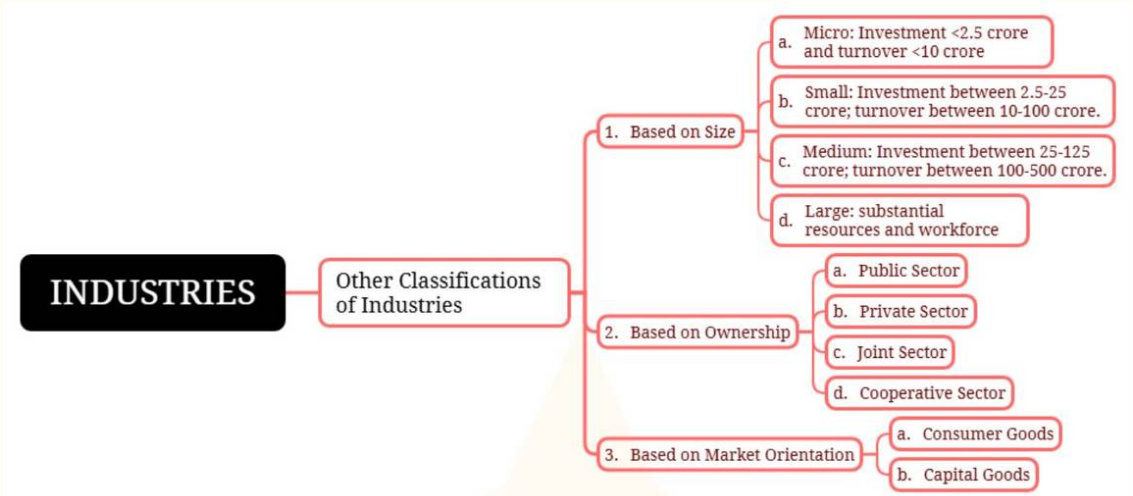

WORLD RESOURCES
Limestone
Composed primarily of; calcium carbonate (CaCO3)
Top Producers: China, India, USA, Russia, Brazil
India
Rajasthan
MP
Andhra Pradesh
Gujarat
Coal
Formed from the remains of plants; process called; Coalification.
Indian coal is of non-coking grade
Top Producers: China, India, USA, Australia, Indonesia.
India
Gondwana; Bituminous); River Valleys
Jharkhand-Bengal Coal Belt
Madhya Pradesh
Chhattisgarh
Odisha
Maharashtra
Andhra Pradesh
Tertiary Coal
Anthracite; Kalakot,
Sub-Bituminous
Assam
Arunachal Pradesh
Meghalaya
Nagaland
Lignite: Tamil Nadu; Gujarat, Puducherry

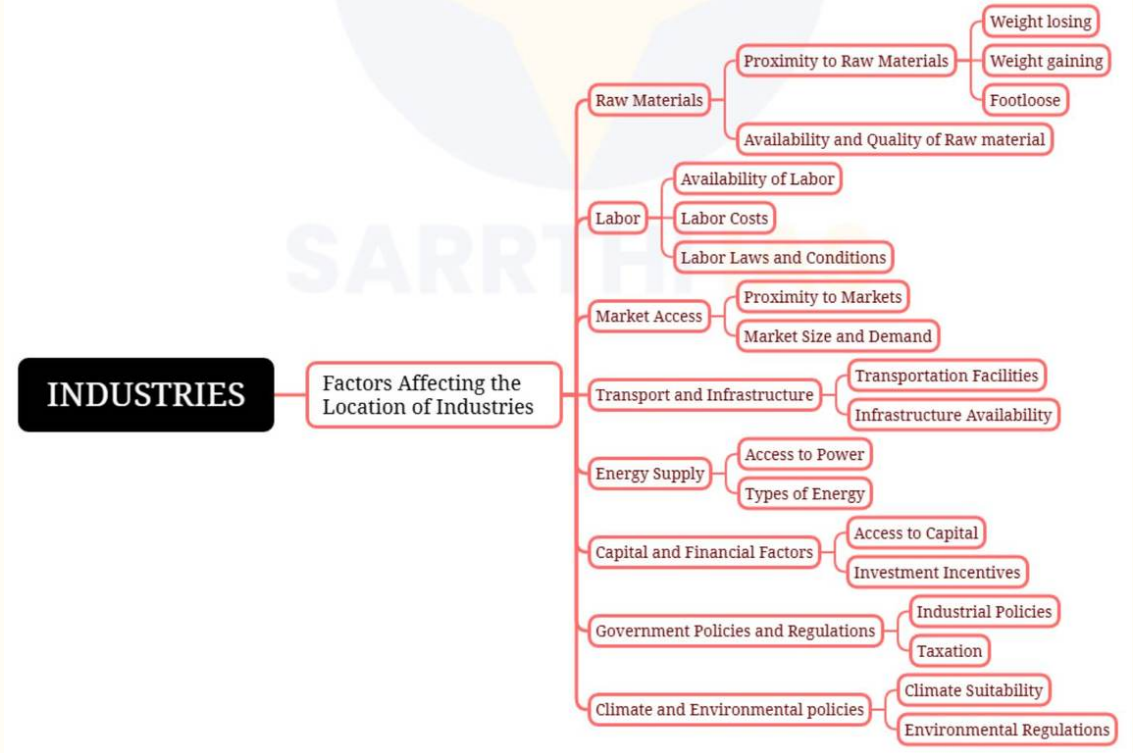

WORLD RESOURCES
Petroleum Products
Fuelling transportation; numerous by-products
Formed through the decomposition of organic matter in sedimentary rocks over millions of years.
Top Producer; USA; Saudi Arabia; Russia; Canada; Iraq
Potential Reserves; Arctic Ocean Basin; South China Sea
India
Assam
Gujarat
Maharashtra
Exploration in KG & Kaveri Basins.
Refining; field-based (Digboi) and market-based (Barauni, Mathura) refineries
Natural Gas
Hydrocarbon mixture primarily composed of methane
Often found in conjunction with Petroleum.
Top Producers; USA; Russia; Iran; China; Canada
India; Mumbai High, Cambay Basin,
Yet to be explored: KG Basin; Assam.
Shale Oil And Gas
Unconventional oil and gas resources trapped within shale rock formations
Extracted through; hydraulic fracturing
Top Producers; USA, China, Argentina, Canada, Russia
India; Promising potential
Problems related to shale oil & gas mining; hydraulic fracturing (fracking),
Water Contamination
Induced Seismicity
Land Disturbance
Air and Climate Impacts

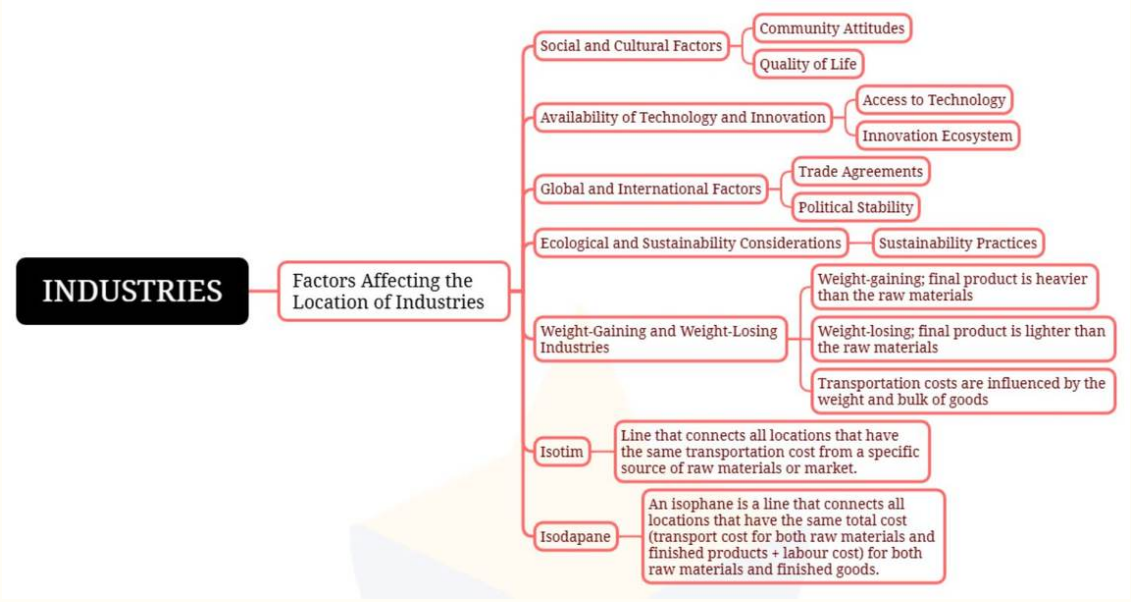

WORLD RESOURCES
Non-Conventional Energy Resources
India's total Renewable Energy Generation:220.10GW
Solar Energy
Thermal & Electric Energy; harnessed
Top Producers; China, USA, India, Japan, Germany
India: target of 280 GW capacity by 2030
By 31st March 2025 (MNRE data), total installed capacity 105.65 GW
Top Producers: Rajasthan, Gujarat, Karnataka, Tamil Nadu, Maharashtra
Problems of covering sub-tropical deserts with Solar Panels
Solar panels; may absorb more heat
High Logistic costs
High maintenance costs
Wind Energy
Onshore and Offshore
In offshore locations,; lack of frictional forces, the speed of the winds is higher
Top Producers: China, USA, Germany, India, Spain
India installed
Top Producers in India: Gujarat, Tamil Nadu, Karnataka, Maharashtra, Rajasthan
No operational offshore
WORLD RESOURCES
Bio-Energy
Waste to Energy
Biogas
Top Producers: USA, Brazil, Indonesia, India, China
In India: Maharashtra, UP, Karnataka, Tamil Nadu, Punjab
Geothermal Energy
USA, Indonesia, Philippines, Turkiye, New Zealand.
India plans to harness 10 GW by 2030 especially at Manikaran, HP.
Currently not tapped.
Singareni Collieries Company Limited; 20 kW pilot geothermal power plant in Manuguru area of Bhadradri Kothagudem district in Telangana.
Tidal Energy
Top Producers: S. Korea, France, Canada, China, Russia
In India: started building; But stopped due to exorbitant prices.

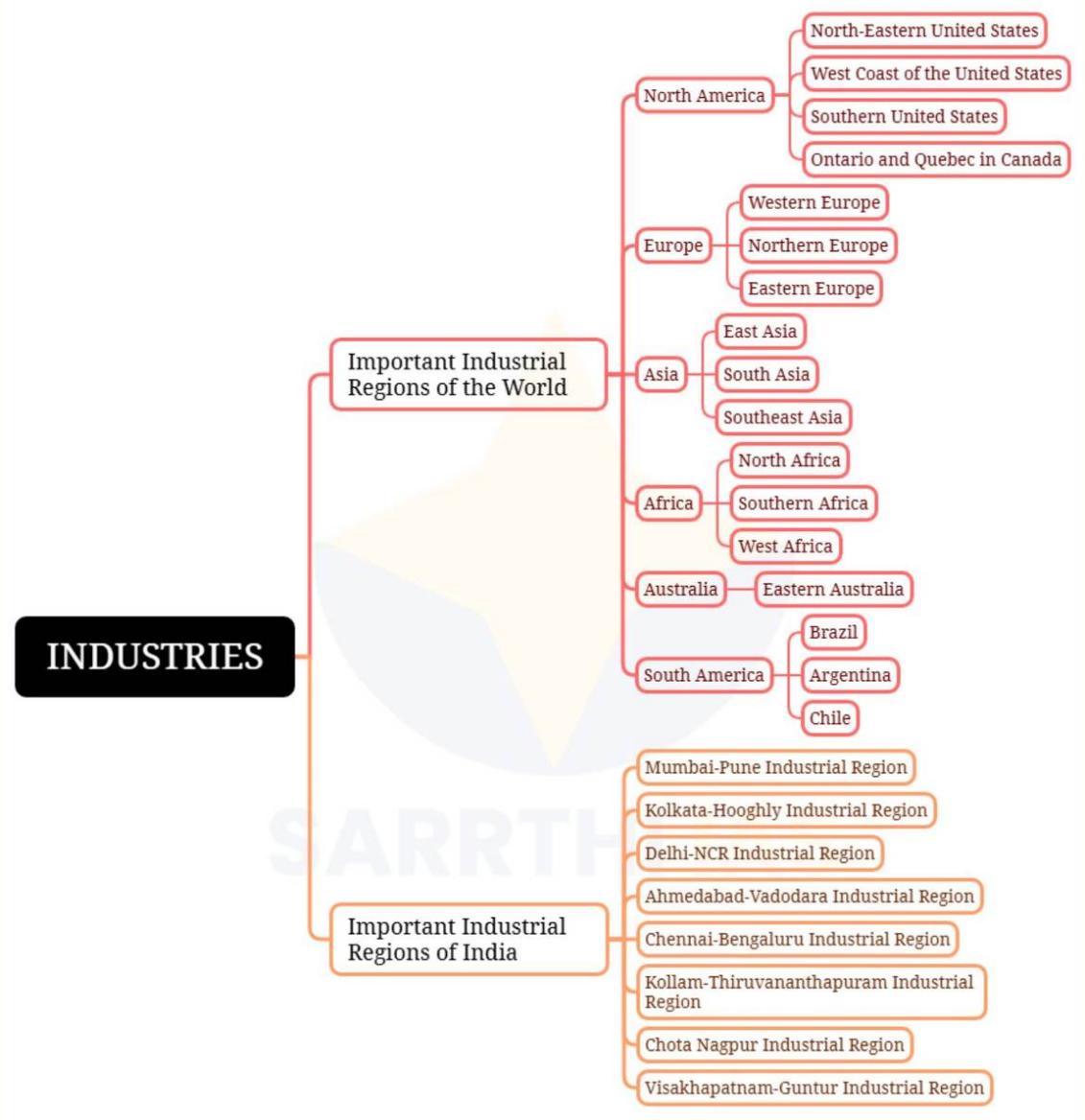

WORLD RESOURCES
Otec (Ocean Thermal Energy Conversion)
Only 2 global operating (pilot-sized) plants
In India, a 65-kW plant at Kavaratti, Lakshadweep
Nuclear Energy
Fusion and Fission.
Top Producers (2021): USA, China, France, Russia, South Korea.
70% Surge in India's Nuclear Power Capacity in the last 10 years
3 Stages of India's Nuclear Power Program
Natural uranium
Fast Breeder Reactors; utilizing plutonium
Advanced nuclear power systems for utilization of thorium


INDUSTRIES
Types of Industries
Primary Industries
Agriculture
Fishing
Forestry
Mining and Quarrying
Secondary Industries
Manufacturing
Light Industry
Heavy Industry
Construction
Tertiary Industries (Service Sector)
Retail and Wholesale Trade
Hospitality and Tourism
Transport and Logistics
Financial Services
Healthcare
Education
Information Technology
Professional Services
Quaternary Industries
Intellectual activities
Research and Development (R&D)
Information Services
Education and Training
Quinary Industries
Highest levels of decision-making and societal advancement
Government
Nonprofit Organizations
Culture and Arts


INDUSTRIES
Other Classifications of Industries
1. Based on Size
a. Micro: Investment <2.5 crore and turnover <10 crore
b. Small: Investment between 2.5-25 crore; turnover between 10-100 crore.
c. Medium: Investment between 25-125 crore; turnover between 100-500 crore.
d. Large: substantial resources and workforce
2. Based on Ownership
a. Public Sector
b. Private Sector
c. Joint Sector
d. Cooperative Sector
3. Based on Market Orientation
a. Consumer Goods
b. Capital Goods

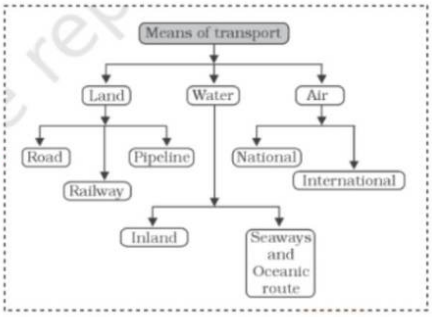

INDUSTRIES
Factors Affecting the Location of Industries
Raw Materials
Proximity to Raw Materials
Weight losing
Weight gaining
Footloose
Availability and Quality of Raw material
Labor
Availability of Labor
Labor Costs
Labor Laws and Conditions
Market Access
Proximity to Markets
Market Size and Demand
Transport and Infrastructure
Transportation Facilities
Infrastructure Availability
Energy Supply
Access to Power
Types of Energy
Capital and Financial Factors
Access to Capital
Investment Incentives
Government Policies and Regulations
Industrial Policies
Taxation
Climate and Environmental policies
Climate Suitability
Environmental Regulations


INDUSTRIES
Factors Affecting the Location of Industries
Social and Cultural Factors
Community Attitudes
Quality of Life
Availability of Technology and Innovation
Access to Technology
Innovation Ecosystem
Global and International Factors
Trade Agreements
Political Stability
Ecological and Sustainability Considerations
Sustainability Practices
Weight-Gaining and Weight-Losing Industries
Weight-gaining; final product is heavier than the raw materials
Weight-losing; final product is lighter than the raw materials
Transportation costs are influenced by the weight and bulk of goods
Isotim
Line that connects all locations that have the same transportation cost from a specific source of raw materials or market.
Isodapane
An isophane is a line that connects all locations that have the same total cost (transport cost for both raw materials and finished products + labour cost) for both raw materials and finished goods.


INDUSTRIES
Important Industrial Regions of the World
North America
North-Eastern United States
West Coast of the United States
Southern United States
Ontario and Quebec in Canada
Europe
Western Europe
Northern Europe
Eastern Europe
Asia
East Asia
South Asia
Southeast Asia
Africa
North Africa
Southern Africa
West Africa
Australia
Eastern Australia
South America
Brazil
Argentina
Chile
Important Industrial Regions of India
Mumbai-Pune Industrial Region
Kolkata-Hooghly Industrial Region
Delhi-NCR Industrial Region
Ahmedabad-Vadodara Industrial Region
Chennai-Bengaluru Industrial Region
Kollam-Thiruvananthapuram Industrial Region
Chota Nagpur Industrial Region
Visakhapatnam-Guntur Industrial Region

TRANSPORT AND COMMUNICATION

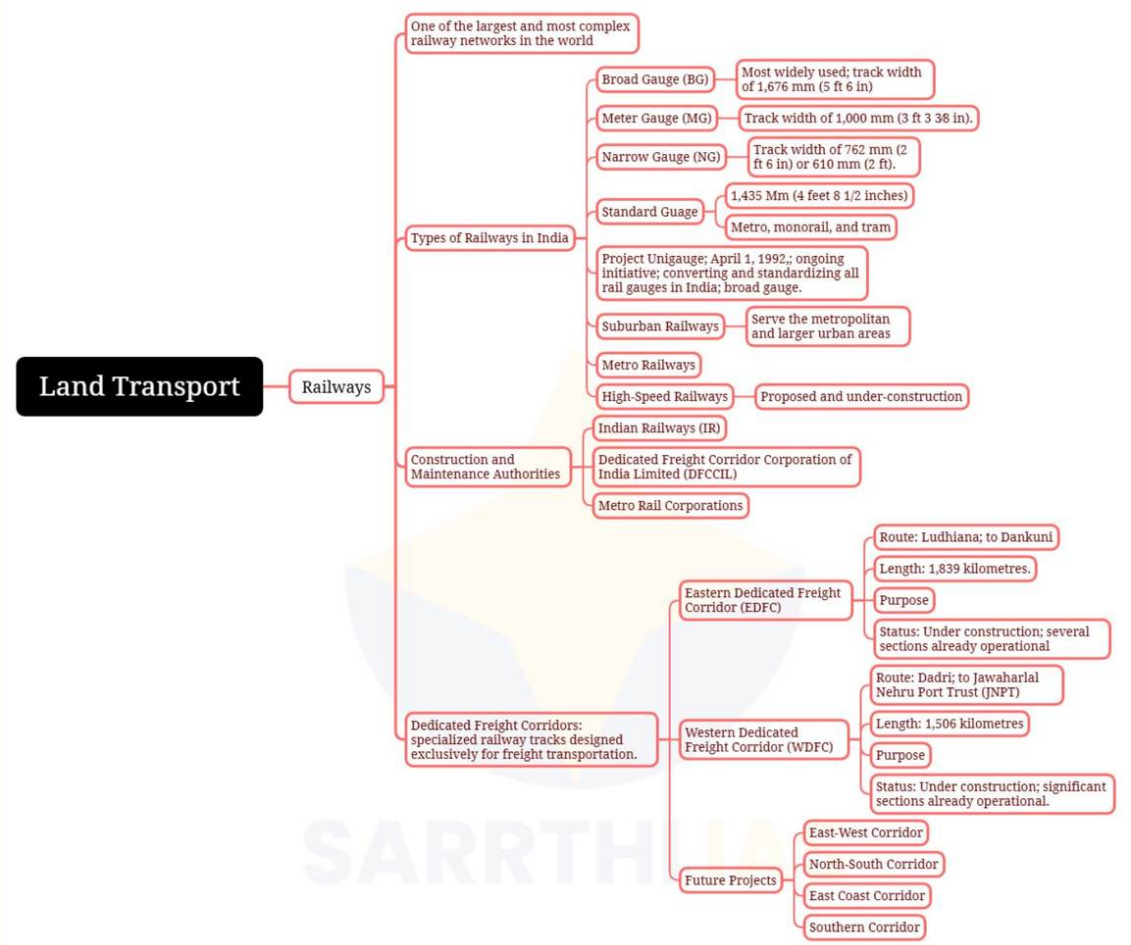
Means Of Transportation

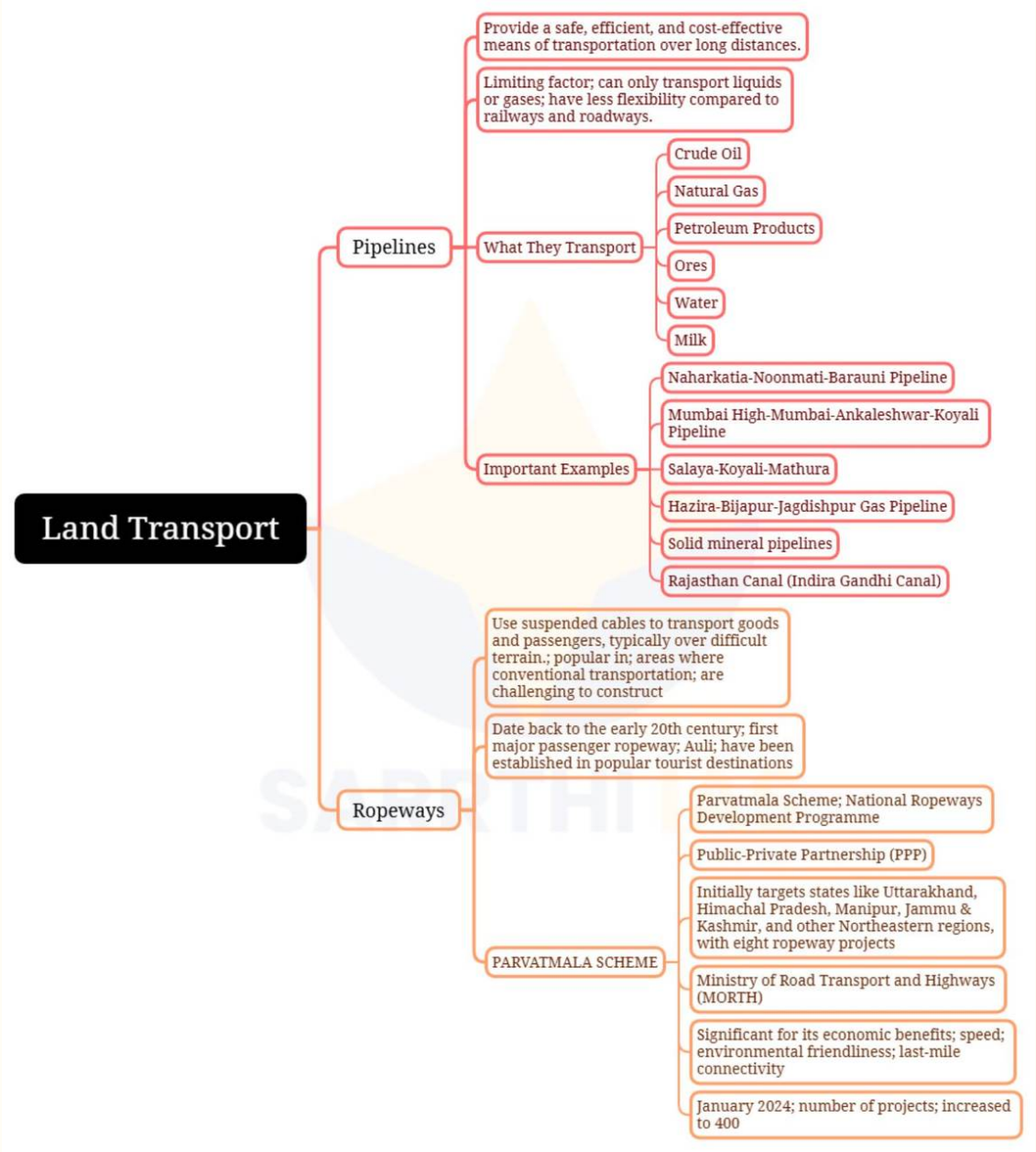


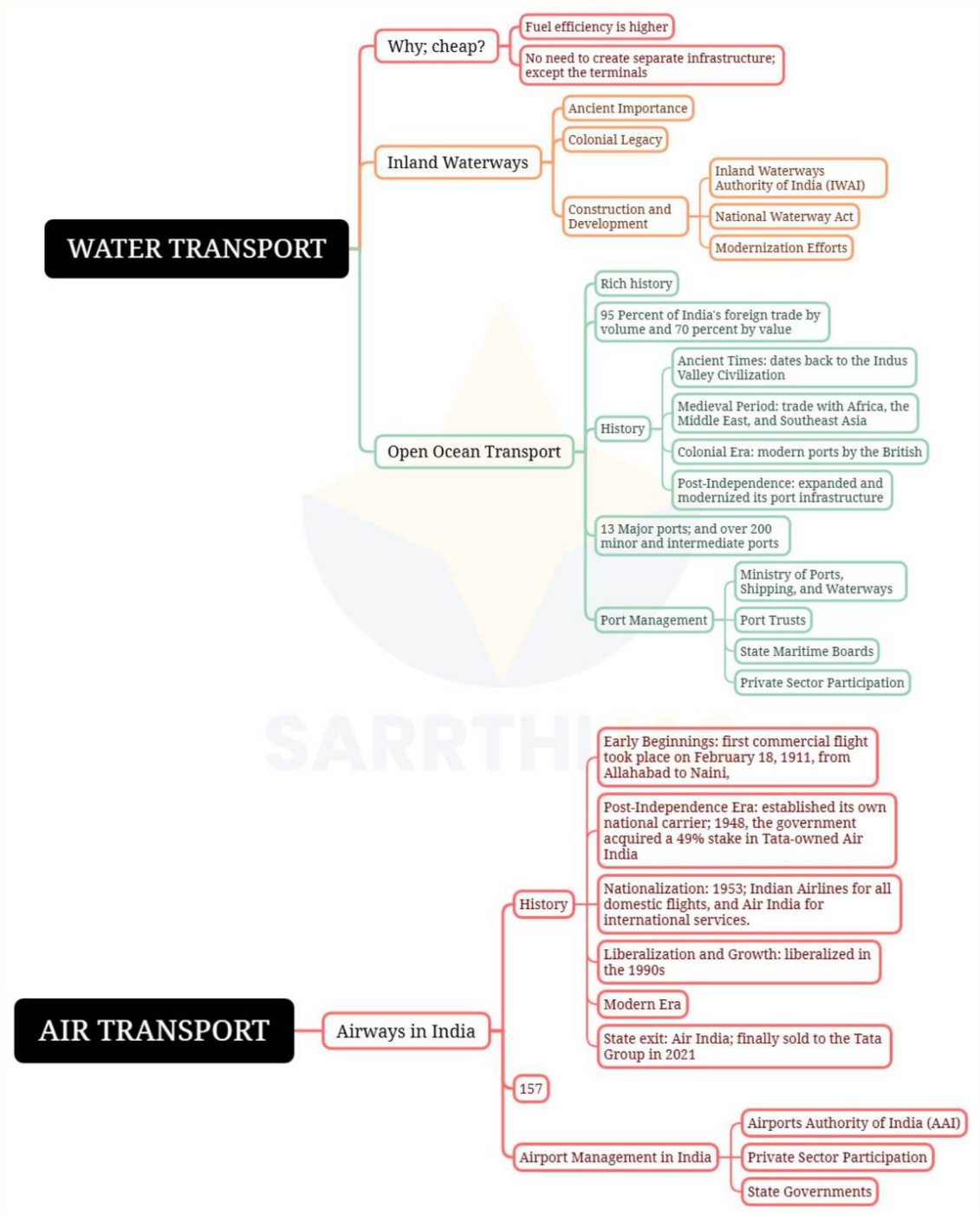
Land Transport

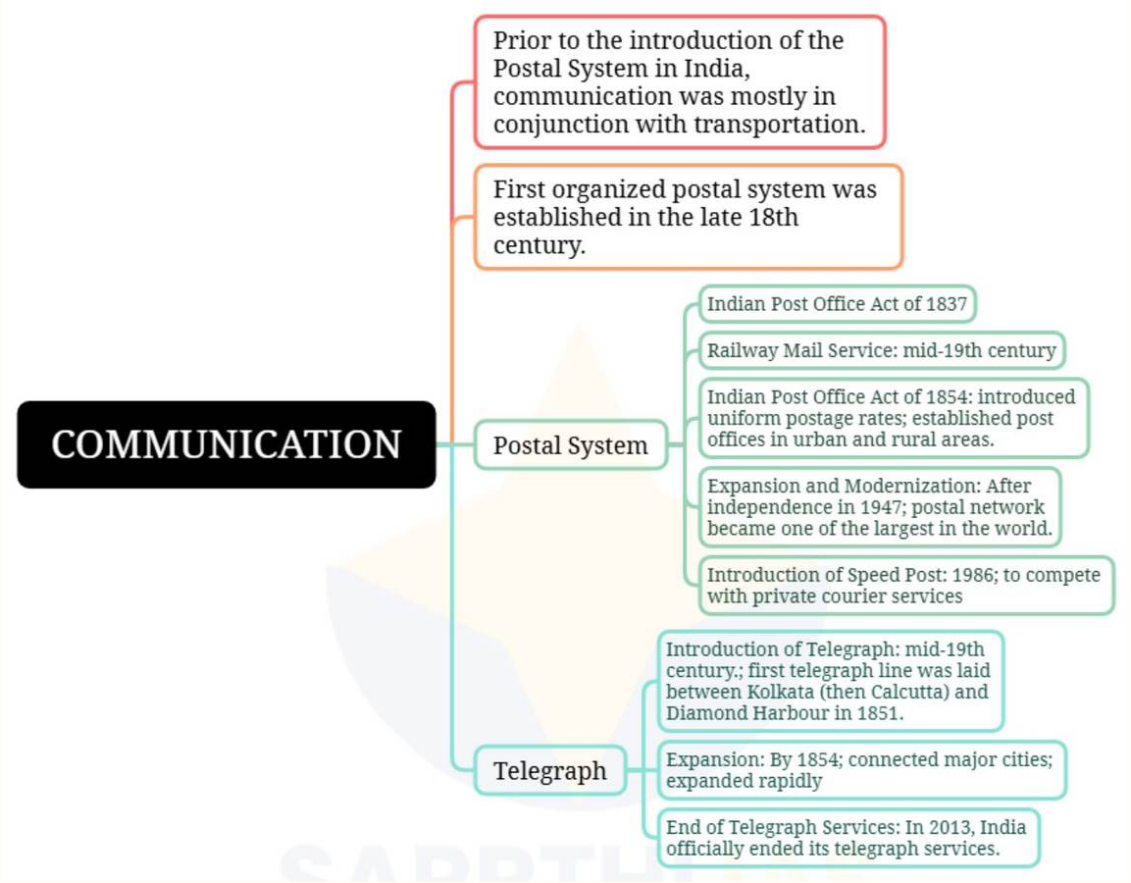
Roadways

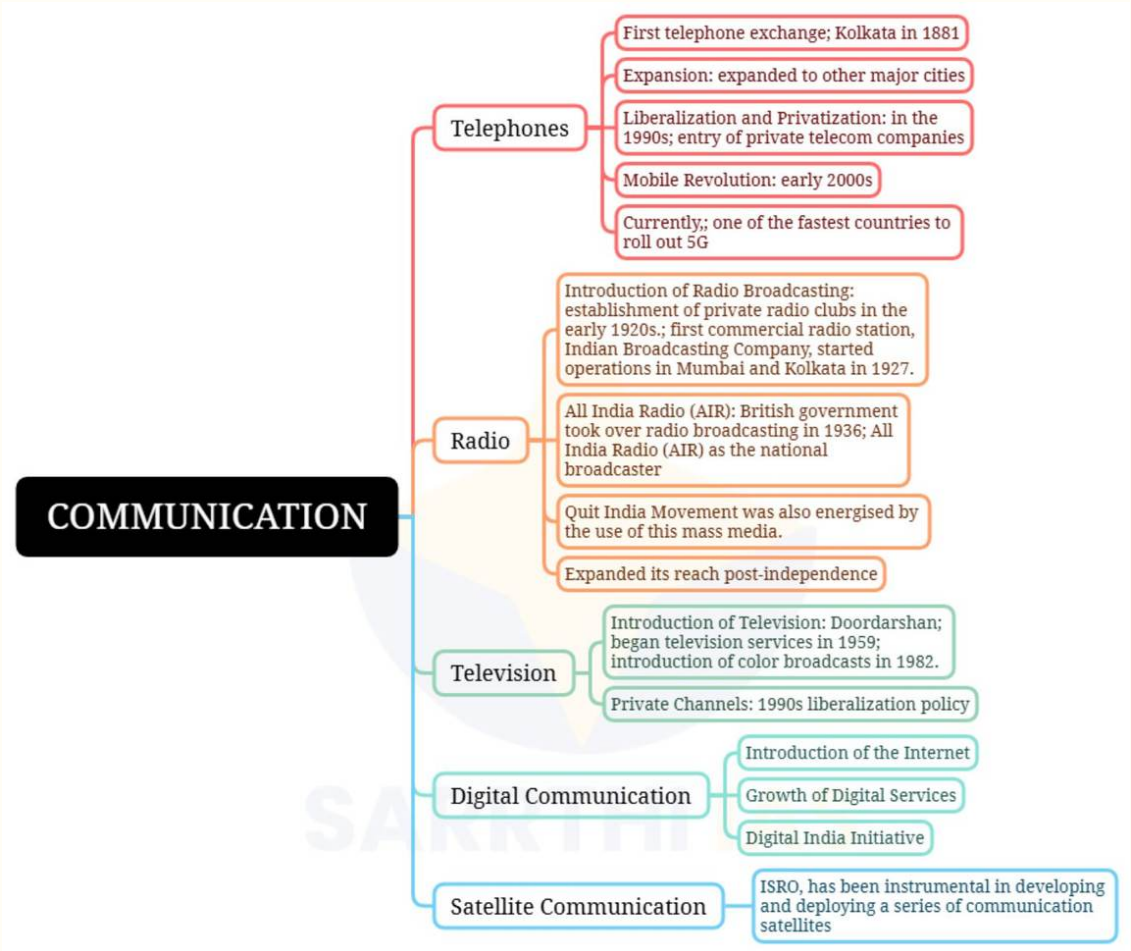
- India's roadway; one of the largest in the world
- Types of Roadways in India
  - National Highways (NH)
    - Primary roadways connecting major cities, ports, state capitals, and large industrial areas.
    - Only 2% of the total road network
    - Authority: National Highways Authority of India (NHAI)
  - State Highways (SH)
    - Connect important cities within a state and provide links to the national highways and other state highways.
    - 4% Of the total Road network.
    - Authority: State Public Works Departments (PWDs).
  - District Roads
    - Connect district headquarters with other places in the district.; classified into Major District Roads (MDR) and Minor District Roads
    - 14% Of the total road network
    - Authority: Zilla Parishads (District Councils) and the District PWDs.
  - Rural Roads
    - Roads connecting rural areas and villages to nearby towns or district roads.
    - Authority: local bodies (Gram Panchayats) and under the Pradhan Mantri Gram Sadak Yojana (PMGSY).
  - Urban Roads
    - Found within the limits of towns and cities.
    - Authority: Urban Local Bodies (Municipal Corporations or Municipalities).
- Naming of National Highways in India
  - Up until 2010; on the basis of the National Highways Act of 1956.
  - Did not give any indication of its location and direction.
  - North-South Highways: even numbers
    - Numbering increases from east to west
  - East-West Highways: odd numbers
    - Numbering increases from north to south
  - Sub-Highways
    - Highways that branch off from primary highways; prefix is added
- New way to calculate roads


Land Transport
Railways
One of the largest and most complex railway networks in the world
Types of Railways in India
Broad Gauge (BG)
Most widely used; track width of 1,676 mm (5 ft 6 in)
Meter Gauge (MG)
Track width of 1,000 mm (3 ft 3 38 in).
Narrow Gauge (NG)
Track width of 762 mm (2 ft 6 in) or 610 mm (2 ft).
Standard Guage
1,435 Mm (4 feet 8 1/2 inches)
Metro, monorail, and tram
Project Unigauge; April 1, 1992,; ongoing initiative; converting and standardizing all rail gauges in India; broad gauge.
Suburban Railways
Serve the metropolitan and larger urban areas
Metro Railways
High-Speed Railways
Proposed and under-construction
Construction and Maintenance Authorities
Indian Railways (IR)
Dedicated Freight Corridor Corporation of India Limited (DFCCIL)
Metro Rail Corporations
Dedicated Freight Corridors: specialized railway tracks designed exclusively for freight transportation.
Eastern Dedicated Freight Corridor (EDFC)
Route: Ludhiana; to Dankuni
Length: 1,839 kilometres.
Purpose
Status: Under construction; several sections already operational
Western Dedicated Freight Corridor (WDFC)
Route: Dadri; to Jawaharlal Nehru Port Trust (JNPT)
Length: 1,506 kilometres
Purpose
Status: Under construction; significant sections already operational.
Future Projects
East-West Corridor
North-South Corridor
East Coast Corridor
Southern Corridor


Land Transport
Pipelines
Provide a safe, efficient, and cost-effective means of transportation over long distances.
Limiting factor; can only transport liquids or gases; have less flexibility compared to railways and roadways.
What They Transport
Crude Oil
Natural Gas
Petroleum Products
Ores
Water
Milk
Important Examples
Naharkatia-Noonmati-Barauni Pipeline
Mumbai High-Mumbai-Ankaleshwar-Koyali Pipeline
Salaya-Koyali-Mathura
Hazira-Bijapur-Jagdishpur Gas Pipeline
Solid mineral pipelines
Rajasthan Canal (Indira Gandhi Canal)
Ropeways
Use suspended cables to transport goods and passengers, typically over difficult terrain.; popular in; areas where conventional transportation; are challenging to construct
Date back to the early 20th century; first major passenger ropeway; Auli; have been established in popular tourist destinations
PARVATMALA SCHEME
Parvatmala Scheme; National Ropeways Development Programme
Public-Private Partnership (PPP)
Initially targets states like Uttarakhand, Himachal Pradesh, Manipur, Jammu & Kashmir, and other Northeastern regions, with eight ropeway projects
Ministry of Road Transport and Highways (MORTH)
Significant for its economic benefits; speed; environmental friendliness; last-mile connectivity
January 2024; number of projects; increased to 400


WATER TRANSPORT
Why; cheap?
Fuel efficiency is higher
No need to create separate infrastructure; except the terminals
Inland Waterways
Ancient Importance
Colonial Legacy
Construction and Development
Inland Waterways Authority of India (IWAI)
National Waterway Act
Modernization Efforts
Open Ocean Transport
Rich history
95 Percent of India's foreign trade by volume and 70 percent by value
History
Ancient Times: dates back to the Indus Valley Civilization
Medieval Period: trade with Africa, the Middle East, and Southeast Asia
Colonial Era: modern ports by the British
Post-Independence: expanded and modernized its port infrastructure
13 Major ports; and over 200 minor and intermediate ports
Port Management
Ministry of Ports, Shipping, and Waterways
Port Trusts
State Maritime Boards
Private Sector Participation
AIR TRANSPORT
Airways in India
History
Early Beginnings: first commercial flight took place on February 18, 1911, from Allahabad to Naini,
Post-Independence Era: established its own national carrier; 1948, the government acquired a 49% stake in Tata-owned Air India
Nationalization: 1953; Indian Airlines for all domestic flights, and Air India for international services.
Liberalization and Growth: liberalized in the 1990s
Modern Era
State exit: Air India; finally sold to the Tata Group in 2021
157
Airport Management in India
Airports Authority of India (AAI)
Private Sector Participation
State Governments


COMMUNICATION
Prior to the introduction of the Postal System in India, communication was mostly in conjunction with transportation.
First organized postal system was established in the late 18th century.
Postal System
Indian Post Office Act of 1837
Railway Mail Service: mid-19th century
Indian Post Office Act of 1854: introduced uniform postage rates; established post offices in urban and rural areas.
Expansion and Modernization: After independence in 1947; postal network became one of the largest in the world.
Introduction of Speed Post: 1986; to compete with private courier services
Telegraph
Introduction of Telegraph: mid-19th century.; first telegraph line was laid between Kolkata (then Calcutta) and Diamond Harbour in 1851.
Expansion: By 1854; connected major cities; expanded rapidly
End of Telegraph Services: In 2013, India officially ended its telegraph services.


COMMUNICATION
Telephones
First telephone exchange; Kolkata in 1881
Expansion: expanded to other major cities
Liberalization and Privatization: in the 1990s; entry of private telecom companies
Mobile Revolution: early 2000s
Currently,; one of the fastest countries to roll out 5G
Radio
Introduction of Radio Broadcasting: establishment of private radio clubs in the early 1920s.; first commercial radio station, Indian Broadcasting Company, started operations in Mumbai and Kolkata in 1927.
All India Radio (AIR): British government took over radio broadcasting in 1936; All India Radio (AIR) as the national broadcaster
Quit India Movement was also energised by the use of this mass media.
Expanded its reach post-independence
Television
Introduction of Television: Doordarshan; began television services in 1959; introduction of color broadcasts in 1982.
Private Channels: 1990s liberalization policy
Digital Communication
Introduction of the Internet
Growth of Digital Services
Digital India Initiative
Satellite Communication
ISRO, has been instrumental in developing and deploying a series of communication satellites